

# THE SPHERE PACKING PROBLEM IN DIMENSION 4 AND THE TWENTY-FOUR-CELL CONJECTURE

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**ABSTRACT.** The twenty-four-cell conjecture asserts that every Voronoi cell of a unit-ball packing of  $\mathbb{R}^4$  has volume at least 8, with equality only at the  $D_4$  configuration, and it gives the density bound  $\pi^2/16$ . We prove it for contact configurations, the directions to the neighbours at distance 2. The proof runs through the number of contacts. Up to twenty-two a covering estimate suffices; at twenty-three the cell is bounded through an exact volume identity inside a ball and a semidefinite certificate for one inequality between pair angles; at twenty-four the configuration is the root system, which we prove from the second level of the semidefinite hierarchy with its equality case, the positive kernel verified in exact arithmetic. For the Voronoi cell of an arbitrary packing we prove the bound whenever every neighbour closer than  $2\sqrt{2}$  touches the centre, and more generally whenever the distances of those neighbours pass a covering criterion that does not look at their directions. We also prove it whenever at most twenty-three neighbours lie within  $\sqrt{6}$ , at any distances, by carrying the semidefinite certificate from contacts to centres at mixed distances; whenever the contacts contain most of a root system; and whenever no neighbour lies at a distance between  $2 + \varepsilon_0$  and  $\sqrt{6}$ , for an absolute constant  $\varepsilon_0 > 0$ , which makes the  $D_4$  configuration a strict local minimum of the cell volume among all packings. The case left open is a centre with at least twenty-four neighbours within  $\sqrt{6}$ , one of them at an intermediate distance, for which we give numerical evidence.

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## 1. INTRODUCTION

The four-dimensional sphere packing problem asks for the supremum density  $\Delta_4$  of a packing of unit balls in  $\mathbb{R}^4$ . The  $D_4$  root lattice, scaled so that nearest centres lie at distance 2, packs balls at density  $\pi^2/16 = 0.6169\dots$  and is the densest lattice packing in dimension 4, by a theorem of Korkine and Zolotareff [KZ1872]. Whether it is densest among all packings, periodic or not, has been an open question, and it remains one. The present paper proves a local statement that answers it for a large class of packings and reduces the rest to one precisely stated case, and it proves within the text every ingredient the argument uses, the one classification it needs included. It supersedes the preprint [BBB26] of the first, second and fourth authors, whose additivity argument is not true in the form stated there (Remark 7.2) and is not used here.

We say at the outset what that statement is and where it stood. The statement is that every Voronoi cell of a unit-ball packing of  $\mathbb{R}^4$  has volume at least 8, the volume of the regular 24-cell circumscribed about the unit ball; it is known as the twenty-four-cell conjecture, it implies the density bound  $\pi^2/16$  at once, and to our knowledge no case of it in which a contact direction leaves a root of  $D_4$  had been proved before: the general estimates of Blichfeldt and Rogers stop short of 8, the linear programming bound stalls above the density of  $D_4$ , and the second-order analysis of the cell degenerates exactly at the dense configurations that matter (Section 15). For contact configurations, where every neighbour that can cut the cell touches the centre, three theorems below cover it. When at most one contact direction leaves a root the bound is proved by elementary means, in exact arithmetic, with no restriction on the deviation (Theorem 1.1). When there are at most 22

contacts it is proved, with room to spare and with any number of deviations, by a covering estimate on the sphere (Theorem 1.3). When there are exactly 23 contacts, which is the case that had no approach at all, since it is neither small enough for a covering estimate nor rigid enough for a classification, it is proved by an identity for the volume of the cell inside a ball, which turns the bound into an inequality between pair angles, and by a semidefinite certificate for that inequality verified in exact and in interval arithmetic (Theorem 7.40). What remains is the case of 24 contacts, and there a classification leaves nothing to prove: a configuration of 24 contacts is a copy of the root system and its cell is the 24-cell itself. The classification was first proved by de Laat, Leijenhorst and de Muinck Keizer [LLM24], by a semidefinite computation, and it is proved here in full (Theorem 7.30): the positive kernel their computation found is redistributed with this paper, its properties are verified in exact arithmetic, and the bound it gives, with the equality case that identifies the configuration, is proved in the text. So the twenty-four-cell conjecture for contact configurations rests on no statement outside this paper and on no computation that has not been checked; the theorem for 23 contacts is the piece that was missing there. What separates it from the conjecture for Voronoi cells is the neighbours that do not touch the centre. Section 2.7 proves the bound for every centre whose neighbours within  $2\sqrt{2}$  all touch it, or whose neighbour distances pass a covering criterion, and isolates the one case that remains, a centre crowded by near-contacts (Conjecture 1.6).

Let  $\mathcal{C} \subset \mathbb{R}^4$  be the centre set of a packing of unit balls and let  $c \in \mathcal{C}$ . The Voronoi cell

$$V_c = \{z \in \mathbb{R}^4 : |z - c| \leq |z - c'| \text{ for all } c' \in \mathcal{C}\}$$

of the  $D_4$  lattice at distance 2 is a regular 24-cell of volume 8, and a density bound of  $\pi^2/16$  for a periodic packing follows from the statement that no Voronoi cell has volume less than 8. That statement is what is at issue. It is false for the analogous cell volume in most dimensions, and in dimension 4 it has resisted proof because the 24-cell is rigid in some directions and very soft in others. A single facet hyperplane can be tilted a long way without the cell losing any volume at all, and when several tilt at once the first-order and second-order changes can cancel exactly, leaving the sign of the total to be decided at fourth order.

We first treat the case in which the local configuration departs from the lattice in a single direction. Call the centres within distance  $2\sqrt{2}$  of  $c$  its *active neighbours*, and write  $V_c^{\text{act}} \supseteq V_c$  for the cell they alone cut out.

**Theorem 1.1** (Local cell bound, single deviation). *Let  $c$  be a centre of a unit-ball packing of  $\mathbb{R}^4$  whose active neighbours lie in root directions of a copy of the  $D_4$  root system  $R$  except for at most one, and, if there is one, suppose that some root direction of  $R$  is taken by no active neighbour. Then  $\text{vol}(V_c^{\text{act}}) \geq 8$ , with equality if and only if the active neighbours of  $c$  form a copy of  $\sqrt{2}R$  centred at  $c$ . Nothing is assumed about the deviating direction or about the size of its deviation.*

The cell  $V_c$  itself is  $V_c^{\text{act}}$  cut by the centres beyond  $2\sqrt{2}$ , which do not meet the ball of radius  $\sqrt{2}$  about  $c$  (Lemma 2.7) but can cut  $V_c^{\text{act}}$  outside it. When every active neighbour touches  $c$ ,  $\text{vol}(V_c) \geq 8$  holds as well, by Theorem 1.5 below.

**Corollary 1.2** (Density bound, single deviation). *Every periodic packing each of whose centres satisfies the hypothesis of Theorem 1.1, with every active neighbour in contact, has density at most  $\pi^2/16$ , with equality for the  $D_4$  lattice.*

The mechanism behind Theorem 1.1 is a change of viewpoint, and it is the part of this paper we would point to first. Fix the deviating contact direction and let  $Q$  be the

intersection of the half-spaces of the twenty-three contact directions that remain at roots. The cell at the all-contact corner is  $Q$  cut by one further half-space at unit distance from the centre, and  $\text{vol}(Q) = \frac{25}{3}$ ; so the cell has volume at least 8 precisely when that half-space cuts at most  $\frac{1}{3}$  off  $Q$  (Proposition 13.2). The deviation parameters have then left the body altogether and sit only in the cutting half-space, and the question has become an extremal problem for one fixed polytope.

That problem is solved by enclosing  $Q$  in a cross-polytope. The sixteen roots making an angle of  $60^\circ$  or  $120^\circ$  with the deviating root turn out to cut out exactly

$$B = \{x : |x_0| + |x_1| + |x_2| + |x_3| \leq 2\}$$

in the orthonormal frame supplied by four pairwise orthogonal roots (Lemma 13.3), so  $Q \subseteq B$ ; and the enclosure is lossless in the one direction that matters, since  $\{x_0 \geq 1\}$  cuts exactly  $\frac{1}{3}$  off  $B$  as well as off  $Q$ . What remains is a clean question about the cross-polytope alone, which we answer in Theorem 13.6: among all half-spaces at unit distance from the centre, those normal to a vertex direction cut off the most, namely  $\frac{1}{3}$ . The proof of that inequality rests on an exact formula for the cap volume as a divided difference of the one-variable function  $a(2\sqrt{a} - 1)_+^4$ , together with the fact that this function has nondecreasing third derivative.

A second, independent result covers configurations with several deviating directions at once, at the cost of a bound on how many neighbours there are. It comes from measuring the cell radially instead of facet by facet.

**Theorem 1.3** (Local cell bound, few contacts). *Let  $c$  be a centre of a unit-ball packing of  $\mathbb{R}^4$  at the all-contact corner, so that every active neighbour lies at distance exactly 2, and suppose there are at most 22 of them. Then  $\text{vol}(V_c) \geq 8.0463\dots > 8$ . No hypothesis is placed on the directions of those neighbours beyond the packing condition itself, and in particular any number of them may deviate from roots, by any amount. With no bound on their number the same argument gives  $\text{vol}(V_c) \geq 7.7989\dots$  for every contact configuration whatever.*

The hypothesis that every active neighbour touches the centre cannot be removed by pulling the others in: pulling a neighbour in shrinks the cell, but the configuration it produces need not be a packing (Lemma 2.9). What survives is sharper than the theorem. The same covering argument, applied to every centre within  $2\sqrt{2}$ , each counted by the cap it actually cuts, gives a lower bound that depends on the distances of those centres alone (Proposition 2.10).

The proof is short. Writing  $\delta(\theta)$  for the angular distance from a direction  $\theta$  to the nearest contact direction, the cell has volume  $\frac{1}{4} \int_{S^3} \sec^4 \delta$  (Lemma 7.15); the  $m$  contact directions are  $60^\circ$  apart, so the caps of radius  $r$  about them leave at least  $2\pi^2 - m\pi(2r - \sin 2r)$  of the sphere uncovered, and integrating that estimate against  $d(\sec^4 r)$  gives  $\text{vol}(V_c) \geq \frac{\pi m}{3} \tan^3 r_m$  with  $2r_m - \sin 2r_m = 2\pi/m$  (Theorem 7.16). Since a contact configuration has at most 24 members, by Musin's kissing theorem [Mus08], and the bound decreases in  $m$ , twenty numbers settle the whole range. (For  $m \leq 11$  the angle  $r_m$  exceeds  $45^\circ$ ; stopping the integral there, so that no centre beyond  $2\sqrt{2}$  can enter, leaves the bound above 11.5, Proposition 2.10.)

The case our own arguments do not reach is that of several simultaneous deviations at a centre with 24 neighbours, all of them active. The free-volume identity of Theorem 7.1 holds exactly for any number of deviating directions, and reduces the general local bound to a single comparison of two volumes; whether that comparison goes the right way at such a centre is what our own arguments leave undecided.

**Conjecture 1.4** (Multi-direction positivity; proved in Corollary 7.31). *Let the configuration be at the all-contact corner with a full active set, and let  $D$  be any set of  $m \geq 2$  roots whose directions are deviated, with arbitrary tilts and arbitrary deviation directions. Then, in the notation of Theorem 7.1,*

$$\operatorname{vol}\left(\bigcup_{j \in D} E_j\right) \geq \operatorname{vol}\left(\bigcup_{j \in D} C_j\right).$$

By Theorem 7.1 this is exactly the assertion  $\operatorname{vol}(V_c) \geq 8$  for such a configuration. The two families are overlapping once  $m \geq 2$ , and the corresponding statement with sums of volumes in place of volumes of unions is a different assertion that neither implies it nor follows from it (Remark 7.2); it is the union form above that is at issue.

The top of that range is settled by a classification of the 24-point kissing configurations in  $\mathbb{R}^4$ : any 24 unit vectors with pairwise inner products at most  $\frac{1}{2}$  form a copy of the  $D_4$  root system. That settles Conjecture 1.4 outright, since a full active set of 24 contacts then admits no deviation at all (Corollary 7.31), and it leaves the bound for contact configurations with one case, a centre with exactly 23 contacts whose cell admits no twenty-fourth, which is the subject of the second half of Section 7 and is settled there. The classification rests, as said above, on the computation of [LLM24], a preprint that has not yet been refereed; since it is proved here in full, as Theorem 7.30, we say in a footnote what that proof consists of.<sup>1</sup>

<sup>1</sup>The reader is entitled to know what the classification rests on and how much of it is checked here; Theorem 7.25, Lemma 7.26, Theorem 7.30 and Section 7.5.3 give the account in full, and this note summarises it. The statement is that any 24 unit vectors in  $\mathbb{R}^4$  with pairwise inner products at most  $\frac{1}{2}$  form a copy of the  $D_4$  root system. Its proof has two halves. The first is a semidefinite programming bound: the second level of the hierarchy for spherical codes in the form de Laat and Vallentin [dLV15] give it, which bounds the size of a code by the value  $K(\emptyset, \emptyset)$  of any positive kernel  $K$  on the two-element subsets of the sphere that satisfies a family of linear inequalities. The authors of [LLM24] found such a kernel with value exactly 24, with truncation degrees 14 and 16, reduced by the symmetry of the sphere, solved to forty digits in 256-bit arithmetic over about two weeks on eight cores with 128 gigabytes of memory, and then rounded to an exact rational point; that point is public [LLM24d], under a licence that permits redistribution, and it travels with this paper. Because the bound is exactly 24 and a code of size 24 exists, the equality case of the bound forces every inner product of such a code into the zero set of the two-point polynomial of the kernel, which is  $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$ , the roots being isolated by a Sturm sequence. Both the bound and its equality case are proved in the text (Lemma 7.27), the positivity of the kernel is proved from the form in which it is built (Lemma 7.28), and every property of the rational point that those lemmas consume is verified in exact arithmetic in code of our own (Proposition 7.29): the format of the data, the positive definiteness of all 127 blocks, the nonnegativity structure of the sum-of-squares terms, the construction of the zonal matrices, the four polynomial identities that use them, the value 24 of the bound, and the zeros of the two-point polynomial, the last in Lean’s kernel. The construction of the zonal matrices needs 128 gigabytes of memory as the authors carry it out; taken so that the expansion of the integrand is never held at once, it needs 400 megabytes. The second half is the passage from these four inner products to the root system, and it is classical: scaled by  $\sqrt{2}$  the vectors have norm 2 and integer inner products, so they generate an integral lattice of rank at most 4 among whose norm-2 vectors they lie; the norm-2 vectors of an integral lattice form a simply laced root system and the lattice they span is a root lattice, an orthogonal sum of lattices of type  $A$ ,  $D$ ,  $E$  with a basis of simple roots; and the root lattices of rank at most 4 have 2, 4, 6, 6, 8, 8, 10, 12, 12, 14, 20 and 24 roots, the last being  $D_4$  alone. So the 24 scaled vectors are all the roots of a copy of  $D_4$ . This half is Lemma 7.26, and its finite content is checked by an exact enumeration (`root_lattices_rank4.py`, and in Lean `D4RootLattices.lean`): every positive definite Gram matrix with 2 on the diagonal and  $-1, 0, 1$  off it, of order up to 4, there are 393 of order 4, has at most 24 integer solutions of  $x^T G x = 2$ , with 24 only at determinant 4 and the neighbour counts of  $D_4$ , the next value being 20 at determinant 5, which is  $A_4$ . The published antecedents are the bound 24.10 of Bachoc and Vallentin [BV08] at the first level of the same hierarchy and Musin’s proof [Mus08] that the kissing number in dimension four is 24, which bounds the size of the code but does not identify it. What has not been repeated is the search for the kernel, which is the semidefinite computation itself; but a proof does not depend on how its witness was found, only on the witness being checked, and it is.

**Theorem 1.5** (General local bound). *Let  $V_c$  be the Voronoi cell of a centre  $c$  of a unit-ball packing of  $\mathbb{R}^4$ , and let  $d_1 \leq d_2 \leq \dots$  be the distances from  $c$  of the other centres closer than  $2\sqrt{2}$ . If every  $d_i$  equals 2, or if  $\Phi(d_1, d_2, \dots) > 8$  for the function  $\Phi$  of Proposition 2.10, then  $\text{vol}(V_c) \geq 8$ , with equality if and only if the centres within  $2\sqrt{2}$  of  $c$  form a copy of  $\sqrt{2}R$  about  $c$ .*

**Conjecture 1.6** (Crowded near-contacts). *The conclusion of Theorem 1.5 holds at every centre of every unit-ball packing of  $\mathbb{R}^4$ , also when some centre lies at a distance strictly between 2 and  $2\sqrt{2}$  from it and  $\Phi(d_1, d_2, \dots) \leq 8$ .*

The conjecture is what is left of the twenty-four-cell conjecture once Theorem 1.5 is proved, and Section 2.7 says what it contains: a centre with at least twenty-three neighbours within  $2\sqrt{2}$ , at least one of which does not touch it, the ones that do not being close to contact (with twenty-two contacts and one further neighbour, closer than 2.1676; with twenty-one and two, closer than 2.0705). Section 2.8 narrows it further.

**Theorem 1.7** (Further cases of the local bound). *The conclusion of Theorem 1.5 holds at a centre  $c$  of a unit-ball packing of  $\mathbb{R}^4$  also in each of the following cases:*

- (a) *at most 23 other centres lie within  $\sqrt{6}$  of  $c$ , at any distances;*
- (b) *the contact directions of  $c$  contain all but at most two of the normalised roots of a copy of  $R$ , or more generally one of the three orthogonal pairs of normalised roots at each vertex of its 24-cell;*
- (c) *no centre lies at a distance from  $c$  strictly between  $2 + \varepsilon_0$  and  $\sqrt{6}$ , where  $\varepsilon_0 > 0$  is an absolute constant.*

These are Theorems 2.17 and 2.18 and Corollary 2.14. Case (c) is a local form of the conjecture at the root system, for all packings: the  $D_4$  configuration is a strict local minimum of the cell volume, whatever the centres beyond  $\sqrt{6}$  do. Its constant  $\varepsilon_0$  comes from a compactness argument and is not explicit, so what is left of Conjecture 1.6 is a centre with at least twenty-four other centres within  $\sqrt{6}$ , one of them at an intermediate distance, between  $2 + \varepsilon_0$  and  $\sqrt{6}$ . A numerical search of that case finds nothing below 8 (Section 2.7); that is evidence, not proof.

The case of 23 contacts is not a deviation problem at all. By Proposition 7.38, a cell of circumradius at least 2 is exactly one whose configuration admits a twenty-fourth contact, and in that case the bound follows from Theorem 7.25 together with the monotonicity of the cell in the configuration; what is left is a saturated 23-point configuration, and Theorem 7.40 states, for every configuration of 23 contacts with bounded cell, saturated or not, that its cell has volume above 8. The configuration that comes nearest to the boundary of that statement is the deletion of a single root, whose cell has circumradius exactly 2 and volume  $\frac{25}{3}$ , and that one is rigid: Proposition 7.48 exhibits an integer stress on its eighty-eight tight pairs and concludes that every first-order motion of it is a rotation. It cannot be turned into a saturated configuration by exchanging one or two directions either. Removing two roots from  $R$  leaves the six supports of the roots with a whole couple of complementary index pairs intact, and that alone forces every direction one could add back to be a root (Lemma 7.54); at three removed roots the only escape is a triple pairwise at  $60^\circ$ , and the region it opens holds no two directions as much as  $60^\circ$  apart unless both are removed roots. Hence a contact configuration meeting a root system in twenty-one or more directions lies inside it or has at most twenty-two elements (Propositions 7.55 and 7.56), and a saturated configuration of twenty-three directions shares at most twenty of them with any copy of  $R$  (Corollary 7.57): at least three of its directions are new. The argument is elementary, and its finite half is checked by Lean's kernel.

A contact configuration of  $m$  directions is nothing other than a spherical code of  $m$  points on  $S^3$  of minimal angle at least  $60^\circ$ , and every computation we know of puts the largest minimal angle available to 23 points at exactly  $60^\circ$ : the same as at 24 points, and unlike the  $60.1399\dots^\circ$  available to 22. If that is so, the configurations in question are not a family with interior but extremal codes (Proposition 7.59), and if the deletions of a single root are the only ones, no saturated configuration exists at all (Theorem 7.60). That classification, of a code at one fixed size below the maximum, is not something an upper bound on the size of a code can deliver (Remark 7.61), and we do not prove it;<sup>2</sup> two things can be said about it. Among the 120 vertices of the 600-cell, the largest finite group of unit quaternions and one that contains  $R$  in twenty-five ways, it holds: an exhaustive enumeration, carried out in three independent implementations and checked in Lean, shows that every 23-element subset with pairwise inner products at most  $\frac{1}{2}$  is an inscribed copy of  $R$  with one vertex removed (Proposition 7.63). And the largest inradius of the convex hull of 23 points whose pairwise inner products are at most  $\frac{1}{2} + \delta$  is a function  $h(\delta)$  that decreases to its value at  $\delta = 0$ , the classification being exactly the statement  $h(0) = \frac{1}{2}$  (Lemma 7.65); the computations of Section 7.10 find  $h(\delta)$  close to  $\frac{2}{3}$  for every  $\delta \geq 0.009$ , far above anything a perturbed deletion can reach, and then collapsing to  $\frac{1}{2} + O(\delta)$  from  $\delta = 0.008$  down, with every configuration found at  $\delta = 0$  a deletion.

The proof of Theorem 7.40 takes a different road, on which nothing has to be classified. The volume of the cell inside the ball of radius  $\sqrt{3}/2$  is an exact function of the pairwise angles of the configuration, because no three contact directions fit in a cap of radius below  $\arcsin(1/\sqrt{3})$  (Proposition 7.68), and the whole of Theorem 7.40 follows from the inequality that this truncated volume is at least 8 for every configuration of 23 contacts (Theorem 7.69); the deletion of a root satisfies that inequality with 0.034 to spare, so it is a strict inequality and not a boundary case. The relaxation of it that reads only the pair angles delivers four fifths of the required amount (Proposition 7.44), and the three-point relaxation of Bachoc and Vallentin [BV08], which reads the triples, delivers all of it: Theorem 7.73 exhibits a semidefinite certificate of degree 8, a polynomial and nine positive definite matrices, and verifies in exact rational arithmetic, checked again by the Lean kernel, and in interval arithmetic that it proves  $\sum_{i < j} \omega(\gamma_{ij}) \geq 0.0929$  for every 23 contact directions, where  $0.092855\dots$  is what the truncated volume needs.

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<sup>2</sup>Nor is it needed, and since the point matters for what the paper claims we set the dependence out precisely. The bound for contact configurations is assembled in Corollary 7.39 from three cases by the number  $m$  of contacts, and Theorem 1.5 carries it to Voronoi cells (Section 2.7):  $m \leq 22$  by the covering estimate (Corollary 7.17),  $m = 24$  by Theorem 7.25, and  $m = 23$  by Theorem 7.40. The density bound, Theorem 1.8, is the local bound followed by the averaging argument of Section 6, and it uses nothing else. The proof of Theorem 7.40 is the inequality of Theorem 7.69 together with its certificate, Theorem 7.73: it bounds the cell of every configuration of 23 contacts from below, whether or not the configuration is saturated and whatever its Gram matrix, and at no point does it ask what the configuration looks like. The classification of Theorem 7.60, that every 23-point code of minimal angle  $60^\circ$  is a deletion of one root, would prove something stronger, namely that no saturated configuration of 23 contacts exists at all, so that the case  $m = 23$  with circumradius below 2 would be empty and not merely settled; but the local bound does not distinguish between an empty case and a settled one, and neither does the density bound. The same is true of the twenty-four-cell conjecture in the form in which it is usually stated, that every Voronoi cell of a unit-ball packing of  $\mathbb{R}^4$  has volume at least that of the regular 24-cell circumscribed about the unit ball: that is Theorem 1.5 together with Conjecture 1.6, and the classification of 23-point codes plays no part in it. Proposition 7.63, Lemma 7.65 and the computations of Section 7.10 are therefore reported as what they are, information about the codes of 23 points that is of interest in its own right, and nothing in Theorem 1.5, Theorem 1.8 or Theorem 7.40 depends on them. The one computational ingredient of the two general theorems that is not a computation of our own is the kernel behind Theorem 7.30 for  $m = 24$ , found in [LLM24] and verified here, as the preceding footnote explains.

**Theorem 1.8** (Density bound). *Every periodic unit-ball packing of  $\mathbb{R}^4$  each of whose centres satisfies the hypothesis of Theorem 1.5 has density at most  $\pi^2/16$ , with equality for the  $D_4$  lattice at nearest-neighbour distance 2. If Conjecture 1.6 holds, the supremum density of sphere packings in  $\mathbb{R}^4$  is  $\Delta_4 = \pi^2/16$ .*

Both of those last statements use Theorem 7.25 for the configurations of twenty-four contacts, and Theorem 7.25 is proved in this paper; the one thing in its proof that is not of our making is a positive kernel found by [LLM24], which Section 7.5.3 redistributes and verifies, the bound it gives being proved in Lemma 7.27. It should be said at once what Conjecture 1.4 is. Written in terms of the contact directions alone (Proposition 7.8), the inequality  $\text{vol}(V_c) \geq 8$  is the twenty-four-cell conjecture for contact configurations, the form in which every neighbour that can cut the cell touches the centre; Theorem 1.1 settles the part of it in which at most one contact direction leaves a root, and Conjecture 1.4 is the remainder. Corollary 7.31 settles it through the classification of Theorem 7.25, and nothing below suggests that restating it makes a proof that avoids the classification easier. The conjecture for Voronoi cells asks more, since it allows neighbours that do not touch, and that part is Conjecture 1.6.

Section 15 through Section 19 report what we know about it: the second-order method that settles small configurations degenerates exactly at a family of dense ones; we locate that degeneracy, measure its consequences, and prove the conjecture outright along three highly symmetric finite-angle paths. The obstruction there is of a different nature from anything in the single-deviation problem, and the cap argument of Section 13 does not reach it, for a reason we explain in Section 19. Sections 7.2 and 7.3 give three exact restatements of the remaining case, as a cap inequality for one fixed polytope, as a polar-volume bound, and as a lower bound on the boundary measure of the cell, and use them to close off four otherwise natural lines of attack: term-by-term reduction to the single-deviation theorem, any bound assembled facet by facet from the pairwise contact condition, the convexity bound for the polar volume integral, and the Mahler-type product bound. Each of the four fails at the  $D_4$  configuration itself, which is the one place a sharp argument can afford to lose nothing, and each fails there for a reason that is visible in closed form and not only in a numerical margin.

**1.1. Structure of the argument.** Theorem 1.1 rests on five layers, each proved completely before the next is invoked; a sixth layer, which the theorem does not need, is where the multi-direction case lives, and it is settled by the number of contacts rather than by these layers.<sup>3</sup>

*Layer 1 (shell localisation, Section 2.4).* A centre at distance  $d$  from  $c$  does not cut the ball of radius  $d/2$  about  $c$ . So the centres at distance at least  $2\sqrt{2}$  never cut the reference 24-cell, and if every other centre is that far,  $\text{vol}(V_c) \geq 8$  at once; they can still cut  $V_c$  outside the ball of radius  $\sqrt{2}$ .

---

<sup>3</sup>A word on what a Voronoi cell is doing in a density statement, since the two are often conflated. Let  $\mathcal{C}$  be the centre set of a packing of unit balls, so  $|c - c'| \geq 2$  for distinct  $c, c' \in \mathcal{C}$ . The Voronoi cells  $V_c$  tile  $\mathbb{R}^4$ , each contains the ball of radius 1 about its centre, and each is convex, being an intersection of half-spaces. If  $\mathcal{C}$  is periodic with period lattice  $L$  and  $N$  centres per fundamental domain, then  $\sum_c \text{vol}(V_c) = \det L$ , the sum running over one period, and the density of the packing is  $N \text{vol}(B^4)/\det L$ , where  $\text{vol}(B^4) = \pi^2/2$  is the volume of the unit ball. A uniform lower bound  $\text{vol}(V_c) \geq v$  therefore gives the density bound  $\text{vol}(B^4)/v$  at once, and  $v = 8$  gives  $\pi^2/16$ . The reduction runs only in this direction: a density bound says nothing about any individual cell. Nothing about periodicity is essential either, since a general packing is approximated by periodic ones on large tori with an error tending to zero, which is the argument recalled in Section 6.2; we work with periodic packings only because the volume bookkeeping is finite there.

*Layer 2 (the root-aligned case, Section 2.5).* If every active neighbour direction is an exact root direction of  $D_4$ , the support function of the reference cell equals 1 at each such direction, forcing the reference cell inside every active half-space and hence inside  $V_c$ :  $\text{vol}(V_c) \geq 8$  immediately.

*Layer 3 (radial reduction, Section 2.6).* The Voronoi volume is monotone non-decreasing in each contact radius, so pulling every active neighbour in to distance 2 gives a lower bound. That lower bound is the cell of a contact configuration when every active neighbour already touches the centre, and in general it is not: the pulled-in configuration need not be a packing (Section 2.7).

*Layer 4 (cell positivity, Sections 3 to 5).* At the all-contact corner,  $W(D_4)$  partitions every packing-valid configuration with at most one root-aligned deviation into 176 Weyl-orbit cell types. For each type  $\mathfrak{k}_k$ , an exact free-volume identity reduces the bound  $\text{vol}(V_c) \geq 8$  to the single scalar inequality  $\Delta_k \geq 0$ , established by the two independent exact arguments of Sections 4 and 5.

*Layer 5 (arbitrary deviation direction, Section 13).* Layer 4 leaves the deviation direction constrained to a root-aligned arc. Removing that constraint is where the cap reformulation enters: the defect becomes the deficiency of a cap cut from  $Q$ , the cap is bounded by the corresponding cap of a cross-polytope, and the extremal cap of a cross-polytope is computed once and for all. This layer subsumes layer 4 logically, though not quantitatively, since layer 4 also returns the exact defect polynomial on each cell.

*Layer 6 (simultaneous deviation).* When two or more active directions deviate at once, the free-volume identity still holds exactly, but nothing in the first five layers gives the sign of the resulting difference; that is Conjecture 1.4, settled in Corollary 7.31 through the classification of Theorem 7.25 and discussed from Section 15 onward for what can be said about it without the classification. The six layers nest. Layers 1 and 3 are reductions; between them they leave the all-contact corner, for the cell  $V_c^{\text{act}}$  of Theorem 1.1 and for every centre whose active neighbours all touch it; at that corner layers 2, 4, 5 and 6 are classes of configurations, each containing the one before it, from the undeviated configuration out to simultaneous deviation in several directions. That outermost class is not settled by anything inside it but by the number of contacts: at most 22 by Theorem 1.3, 23 by Theorem 7.40, and 24 by the classification. Centres with active neighbours that do not touch them are the subject of Section 2.7.

The same layers act on the three-dimensional analogue, where the objects can be drawn: the lattice is the face-centred cubic one, the reference cell is the rhombic dodecahedron, and the twenty-four roots become twelve contact directions. Figure 1 follows them there, and the volumes in its captions are computed from the half-space intersections, not quoted.

**1.2. Background.** Determining the supremum density  $\Delta_n$  of a packing of congruent balls in  $\mathbb{R}^n$  is a question whose difficulty varies wildly with  $n$ , and the four-dimensional case sits in an awkward middle range: too large for the exhaustive planar and spatial methods, too small for the arithmetic that resolves dimensions 8 and 24.

In the plane, Thue [Thu10] announced the hexagonal bound and Fejes Tóth [FT42] gave the first complete proof,  $\Delta_2 = \pi/\sqrt{12}$ . In three dimensions Kepler's 1611 assertion that the face-centred cubic arrangement is optimal was proved by Hales, first through the computational programme begun in [Hal97] and completed in [Hal05], then in a machine-checked form [Hal17]. The *lattice* case is much older and much easier. Korkine and Zolotarëff [KZ1872] showed that the quaternary form attached to  $D_4$  minimises the determinant among positive quaternary forms of given minimum, so  $D_4$  is the densest lattice packing

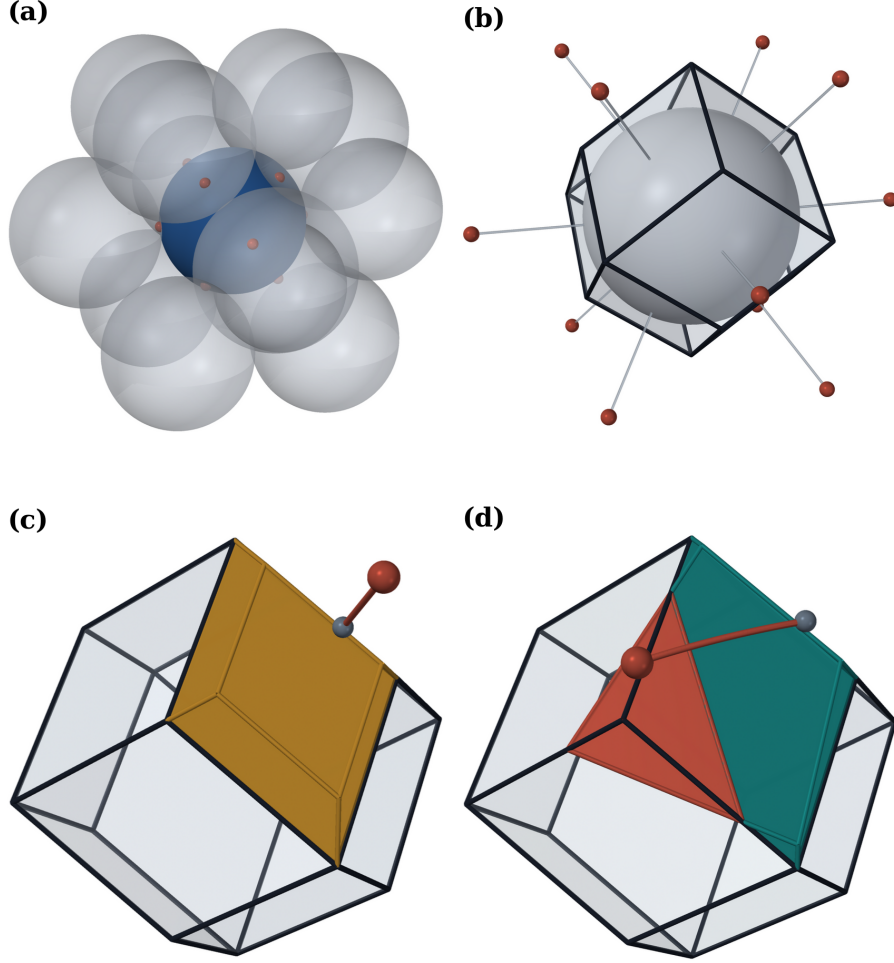


FIGURE 1. (a) The all-contact corner in dimension three: the unit ball about the centre (blue), its twelve contact points (clay), and the twelve unit balls that touch it, centres at distance 2 in the twelve contact directions of the face-centred cubic lattice. (b) The Voronoi cell of that centre, the rhombic dodecahedron, with the unit ball inscribed: one facet per contact, each at distance 1, and volume 5.6569, computed from the half-space intersection and not quoted. (c) Layer 3, radial monotonicity. One neighbour is pushed from distance 2 out to 2.6, its half-space recedes, and the cell grows from 5.6569 to 5.9666; the ochre solid is exactly the volume gained. Pulling the neighbour back in is therefore a lower bound; Section 2.7 explains why that lower bound need not be a packing. (d) Layers 4 and 5, deviation. One contact direction is tilted by  $25^\circ$  at radius 2: teal is the volume gained, clay the volume lost, and teal wins, the cell rising from 5.6569 to 5.7470. That is the defect being positive, with equality only at zero tilt.

of  $\mathbb{R}^4$ ; Voronoi's reduction theory [Vor08], whose polyhedral language we use throughout, placed such classifications on a systematic footing, and Martinet [Mar03] gives the modern account of the perfect-form machinery that grew out of it. Upper bounds valid for all packings, lattice or not, begin with Blichfeldt's second-moment method [Bli29], which yields

$\Delta_4 \leq 0.7797$ , and were sharpened by the simplex bound of Rogers [Rog64]. Gruber and Lekkerkerker [Gru87] survey the classical theory; Conway and Sloane [CS88] is the standard source for  $D_4$ ,  $E_8$ , the Leech lattice, and their Voronoi cells; Bezdek [Bez10] collects the local and combinatorial side of the subject.

The two dimensions in which an exact non-lattice answer is known both lie far above 4. Cohn and Elkies [CE03] recast the upper bound as an infinite-dimensional linear program over radial Schwartz functions whose values and Fourier transforms obey prescribed sign conditions; equality in the resulting duality inequality requires a single function with a prescribed double sequence of zeros. Viazovska [Via17] constructed that function in dimension 8 out of quasimodular forms, and the Leech case followed in [CKMRV17]. Neither construction has a four-dimensional analogue, and the obstruction is not that none has been found. Li [Li25] bounds the optimum of the Cohn–Elkies program from below by reducing it to a finite-dimensional problem, and in dimension 4 that lower bound is a centre density of 0.12914461 against the 0.125 of  $D_4$ . No admissible function can close the remaining three per cent, so the linear programming bound cannot decide dimension 4 at all, and the arithmetic that forces the zeros into place in dimensions 8 and 24 has nothing to be applied to here. Refinements of the linear programming method by Cohn and Zhao [CZ14], which couple the packing bound to spherical-code bounds, and the universal optimality results of Cohn and Kumar [CK07, CK09], likewise leave dimension 4 untouched.

Because the Voronoi cell of  $D_4$  is the regular 24-cell [Cox73], the four-dimensional packing problem is equivalent to the assertion that no Voronoi cell of a unit-ball packing has volume less than that of the 24-cell; this is the 24-cell conjecture. The neighbouring kissing-number question, which asks only how many unit balls can touch one, was settled in dimension 4 by Musin [Mus08], who showed the answer is 24, exactly the number of contacts in  $D_4$ , by adding an auxiliary polynomial constraint to the classical linear programming bound; the corresponding three-dimensional statement is the theorem of Schütte and van der Waerden [SW52]. A kissing bound constrains the contact structure at one centre and says nothing about volume, so it does not imply the packing bound, but it does rule out the cheapest way the 24-cell conjecture could have failed. Musin [Mus18] has since proposed a semidefinite route to the 24-cell conjecture itself, through the contact graph on  $S^3$ .

The route taken here is local and combinatorial. It uses the symmetry of  $D_4$ , the order-192 Weyl group and the self-duality of the 24-cell, together with the polyhedral facts recorded in Ziegler [Zie95] and Grünbaum [Gru03] and the root-system conventions of Humphreys [Hum90], and it uses no Fourier analysis and no modular forms. Semidefinite programming enters at one point, the certificate of Theorem 7.73 for the 23-point case, and there only as a search: what the proof uses is the certificate it found, checked afterwards in exact rational arithmetic, in interval arithmetic and by Lean. Every other certificate below is an exact computation in  $\mathbb{Q}$ ,  $\mathbb{Q}(\sqrt{2})$  or  $\mathbb{Q}(\sqrt{3})$ , and whatever is numerical is labelled at the point of use as evidence and never used as a step in a proof.

## 2. THE $D_4$ ROOT SYSTEM AND ITS VORONOI CELL

**2.1. Lattice and roots.** Write  $D_4 = \{m \in \mathbb{Z}^4 : m_1 + m_2 + m_3 + m_4 \equiv 0 \pmod{2}\}$  for the standard integral  $D_4$  lattice, and set  $\Lambda = \sqrt{2} D_4$ , which has nearest-neighbour distance 2, the natural normalisation for a unit-ball packing. The associated root system is

$$R = \{\pm e_i \pm e_j : 1 \leq i < j \leq 4\}, \quad |R| = 24, \quad |\alpha|^2 = 2 \text{ for every } \alpha \in R.$$

The Weyl group  $W(D_4)$  acts on  $R$  by signed permutations of the coordinates with an even number of sign changes; it has order 192 and acts transitively on  $R$ .

**Lemma 2.1** (Gram values on  $\mathbb{R}$ ). *For distinct  $\alpha, \beta \in \mathbb{R}$ , the normalised inner product  $s_{\alpha\beta} = \langle \alpha, \beta \rangle / 2$  takes one of the four values  $-1, -\frac{1}{2}, 0, \frac{1}{2}$ , and  $s_{\alpha\beta} = -1$  occurs only for  $\beta = -\alpha$ . The three remaining values correspond respectively to  $\alpha + \beta \in \mathbb{R}$  (for  $s = \frac{1}{2}$ ),  $\alpha \perp \beta$  (for  $s = 0$ ), and  $\alpha - \beta \in \mathbb{R}$  (for  $s = -\frac{1}{2}$ , excluding  $\beta = \alpha$ ).*

*Proof.* Since  $\alpha, \beta \in \mathbb{Z}^4$ ,  $\langle \alpha, \beta \rangle \in \mathbb{Z}$ , and Cauchy–Schwarz gives  $|\langle \alpha, \beta \rangle| \leq |\alpha||\beta| = 2$  with equality forcing  $\beta = \pm\alpha$ . For the remaining integer values  $\langle \alpha, \beta \rangle \in \{-1, 0, 1\}$ , expand  $|\alpha \mp \beta|^2 = |\alpha|^2 + |\beta|^2 \mp 2\langle \alpha, \beta \rangle = 4 \mp 2\langle \alpha, \beta \rangle \in \{2, 4, 6\}$ ; the value 2 exactly identifies  $\alpha \mp \beta \in \mathbb{R}$ .  $\blacksquare$

This four-valued Gram structure,  $-1$  (antipodal),  $\frac{1}{2}$  (“acute” pairs),  $0$  (orthogonal pairs),  $-\frac{1}{2}$  (“obtuse” pairs), is the entire combinatorial input to everything that follows: every chamber Hessian, every breakpoint curve, and every certificate in this paper is built, in the end, from these four numbers and nothing else.

## 2.2. The reference cell.

**Definition 2.2.** The *reference cell* is

$$\mathcal{V} = \{z \in \mathbb{R}^4 : \langle z, \alpha \rangle \leq \sqrt{2} \text{ for all } \alpha \in \mathbb{R}\}.$$

**Lemma 2.3.**  $\mathcal{V}$  has exactly 24 vertices, all of norm  $\sqrt{2}$ :

$$\text{vert}(\mathcal{V}) = \{\pm\sqrt{2}e_i : 1 \leq i \leq 4\} \cup \left\{ \frac{1}{\sqrt{2}}(\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4) : \epsilon_i \in \{\pm 1\} \right\},$$

an 8-point axis family and a 16-point sign family.  $\text{vol}(\mathcal{V}) = 8$ , and the symmetry group of  $\mathcal{V}$  contains  $W(D_4)$  as an index-3 normal subgroup, the outer factor being the triality automorphism group unique to the  $D_4$  Dynkin diagram.

The proof is a feasibility, extremality and completeness check on the 24 listed points against the 24 defining half-spaces, parallel to the classical description of the regular 24-cell as the Voronoi cell of  $D_4$  ([Cox73, Ch. VIII] and [CS88, Ch. 4] for the polytope’s vertex and facet data); we omit the routine verification, which is also checked by computer algebra in Appendix F.<sup>4</sup>

The polytope itself is drawn four times over, each time from the coordinates and not as a schematic: Figure 2(a) is an orthogonal projection to  $\mathbb{R}^3$ , in which the axis family and the sign family are told apart by colour, Figure 2(b) is the Schlegel diagram from one octahedral facet, Figure 2(c) is the decomposition into the three 16-cells that triality permutes, and Figure 2(d) is the three-dimensional form of the cap inequality of Section 13.

**Definition 2.4** (Support function and angular gap). For  $u \in S^3$ , the support function of  $\mathcal{V}$  is  $h(u) = \max_{z \in \mathcal{V}} \langle z, u \rangle$ , and the *angular gap* is  $\gamma(u) = h(u) - 1$ .

<sup>4</sup>Since the 24-cell is the one object every later section perturbs, its arithmetic is worth having in one place. Take  $\mathbb{R}$  to be the 24 integer vectors  $\pm e_i \pm e_j$ ,  $i < j$ , each of squared length 2; the lattice  $\Lambda = D_4$  they generate is  $\{x \in \mathbb{Z}^4 : \sum x_i \text{ even}\}$ , of determinant 2, with minimum  $\sqrt{2}$  and kissing number 24. Scaling by  $\sqrt{2}$  puts the nearest centres at distance 2, as a unit-ball packing requires, and  $\det(\sqrt{2}\Lambda) = 4 \cdot 2 = 8$ , which is the volume of one Voronoi cell and the number 8 that every bound in this paper is measured against. The cell itself,  $\mathcal{V} = \{x : \langle x, \alpha \rangle \leq 1 \text{ for all } \alpha \in \mathbb{R}/\sqrt{2}\}$ , is the regular 24-cell: 24 vertices, 96 edges, 96 triangular faces, 24 octahedral facets, Schläfli symbol  $\{3, 4, 3\}$ , self-dual, and with each facet centred on one of the 24 contact directions. Its vertices split under  $W(D_4)$  into the 8 points  $\pm\sqrt{2}e_i$  and the 16 points  $(\pm 1, \pm 1, \pm 1, \pm 1)/\sqrt{2}$ , which is the axis and sign division used throughout. Two numbers recur: the support function of  $\mathcal{V}$  equals 1 at every contact direction, which is Layer 2 of Section 1.1 in one line; and the circumradius is  $\sqrt{2}$ , so the inradius-to-circumradius ratio is  $1/\sqrt{2}$ , the slack that lets a facet hyperplane tilt without the cell losing volume. The density  $\pi^2/16$  is  $\text{vol}(B^4)/8 = (\pi^2/2)/8$ . Nothing beyond these facts and the four-valued Gram structure of Lemma 2.1 is used anywhere below.

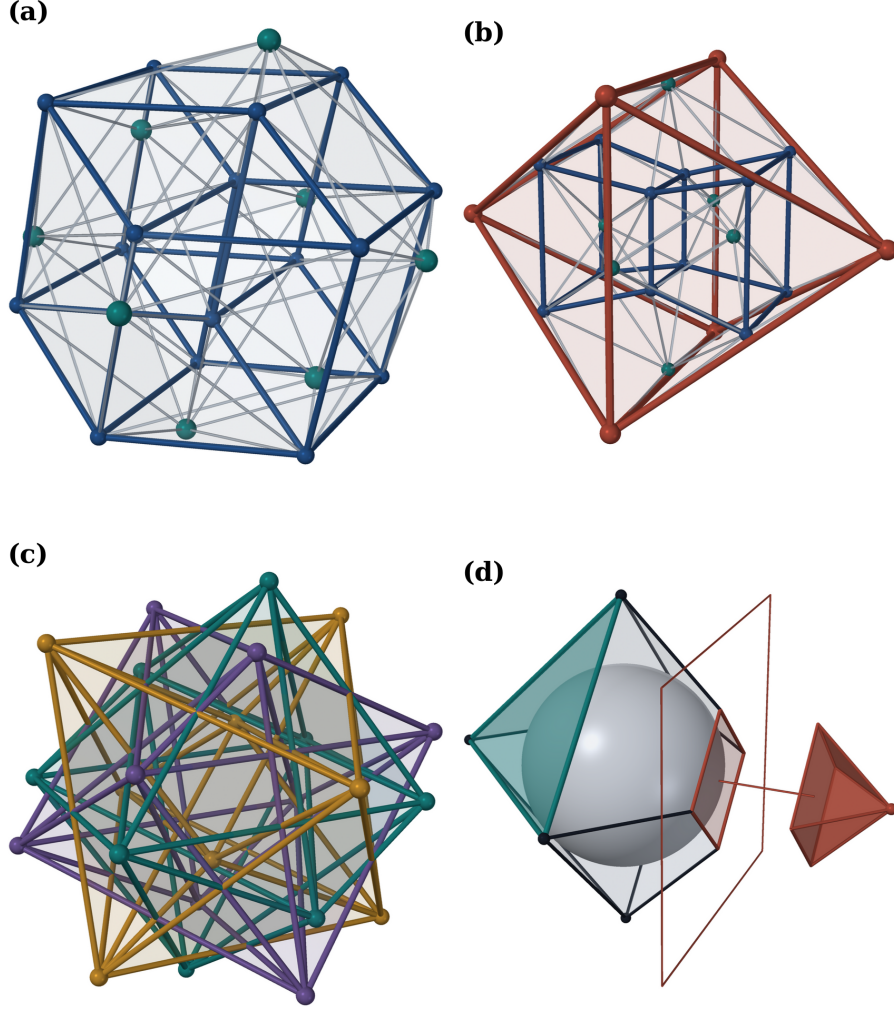


FIGURE 2. (a) The reference cell, the regular 24-cell  $\mathcal{V}$ , in an orthogonal projection to three dimensions: the 8 vertices of the axis family (teal), the 16 of the sign family (blue), the 32 edges inside the sign family and the 64 joining the two families (grey). (b) The Schlegel diagram of the same cell from one octahedral facet (clay): the remaining 18 vertices and 84 edges fill the interior of that facet with the other 23 octahedra. (c) The same 24 vertices read as three 16-cells, which triality permutes, each drawn with its own 24 edges and its own body: the axis family (teal) and the sign family split by the parity of the number of minus signs (ochre, violet). (d) The cap inequality one dimension down. The octahedron of inradius 1 carries its inscribed ball; the half-space at unit distance normal to a vertex cuts off the clay pyramid, drawn displaced along its normal, and the one normal to a facet touches that facet (teal) and cuts off nothing.

**Lemma 2.5.** *For every  $u \in S^3$ ,  $h(u) \geq 1$ , with equality exactly on the 24 unit root directions  $\alpha/\sqrt{2}$ ,  $\alpha \in \mathbb{R}$ . The maximum  $h(u) = \sqrt{2}$  is attained exactly at the 8 coordinate directions  $\pm e_i$ , so  $\gamma(u) \in [0, \sqrt{2} - 1]$  throughout, vanishing exactly on root directions.*

The argument checks the inner product of each of the 24 vertices of  $\mathcal{V}$  against a generic root direction and against a generic coordinate direction. It is routine, if notationally heavy, and we do not reproduce it in full.

**2.3. Conventions used throughout.** We fix, once and for all, the following conventions.

*Notation 2.6.* A *packing centre* is a point  $c$  in the centre set  $\mathcal{C}$  of a unit-ball packing; its *active neighbours* are those centres  $y \in \mathcal{C}$  with  $|y - c| < 2\sqrt{2}$  (Section 2.4 shows this is the only range that can matter). An active neighbour's *contact direction* is  $u = (y - c)/|y - c| \in S^3$ . The *all-contact corner* is the configuration in which every active radius equals the minimum packing radius 2 exactly. We reserve  $\varphi$  for a deviation (“tilt”) angle away from an undeviated root direction,  $s$  for a position parameter along a one-parameter family of deviation directions, and  $\eta(s)$  for the deviation direction itself, a unit vector in the 3-space orthogonal to the fixed root being deviated from. The volume-defect quantity attached to a combinatorial cell type  $\mathfrak{k}_k$  is written  $\Delta_k$ , and its reference normalisation constant  $\mathcal{N}_k$ .

Appendix A collects every piece of notation used more than once in this paper, symbol by symbol, for reference.

**2.4. Shell localisation.**

**Lemma 2.7** (Shell localisation). *Let  $c$  be a centre of a unit-ball packing and  $R > 0$ . A centre  $y$  with  $|y - c| \geq 2R$  does not cut the ball  $B(c, R)$ : its half-space  $\{z : |z - c| \leq |z - y|\}$  contains  $B(c, R)$ . In particular the centres at distance at least  $2\sqrt{2}$  from  $c$  contain  $B(c, \sqrt{2})$ , and with it the reference cell  $c + \mathcal{V}$ , in their half-spaces; and if no other centre lies within  $2\sqrt{2}$  of  $c$ , then  $V_c \supseteq c + \mathcal{V}$  and  $\text{vol}(V_c) \geq 8$ .*

*Proof.* The half-space of  $y$  is  $\{z : \langle z - c, u \rangle \leq |y - c|/2\}$ , with  $u$  the unit vector from  $c$  towards  $y$ , and  $|y - c|/2 \geq R$ . The reference cell has circumradius  $\sqrt{2}$  (Lemma 2.3). ■

An earlier version of this paper stated more: that only the centres at distance in  $[2, 2\sqrt{2})$  can affect  $\text{vol}(V_c)$  at all. That is true of the part of  $V_c$  inside  $B(c, \sqrt{2})$  and false outside it. A cell that reaches beyond  $\sqrt{2}$ , as the cell of the deletion of a root does, its circumradius being 2, can be cut there by a centre farther than  $2\sqrt{2}$ : with the twenty-three roots of a deletion at distance 2 and one more centre at distance 2.9 on the axis of the deleted root, which is a packing, the cell loses  $\frac{1}{3}(2 - 1.45)^4$  and has volume  $\frac{1328453}{160000} = 8.3028\dots$  instead of  $\frac{25}{3}$  (`shell_reduction.py`). Every argument below that bounds the cell of a packing either stays inside  $B(c, \sqrt{2})$  or says what it assumes about the centres beyond  $2\sqrt{2}$ .

**2.5. The root-aligned case.**

**Lemma 2.8** (Root-aligned configurations). *If every active neighbour direction  $u$  (in the sense of Notation 2.6) is an exact root direction  $\alpha/\sqrt{2}$ ,  $\alpha \in \mathbb{R}$ , then  $\text{vol}(V_c) \geq 8$ .*

*Proof.* By Lemma 2.5,  $h(\alpha/\sqrt{2}) = 1$  for every root direction, so the supporting hyperplane cut by an active neighbour at the minimum packing radius 2 along a root direction passes exactly through  $\mathcal{V}$ 's own bounding hyperplane in that direction, neither cutting into  $\mathcal{V}$  nor leaving room outside it. Hence every active half-space contains  $\mathcal{V}$ , and  $V_c \supseteq \mathcal{V}$ , giving  $\text{vol}(V_c) \geq \text{vol}(\mathcal{V}) = 8$  immediately, with equality exactly at the all-contact, all-root-aligned configuration. ■

This is the base case that every other configuration, deviating by a tilt angle away from a root, or having a contact radius above the minimum, is compared against; Section 2.6 handles the radius freedom, and Sections 4 and 5 handle root deviation.

## 2.6. Radial reduction.

**Lemma 2.9** (Monotonicity in the contact radii). *Let  $u_1, \dots, u_n$  be the directions from  $c$  of its active neighbours and  $d_1, \dots, d_n$  their distances. For fixed directions, the volume of  $\bigcap_i \{z : \langle z - c, u_i \rangle \leq d_i/2\}$  is non-decreasing in each  $d_i$ , so it is at least the volume of the same intersection with every  $d_i$  replaced by 2. When every  $d_i$  equals 2 the directions are pairwise at least  $60^\circ$  apart and form a contact configuration (Definition 7.7); when some  $d_i$  exceeds 2 they need not be, and the intersection with every  $d_i$  set to 2 is then a lower bound that belongs to no packing.*

*Proof.* Moving a bounding hyperplane away from  $c$  can only enlarge the region it helps bound. Two centres at distance 2 from  $c$  and at least 2 from each other make an angle of at least  $60^\circ$  at  $c$ .  $\blacksquare$

An earlier version of this paper drew from the lemma the conclusion that the minimum over packing-valid radii is attained at the all-contact corner, and then treated that corner as a contact configuration. The corner is a lower bound but not a packing. Two centres at distance 2.4 from  $c$  and 2 from each other are  $49.25^\circ$  apart as seen from  $c$ , and pulled in to distance 2 they overlap (Figure 3(a)). Forty-eight centres at distance 2.6131 in the directions of the binary octahedral group, the vertices of two 24-cells in dual position, make a packing of 49 balls whose central cell has volume 19.216; at the corner the 48 directions have inner products up to 0.7071, and the cell has volume 6.594. So the reduction to contact configurations holds when every active neighbour touches the centre, and Section 2.7 deals with the neighbours that do not.

This reduces the search for the worst-case local configuration among centres whose active neighbours all touch, over the continuum of their directions, to the combinatorics of the active-direction pattern (Section 3) and the deviation parameters  $(\varphi, s)$  (Sections 4 and 5 for the root-aligned single-deviation case; Sections 7 and 15 onward for the general case).

**2.7. Neighbours that do not touch.** Call  $c$  *shell-free* if every centre within  $2\sqrt{2}$  of it lies at distance exactly 2. The volume of a cell inside a ball is read off the sphere by the radial form of Lemma 7.15; the version needed here allows the centres to be at any distance.

**Proposition 2.10** (The distance criterion). *Let  $c$  be a centre of a unit-ball packing of  $\mathbb{R}^4$  and  $d_1 \leq d_2 \leq \dots$  the distances from  $c$  of the other centres within  $2\sqrt{2}$ . Then*

$$\text{vol}(V_c) \geq \text{vol}(V_c \cap B(c, \sqrt{2})) \geq \Phi(d_1, d_2, \dots),$$

$$\Phi(d_1, d_2, \dots) = \frac{1}{4} \left[ 2\pi^2 + \int_0^{\pi/4} \left( 2\pi^2 - \sum_i C(\arccos(\frac{d_i}{2} \cos r)) \right)_+ d(\sec^4 r) \right],$$

where  $C(\rho) = \pi(2\rho - \sin 2\rho)$  is the measure of a cap of angular radius  $\rho$  on  $S^3$  and the  $i$ -th term is omitted when  $\frac{d_i}{2} \cos r \geq 1$ .  $\Phi$  depends on the distances alone and is non-decreasing in each of them. For  $m$  contacts and no other centre within  $2\sqrt{2}$ ,  $\Phi > 8.044$  for every  $m \leq 22$ .

*Proof.* Put  $c = 0$ . The cell is star-shaped about 0, with radial function  $\rho(\theta) = \min_y (|y|/2) / \langle \theta, u_y \rangle$  over the centres  $y$  with  $\langle \theta, u_y \rangle > 0$ , and  $\rho \geq 1$ . A centre beyond  $2\sqrt{2}$  contributes a value at least  $\sqrt{2}$ , so  $\min(\sqrt{2}, \rho)$  is determined by the centres within  $2\sqrt{2}$ , and  $\text{vol}(V_c \cap B(\sqrt{2})) = \frac{1}{4} \int_{S^3} \min(\sqrt{2}, \rho)^4 = \frac{1}{4} [2\pi^2 + \int_0^{\pi/4} U(r) d(\sec^4 r)]$ ,  $U(r)$  being the measure of  $\{\theta : \rho(\theta) > \sec r\}$ . The complement of that set is the union over the centres of  $\{\theta : \langle \theta, u_i \rangle \geq \frac{d_i}{2} \cos r\}$ , a cap of angular radius  $\arccos(\frac{d_i}{2} \cos r) \leq r$ , and so

TABLE 1. The distance criterion for  $m$  contacts and  $k$  further centres at a common distance  $d$ :  $\Phi > 8$ , and so  $\text{vol}(V_c) > 8$ , as soon as  $d$  is at least the value shown (ball arithmetic, `shell_reduction.py`).

| $m$ | $k$ | $d \geq$ | $m$ | $k$ | $d \geq$ |
|-----|-----|----------|-----|-----|----------|
| 22  | 1   | 2.1676   | 22  | 2   | 2.2340   |
| 21  | 2   | 2.0705   | 21  | 3   | 2.1267   |
| 20  | 3   | 2.0451   | 18  | 6   | 2.0560   |
| 19  | 4   | 2.0332   | 12  | 12  | 2.0267   |
| 23  | 1   | never    |     |     |          |

$U(r) \geq (2\pi^2 - \sum_i C(\cdot))_+$ . A cap shrinks as its centre moves out, which is the monotonicity. For  $m$  contacts the bound is the covering estimate of Theorem 7.16 stopped at  $45^\circ$ , which changes it only for  $m \leq 11$ , where it stays above 11.5; its value at  $m = 22$  is evaluated in ball arithmetic by `shell_reduction.py`. ■

**Lemma 2.11** (Twenty-four contacts leave no room). *A centre with 24 contacts has no other centre within  $2\sqrt{2}$ , and its Voronoi cell is the 24-cell.*

*Proof.* By Theorem 7.25 the contact directions are the normalised roots of a copy of  $R$ . For a unit vector  $\theta$  whose absolute coordinates in the frame of Section 2 are  $a \geq b \geq c' \geq d'$ , the largest inner product with a normalised root is  $(a+b)/\sqrt{2}$ , and  $(a+b)^2 \geq a^2 + b^2 + c'^2 + d'^2 = 1$  because  $2ab \geq 2b^2 \geq c'^2 + d'^2$ ; so every direction is within  $45^\circ$  of a contact direction, which is to say that the covering radius of the root directions is  $45^\circ$ . A centre  $y$  at distance  $d$  from  $c$  and at an angle of at most  $45^\circ$  from the contact  $2w$  has  $|y - 2w|^2 \leq d^2 + 4 - 2\sqrt{2}d$ , which is at least 4 only when  $d \geq 2\sqrt{2}$ . The contacts cut out the 24-cell, of circumradius  $\sqrt{2}$ , and by Lemma 2.7 no centre beyond  $2\sqrt{2}$  cuts it. ■

*Proof of Theorem 1.5.* If  $\Phi(d_1, d_2, \dots) > 8$  this is Proposition 2.10, with strict inequality. Let  $c$  be shell-free with  $m$  contacts, whose directions form a contact configuration  $W$ ; by Lemma 2.7,  $V_c \cap B(c, \sqrt{2}) = V_c(W) \cap B(c, \sqrt{2})$ . If  $m \leq 22$ , Proposition 2.10 gives  $\text{vol}(V_c) > 8.044$ . If  $m = 24$ , Lemma 2.11 makes  $V_c$  the 24-cell, of volume 8. Let  $m = 23$ . The cell  $V_c(W)$  is bounded, since a direction  $\theta$  with  $\langle \theta, w \rangle \leq 0$  for every  $w \in W$  would make  $W \cup \{\theta\}$  a contact configuration of 24 directions, hence a copy of the root directions by Theorem 7.25, whose deletions have bounded cells. The proof of Theorem 7.40 bounds the part of that cell inside the smaller ball,  $\text{vol}(V_c(W) \cap B(c, \sqrt{3/2})) > 8$  (Theorem 7.73), whatever the circumradius, and  $V_c \cap B(c, \sqrt{3/2}) = V_c(W) \cap B(c, \sqrt{3/2})$ ; so  $\text{vol}(V_c) > 8$ . Equality occurs only for  $m = 24$ , where the contacts form a copy of  $\sqrt{2}R$ . ■

Theorem 1.5 leaves Conjecture 1.6, a centre that is not shell-free and whose distances fail the criterion, and Proposition 2.10 places that case precisely. At most twenty-two centres within  $2\sqrt{2}$  never fail it, and neither does any profile in which the centres that do not touch are far enough out; Table 1 gives the thresholds for a few profiles, certified in ball arithmetic by `shell_reduction.py`. Twenty-three contacts and any further centre within  $2\sqrt{2}$  always fail it, since the criterion counts caps and the case of 23 contacts needs the pair terms of Section 7.11. The open case is therefore a centre crowded by at least twenty-three neighbours within  $2\sqrt{2}$ , among them near-contacts, centres a little farther than 2. Both examples above are settled by the criterion, the 49 balls with  $\Phi > 18.33$ .

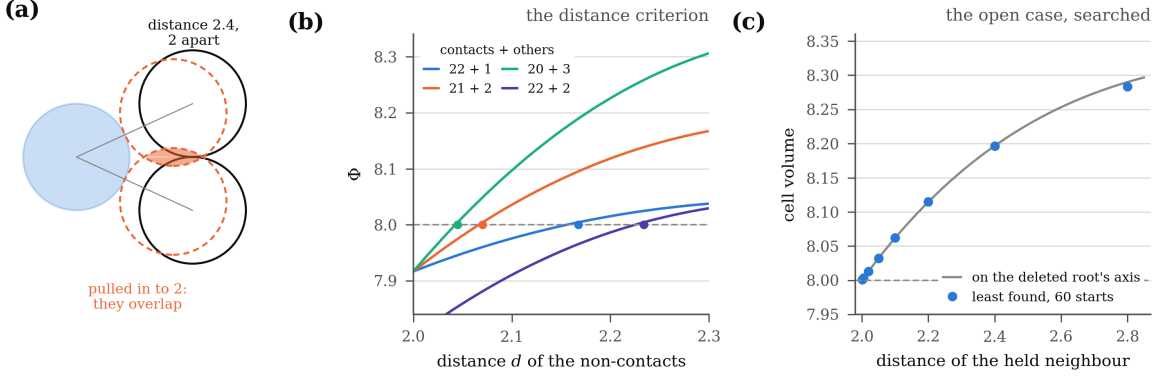


FIGURE 3. Neighbours that do not touch. (a) A two-dimensional section, to scale: two centres at distance 2.4 from the central one and 2 from each other, a packing; pulled in to distance 2 they overlap (shaded), so the configuration at the all-contact corner is not a packing and the earlier form of Lemma 2.9 fails. (b) The distance criterion of Proposition 2.10 for  $m$  contacts and  $k$  further centres at a common distance  $d$ ; it passes 8 at the thresholds of Table 1 (dots). (c) The open case, searched: with one neighbour held at distance  $2 + \delta$ , the least cell volume found over 60 starts (`shell_neighbour_search.py`), against the exact volume  $\frac{25}{3} - \frac{1}{3}(1 - \frac{\delta}{2})^4$  of the deletion of a root with that neighbour on the deleted root's axis. Nothing falls below 8.

*The open case, searched.* `shell_neighbour_search.py` minimises the exact volume of the cell over configurations of 22 to 26 centres around a fixed one, with the packing constraints imposed exactly and the exact gradient of the volume, from starts that crowd the centre with near-contacts: the root system with every root pushed out and turned a little and centres added in its holes, deletions of a root with up to three centres near the deleted one, and random configurations in the shell. Without further constraint, over 240 starts, 224 of the 239 feasible endpoints brought their non-contacts into contact or moved them beyond  $2\sqrt{2}$ , the least volume found being 8, at the root system; the other 15 kept a centre in the shell, and the least volume among them was 8.3362. With one neighbour held at a prescribed distance  $2 + \delta$ , for eight values of  $\delta$  from 0.001 to 0.8 and 60 starts each, the least volume found was, for every  $\delta \leq 0.4$ , that of the deletion of a root with the held neighbour on the deleted root's axis,  $\frac{25}{3} - \frac{1}{3}(1 - \frac{\delta}{2})^4$ , to nine digits, and at  $\delta = 0.8$  it was 8.2837, with 22 contacts, a little below that value; the least over all eight was 8.00067, at  $\delta = 0.001$ , and no feasible endpoint fell below 8 (Figure 3(c)). As  $\delta$  tends to 0 these values tend to 8 and the configuration to the root system, so the open case is tight: a proof of Conjecture 1.6 has to be sharp near the 24-cell, as the proof for contact configurations is. This is evidence and not proof. Closing the case would take, for caps of unequal radii, what the proof of Theorem 7.40 does for equal ones, an exact account of the pairs, with the three-point method of [BV08] carried from the sphere to a shell.

**2.8. Further reductions, and the conjecture near the contact regime.** This subsection narrows Conjecture 1.6 in three directions. It removes every centre with at most twenty-three others within  $\sqrt{6}$ , at any distances; every centre whose contacts contain most of a root system; and every centre around which no other centre lies at a distance between  $2 + \varepsilon_0$  and  $\sqrt{6}$ , for an absolute constant  $\varepsilon_0 > 0$ , which is a local form of the twenty-four-cell conjecture at the root system, for all packings. The tools are three elementary facts about

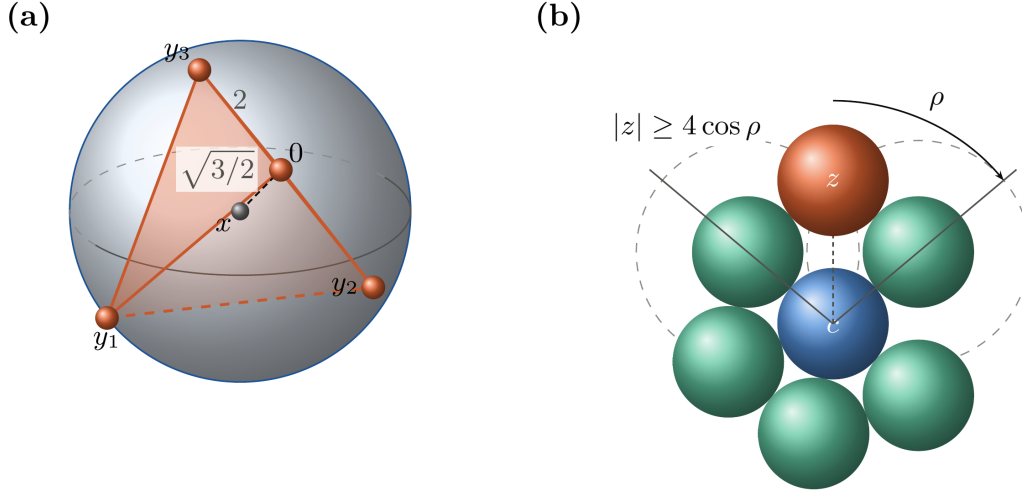


FIGURE 4. Two elementary facts behind Section 2.8, to scale. (a) Lemma 2.15: four points pairwise at least 2 apart do not fit in a ball of radius less than  $\sqrt{3}/2$ , and the regular tetrahedron of edge 2 in the ball of radius  $\sqrt{3}/2$  is the extreme case; so no point  $x$  of the open ball  $B(\sqrt{3}/2)$  about the centre 0 is closer to three other centres than to 0. (b) Lemma 2.12 in a plane section: the balls of the contacts (green) leave a hole of angular radius  $\rho$ , here  $50^\circ$ , and a further centre  $z$  can enter it only at a distance  $|z| \geq 4 \cos \rho$  from  $c$ , where it meets the circles of radius 2 (dashed) about the two contacts at the rim.

packings that hold whatever the distances, and, for the first, the certificate of Theorem 7.73 carried from contacts to centres at any distance. Throughout,  $c = 0$ , and the half-space of a centre  $z$  is  $H_z = \{x : \langle x, z \rangle \leq |z|^2/2\}$ . The identities in the proofs of Lemmas 2.12 and 2.15 and Proposition 2.13 and the combinatorics of Corollary 2.14 are checked twice, by `closure_lemmas.py` and in Lean's kernel by `lean/D4Closure.lean` (Section 23).

**Lemma 2.12** (Holes). *Let  $W$  be the set of contact directions of  $c$ , so that  $2w$  is a centre for every  $w \in W$ . Every other centre  $z$  satisfies  $|z| \geq 4\langle z/|z|, w \rangle$  for every  $w \in W$ , and so  $|z| \geq 4g(W)$ , where  $g(W) = \min_{\theta \in S^3} \max_{w \in W} \langle \theta, w \rangle$  is the cosine of the covering radius of  $W$  on  $S^3$ .*

*Proof.*  $4 \leq |z - 2w|^2 = |z|^2 - 4\langle z, w \rangle + 4$ . ■

For the root system  $g = 1/\sqrt{2}$ , and the lemma returns the bound  $2\sqrt{2}$  of Lemma 2.11. A centre that does not touch  $c$  can come close to it only in a deep hole of its contacts: at distance  $2 + \delta$  only where the contact directions leave a cap of angular radius  $\arccos(\frac{1}{2} + \frac{\delta}{4})$  empty (Figure 4(b)).

**Proposition 2.13** (The inversion hull). *Let  $Y$  be any set of centres other than  $c$ , and let  $K(Y)$  be the convex hull of 0 and the points  $4y/|y|^2$ ,  $y \in Y$ , the images of the centres under inversion in the sphere of radius 2. Every centre  $z \notin Y \cup \{0\}$  has  $K(Y) \subseteq H_z$ . Consequently*

$$V_c \supseteq V(Y) \cap K(Y), \quad V(Y) = \bigcap_{y \in Y} H_y.$$

*Proof.*  $H_z$  is convex and contains 0, so it suffices to test the points  $q = 4y/|y|^2$ , and for these

$$\frac{|z|^2}{2} - \langle q, z \rangle = \frac{(|z|^2 - 4)(|y|^2 - 4) + 4(|y - z|^2 - 4)}{2|y|^2} \geq 0,$$

since centres are at least 2 apart and at least 2 from 0.  $\blacksquare$

The bound depends on  $Y$  alone: whatever the rest of the packing does, it does not cut into  $V(Y) \cap K(Y)$  (Figure 5(b) shows the three-dimensional analogue). With  $Y$  the twenty-four neighbours of a centre near the root system it is sharp to first order (`near_contact_probe.py`).

**Corollary 2.14** (Contacts that contain most of a root system). *Let  $U = \{\alpha/\sqrt{2} : \alpha \in R'\}$  be the normalised roots of a copy  $R'$  of  $R$ , and  $Q = \{x : \langle x, u \rangle \leq 1, u \in U\}$  its 24-cell. Every vertex of  $Q$  is the sum  $u + u'$  of exactly three pairs of orthogonal elements of  $U$ . If the contact directions  $W$  of  $c$  contain, for every vertex of  $Q$ , one of its three pairs, then  $Q \subseteq V_c$ , so  $\text{vol}(V_c) \geq 8$ , with equality only if  $W = U$ . The hypothesis holds whenever  $W$  contains all but at most two elements of  $U$ , and whenever it contains all but three, unless the three missing ones are pairwise at  $60^\circ$  with a common vertex (96 of the 2024 triples).*

*Proof.* A vertex  $v = u + u'$  with  $u, u' \in W$  is the midpoint of the points  $2u$  and  $2u'$ , which  $K(2W)$  contains, so  $Q$ , the convex hull of its vertices, lies in  $K(2W)$ , and by Proposition 2.13 in the half-space of every centre that is not a contact. It lies in the half-space  $\{\langle x, w \rangle \leq 1\}$  of every contact too: for  $w \in U$  because  $v \in Q$ , and for  $w \notin U$  because  $\langle v, w \rangle = \langle u, w \rangle + \langle u', w \rangle \leq \frac{1}{2} + \frac{1}{2}$ , the contact directions being pairwise at least  $60^\circ$  apart. So  $Q \subseteq V_c$ . If some  $u \in U$  is not in  $W$ , then  $\langle u, w \rangle < 1$  for every contact direction  $w$ , and a centre  $z$  with  $|z| > 2$  has  $\langle (1+t)u, z \rangle \leq (1+t)|z| < |z|^2/2$  as soon as  $2(1+t) < |z|$ ; only finitely many centres lie within 3 of  $c$ , so for small  $t > 0$  a neighbourhood of  $(1+t)u$  lies in  $V_c$  and outside  $Q$ , and  $\text{vol}(V_c) > 8$ . If  $W \supseteq U$ , then  $W = U$  by the kissing number and  $V_c = Q$  by Lemma 2.11. The combinatorial statements are finite and are checked in integer arithmetic by `closure_lemmas.py`: the 72 orthogonal pairs, three at each of the 24 vertices; no set of one or two roots meets all three pairs at a vertex; exactly 96 sets of three do, the triples of pairwise  $60^\circ$  roots at a common vertex, which are also the triples whose deletion raises the circumradius of the cell to  $\sqrt{6}$  (`root_deletions_exact.py`, Appendix F). The same statements are proved again in Lean's kernel by `lean/D4Closure.lean`. Figure 5(a) shows one vertex and its three pairs.  $\blacksquare$

**Lemma 2.15** (No three centres cut the same point of  $B(\sqrt{3/2})$ ). *If  $|x| < \sqrt{3/2}$ , at most two centres  $y$  satisfy  $|x - y| < |x|$ . Hence, for  $R \leq \sqrt{3/2}$ ,*

$$\text{vol}(V_c \cap B(R)) = \text{vol } B(R) - \sum_y S_R(|y|) + \sum_{\{y, y'\}} \text{vol}(B(R) \setminus (H_y \cup H_{y'})),$$

where  $S_R(d)$  is the volume of the part of  $B(R)$  beyond a hyperplane at distance  $d/2$  from 0, and the sums run over the centres other than  $c$ , only those closer than  $2R$  contributing.

*Proof.* If three centres  $y_1, y_2, y_3$  did, the four points  $0, y_1, y_2, y_3$ , pairwise at least 2 apart, would lie in the closed ball of radius  $|x|$  about  $x$  (Figure 4(a)). For four points  $p_i$  with centroid  $g$ ,  $\sum_{i < j} |p_i - p_j|^2 = 4 \sum_i |p_i - g|^2 \leq 4 \sum_i |p_i - x|^2$ , so  $24 \leq 16|x|^2$ , against  $|x| < \sqrt{3/2}$ . The formula is inclusion-exclusion, whose terms of order three vanish.  $\blacksquare$

For contact configurations this is the absence of triple overlaps on which Theorem 7.73 rests; the lemma says that it holds for every packing.

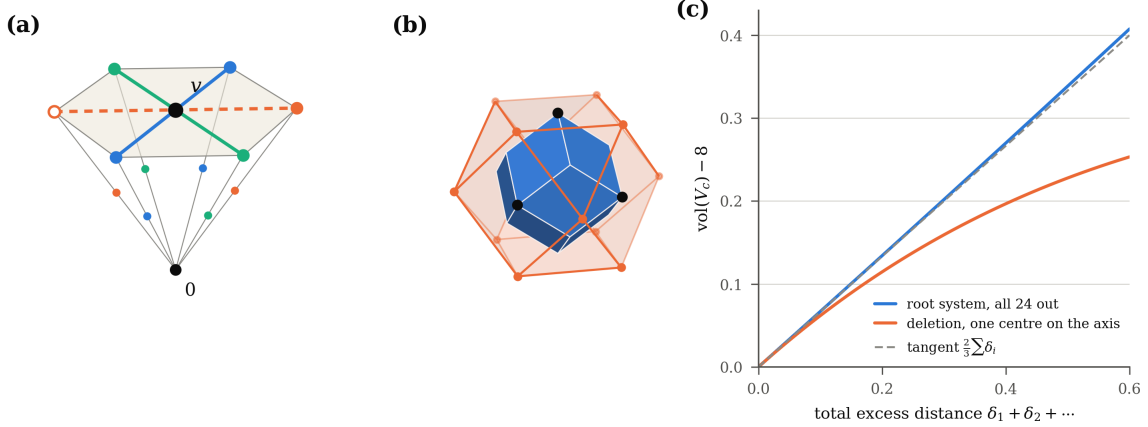


FIGURE 5. (a) Corollary 2.14: a vertex  $v$  of the 24-cell and its three orthogonal pairs of normalised roots, in the linear projection of  $\mathbb{R}^4$  that keeps the vertex direction vertical and sends the other three coordinate directions to horizontal unit vectors at  $120^\circ$ . The six roots at  $v$  lie on a hexagon, the points  $2u$  on a larger one, and  $v$  is the midpoint of the chord from  $2u$  to  $2u'$  of each pair, so it lies in the inversion hull of any contact set that keeps one pair. One root is missing (hollow), its chord is lost (dashed), and  $v$  keeps the other two. (b) Proposition 2.13 in three dimensions: the Voronoi cell of the twelve contacts of the face-centred cubic packing, a rhombic dodecahedron (blue), lies in the convex hull of the points  $2u$  (orange) and touches its boundary only at its six four-valent vertices (black), as the 24-cell touches the hull of the points  $2u$  at its 24 vertices; every other centre leaves that hull whole. (c) Theorem 2.18 to first order: the exact volume excess of the root system pushed out uniformly,  $8(1 + \delta/2)^4 - 8$  with  $\sum \delta_i = 24\delta$ , and of the deletion with one centre at  $2 + \delta$  on the deleted axis,  $\frac{1}{3} - \frac{1}{3}(1 - \frac{\delta}{2})^4$ , against the total excess distance; both are tangent at the root system to  $\frac{2}{3} \sum \delta_i$ .

**Corollary 2.16** (At most twenty-two centres within  $\sqrt{6}$ ). Put  $S = S_{\sqrt{3/2}}$  and  $\Psi(d_1, d_2, \dots) = \frac{9}{8}\pi^2 - \sum_i S(d_i)$ . For every centre,  $\text{vol}(V_c) \geq \Psi(d_1, d_2, \dots)$ . In particular, if at most 22 centres other than  $c$  lie within  $\sqrt{6}$  of  $c$ , then  $\text{vol}(V_c) > 8.046$ .

*Proof.* Take  $R = \sqrt{3/2}$  in Lemma 2.15 and drop the pair terms, which are nonnegative.  $S$  decreases and vanishes for  $d \geq \sqrt{6}$ , so at most twenty-two centres within  $\sqrt{6}$  give  $\text{vol}(V_c) \geq \frac{9}{8}\pi^2 - 22S(2) = 8.04610\dots$ , in ball arithmetic by `closure_lemmas.py`. ■

$\Psi$  is a second distance criterion beside  $\Phi$ , with thresholds within 0.003 of those of Table 1 (the script prints both). What is new is where the count is taken: within  $\sqrt{6}$ , and not within  $2\sqrt{2}$ , so that any number of centres between  $\sqrt{6}$  and  $2\sqrt{2}$  is allowed.

**Theorem 2.17** (At most twenty-three centres within  $\sqrt{6}$ ). If at most 23 centres other than  $c$  lie within  $\sqrt{6}$  of  $c$ , then  $\text{vol}(V_c) > 8$ , whatever their distances from  $c$  and from one another.

With all twenty-three at distance 2 this is Theorem 7.40; what is new is that they need not touch  $c$ . A centre that moves out from distance 2 to  $d$  hands back to  $B(\sqrt{3/2})$  the part  $S(2) - S(d)$  of its cap, and it takes from each pair term it shares only the part of the moving section that the other cap covers. The proof lets the certificate of Theorem 7.73

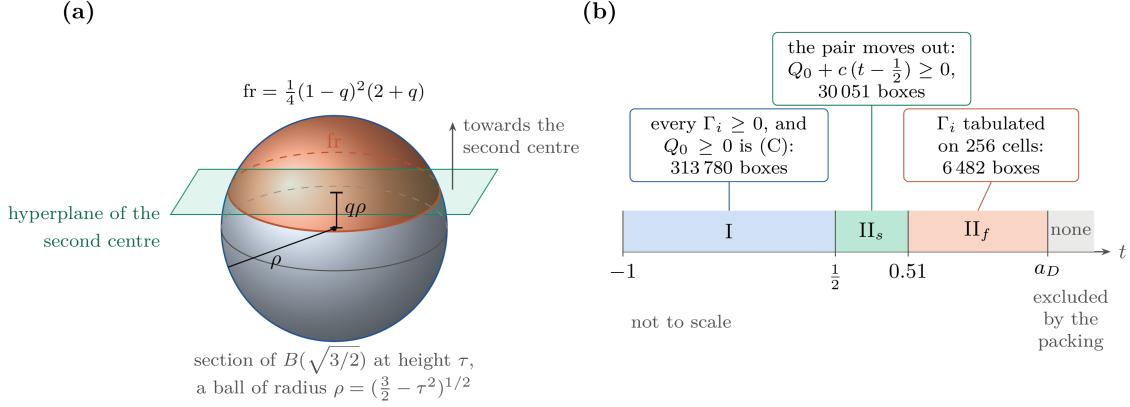


FIGURE 6. The proof of Theorem 2.17. (a) The share  $fr$ : the section of  $B(\sqrt{3/2})$  by the hyperplane of one centre is a ball of dimension three, and the hyperplane of a second centre cuts from it a cap; moving the first hyperplane out removes from the pair term exactly this cap, and from the cap term of the centre the whole section. (b) The three ranges of the largest inner product  $t$  of three directions, with what settles each; beyond  $a_D$  the packing leaves no room.

spend one eleventh of what is handed back in each triple of centres, and shows that this pays for everything the certificate loses when two directions come closer than  $60^\circ$ , which the packing allows only to centres that have moved out (Figures 6 and 7).

*Proof.* With at most 22 such centres this is Corollary 2.16. Let there be exactly 23,  $y_1, \dots, y_{23}$ , and write  $d_i = |y_i| \in [2, \sqrt{6})$ ,  $w_i = y_i/d_i$ ,  $h_i = d_i/2$  and  $u_{ij} = \langle w_i, w_j \rangle$ . Put  $R = \sqrt{3/2}$ , let  $S = S_R$  be as in Corollary 2.16, and let  $s(d) = S(2) - S(d)$ , which vanishes at  $d = 2$  and increases. For  $h, h' \in [1, R)$  and  $-1 \leq x < 1$  let  $\Pi(h, h', x)$  be the volume of the part of  $B(R)$  beyond two hyperplanes at distances  $h$  and  $h'$  from 0 whose unit normals have inner product  $x$ . The pair term of  $y_i$  and  $y_j$  in Lemma 2.15 is  $\Pi(h_i, h_j, u_{ij})$ , and  $\Pi(1, 1, u) = \omega(u)$  is the weight of Section 7.12. With  $A_* = \frac{9}{8}\pi^2 - 23S(2)$ , Lemma 2.15 reads

$$\text{vol}(V_c) \geq \text{vol}(V_c \cap B(R)) = A_* + \sum_i s(d_i) + \sum_{i < j} \Pi(h_i, h_j, u_{ij}). \quad (2.1)$$

*Step 1: a centre far out.* Let  $D = 2.1648$ . In ball arithmetic  $s(D) = 0.0928820\dots$ , which exceeds  $8 - A_* = 0.0928555\dots$ ; so if some  $d_i \geq D$ , the right side of (2.1) exceeds 8. From now on every  $d_i < D$ . The packing condition  $|y_i - y_j| \geq 2$  reads

$$u_{ij} \leq a(d_i, d_j) := \frac{d_i^2 + d_j^2 - 4}{2d_i d_j} \leq a(D, D) = 0.573229\dots =: a_D, \quad (2.2)$$

and  $a$  increases in each argument on  $[2, D]^2$ .

*Step 2: a certificate for centres at mixed distances.* Let  $P$  and  $B$  be the polynomial and the bound of Theorem 7.73, so that (7.47) reads  $\omega(u) + \omega(v) + \omega(t) \geq P(u, v, t)$ , and for three of the centres put

$$\begin{aligned} Q_{ijk} &= \Pi(h_i, h_j, u_{ij}) + \Pi(h_i, h_k, u_{ik}) + \Pi(h_j, h_k, u_{jk}) \\ &\quad + \frac{1}{11}(s(d_i) + s(d_j) + s(d_k)) - P(u_{jk}, u_{ik}, u_{ij}). \end{aligned}$$

Each pair of the 23 centres lies in 21 triples and each centre in  $231 = 21 \cdot 11$ ,<sup>5</sup> so the sum of the  $Q_{ijk}$  over all triples is  $21(\sum_i s(d_i) + \sum_{i < j} \Pi(h_i, h_j, u_{ij})) - \sum_{i < j < k} P(u_{jk}, u_{ik}, u_{ij})$ . The proof of Lemma 7.72 uses of the directions only that they are  $N = 23$  unit vectors: Schoenberg's theorem for the terms in  $f$  and Lemma 7.71 for the terms in  $F$  give  $\sum_{i < j < k} P(u_{jk}, u_{ik}, u_{ij}) \geq (N - 2)B = 21B$  for any such vectors. Hence, if every  $Q_{ijk} \geq 0$ , then  $\sum_i s(d_i) + \sum_{i < j} \Pi(h_i, h_j, u_{ij}) \geq B$ , and (2.1) gives  $\text{vol}(V_c) \geq A_* + B > 8$ .

*Step 3: three variables.*  $Q_{ijk}$  is symmetric in the three centres, because  $P$  is; number them 1, 2, 3 so that  $t = u_{12} \geq v = u_{13} \geq u = u_{23}$ . For  $1 \leq \tau < R$  the section  $\{x \in B(R) : \langle x, w \rangle = \tau\}$  is a ball of dimension three and radius  $\rho = (R^2 - \tau^2)^{1/2}$ , of volume  $A(\tau) = \frac{4\pi}{3}\rho^3$ . A second hyperplane  $\{x : \langle x, w' \rangle = h'\}$  with  $\langle w, w' \rangle = x$  meets it in a plane at signed distance  $q\rho$  from its centre,  $q = (h' - \tau x)/((1 - x^2)^{1/2}\rho)$ , so the share of the section beyond the second hyperplane is

$$\text{fr}(\tau, h', x) = \frac{1}{4}(1 - q)^2(2 + q) \quad \text{if } q < 1, \quad \text{fr}(\tau, h', x) = 0 \quad \text{if } q \geq 1,$$

the volume  $\frac{\pi}{3}\eta^2(3\rho - \eta)$  of a cap of height  $\eta = (1 - q)\rho$  divided by that of the ball (Figure 6(a)). By Fubini's theorem  $\partial_h \Pi(h, h', x) = -A(h) \text{fr}(h, h', x)$ , and  $\frac{d}{dh} s(2h) = A(h)$ . On the range that occurs,  $h, h' \in [1, D/2]$  and  $x \leq a_D$ , the share decreases in  $h'$  and in  $\tau$  and increases in  $x$ :  $q$  increases in  $h'$ ; its derivative in  $x$  has the sign of  $x - \tau < 0$ ; and  $\frac{d}{d\tau} \log q = -\frac{x}{h' - \tau x} + \frac{\tau}{R^2 - \tau^2} > 0$ , since  $\frac{x}{1 - \tau x} \leq \frac{a_D}{1 - (D/2)a_D} = 1.5103\dots < 2 \leq \frac{\tau}{R^2 - \tau^2}$ . Raising first  $h_i$  and then  $h_j$  from 1, and bounding each share by its value with the other height at 1, gives  $\Pi(h_i, h_j, x) \geq \omega(x) - \varphi(h_i, x) - \varphi(h_j, x)$  with  $\varphi(h, x) = \int_1^h A(\tau) \text{fr}(\tau, 1, x) d\tau$ . As  $s(d_i) = \int_1^{h_i} A$ , it follows that

$$Q_{123} \geq Q_0(u, v, t) + \Gamma_1(h_1) + \Gamma_2(h_2) + \Gamma_3(h_3), \quad (2.3)$$

where  $Q_0 = \omega(u) + \omega(v) + \omega(t) - P(u, v, t)$  is the quantity that (7.47) asserts to be nonnegative,

$$\Gamma_1(h) = \int_1^h A(\tau) \left( \frac{1}{11} - \text{fr}(\tau, 1, t) - \text{fr}(\tau, 1, v) \right) d\tau,$$

and  $\Gamma_2$  and  $\Gamma_3$  are the same with  $(t, u)$  and  $(v, u)$ , the inner products at the centre concerned, in place of  $(t, v)$ . It remains to show that the right side of (2.3) is nonnegative for all heights that (2.2) allows. By (2.2),  $t \leq a_D$ , and there are three ranges of  $t$  (Figure 6(b)).

(I)  $t \leq \frac{1}{2}$ . Every share is at most  $\text{fr}(1, 1, \frac{1}{2}) = 0.023710\dots < \frac{1}{22}$ , so every  $\Gamma_i \geq 0$ , and  $Q_0 \geq 0$  is (7.47), which Theorem 7.73 proves.

(II<sub>s</sub>)  $\frac{1}{2} < t \leq 0.51$ . The integrands are now at least  $r = \frac{1}{11} - 2 \text{fr}(1, 1, 0.51) \geq 0.037900$ , so  $\Gamma_1 + \Gamma_2 + \Gamma_3 \geq r(s(d_1) + s(d_2))$ . The identity

$$\frac{1}{2} + \frac{1}{4}(d_1 + d_2 - 4) - a(d_1, d_2) = \frac{(d_1 - 2)(d_2 - 2)(d_1 + d_2 + 2)}{4d_1 d_2}$$

turns (2.2) for the pair  $y_1, y_2$  into  $(d_1 - 2) + (d_2 - 2) \geq 4(t - \frac{1}{2})$ . The function  $x \mapsto s(2 + x)$  vanishes at 0 and is concave, its derivative  $\frac{1}{2}A(1 + \frac{x}{2})$  decreasing; so it is subadditive and lies above its chords, and  $s(d_1) + s(d_2) \geq s(2 + 4(t - \frac{1}{2})) \geq 4\kappa(t - \frac{1}{2})$  with  $\kappa = s(2.04)/0.04 \geq 0.696362$ . Hence  $Q_{123} \geq Q_0(u, v, t) + c(t - \frac{1}{2})$  with  $c = 4r\kappa \geq 0.105569$ . The branch and bound of the proof of Theorem 7.73, with this linear term added to the Taylor form and

<sup>5</sup>This fixes the weight  $\frac{1}{11}$ : it is the one with which each centre's term  $s(d_i)$  enters the sum over triples with the same factor, 21, as each pair term.

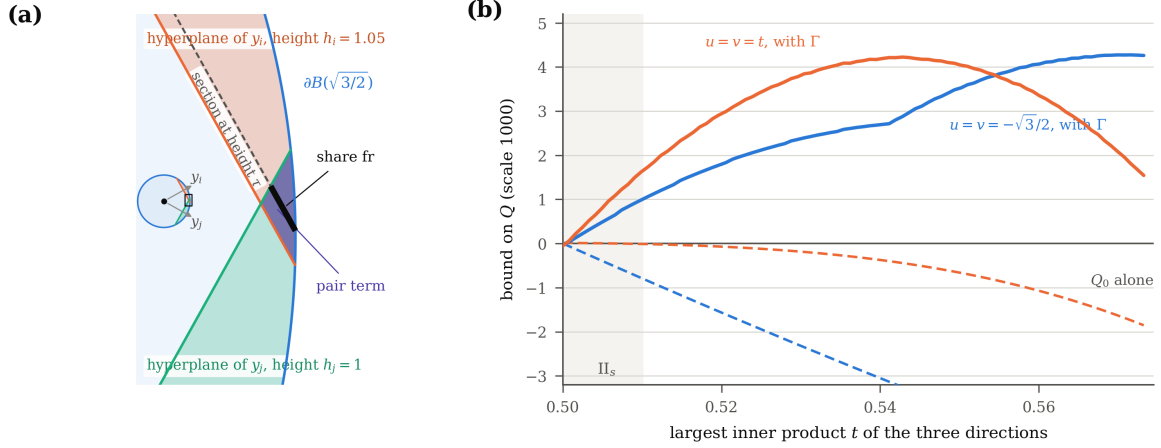


FIGURE 7. (a) A plane section through  $c$  and two centres at the packing boundary  $|y_i - y_j| = 2$ , with  $d_i = 2.10$  and  $d_j = 2$ , so that the cosine  $a(d_i, d_j) = 0.525$  of their angle exceeds  $\frac{1}{2}$ , which two contacts cannot do; the inset shows the whole section and the window. Beyond the two hyperplanes lie the caps (orange, green) and their overlap, the pair term (violet). The section of  $B(\sqrt{3}/2)$  at height  $\tau$  (dashed) meets the second cap in the share fr (thick). To scale. (b) The bound along two lines past the face  $t = \frac{1}{2}$  where contacts end:  $Q_0$  alone (dashed) turns negative, while  $Q_0$  plus the lower bound for  $\Gamma_1 + \Gamma_2 + \Gamma_3$  that the proof uses (solid) stays positive. Blue:  $u = v = -\frac{\sqrt{3}}{2}$ , where (7.47) comes nearest to equality on the face; orange:  $u = v = t$ . The shaded strip is the range  $\Pi_s$ . The scale is that of the computation, 1000 times the volume.

$\omega$  tabulated up to  $u = a_D$ , shows the right side nonnegative on the slab: it verifies 30 051 boxes and discards 3 408 outside the domain.<sup>6</sup>

(II<sub>f</sub>)  $0.51 \leq t \leq a_D$ . The branch and bound bounds  $Q_0$  on each box as before, and  $\Gamma_1 + \Gamma_2 + \Gamma_3$  from below by a table: each  $\Gamma_i$  is integrated in ball arithmetic over 256 cells of  $h \in [1, D/2]$ , with the share taken at the lower end of the cell and at the upper ends of the box's inner products; it is replaced by its right-hand minimum  $\min_{h' \geq h} \Gamma_i(h')$ , which is nondecreasing in  $h$ , and then minimised over the triples of cells whose upper ends satisfy (2.2) at the lower ends of the box's inner products. It verifies 6 482 boxes and discards 1 242 outside the domain.

So every  $Q_{ijk} \geq 0$ , and Step 2 concludes. The constants of Steps 1 and 3, the two branch and bounds and a rerun of the verification of (7.47) are `labelled_certificate_check.py`; the identity for  $a$  is also proved in Lean, in `D4Closure.lean` (Section 23). ■

A centre with 23 contacts may have further centres near it. If the contact directions have covering radius at most  $\arccos(\sqrt{6}/4) = 52.2388^\circ$ , every other centre is at distance at least  $4 \cos 52.2388^\circ = \sqrt{6}$  by Lemma 2.12, and Theorem 2.17 applies; a deletion of a root is Corollary 2.14. A contact configuration of 23 directions whose cell has circumradius at least 2 is a deletion (Proposition 7.38 and Theorem 7.25). So a centre with 23 contacts and a

<sup>6</sup>The certificate alone does not reach past the face  $t = \frac{1}{2}$ . At  $u = v = -\frac{\sqrt{3}}{2}$ , where (7.47) comes nearest to equality on that face,  $Q_0$  falls by about 0.080 per unit of  $t$  beyond it (Figure 7(b)). Two directions closer than  $60^\circ$  belong to centres that have moved out, and that gains at least  $c = 0.1056$  per unit of  $t$ .

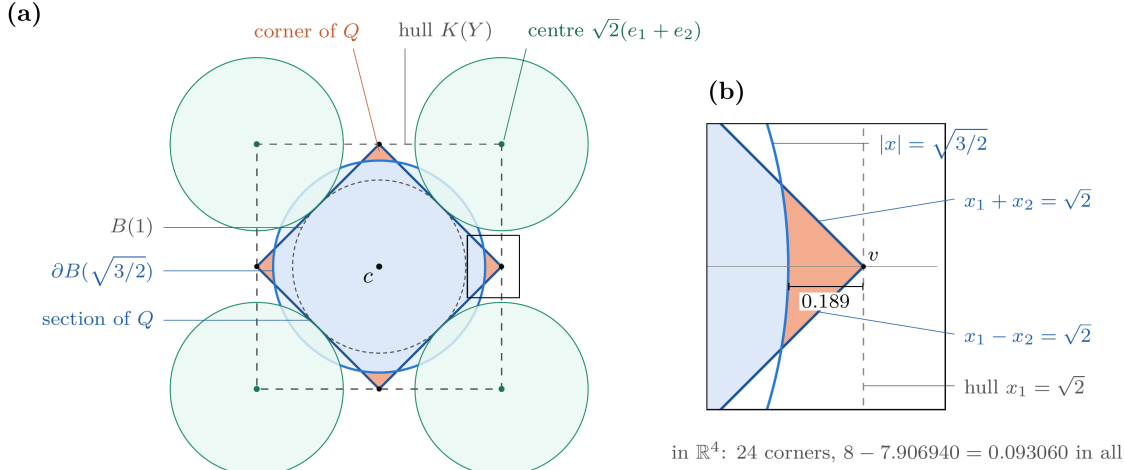


FIGURE 8. Why the pair terms stop at twenty-three centres, to scale. (a) The section of the 24-cell  $Q$  by the plane  $x_3 = x_4 = 0$ : the square  $|x_1| + |x_2| \leq \sqrt{2}$ , whose sides touch the unit ball and whose corners are vertices of  $Q$  at radius  $\sqrt{2}$ . The ball  $B(\sqrt{3/2})$ , inside which Lemma 2.15 makes the cell an exact function of the pairs, cuts the corners off (orange). The four centres of the plane,  $\sqrt{2}(\pm e_1 \pm e_2)$  at distance 2, with their unit balls, and their inversion hull (dashed), which meets  $Q$  only at its vertices. (b) One corner, magnified: the vertex  $v = \sqrt{2}e_1$ , the two sides through it and the side  $x_1 = \sqrt{2}$  of the hull; the corner is  $\sqrt{2} - \sqrt{3/2} = 0.189$  deep. In  $\mathbb{R}^4$  the 24 corners hold  $8 - 7.906940 = 0.093060$  of the volume of  $Q$ .

cell of volume at most 8 would need a configuration that is not a deletion, whose existence is itself open (Theorem 7.60), with a hole of angular radius between  $52.24^\circ$  and  $60^\circ$  holding a twenty-fourth centre closer than  $\sqrt{6}$ .

*Why the count stops at twenty-three.* At the root system twenty-four centres lie within  $\sqrt{6}$ , and the right side of Lemma 2.15 is  $\frac{9}{8}\pi^2 - 24S(2) + 96\omega(\frac{1}{2}) = 7.906940\dots$ , the volume of the 24-cell inside  $B(\sqrt{3/2})$ : the cell reaches out to radius  $\sqrt{2}$ , and the ball misses 0.093 of it (Figure 8).<sup>7</sup> So with twenty-four or more centres within  $\sqrt{6}$  no lower bound for  $\text{vol}(V_c \cap B(\sqrt{3/2}))$  reaches 8, and a proof has to look beyond that ball. Near the root system the next theorem does, through the inversion hull.

**Theorem 2.18** (The conjecture near the contact regime). *There is an absolute constant  $\varepsilon_0 > 0$  with the following property. If no centre of a unit-ball packing of  $\mathbb{R}^4$  lies at a distance from  $c$  strictly between  $2 + \varepsilon_0$  and  $\sqrt{6}$ , then  $\text{vol}(V_c) \geq 8$ , with equality only if the contacts of  $c$  form a copy of  $\sqrt{2}\mathbb{R}$  and  $V_c$  is the regular 24-cell they cut out.*

In particular the  $D_4$  configuration is a strict local minimum of the cell volume among all packings: every packing whose centres within  $\sqrt{6}$  of  $c$  are close enough to  $\sqrt{2}\mathbb{R}'$ , for some copy  $\mathbb{R}'$  of  $\mathbb{R}$ , and whose other centres are unrestricted, has  $\text{vol}(V_c) \geq 8$ , with equality only

<sup>7</sup>The 96 is the number of pairs of roots at  $60^\circ$ , eight at each of the 24 roots; pairs at  $90^\circ$  or more have disjoint caps in  $B(\sqrt{3/2})$ , since the caps of two contacts overlap only when their angle is below  $2\arctan(1/\sqrt{2}) = 70.53^\circ$ . The missing volume sits at the 24 vertices of the 24-cell, the points  $r + r'$  at radius  $\sqrt{2}$ , in 24 congruent corners of volume 0.003878 each; in the plane of the two orthogonal vertices  $\sqrt{2}e_1$  and  $\sqrt{2}e_2$  each corner is a right-angled triangle of depth  $\sqrt{2} - \sqrt{3/2} = 0.189$  with its tip cut off by an arc.

at  $\sqrt{2}R'$  itself, and to first order the excess is  $\frac{2}{3}\sum\delta_i$  (Figure 5(c)). The constant  $\varepsilon_0$  comes from a compactness argument and is not explicit.

*Proof. Step 1: a centre with  $\text{vol}(V_c) \leq 8$  in this regime is close to the root system.* We show that for every  $\eta > 0$  there is  $\varepsilon > 0$  such that, if no centre lies at a distance from  $c$  in  $(2 + \varepsilon, \sqrt{6})$  and  $\text{vol}(V_c) \leq 8$ , then the centres within  $\sqrt{6}$  of  $c$  can be matched with the points of  $\sqrt{2}R'$ , for some copy  $R'$  of  $R$ , each within  $\eta$  of its partner. If not, there are  $\eta > 0$  and packings  $\mathcal{P}_n$  with a centre at 0, no centre at a distance in  $(2 + 1/n, \sqrt{6})$ ,  $\text{vol}(V^{(n)}) \leq 8$  for the cell of 0, and no such matching. The packing condition bounds the number of centres in every ball independently of  $n$ , so after passing to a subsequence the centres within  $B(\rho)$  converge, as finite sets, to those of a packing  $\mathcal{P}$ , for every  $\rho$  outside the countable set of distances at which  $\mathcal{P}$  has a centre (a diagonal argument). In  $\mathcal{P}$  every centre is at distance 2 from 0 or at distance at least  $\sqrt{6}$ . For each such  $\rho \geq 3$  the set  $V^{(n)} \cap B(\rho/2)$  is cut out by the finitely many centres within  $\rho$  and converges in the Hausdorff metric to  $V \cap B(\rho/2)$ ,  $V$  the cell of 0 in  $\mathcal{P}$ , which contains  $B(1)$ ; so  $\text{vol}(V \cap B(\rho/2)) \leq \liminf \text{vol}(V^{(n)}) \leq 8$  for all these  $\rho$ , and  $\text{vol}(V) \leq 8$ . Let  $m$  be the number of contacts of 0 in  $\mathcal{P}$ . If  $m \leq 23$ , the only centres of  $\mathcal{P}$  within  $\sqrt{6}$  of 0 are its  $m$  contacts, and Theorem 2.17 gives  $\text{vol}(V) > 8$ . So  $m = 24$ , the contacts form a copy of  $\sqrt{2}R'$  (Theorem 7.25), and the centres of  $\mathcal{P}_n$  within  $\sqrt{6}$ , which converge to them, can be matched with them within  $\eta$  for large  $n$ , a contradiction.

*Step 2: the root system is a strict local minimum.* Let  $Y$  be the set of centres within  $\sqrt{6}$  of  $c$ , matched as in Step 1 with the points of  $\sqrt{2}R'$  within a small  $\eta$ . Rotate so that  $R' = R$ , write  $u_i$  for the normalised roots and  $y_i = d_i w_i$  for the partner of  $2u_i$ , with  $|w_i| = 1$  and  $d_i = 2 + \delta_i$ ,  $0 \leq \delta_i \leq \varepsilon$ , and choose the rotation and the labelling that attain the minimum  $d(W)$  of Theorem 7.51 for  $W = \{w_i\}$ . Put  $\varepsilon_i = w_i - u_i$ ,  $e_i = |\varepsilon_i|$  and  $\|\varepsilon\|^2 = \sum_i e_i^2$ , so that  $\|\varepsilon\| \rightarrow 0$  as  $\eta \rightarrow 0$ . Three estimates hold once  $\eta$  and  $\varepsilon$  are small.

(a) *Rigidity with slack:*  $\|\varepsilon\| \leq 6\sum_i \delta_i$ . The proof of Theorem 7.51 goes through with the contact condition  $\langle w_i, w_j \rangle \leq \frac{1}{2}$  on the 96 pairs of  $T$  replaced by the packing condition  $|y_i - y_j| \geq 2$ , which reads  $\langle w_i, w_j \rangle \leq \frac{1}{2} + \kappa_{ij}$  with  $\kappa_{ij} = (d_i^2 + d_j^2 - d_i d_j - 4)/(2d_i d_j) \leq \frac{1}{4}(\delta_i + \delta_j) + \frac{1}{8}(\delta_i + \delta_j)^2$ . The positive parts of  $\Lambda\tau$  then sum to at most  $6\|\varepsilon\|^2 + \kappa$ ,  $\kappa = \sum_T \kappa_{ij}$ ; the constant stress still makes the entries of  $\Lambda\tau$  sum to 0, so  $\|\tau\|_2 \leq \|\Lambda\tau\|_1 \leq 12\|\varepsilon\|^2 + 2\kappa$  by (7.35), and with  $\sum_i \rho_i^2 \leq \frac{1}{4}\|\varepsilon\|^4$ ,  $\|\varepsilon\| \leq \frac{1}{4}\|\varepsilon\| + 2\kappa + \frac{1}{2}\|\varepsilon\|^2$ . For  $\|\varepsilon\| \leq 1/48$  this gives  $\|\varepsilon\| \leq \frac{192}{71}\kappa$ , and every index lies in eight pairs of  $T$ , so  $\kappa \leq 2(1 + \varepsilon)\sum_i \delta_i$ .

(b) *The volume to first order:*  $\text{vol}(V(Y)) \geq 8 + \frac{2}{3}\sum_i \delta_i - C(\|\varepsilon\| + \sum_i \delta_i)^2$ . Move the data linearly,  $w_i(t) = u_i + t\varepsilon_i$  and  $h_i(t) = 1 + t\delta_i/2$  for  $0 \leq t \leq 1$ , and let  $P(t) = \{x : \langle x, w_i(t) \rangle \leq h_i(t)\}$ , so that  $P(0) = Q$  and  $P(1) = V(Y)$ . Each facet of  $P(t)$  moves with normal velocity  $(\dot{h}_i - \langle x, \dot{w}_i \rangle)/|w_i|$ , so

$$\frac{d}{dt} \text{vol } P(t) = \sum_i \frac{1}{|w_i(t)|} \left( \frac{\delta_i}{2} A_i(t) - \langle \varepsilon_i, M_i(t) \rangle \right),$$

$A_i$  the 3-volume of the  $i$ -th facet and  $M_i = \int x dA$  over it. At  $t = 0$  every facet is a regular octahedron of volume  $\frac{4}{3}$  centred at  $u_i$ , so the right side is  $\frac{2}{3}\sum_i \delta_i - \frac{4}{3}\sum_i \langle \varepsilon_i, u_i \rangle = \frac{2}{3}\sum_i \delta_i + \frac{2}{3}\sum_i e_i^2$ , by (7.36). Each facet of  $P(t)$  is the solution set, in its hyperplane, of a system of linear inequalities whose coefficients depend smoothly on  $t$ , and at  $t = 0$  its centre satisfies every inequality with a margin of at least  $\frac{1}{2}$ ; shrinking a solution set towards such an interior point by the factor  $1 - \lambda$  keeps it feasible after a perturbation of size  $O(\lambda)$ , so the facet moves by  $O(\|\varepsilon\| + \max_i \delta_i)$  in the Hausdorff metric, uniformly for  $t \leq 1$ , and so do  $A_i$ ,  $M_i$  and  $|w_i|$ . Integrating the derivative gives (b).

(c) *The hull costs only a fourth-order amount:*  $\text{vol}(V(Y) \setminus K(Y)) \leq C(\|\varepsilon\| + \max_i \delta_i)^4$ . For  $x \in V(Y)$ ,  $\langle x, u_i \rangle \leq h_i + |x|e_i$ , so  $V(Y) \subseteq (1 + \theta)Q$  with  $\theta = O(\|\varepsilon\| + \max_i \delta_i)$ . The points  $4w_i/d_i$  spanning  $K(Y)$  lie within  $\delta_i + 2e_i$  of  $2u_i$ , and the convex hull of the  $2u_i$  has inradius  $\sqrt{2}$ , so  $K(Y) \supseteq (1 - \theta')\text{conv}\{2u_i\}$  with  $\theta'$  of the same order.  $Q$  lies in  $\text{conv}\{2u_i\}$  and meets its boundary only at its 24 vertices, each of which is the unique point of  $Q$  on the supporting hyperplane there, because  $v = r + r'$  is a positive combination of the six roots at  $v$  and so lies in the interior of the normal cone of  $Q$  at  $v$ . Hence  $(1 + \theta)Q \setminus (1 - \theta')\text{conv}\{2u_i\}$  consists of 24 caps of height  $O(\theta + \theta')$  cut from the vertex cones of  $Q$ , whose volume is  $O((\theta + \theta')^4)$ .

By Proposition 2.13, (b), (c) and (a),

$$\text{vol}(V_c) \geq \text{vol}(V(Y) \cap K(Y)) \geq 8 + \frac{2}{3} \sum_i \delta_i - C' \left( \sum_i \delta_i \right)^2,$$

which is at least 8 once  $\sum_i \delta_i \leq 2/(3C')$ , with equality only if every  $\delta_i = 0$ . Then  $\kappa = 0$ , so  $\varepsilon = 0$  by (a),  $Y = \sqrt{2}R$ , and  $V_c \supseteq V(Y) \cap K(Y) = Q$  with  $\text{vol}(V_c) = 8$ , so  $V_c = Q$ . Choosing  $\eta$  and then  $\varepsilon$  in Step 1 small enough for (a), (b) and (c) gives  $\varepsilon_0$ .  $\blacksquare$

*What is left, and the evidence.* Together with Proposition 2.10, the results above confine Conjecture 1.6 to a centre  $c$  with all of the following: at least 24 other centres within  $\sqrt{6}$ ; at least one of them at a distance between  $2 + \varepsilon_0$  and  $\sqrt{6}$ ;  $\Phi \leq 8$  and  $\Psi \leq 8$ ; and contacts that do not contain one orthogonal pair of roots at each vertex of a 24-cell. Near the root system the inversion hull is sharp to first order: `near_contact_probe.py` finds  $\text{vol}(V(Y) \cap K(Y)) - 8$  and  $\text{vol}(V_c) - 8$  equal to  $\frac{2}{3} \sum \delta_i$  to first order on the root system pushed out, and over 200 minimisations of the volume near the root system with one centre held out,  $\|\varepsilon\|/\kappa$  at most 0.0003 (estimate (a) allows 2.71) and the quantity  $(\text{vol}(V_c) - 8 - \frac{2}{3} \sum \delta_i)/(\|\varepsilon\|^2 + \sum \delta_i^2)$  at least  $-\frac{1}{2}$ , the value on the deleted root's axis. Where the pair terms of Lemma 2.15 fail is a question for numerics too: `truncated_search.py` minimises the right side of Lemma 2.15 over configurations of exactly  $M$  centres within  $\sqrt{6}$ , from twelve starts each. For  $M = 24$  every run ends at the root system, at 7.906940...; for  $M = 25, 26$  and 27 every local minimum it finds lies above 8.26. If that is the whole truth, the pair terms fail only at twenty-four centres near the root system, and twenty-five or more could be settled as in Theorem 2.17, with a certificate for each count up to the largest that fits within  $\sqrt{6}$ . Closing what is left would take an explicit  $\varepsilon_0$ , which needs a quantitative form of Theorem 7.25, and, at intermediate distances, a lower bound for the part of the cell outside  $B(\sqrt{3/2})$ , which the pair terms of Lemma 2.15 do not see.

### 3. THE COMBINATORIAL CELL REGISTER

**3.1. Cell types under the Weyl group.** At the all-contact corner, a packing-valid local configuration is determined, up to the action of  $W(D_4)$ , by which subset of  $R$  is “active”, realised as an actual contact direction, and, for the single direction (if any) permitted to deviate from an exact root by Theorem 1.1, by the deviation parameters  $(\varphi, s)$ . Enumerating the combinatorially distinct active-direction patterns that can occur as facet sets of a bounded, packing-valid local Voronoi region, and quotienting by  $W(D_4)$ , produces exactly 176 orbit representatives, which we call *cells* and index  $\mathfrak{k}_1, \dots, \mathfrak{k}_{176}$ .

This count and the explicit orbit representatives are obtained by an exact combinatorial enumeration (not by sampling): the active sets that can occur are exactly the vertex-figure patterns of  $\mathcal{V}$  restricted to a single deviating cap direction, and  $W(D_4)$ 's order-192 action

partitions these into orbits whose sizes are computed exactly from stabiliser subgroups.<sup>8</sup> The complete pivot data for all 176 cells is tabulated in Appendix B; here we describe the structure that makes the count tractable.

**Lemma 3.1** (Four denominator classes). *Every cell  $\mathfrak{k}_k$  falls into exactly one of four denominator classes, according to the reference constant  $N_k \in \{192, 96, 64, 48\}$  arising from the volume of the simplex spanned by the cell’s boundary structure at the origin. The class is determined entirely by the stabiliser (in  $W(D_4)$ ) of the cell’s active-direction pattern.*

The four classes, written a, b, c, d below, account for all 176 cells, with multiplicities recorded in Appendix B. Class a (reference constant 192) consists of cells whose active pattern has trivial stabiliser; class d (reference constant 48) consists of the most symmetric cells, whose active pattern is fixed by a stabiliser of order 4.

**3.2. Cramer’s vertex formula and the volume-defect polynomial.** For a cell  $\mathfrak{k}_k$  with a single deviating direction at parameters  $(\varphi, s)$ , every vertex of the corresponding local Voronoi region that involves the deviating cap is the unique solution of 4 linear equations (the deviating cap’s own supporting hyperplane together with 3 of the cell’s other active facets), solved exactly by Cramer’s rule over the field generated by  $\sqrt{2}$  and the trigonometric functions of  $\varphi, s$ . Collecting these vertex solutions into a simplicial decomposition of the local region and summing signed simplex volumes produces a single rational function

$$\Delta_k(\varphi, s) = \frac{N_k(\varphi, s)}{D_k(\varphi, s)}$$

in the trigonometric functions of  $\varphi$  and  $s$  (equivalently, a rational function of the Weierstrass variables  $u = \tan(\varphi/2)$ ,  $v = \tan(s/2)$  once the substitution is made), which we call the cell’s *volume-defect polynomial*. Sections 4 and 5 give two independent exact arguments that  $\Delta_k \geq 0$  throughout the packing-valid domain of  $(\varphi, s)$ , for every one of the 176 cells.

*Example 3.2* (Cell  $\mathfrak{k}_1$ ). The most symmetric class-a cell has reference constant  $N_1 = 192$  and, at the root-aligned point  $\varphi = 0$ , volume-defect value  $\Delta_1(0, \cdot) = 0$  exactly (the equality case of Theorem 1.1). Its full Cramer vertex data is given in Appendix B, and its explicit  $\Delta_1(\varphi, s)$  is worked out in full in Section 5.

### 3.3. Degree bound and the reference normalisation.

**Lemma 3.3** (Degree bound). *For every cell  $\mathfrak{k}_k$ , the numerator  $N_k$  and denominator  $D_k$  of  $\Delta_k$ , expressed as polynomials in the Weierstrass variables  $u, v$ , have total degree at most 12 in each variable separately. This bound follows from the fact that at most 6 of the cell’s boundary facets can be simultaneously active at any vertex involving the deviating cap (a consequence of the 4-dimensional ambient space and the combinatorics of  $R$ ), each contributing at most a linear factor in  $u$  or  $v$  once Cramer’s rule is expanded.*

This degree bound is what makes an exact Bernstein-basis sign certificate (Section 5) a finite computation: a polynomial of bounded degree has finitely many Bernstein coefficients over any rectangular domain, and checking the sign of each one is an exact rational (or  $\mathbb{Q}(\sqrt{2})$ -rational) comparison.

<sup>8</sup>The 176 is not a “magic number” but a consequence of  $D_4$ ’s exceptional symmetry: the Voronoi cell of  $D_4$  is the 24-cell, which has 24 vertices and 24 facets, producing a richer combinatorial structure than the analogous objects for  $A_n$  or  $D_n$ ,  $n \neq 4$ . For  $D_3 \cong A_3$  the analogous count is 6 orbit types; for  $D_5$  the count exceeds  $10^3$ .

## 4. STAGE I: THE RADIAL ESTIMATE

**4.1. The Hessian at the all-contact corner.** At the all-contact corner, small deviations of a single active direction away from an exact root can be analysed to second order by a Hessian whose entries are determined entirely by the Gram-value structure of Lemma 2.1.

**Definition 4.1** (Cell Hessian). For a cell  $\mathfrak{k}_k$  with active roots  $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ , the *cell Hessian*  $H_k \in \mathbb{Q}^{3 \times 3}$  (acting on the 3-dimensional tangent space orthogonal to the deviating root) has entries determined by the signed Gram-adjacency pattern of  $\alpha_1, \dots, \alpha_n$ : a uniform formula assigns each pair of active roots a contribution of  $+\frac{1}{3}$ , 0, or  $-\frac{1}{3}$  according to whether their Gram value is  $\frac{1}{2}$ , 0, or  $-\frac{1}{2}$  respectively (Lemma 2.1), summed over all active pairs sharing the tangent direction in question.

**Lemma 4.2** (Uniform structure of  $H_k$ ).  $H_k$  depends only on the Gram-adjacency pattern of  $\mathfrak{k}_k$ 's active roots, not on any other feature of the cell. In particular,  $H_k$  is independent of the choice of orbit representative within a  $W(D_4)$  orbit, and is computed once per orbit, not once per cell.

**4.2. Exact  $LDL^\top$  decomposition and Sylvester's criterion.**

**Proposition 4.3** (Exact positivity via  $LDL^\top$ ). For every one of the 176 cells  $\mathfrak{k}_k$ , the Hessian  $H_k$  admits an exact rational  $LDL^\top$  decomposition  $H_k = L_k D_k L_k^\top$  with  $L_k$  unit lower-triangular and  $D_k$  diagonal with strictly positive rational entries. Consequently  $H_k$  is positive definite, and by Sylvester's criterion every leading principal minor of  $H_k$  is strictly positive.

This is verified once per denominator class (a, b, c, d of Lemma 3.1) by exact rational arithmetic, `Fractions.Fraction` in one independent implementation and `sympy`'s rational number type in a second, completely separate code path, agreeing to the last digit on every pivot, and the result is inherited by every cell in the class.<sup>9</sup> The complete  $LDL^\top$  pivot table for all 176 cells is given in Appendix B.

**4.3. Minimum eigenvalue bounds.**

**Lemma 4.4** (Minimum eigenvalue). For every cell  $\mathfrak{k}_k$ , the minimum eigenvalue of  $H_k$  satisfies  $\lambda_{\min}(H_k) \geq \frac{1}{12}$ , with equality attained exactly on denominator class d (the most symmetric cells).

Combined with an exact quadratic lower bound on  $\Delta_k(\varphi, \cdot)$  near  $\varphi = 0$  coming from  $H_k$ , this gives an exact rational region around the root-aligned point on which  $\Delta_k \geq 0$  is certified purely from the second-order data, the *radial* part of the two-stage argument. It does not, by itself, cover the entire packing-valid range of  $\varphi$ ; the remainder is the subject of Section 5.

**4.4. Worked example: cell  $\mathfrak{k}_1$ .** For the class-a cell  $\mathfrak{k}_1$  (introduced in Example 3.2), which has 9 acute ( $+\frac{1}{2}$ ) and 2 obtuse ( $-\frac{1}{2}$ ) active roots beyond the deviating direction itself (Appendix B), the exact  $LDL^\top$  decomposition of  $H_1$  has six pivots, with minimum pivot value  $3936/77 \approx 51.13$  and reference constant  $\mathcal{N}_1 = 192$ ; the corresponding chamber-level

<sup>9</sup>The  $LDL^\top$  decomposition is computed symbolically, not numerically, precisely because the cell Hessians  $H_k$  have entries in  $\mathbb{Q}$ : a numerical float would introduce rounding, making the “positivity” of the pivots a floating-point artefact and not a mathematical fact. The choice of `Fractions.Fraction` (arbitrary-precision exact rationals in Python's standard library) means the computation is slow but its output is exact. The two independent implementations (one using only built-in Python, one using `sympy`) serve as cross-validation, ruling out implementation bugs in either single path.

positivity check (Appendix B) passes exactly, by exact rational arithmetic, with no floating-point step. We record the actual computed pivot value here, and not a simplified illustrative number, precisely because  $\mathfrak{k}_1$ 's Hessian is not the isolated single-direction case (that simpler case,  $H = \frac{2}{3}I_3$ , is recorded separately in Lemma 15.2 for the multi-direction setting, a related but distinct construction that does not include the cross-terms from a cell's other active, non-deviating roots). The exact computation for every other cell follows the identical procedure and is tabulated in full in Appendix B.

## 5. STAGE II: THE ANGULAR ESTIMATE

**5.1. Setup and reduction to a single function.** The radial estimate of Section 4 certifies  $\Delta_k \geq 0$  only in a neighbourhood of the root-aligned point  $\varphi = 0$ ; establishing it for the full packing-valid range of  $\varphi$  (up to the point where the combinatorial cell type itself changes) requires a separate, angular argument. Writing  $g_k(\varphi) = \Delta_k(\varphi, s)$  for a fixed root-aligned arc position  $s$ , the radial estimate gives an exact lower bound on  $g_k''(0)$ ; STAGE II supplies an exact upper bound on  $|g_k'''(\varphi)|$  throughout the relevant range, from which a rigorous Taylor remainder bound follows.

**Lemma 5.1** (Taylor coefficient bound). *For every cell  $\mathfrak{k}_k$  and every root-aligned arc, the third derivative of  $g_k$  satisfies an explicit bound  $|g_k'''(\varphi)| \leq M_k$  for an exact rational constant  $M_k$ , computed directly from the cell's volume-defect polynomial by exact differentiation and an exact rational bound on the resulting expression over the relevant  $\varphi$ -range.*

**Proposition 5.2** (Positivity on the full range). *Combining Lemmas 4.4 and 5.1 by a third-order Taylor expansion with exact remainder gives  $g_k(\varphi) \geq 0$  throughout the entire packing-valid range of  $\varphi$  for the arc in question, whenever  $3M_k < \lambda_{\min}(H_k) \cdot c_k$ , where  $c_k$  is an explicit constant depending only on the range length of the arc, a condition verified to hold, by exact rational computation, for every one of the 176 cells and every root-aligned arc.*

Where the direct Taylor bound of Proposition 5.2 is not by itself strong enough on some sub-range (this happens on a small number of cells with an unusually long root-aligned arc), an independent monotonicity argument, tracking the sign of  $g_k'$  directly via the exact breakpoint structure of the cell's active-facet pattern, closes the remaining sub-range exactly. Both arguments are exact over  $\mathbb{Q}$ ; no numerical tolerance is introduced at any point. Figure 9 shows where each of the two takes over.

**5.2. Exact breakpoint values.** The root-aligned arcs themselves are finite: a deviating direction  $\varphi \mapsto \cos \varphi \alpha_0 / \sqrt{2} + \sin \varphi \eta_0$  (for a fixed root-aligned direction  $\eta_0$ ) changes the active-facet pattern of the cell only at finitely many exact values of  $\varphi$ , each characterised as the point where some other root direction achieves the same support-hyperplane value as the deviating cap. These breakpoints are located exactly (not numerically) by solving  $\langle \beta, \text{cap direction} \rangle = 1$  for each candidate root  $\beta$ , an equation that is polynomial once  $\varphi$  is eliminated via the Weierstrass substitution, hence solvable in exact radicals.

**5.3. Worked example: cell  $\mathfrak{k}_1$  on its own root-aligned arc.** Continuing Example 3.2, the volume-defect function for  $\mathfrak{k}_1$  restricted to its root-aligned arc is an explicit rational function of  $\tan(\varphi/2)$  with denominator strictly positive throughout the packing-valid range and numerator vanishing to exactly second order at  $\varphi = 0$  and strictly positive elsewhere on the range, verified both by the exact Taylor argument above and, independently, by an exact sign check on the numerator's Bernstein coefficients over the range, the same technique

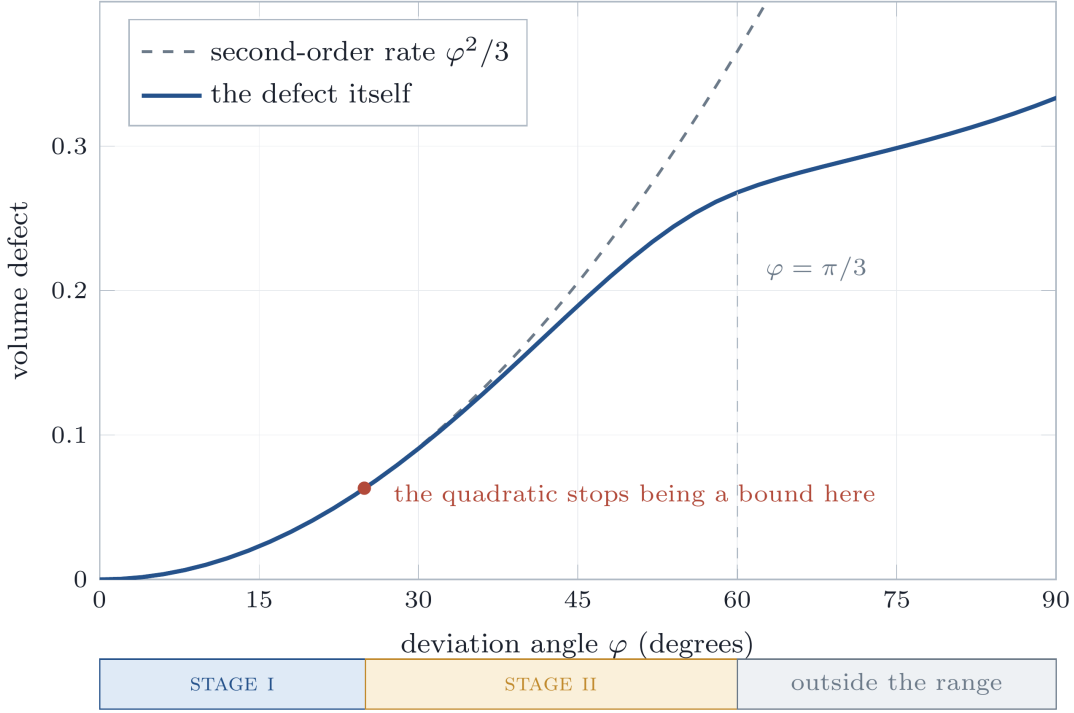


FIGURE 9. The computed volume defect for a root-aligned deviation, against the second-order rate  $\varphi^2/3$  that STAGE I supplies. The two agree to plotting accuracy up to about  $25^\circ$  and part company after it, the quadratic overtaking the defect and ceasing to be a lower bound; this is why the second-order argument alone does not reach the end of the range and STAGE II is needed. The strip beneath marks which argument covers which part of the range. The packing-valid range for a single deviation ends at  $\varphi = \pi/3$ , where a second root becomes the nearest one.

used at much greater length in Sections 10 and 12 for the harder, non-root-aligned deviation directions.<sup>10</sup>

<sup>10</sup>The certificate rests on one property of the Bernstein basis, which we state here so that no later section has to appeal outside the paper. Write a polynomial of degree  $n$  on  $[0, 1]$  as  $p(t) = \sum_{i=0}^n b_i \binom{n}{i} t^i (1-t)^{n-i}$ ; the  $b_i$  are its Bernstein coefficients, and they are obtained from the monomial coefficients by an exactly invertible integer linear change of basis, so if  $p$  has rational coefficients then so do the  $b_i$ , with no rounding anywhere. Because the basis functions are nonnegative on  $[0, 1]$  and sum to 1,  $\min_i b_i \leq p(t) \leq \max_i b_i$  there; in particular  $b_i \geq 0$  for all  $i$  implies  $p \geq 0$  on  $[0, 1]$ . The converse fails, which is the whole reason a subdivision is needed: a polynomial can be positive on  $[0, 1]$  and still have a negative Bernstein coefficient. What rescues the method is that the bound is asymptotically tight under subdivision. Splitting  $[0, 1]$  at a rational point and re-expanding on each half, which de Casteljau's algorithm does in exact arithmetic, produces coefficients whose minimum converges to  $\min p$  quadratically in the width of the subinterval [Far12]. So a polynomial that is strictly positive on a compact interval is certified after finitely many bisections, and a polynomial with a zero in the interior is never certified, however fine the subdivision. That dichotomy is exactly the one recorded in the algorithm of Appendix F, and it is why the regions in Section 10 that contain a zero of the defect, on the boundary or inside, carry an excluded margin rather than a zero-margin certificate. In several variables the same statements hold on a box, with the tensor-product basis and bisection along one coordinate at a time. Appendix I works this mechanism through on a small polynomial chosen to also

**5.4. Proof of Theorem 1.1.** Combining Section 4.1 and Proposition 5.2 with the four-layer reduction of Section 1.1: the statement concerns  $V_c^{\text{act}}$ , so the centres beyond  $2\sqrt{2}$  play no part; the root-aligned case (Section 2.5) handles undeviated configurations directly; radial reduction (Section 2.6) replaces every active distance by 2, which only shrinks the cell, and since the deviating direction replaces a root that no active neighbour takes, the result is the cell  $Q$  of at most twenty-three roots cut by one more half-space at unit distance, which Section 1.3 bounds for every direction of that half-space whether or not the configuration is a packing; and cell positivity, established here for every one of the 176 cells and every root-aligned deviation arc by the two independent exact arguments of STAGE I and STAGE II, gives  $\Delta_k \geq 0$  throughout. Summing the free-volume identity of Section 2 over the active cell type completes the proof of Theorem 1.1. ■

## 6. FROM THE LOCAL BOUND TO THE DENSITY BOUND

### 6.1. Periodic packings.

*Proof of Corollary 1.2.* For a periodic packing with fundamental domain  $\mathcal{F}$ , the density equals  $\text{vol}(B^4)/\bar{v}$ , where  $\bar{v}$  is the average Voronoi cell volume over  $\mathcal{F}$  and  $\text{vol}(B^4) = \pi^2/2$ . Every centre is shell-free, so by Theorem 1.5 every cell in  $\mathcal{F}$  has volume at least 8, so  $\bar{v} \geq 8$  and the density is at most  $(\pi^2/2)/8 = \pi^2/16$ . The  $D_4$  lattice attains  $\bar{v} = 8$  exactly (Lemma 2.3), so this bound is sharp. The same argument proves the first statement of Theorem 1.8. ■

**6.2. General packings and the compactness argument.** For a general packing  $\mathcal{C} \subset \mathbb{R}^4$  of unit balls, with no assumed translational symmetry, the density is the upper limiting density

$$\delta(\mathcal{C}) = \limsup_{R \rightarrow \infty} \frac{\text{vol}(B^4) \cdot \#\{c \in \mathcal{C} : |c| \leq R\}}{\text{vol}(B_R^4)}.$$

The passage from periodic packings to these uses a compactness argument that is standard in the subject, in the form given by Rogers [Rog64]; see also [Gru87] for the general theory of packing densities in this local, finite-ball formulation. We recall it, since everything the argument needs has already appeared above.

Topologise the space of packing configurations by local Hausdorff convergence on increasing balls:  $\mathcal{C}_n \rightarrow \mathcal{C}$  when, for every  $R$ , the points of  $\mathcal{C}_n$  within distance  $R$  of the origin converge as a finite point set to those of  $\mathcal{C}$ . This space is compact. The packing constraint bounds the number of centres in any fixed ball uniformly over all packings, so a diagonal argument over an increasing sequence of balls extracts a convergent subsequence from any sequence of packings. The finite-ball density functional is upper semicontinuous for this topology, since a configuration cannot gain centres in the limit, only lose them to infinity.

Given  $\mathcal{C}$  and  $\varepsilon > 0$ , take a periodic packing  $\mathcal{C}'$  agreeing with  $\mathcal{C}$  to within  $\varepsilon$  on the ball of radius  $1/\varepsilon$ . The Voronoi cells of  $\mathcal{C}'$  at centres well inside that ball differ from the corresponding cells of  $\mathcal{C}$  by an amount that tends to zero as the radius grows, because a cell is determined by the centres within twice its circumradius. If the bound  $\text{vol}(V_c) \geq 8$  holds at every centre of every periodic packing, it passes to the limit and bounds  $\delta(\mathcal{C})$  by  $\pi^2/16$ ; that is how the second statement of Theorem 1.8 follows from Theorem 1.5 and Conjecture 1.6. A hypothesis on the centres of  $\mathcal{C}$  alone does not pass to the approximants, whose centres near the boundary of the ball can have other neighbourhoods, which is why the unconditional density statements above are made for periodic packings. The argument

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exhibit the boundary-zero difficulty just described, for a reader who wants to see the whole computation before meeting it at the scale of an actual cell.

is the mechanism, with the appropriate substitutions, by which every sphere-packing upper bound from Blichfeldt [Bli29] onward turns a local statement about individual cells into a global statement about density.

**6.3. The general case, and what it rests on.** Theorem 1.5 rests on the bound for contact configurations, which is Corollary 7.39 with Theorem 7.40, and on Section 2.7, which carries that bound to the Voronoi cell of a packing when every active neighbour touches the centre or the distances pass the criterion of Proposition 2.10. The free-volume identity behind everything here, stated as Theorem 7.1 in Section 7, holds for every packing configuration; what Theorem 1.1 adds to it is the sign, and Conjecture 1.4 is the assertion that the sign survives when several directions deviate at once at a centre whose twenty-four active neighbours all touch it. That is settled in Corollary 7.31 by the classification of Theorem 7.25, and the case of 23 contacts, Theorem 7.40, is proved in Sections 7.11 and 7.12; beyond the arguments of this paper these rest on the one verified kernel of Section 7.5.3 alone, and say so wherever they appear. What remains of the twenty-four-cell conjecture is Conjecture 1.6, which Section 2.8 narrows to centres crowded by neighbours at intermediate distances, and the density bound  $\Delta_4 = \pi^2/16$  for all packings waits on it.

**6.4. Remark: what a proof that did not use the classification would need.** Theorem 1.5 is proved above, and it is proved using the classification of Theorem 7.25 at the configurations of twenty-four contacts, that is, using a semidefinite certificate. This subsection is about the other thing, a proof of the same statement that uses no classification and no certificate, and we set out what such a proof would have to supply, since it is not what the present one supplies.

It would require  $\Delta_k \geq 0$ , equivalently the volume comparison of Conjecture 1.4, uniformly over *every* packing-valid configuration of active directions, of *every* size  $m \geq 1$ , with *every* deviating direction ranging over its full 3-sphere of possibilities, not merely the single-direction deviations covered by Theorem 1.1. Sections 15 to 18 explain in detail why the natural perturbative (Hessian-based) technique that succeeds in the root-aligned case cannot, by itself, resolve the general multi-direction case at every configuration, not because the relevant computations are difficult, but because the technique's own governing quantity is exactly zero at specific configurations that are themselves well within the scope the conjecture must cover. Such a proof will therefore need higher-order information at those configurations, and locating that obstruction precisely is a contribution of this paper separate from Theorem 1.1.

## 7. THE GENERAL LOCAL PROBLEM

**7.1. The free-volume identity in general.** Throughout this subsection the configuration is at the all-contact corner with a full active set: the active directions are  $w_1, \dots, w_{24}$ , one for each root, with  $w_i = u_i$  except on a subset  $D$  of deviating indices, where  $w_j = \cos \varphi_j u_j + \sin \varphi_j \eta_j$ . A configuration with a proper active subset is covered a fortiori, since dropping a constraint only enlarges the cell. Write

$$V_c = \bigcap_{i=1}^{24} \{\langle x, w_i \rangle \leq 1\}, \quad C_j = \mathcal{V} \cap \{\langle x, w_j \rangle \geq 1\}, \quad E_j = V_c \cap \{\langle x, u_j \rangle \geq 1\},$$

for  $j \in D$ . Thus  $C_j$  is what the  $j$ -th half-space cuts off the reference cell as it tilts, and  $E_j$  is what the same motion frees beyond the  $j$ -th facet.

**Theorem 7.1** (Free-volume identity). *For every  $D$  and every choice of tilts and deviation directions,*

$$\text{vol}(V_c) = \text{vol}(\mathcal{V}) - \text{vol}\left(\bigcup_{j \in D} C_j\right) + \text{vol}\left(\bigcup_{j \in D} E_j\right). \quad (7.1)$$

*The identity is exact, with no approximation of any kind. In particular  $\text{vol}(V_c) \geq 8$  if and only if  $\text{vol}(\bigcup_j E_j) \geq \text{vol}(\bigcup_j C_j)$ .*

*Proof.*  $V_c$  and  $\mathcal{V}$  are intersections of half-spaces that agree in the  $24 - |D|$  constraints indexed outside  $D$ . A point of  $\mathcal{V}$  fails to lie in  $V_c$  exactly when  $\langle x, w_j \rangle > 1$  for some  $j \in D$ , so  $\mathcal{V} \setminus V_c = \bigcup_{j \in D} C_j$  up to a null set; a point of  $V_c$  fails to lie in  $\mathcal{V}$  exactly when  $\langle x, u_j \rangle > 1$  for some  $j \in D$ , so  $V_c \setminus \mathcal{V} = \bigcup_{j \in D} E_j$ . Both sets are finite unions of polytopes, hence measurable, and  $\text{vol}(V_c) = \text{vol}(\mathcal{V}) - \text{vol}(\mathcal{V} \setminus V_c) + \text{vol}(V_c \setminus \mathcal{V})$ .  $\blacksquare$

For  $|D| \leq 1$  the two unions are single sets and (7.1) reduces to  $\text{vol}(V_c) = 8 - \text{vol}(C) + \text{vol}(E)$ , which is the form used throughout Sections 4, 5, 10 and 12 and again, in the cap form of Proposition 13.2, in Section 13. For  $|D| \geq 2$  the unions matter, and we are exact here about how.

*Remark 7.2* (The sums are not the unions). Neither family is disjoint once  $|D| \geq 2$ . The sets  $E_j$  overlap whenever two deviating roots are adjacent, that is when  $\langle u_j, u_k \rangle = \frac{1}{2}$ : the pyramid  $\text{Pyr}_j$  of Lemma 13.1 has one of its eight lateral facets on the hyperplane  $\{\langle x, u_k \rangle = 1\}$ , so removing the constraint at  $u_k$  opens  $E_j$  out past that facet and into the region where  $\langle x, u_k \rangle > 1$ , which is where  $E_k$  lives. At an adjacent pair with both tilts  $30^\circ$  the overlap has volume  $0.0137\dots$ , against a defect of  $0.1845\dots$ , so it is not a small correction. The sets  $C_j$  can also overlap, though only at large tilts and only for adjacent pairs; the largest overlap we have found in a search over random packing-valid configurations of two to four deviating directions is  $1.6 \times 10^{-3}$ .

Consequently the quantity  $\sum_j \text{vol}(E_j) - \sum_j \text{vol}(C_j)$  is *not* equal to  $\text{vol}(V_c) - 8$ , and its sign is neither implied by nor implies the cell bound: in samples it runs both above the defect (for instance  $0.1798$  against  $0.1570$ ) and below it (for instance  $0.20945$  against  $0.20971$ ). Only the union form (7.1) is an identity, and it is the union form that Conjecture 1.4 asserts.

*Remark 7.3* (Why the single-deviation theorem does not reduce the general case). The natural hope is to split the comparison of Conjecture 1.4 into  $m$  separate ones, each settled by Theorem 13.7. Two things stand in the way, and both are structural.

The first is the overlapping just described. It is not fatal on its own: Bonferroni's inequality gives

$$\text{vol}\left(\bigcup_j E_j\right) \geq \sum_j \text{vol}(E_j) - \sum_{j < k} \text{vol}(E_j \cap E_k),$$

and together with  $\text{vol}(\bigcup_j C_j) \leq \sum_j \text{vol}(C_j)$  this reduces the conjecture to

$$\sum_j (\text{vol}(E_j) - \text{vol}(C_j)) \geq \sum_{j < k} \text{vol}(E_j \cap E_k). \quad (7.2)$$

The overlaps on the right are second order in the tilts, the same order as each term on the left, so (7.2) is not obviously out of reach.

The second obstruction is what closes the route. The  $E_j$  appearing in (7.2) are not the sets the single-deviation theorem controls. With one root deviating,  $E_j$  is cut out by the other twenty-three root half-spaces, and Lemma 13.1 identifies it inside the pyramid  $\text{Pyr}_j$ . With  $m$  roots deviating it is cut out by the  $24 - m$  fixed half-spaces together with the  $m - 1$  other tilted ones, and the exchange is not benign. The eight lateral facets of  $\text{Pyr}_j$  lie on

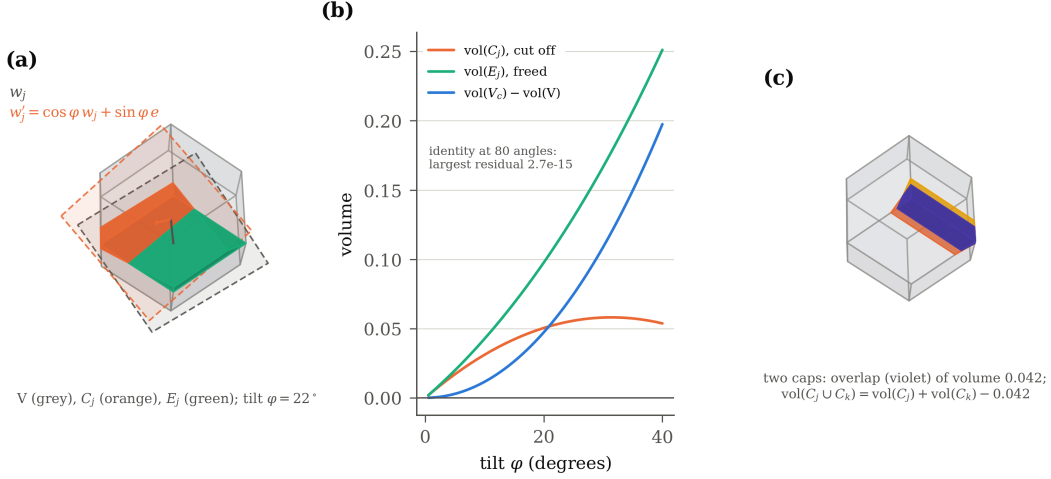


FIGURE 10. The free-volume identity of Theorem 7.1, drawn on real geometry. Because the identity holds in every dimension, it is drawn in three, where its pieces can be seen: the reference cell is the Voronoi cell of the face-centred cubic contacts, the rhombic dodecahedron  $\{x : \langle x, w_i \rangle \leq 1\}$  over the twelve directions  $(\pm 1, \pm 1, 0)/\sqrt{2}$  and permutations. (a) One direction  $w_j$  tilted by  $22^\circ$  to  $w'_j$ ; the cap  $C_j$  is what the tilted half-space cuts off the cell and  $E_j$  is what the motion frees beyond the old facet, both computed as half-space intersections and drawn to scale. (b) The two volumes and their difference against the tilt, each piece computed as a polytope of its own; the identity holds at every angle to the precision of the arithmetic. (c) Two directions tilted towards each other: the caps overlap, so for  $m \geq 2$  the corrections are volumes of unions and not sums of volumes (Remark 7.2).

the hyperplanes of the eight roots at  $60^\circ$  to  $\alpha_j$ , so for an adjacent  $k$  the pyramid meets  $\{\langle x, w_k \rangle = 1\}$  in a facet and not transversally: dropping that constraint opens  $E_j$  out, and reimposing it in the tilted position  $\{\langle x, w_k \rangle \leq 1\}$  slices back in along a whole face. The two effects do not cancel, and neither dominates. In a search over packing-valid configurations with  $2 \leq m \leq 6$  and tilts up to 0.9 we found  $\text{vol}(E_j)$  both exceeding its single-deviation value and falling short of it, in the latter case by more than 0.1, against defects of order 0.3. The search is reproduced by `multi_cap_reformulation.py` in the supplement. So Theorem 13.7 gives no bound at all on  $\text{vol}(E_j) - \text{vol}(C_j)$  once  $m \geq 2$ , and (7.2) cannot be reached term by term.

What would suffice is a replacement for  $\text{Pyr}_j$ : a body, depending on the whole configuration rather than on one root, that contains each  $E_j$ , is cut by the tilted half-spaces in a controlled way, and for which the analogue of Theorem 13.6 can be proved. We do not have one.

**Corollary 7.4.** *In the setting above with  $|D| = m \geq 2$ ,  $\text{vol}(V_c) \geq 8$  if and only if*

$$\text{vol}\left(\bigcup_{j \in D} E_j\right) \geq \text{vol}\left(\bigcup_{j \in D} C_j\right).$$

Figure 10 draws the three terms of (7.1).

**7.2. A fixed-body form of the multi-direction problem.** Remark 7.3 says what does not work. We record what the cap reformulation of Section 13 does become for  $m \geq 2$ ,

since the answer is a single extremal problem about a single fixed polytope, exactly as in the case  $m = 1$ , and not the sum of  $m$  problems.

Let  $D$  be the set of deviating roots,  $|D| = m$ , and put

$$Q_D = \bigcap_{i \notin D} \{x \in \mathbb{R}^4 : \langle x, u_i \rangle \leq 1\}, \quad \text{Cap}_{Q_D}(w) = \{x \in Q_D : \langle x, w \rangle \geq 1\}.$$

For  $m = 1$  this is the polytope  $Q$  of Section 13.1.

**Proposition 7.5** (Multi-cap form). *Suppose  $Q_D$  is bounded. Then, for every choice of tilts and deviation directions,*

$$\text{vol}(V_c) = \text{vol}(Q_D) - \text{vol}\left(\bigcup_{j \in D} \text{Cap}_{Q_D}(w_j)\right), \quad (7.3)$$

and therefore  $\text{vol}(V_c) \geq 8$  if and only if

$$\text{vol}\left(\bigcup_{j \in D} \text{Cap}_{Q_D}(w_j)\right) \leq \text{vol}(Q_D) - 8. \quad (7.4)$$

The right-hand side depends only on  $D$ , not on the tilts, and equals  $\text{vol}(\bigcup_{j \in D} \text{Cap}_{Q_D}(u_j))$ , so (7.4) holds with equality at the undeviated configuration. It equals  $m/3$  exactly when no two roots of  $D$  are adjacent, and exceeds  $m/3$  otherwise.

*Proof.*  $V_c = Q_D \cap \bigcap_{j \in D} \{\langle x, w_j \rangle \leq 1\}$  by definition, so  $Q_D \setminus V_c = \bigcup_{j \in D} \text{Cap}_{Q_D}(w_j)$  up to a null set, which is (7.3). Taking  $w_j = u_j$  makes  $V_c = \mathcal{V}$  and gives  $\text{vol}(\bigcup_j \text{Cap}_{Q_D}(u_j)) = \text{vol}(Q_D) - 8$ . If no two roots of  $D$  are adjacent then  $\text{Cap}_{Q_D}(u_j) = \text{Pyr}_j$  for each  $j$ , since by the proof of Lemma 13.1 only the eight roots at  $60^\circ$  to  $\alpha_j$  are active on  $\{\langle x, u_j \rangle \geq 1\}$  and none of them lies in  $D$ ; those pyramids are pairwise disjoint, again by Lemma 13.1, so the union has volume  $m/3$ . If two roots of  $D$  are adjacent, the corresponding  $\text{Cap}_{Q_D}(u_j)$  strictly contains  $\text{Pyr}_j$ , since the constraint carrying one of its lateral facets has been removed. ■

So the multi-direction problem is again an extremal problem for caps of one fixed polytope, with all the parameters in the cutting half-spaces; what has changed from  $m = 1$  is that the quantity to bound is the volume of a *union* of  $m$  caps rather than of one cap, and Remark 7.3 shows the union cannot be traded for a sum. The inequality itself holds, by Corollary 7.31 and therefore by the classification of Theorem 7.25; what the present section and Section 15 to Section 19 are after is a proof of it that does not go through a classification, and we state that question as it stands.

**Open Problem 7.6** (Multi-cap inequality, by direct means). Prove (7.4) for every  $D$  with  $Q_D$  bounded and every packing-valid choice of  $w_j$ , with equality only at  $w_j = u_j$ , without appeal to the classification of the 24-point contact configurations.

The boundedness hypothesis is mild and its exact range is known.  $Q_D$  is bounded precisely when the  $24 - m$  undeviated root directions positively span  $\mathbb{R}^4$ , and an exhaustive enumeration over all packing-valid sets  $D$  gives the following: for  $m \leq 5$  it is bounded in every case, all 24, 276, 2024, 10626 and 42504 of them respectively; at  $m = 6$  it fails for exactly 24 of the 134596 packing-valid sets, the first in lexicographic order being the six roots  $e_1 \pm e_2$ ,  $e_1 \pm e_3$ ,  $e_1 \pm e_4$ , whose complement lies in the closed half-space  $\{x_1 \leq 0\}$  and therefore leaves  $Q_D$  unbounded in the direction  $e_1$ . Beyond  $m = 6$  unbounded cases become common, and the dense configurations of Section 15.4, at  $m$  in the high teens, are almost all of that kind. For those a bounded substitute for  $Q_D$  would have to come first; (7.4) as stated does not reach them.

**7.3. Polar and boundary forms of the same question.** The multi-cap form keeps a fixed body and moves the cutting half-spaces. There is a third way of writing the same question, which discards the reference cell altogether and states the problem directly in terms of the contact configuration. It costs nothing to set up, it makes the relation to the 24-cell conjecture explicit, and it gives the sharpest available statement of what a proof of Conjecture 1.4 cannot look like.

**Definition 7.7.** A *contact configuration* is a finite set  $W \subset S^3$  with  $\langle w, w' \rangle \leq \frac{1}{2}$  for all distinct  $w, w' \in W$ . Its *cell* is

$$V_c(W) = \{x \in \mathbb{R}^4 : \langle x, w \rangle \leq 1 \text{ for all } w \in W\}.$$

The condition is exactly packing validity.<sup>11</sup> Every local configuration considered in this paper, deviated or not, is a contact configuration, and  $V_c$  is its cell.

**Proposition 7.8** (Polar form). *For every contact configuration  $W$ ,*

$$V_c(W) = \text{conv}(W)^\circ, \quad (7.5)$$

*the polar dual of the convex hull of  $W$ . Consequently*

- (i)  $\text{vol}(V_c(W))$  depends on  $W$  only through  $\text{conv}(W)$ ;
- (ii)  $W \subseteq W'$  implies  $V_c(W') \subseteq V_c(W)$ , so enlarging a contact configuration never increases the volume of its cell;
- (iii)  $|W| \leq 24$ , by Musin's kissing bound [Mus08].

Conjecture 1.4 is equivalent to the assertion  $\text{vol}(\text{conv}(W)^\circ) \geq 8$  for every contact configuration whose cell is bounded, with equality only when  $W$  is the set of 24 root directions.

*Proof.* The polar of a compact convex set  $K$  containing the origin is  $K^\circ = \{x : \langle x, p \rangle \leq 1 \text{ for all } p \in K\}$ , and a linear functional attains its maximum over  $\text{conv}(W)$  at a point of  $W$ , so  $\text{conv}(W)^\circ$  is cut out by the constraints indexed by  $W$  alone, which is (7.5). Item (i) is immediate from (7.5) and (ii) from the definition. Item (iii) is the kissing-number theorem in dimension four, since a contact configuration is precisely a set of directions to mutually non-overlapping unit balls touching a fixed one. The final sentence is Theorem 7.1 together with the observation that a packing configuration at the all-contact corner, every active neighbour at distance exactly 2, is a contact configuration of exactly the shape in Definition 7.7: two centres at distance 2 from  $c$  and at least 2 apart make an angle of at least  $60^\circ$  at  $c$ . A configuration obtained by pulling neighbours that do not touch in to distance 2 need not be one (Section 2.7). ■

**Corollary 7.9** (A worst configuration exists, and may be taken maximal). *The infimum of  $\text{vol}(V_c(W))$  over all contact configurations with bounded cell is attained, and it is attained at a configuration to which no further direction can be added.*

*Proof.* By Proposition 7.8(iii) a contact configuration has at most 24 elements, so for each  $m \leq 24$  the set  $\mathcal{C}_m = \{(w_1, \dots, w_m) \in (S^3)^m : \langle w_i, w_j \rangle \leq \frac{1}{2} \text{ (} i \neq j \text{)}\}$  is a closed subset of a compact space, hence compact, and its points have pairwise distinct entries. The volume functional  $f(W) = \text{vol}(V_c(W)) \in (0, \infty]$  is lower semicontinuous on  $\mathcal{C}_m$ : if  $S$  is a compact

<sup>11</sup>If unit balls centred at  $2w$  and  $2w'$  are to have disjoint interiors then  $|2w - 2w'| \geq 2$ , that is  $|w - w'| \geq 1$ , that is  $2 - 2\langle w, w' \rangle \geq 1$ ; so the pairwise bound  $\frac{1}{2}$  is the whole of the packing constraint on the directions, and the bisector of the segment from 0 to  $2w$  is the hyperplane  $\langle x, w \rangle = 1$ , which is the constraint defining  $V_c(W)$ . The directions of the  $D_4$  roots,  $u_i = \alpha_i/\sqrt{2}$ , form a contact configuration with the pairwise inner products taking the four values  $\pm\frac{1}{2}$ , 0 and  $-1$ , so the bound is attained but never exceeded; this is the  $g$ -valued statement of Lemma 2.1.

subset of the interior of  $V_c(W)$  then  $\langle x, w \rangle < 1$  on  $S \times W$ , hence the same holds for every configuration near  $W$ , so  $S \subseteq V_c(W')$  and  $f(W') \geq \text{vol}(S)$ ; taking the supremum over such  $S$  and using inner regularity gives  $\liminf_{W' \rightarrow W} f(W') \geq f(W)$ . A lower semicontinuous function on a compact set attains its infimum, so  $f$  attains its infimum on each  $\mathcal{C}_m$ , and therefore on the union of the finitely many  $\mathcal{C}_m$  with  $m \leq 24$ . If a minimiser  $W^*$  admits an additional admissible direction, extend it, repeatedly if necessary; the process stops after at most  $24 - |W^*|$  steps and each step leaves the volume no larger by Proposition 7.8(ii), so the resulting maximal configuration is again a minimiser.  $\blacksquare$

So the conjecture is a statement about maximal contact configurations only, and there is a genuine extremal configuration to identify. That is as far as soft arguments take it: what they leave is the question of whether the maximal configurations of 24 directions are the root system alone, which is the rigidity half of the kissing problem in dimension four and is settled in Theorem 7.25 below, by the semidefinite certificate of Theorem 7.30.

The second form replaces the volume by the boundary measure.

**Lemma 7.10** (Boundary form). *Let  $W$  be a contact configuration with bounded cell, and let  $F_w = V_c(W) \cap \{\langle x, w \rangle = 1\}$  for  $w \in W$ . Then*

$$\text{vol}(V_c(W)) = \frac{1}{4} \sum_{w \in W} \text{vol}_3(F_w) = \frac{1}{4} \text{area}(\partial V_c(W)), \quad (7.6)$$

so  $\text{vol}(V_c(W)) \geq 8$  if and only if the boundary 3-volume of the cell is at least 32.

*Proof.* The origin is interior to  $V_c(W)$ , so the cell is the union over its facets of the pyramids  $\text{conv}(\{0\} \cup F)$ , with pairwise disjoint interiors. Each facet lies on a hyperplane  $\langle x, w \rangle = 1$  with  $|w| = 1$ , at distance exactly 1 from the origin, so the corresponding pyramid has 4-volume  $\frac{1}{4} \text{vol}_3(F)$ . Summing gives (7.6); the facets of  $V_c(W)$  are among the  $F_w$ , and any  $F_w$  that is not a facet is lower-dimensional and contributes nothing.  $\blacksquare$

At the root configuration all 24 sets  $F_w$  are facets, each a regular octahedron of 3-volume  $\frac{4}{3}$ , and (7.6) reads  $8 = \frac{1}{4} \cdot 32$ . Why every facet is an octahedron of exactly that size is worth isolating, since the same computation is what limits the method.

**Lemma 7.11** (Facet shape and inball). *Let  $W$  be a contact configuration,  $w \in W$ , and write  $g_{ww'} = \langle w, w' \rangle$  and  $\widehat{w'} = (w' - g_{ww'}w)/\sqrt{1 - g_{ww'}^2}$  for the unit vector along the orthogonal projection of  $w'$  onto  $w^\perp$ , defined whenever  $g_{ww'} \neq \pm 1$ . Then, inside the 3-space  $w^\perp$ ,*

$$F_w - w = \left\{ y \in w^\perp : \langle y, \widehat{w'} \rangle \leq \rho(g_{ww'}) \text{ for all } w' \in W \setminus \{w\} \right\}, \quad \rho(g) = \sqrt{\frac{1-g}{1+g}}, \quad (7.7)$$

with the convention that  $w' = -w$  imposes no constraint. The function  $\rho$  is strictly decreasing on  $(-1, 1]$  and  $\rho(\frac{1}{2}) = 1/\sqrt{3}$ , so  $F_w - w$  contains the 3-ball of radius  $1/\sqrt{3}$  about the origin of  $w^\perp$  and

$$\text{vol}_3(F_w) \geq \frac{4\pi}{3} \left( \frac{1}{\sqrt{3}} \right)^3 = \frac{4\pi}{9\sqrt{3}} = 0.806133\dots \quad (7.8)$$

*Proof.* Write  $x = w + y$  with  $y \in w^\perp$ . Then  $\langle x, w \rangle = 1$  automatically, and for  $w' \neq w$  the constraint  $\langle x, w' \rangle \leq 1$  reads  $g_{ww'} + \langle y, w' \rangle \leq 1$ , that is  $\langle y, w' - g_{ww'}w \rangle \leq 1 - g_{ww'}$ , since  $y \perp w$ . Dividing by  $|w' - g_{ww'}w| = \sqrt{1 - g_{ww'}^2}$  turns this into  $\langle y, \widehat{w'} \rangle \leq (1 - g)/\sqrt{1 - g^2} = \rho(g)$  at  $g = g_{ww'}$ , which is (7.7). On  $(-1, 1]$  the quotient  $(1 - g)/(1 + g)$  is strictly decreasing, hence so is  $\rho$ , and  $\rho(\frac{1}{2}) = \sqrt{(1/2)/(3/2)} = 1/\sqrt{3}$ . Since  $g_{ww'} \leq \frac{1}{2}$  for every  $w' \neq w$ , every

constraint in (7.7) sits at distance at least  $1/\sqrt{3}$  from the origin of  $w^\perp$ , so the ball of that radius is contained in  $F_w - w$ , and (7.8) is its volume.  $\blacksquare$

At the root configuration the neighbours with  $g = \frac{1}{2}$  are the eight roots at  $60^\circ$  to  $\alpha_w$ ; their projections into  $w^\perp$  have pairwise inner products in  $\{-1, -\frac{1}{3}, \frac{1}{3}\}$ , so they are the eight vertex directions of a cube, and the eight constraints  $\langle y, w' \rangle \leq 1/\sqrt{3}$  cut out exactly the regular octahedron  $\{y : |y_1| + |y_2| + |y_3| \leq 1\}$  of 3-volume  $\frac{4}{3}$ , the remaining sixteen constraints being inactive. The bound (7.8) is therefore attained in shape but not in size: the octahedron circumscribes the ball of radius  $1/\sqrt{3}$  and has  $1.654\dots$  times its volume.

*Remark 7.12* (Why no facet-by-facet argument can succeed). Combining (7.6) and (7.8) with  $|W| \leq 24$  gives, for every contact configuration,

$$\text{vol}(V_c(W)) \geq \frac{|W|}{4} \cdot \frac{4\pi}{9\sqrt{3}} = \frac{|W|\pi}{9\sqrt{3}},$$

which at the largest possible value  $|W| = 24$  is  $8\pi/(3\sqrt{3}) = 4.836798\dots$ , short of 8 by more than 3. The inequality  $V_c(W) \supseteq B^4$ , which follows from  $|w| = 1$  for every  $w$  and is the same computation run in one step instead of facet by facet, does slightly better at  $\pi^2/2 = 4.934802\dots$ , and no better.

The conclusion is not that these two bounds happen to be weak. It is that the whole class they belong to is too small, and one example settles it. An argument that uses only the pairwise condition  $g_{ww'} \leq \frac{1}{2}$ , and nothing about how the rest of the configuration is arranged, can produce only a bound of the form  $\text{vol}_3(F_w) \geq \beta$  with  $\beta$  a universal constant; the largest such constant is  $\beta^* = \inf \text{vol}_3(F_w)$ , the infimum being over all contact configurations and all  $w$ . The root configuration gives  $\beta^* \leq \frac{4}{3}$ , and one may do better. Keep all eight tight neighbours but replace the cube by a square antiprism: in an orthonormal frame with  $w = e_0$  and  $w^\perp = \langle e_1, e_2, e_3 \rangle$ , take

$$w'_j = \frac{1}{2} e_0 + \frac{\sqrt{3}}{2} \hat{n}_j, \quad \hat{n}_j \in \frac{1}{\sqrt{3}} \{(\pm\sqrt{2}, 0, 1), (0, \pm\sqrt{2}, 1), (\pm 1, \pm 1, -1)\}. \quad (7.9)$$

Each  $\hat{n}_j$  is a unit vector, so  $\langle w, w'_j \rangle = \frac{1}{2}$ ; and the  $\hat{n}_j$  have pairwise inner products in  $\{\frac{1}{3}, -\frac{1}{3}, \frac{\sqrt{2}-1}{3}, -\frac{\sqrt{2}+1}{3}\}$ , so  $\langle w'_i, w'_j \rangle = \frac{1}{4} + \frac{3}{4} \langle \hat{n}_i, \hat{n}_j \rangle \leq \frac{1}{2}$  and the configuration is admissible. By (7.7) the resulting facet is  $\{y : \langle y, \sqrt{3} \hat{n}_j \rangle \leq 1\}$ , and integrating its height over the square  $|y_1|, |y_2|$  gives its volume in closed form:

$$\text{vol}_3(F_w) = 16\sqrt{2} - \frac{64}{3} = 1.2940836\dots < \frac{4}{3}.$$

Feeding  $\beta^*$  into (7.6) together with  $|W| \leq 24$ ,

$$\text{vol}(V_c(W)) \geq \frac{|W|}{4} \beta^* \leq 6\beta^* \leq 96\sqrt{2} - 128 = 7.764502\dots$$

So no bound of this class can reach 8 at all, whatever it is, and its ceiling is below even the crude covering estimate of Theorem 7.16. For a configuration with  $|W| \leq 23$  the same computation returns at most  $7.441\dots$ , so the method fails first exactly where the statement is least delicate.

What is missing is the interaction between facets: the direction  $w'$  that makes  $F_w$  small is itself a direction whose own facet  $F_{w'}$  is then constrained, and (7.6) becomes usable only if that trade is accounted for across the configuration. This is the same phenomenon that Remark 7.3 records in the free-volume language, seen from the boundary instead.

*Remark 7.13* (Two global routes that do not reach the equality case). Two standard tools apply to (7.5) and both fail at the configuration where the conjecture is tight, which is the decisive test for a method that must produce a sharp constant.

The polar volume has the integral form  $\text{vol}(K^\circ) = \frac{1}{4} \int_{S^3} h_K(\theta)^{-4} d\theta$ , where  $h_K$  is the support function and  $d\theta$  is the surface measure of total mass  $2\pi^2$ . Applying Jensen's inequality to the convex function  $t \mapsto t^{-4}$  and the normalised measure gives  $\text{vol}(K^\circ) \geq (\pi^2/2) \bar{h}^{-4}$  with  $\bar{h}$  the mean of  $h_K$  over the sphere, and the target 8 would follow from  $\bar{h} \leq (\pi^2/16)^{1/4} = 0.886226\dots$ . For  $K = \text{conv}(W)$  at the root configuration,  $h_K(\theta)$  is  $2^{-1/2}$  times the sum of the two largest absolute coordinates of  $\theta$ , and  $\bar{h} = 0.89372\dots$ , evaluated numerically at four million sample directions with a standard error of  $2 \times 10^{-5}$ , so the criterion fails by a small margin. The failure is not an artefact of the estimate being loose in a repairable way: Jensen is an equality only when  $h_K$  is constant, that is only for a ball, so at any polytope the bound is strictly below the true volume, and at the root configuration it returns  $7.7351\dots$  against the true 8. No inequality of this shape can be sharp where the conjecture is.

The Mahler-type route fares worse. The conjectured lower bound  $\text{vol}(K) \text{vol}(K^\circ) \geq 4^n/n!$ , itself open in dimension four, would give  $\text{vol}(K) \text{vol}(K^\circ) \geq \frac{32}{3}$ ; but at the root configuration  $\text{conv}(W)$  is the 24-cell of circumradius 1, of volume 2, and the product is 16, so the inequality carries a factor  $\frac{3}{2}$  of slack exactly where none can be afforded, and dividing by  $\text{vol}(K) \leq 2$  returns only  $\frac{16}{3}$ . Using instead the crude  $\text{vol}(K) \leq \text{vol}(B^4) = \pi^2/2$  returns  $64/(3\pi^2) = 2.16\dots$

Both evaluations, together with the facet-local bound of Remark 7.12, the boundary identity (7.6) and the facet description (7.7), are reproduced by `polar_surface_reformulation.py` in the supplement, and collected in Figure 11.

*Remark 7.14* (What the multi-direction case is). Proposition 7.8 identifies Conjecture 1.4 with a named problem. The assertion that every Voronoi cell of a unit-ball packing of  $\mathbb{R}^4$  has volume at least that of the regular 24-cell, with equality only at the  $D_4$  configuration, is the 24-cell conjecture. For contact configurations, where every neighbour that can cut the cell touches the centre, Theorem 1.1 settles the part of it in which at most one contact direction leaves a root, and Conjecture 1.4 is the remainder, settled in Corollary 7.31 by the classification of Theorem 7.25 and by nothing else in this paper; what the conjecture asks beyond contact configurations is Conjecture 1.6. We record this plainly. The reformulations of this subsection do not diminish the problem of proving it directly, and we do not claim otherwise; what they do is fix the exact shape of the obstacle and remove three families of approach from consideration, a fourth having gone in Remark 7.3. Partial results towards the conjecture, by a different route, are in [Mus18].

**7.4. A bound from the contact caps.** The boundary form measures the cell from outside, facet by facet. There is a third measurement, from the centre outward, and it is the one that gives a bound for every configuration at once, however many of its contact directions deviate.

**Lemma 7.15** (Radial form). *Let  $W$  be a contact configuration and let*

$$\delta(\theta) = \min_{w \in W} \arccos\langle \theta, w \rangle, \quad \theta \in S^3,$$

*be the angular distance from  $\theta$  to the nearest contact direction. Then  $V_c(W)$  is bounded if and only if  $\delta < \pi/2$  everywhere, and in that case*

$$\text{vol}(V_c(W)) = \frac{1}{4} \int_{S^3} \sec^4(\delta(\theta)) d\theta, \quad (7.10)$$

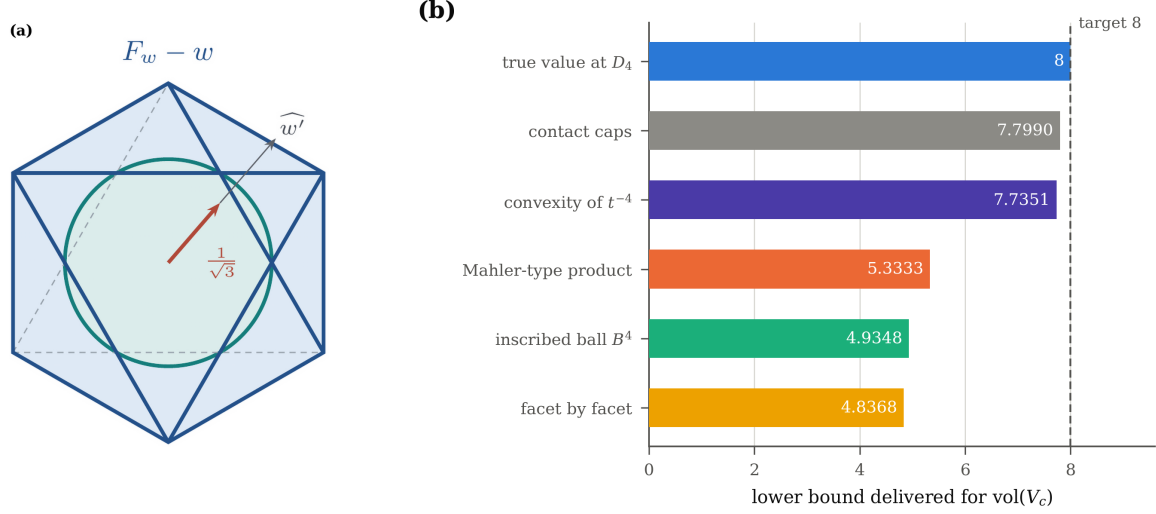


FIGURE 11. (a) A single facet of the reference cell, drawn inside the 3-space orthogonal to its own contact direction. At the root configuration the eight tight neighbours cut out the regular octahedron (Lemma 7.11); the pairwise contact condition on its own guarantees no more than the inscribed ball, and the gap between the two is what the boundary form (7.6) has to absorb. (b) The bounds the boundary form and the other global routes actually deliver, all of them evaluated at the  $D_4$  configuration where the conjecture is tight (Remarks 7.12 and 7.13).

where  $d\theta$  is the surface measure on  $S^3$ , of total mass  $2\pi^2$ .

*Proof.* In polar coordinates the volume of a star body with radial function  $\rho$  is  $\frac{1}{4} \int_{S^3} \rho^4$ . The radial function of a polar dual is the reciprocal of the support function of the body, so by Proposition 7.8,  $\rho_{V_c(W)}(\theta) = 1/h_{\text{conv}(W)}(\theta)$  with  $h_{\text{conv}(W)}(\theta) = \max_w \langle \theta, w \rangle$ . Since every  $w$  is a unit vector, that maximum is  $\max_w \cos \angle(\theta, w) = \cos \delta(\theta)$ , which is positive exactly when  $\delta(\theta) < \pi/2$ ; where it fails the cell contains the ray through  $\theta$  and is unbounded. Substituting  $\rho = 1/\cos \delta$  gives (7.10). ■

So the problem is to show that a set of  $m \leq 24$  points on  $S^3$  with pairwise angular distance at least  $60^\circ$  cannot be so evenly spread that the mean of  $\sec^4 \delta$  drops below  $32/(2\pi^2)$ . Written this way, a crude covering estimate already gives most of what is wanted.

**Theorem 7.16** (Covering bound). *Let  $W$  be a contact configuration with  $|W| = m$  and bounded cell, and let  $r_m$  be the unique solution in  $(0, \pi/2)$  of*

$$2r - \sin 2r = \frac{2\pi}{m}. \quad (7.11)$$

Then

$$\text{vol}(V_c(W)) \geq \frac{\pi m}{3} \tan^3 r_m. \quad (7.12)$$

*Proof.* A geodesic cap of angular radius  $r$  in  $S^3$  has surface measure

$$C(r) = \int_0^r (4\pi \sin^2 t) dt = \pi (2r - \sin 2r),$$

the integrand being the area of the 2-sphere of radius  $\sin t$  that the cap meets at angular distance  $t$  from its centre; in particular  $C(\pi) = 2\pi^2$ , the whole measure. The set  $\{\delta \leq r\}$  is

the union of the  $m$  caps of radius  $r$  about the points of  $W$ , so

$$\sigma(\{\delta > r\}) \geq S(r) := 2\pi^2 - m\pi(2r - \sin 2r),$$

where  $\sigma$  denotes surface measure. The function  $2r - \sin 2r$  has derivative  $2 - 2\cos 2r \geq 0$ , vanishing only at  $r = 0$ , and increases from 0 to  $\pi$  on  $[0, \pi/2]$ ; since  $m \geq 5$  forces  $2\pi/m < \pi$ , equation (7.11) has exactly one root  $r_m$  there, and  $S > 0$  on  $[0, r_m)$ ,  $S(r_m) = 0$ .

Now  $\sec^4 \delta \geq 1$ , so by the layer-cake formula and the substitution  $s = \sec^4 r$ , under which  $ds = 4\sec^4 r \tan r dr = d(\sec^4 r)$ ,

$$\begin{aligned} \int_{S^3} \sec^4 \delta d\theta &= 2\pi^2 + \int_1^\infty \sigma(\{\sec^4 \delta > s\}) ds \\ &= 2\pi^2 + \int_0^{\pi/2} \sigma(\{\delta > r\}) d(\sec^4 r). \end{aligned}$$

The integrand is nonnegative, so restricting the range to  $[0, r_m]$  and replacing  $\sigma(\{\delta > r\})$  by the smaller  $S(r)$  only decreases the value. Integrating by parts, with  $S(0) = 2\pi^2$ ,  $S(r_m) = 0$  and  $S'(r) = -4\pi m \sin^2 r$ ,

$$\begin{aligned} \int_0^{r_m} S(r) d(\sec^4 r) &= [S(r) \sec^4 r]_0^{r_m} + 4\pi m \int_0^{r_m} \sin^2 r \sec^4 r dr \\ &= -2\pi^2 + 4\pi m \int_0^{r_m} \tan^2 r \sec^2 r dr, \end{aligned}$$

and the last integral is  $\frac{1}{3} \tan^3 r_m$ . The two occurrences of  $2\pi^2$  cancel, leaving  $\int_{S^3} \sec^4 \delta \geq \frac{4\pi m}{3} \tan^3 r_m$ , which with (7.10) is (7.12).  $\blacksquare$

The right-hand side of (7.12) decreases as  $m$  grows, since a larger  $m$  both multiplies by more and shrinks  $r_m$ ; the second effect wins. The values for the whole admissible range  $5 \leq m \leq 24$ , a bounded cell in  $\mathbb{R}^4$  needing at least five facets and Proposition 7.8(iii) allowing at most 24, are these.

| $m$ | $r_m$ (degrees) | bound     | $m$ | $r_m$ (degrees) | bound           |
|-----|-----------------|-----------|-----|-----------------|-----------------|
| 5   | 60.536972       | 29.042559 | 15  | 40.255989       | 9.536151        |
| 6   | 56.406799       | 21.440235 | 16  | 39.340721       | 9.227382        |
| 7   | 53.197354       | 17.503703 | 17  | 38.502427       | 8.962031        |
| 8   | 50.601922       | 15.119204 | 18  | 37.730497       | 8.731395        |
| 9   | 48.441431       | 13.525786 | 19  | 37.016289       | 8.528959        |
| 10  | 46.603050       | 12.387172 | 20  | 36.352679       | 8.349750        |
| 11  | 45.011524       | 11.533082 | 21  | 35.733735       | 8.189905        |
| 12  | 43.614387       | 10.868477 | 22  | 35.154469       | <b>8.046376</b> |
| 13  | 42.373749       | 10.336286 | 23  | 34.610650       | 7.916728        |
| 14  | 41.261436       | 9.900225  | 24  | 34.098663       | 7.798989        |

TABLE 2. The covering bound (7.12) over the whole admissible range of  $m$ . It exceeds 8 for every  $m \leq 22$  and falls below 8 at  $m = 23$ . Each  $r_m$  is the root of (7.11), computed to the digits shown by bisection on an exactly evaluated monotone function.

**Corollary 7.17** (The local bound for at most 22 contacts). *Every contact configuration  $W$  with  $|W| \leq 22$  and bounded cell satisfies  $\text{vol}(V_c(W)) \geq 8.0463\dots > 8$ , with no hypothesis on how many of its directions deviate from roots and no hypothesis on the size of those deviations. Consequently*

$$\text{vol}(V_c) \geq \frac{\pi \cdot 24}{3} \tan^3 r_{24} = 7.798989\dots$$

*holds for every contact configuration whatever, and the local bound  $\text{vol}(V_c) \geq 8$  is left undecided only for configurations with exactly 23 or exactly 24 contacts.*

*Proof.* Immediate from Theorem 7.16 and Table 2, using  $|W| \leq 24$  from Proposition 7.8(iii) and the monotonicity visible in the table, which over the finite range  $5 \leq m \leq 24$  is a comparison of twenty numbers.  $\blacksquare$

Corollary 7.17 is Theorem 1.3 of the introduction, and it settles one point that the deviation picture leaves open. A configuration in which several of the twenty-four contacts are absent and not merely deviated is not a perturbation of the root configuration at all, and the argument of Section 13 says nothing about it. Corollary 7.17 disposes of every such configuration with at most twenty-two contacts outright, whatever its directions.

**Proposition 7.18** (The covering bound is the best area-based bound). *Let  $W$  be a contact configuration with bounded cell and let  $\Omega_w = \{\theta \in S^3 : \langle \theta, w \rangle \geq \langle \theta, w' \rangle \text{ for all } w' \in W\}$  be its spherical Voronoi cells, so that  $\delta(\theta)$  is the distance to  $w$  on  $\Omega_w$  and  $\sum_w \sigma(\Omega_w) = 2\pi^2$ . Write  $\rho_w$  for the radius of the cap of the same measure,  $C(\rho_w) = \sigma(\Omega_w)$ . Then*

$$\text{vol}(V_c(W)) \geq \frac{\pi}{3} \sum_{w \in W} \tan^3 \rho_w, \quad (7.13)$$

*and the function  $\phi(s) = \tan^3(C^{-1}(s))$  is strictly convex, with  $\phi'(s) = \frac{3}{4\pi} \sec^4(C^{-1}(s))$ . Consequently the right-hand side of (7.13) is smallest, over all ways of distributing the total measure  $2\pi^2$  among  $m$  cells, exactly when the cells have equal measure, and its value there is  $\frac{\pi m}{3} \tan^3 r_m$ : the bound of Theorem 7.16.*

*Proof.* On  $\Omega_w$  the integrand of (7.10) is  $\sec^4 d(\theta, w)$ , an increasing function of the distance to  $w$ , so its sublevel sets are the caps about  $w$ . By the bathtub principle, among all sets of measure  $\sigma(\Omega_w)$  the integral is least on the sublevel set of that measure, which is the cap of radius  $\rho_w$ ; and

$$\int_{\{d(\theta, w) \leq \rho\}} \sec^4 d(\theta, w) d\theta = \int_0^\rho \sec^4 t \cdot 4\pi \sin^2 t dt = 4\pi \int_0^\rho \tan^2 t \sec^2 t dt = \frac{4\pi}{3} \tan^3 \rho.$$

Summing over  $w$  and dividing by 4 gives (7.13). For the convexity,  $C'(\rho) = 4\pi \sin^2 \rho$ , so

$$\phi'(s) = \frac{3 \tan^2 \rho \sec^2 \rho}{4\pi \sin^2 \rho} = \frac{3 \sec^4 \rho}{4\pi}, \quad \rho = C^{-1}(s),$$

which is strictly increasing in  $\rho$  and hence in  $s$ . Jensen's inequality for a convex function then gives  $\sum_w \phi(\sigma(\Omega_w)) \geq m \phi(2\pi^2/m)$ , with equality only when all the measures agree, and  $\phi(2\pi^2/m) = \tan^3 r_m$  by the definition (7.11) of  $r_m$ .  $\blacksquare$

So Theorem 7.16 is not one estimate among several: it is the strongest conclusion that the areas of the spherical Voronoi cells can support, and at the root configuration those areas are all equal, so no redistribution argument can improve it there. What the estimate gives away is shape. It replaces each cell by the round set of the same measure, and a round set is exactly the worst case for an integrand that grows with the distance. At the root configuration each cell is a spherical polytope with eight facets and circumradius  $45^\circ$ ,

while the cap of the same measure has radius  $34.0986\dots^\circ$ ; that difference is the whole of the remaining  $0.2011$ . Any improvement must therefore begin by proving that a spherical Voronoi cell of a  $60^\circ$ -separated set cannot be round, which is a statement about the shape of the cells and not about their areas.

*Remark 7.19* (The shape is not local to a cell). The natural way to get at that shape is one cell at a time, and it fails in both of the decompositions the configuration provides.

On  $\Omega_w$  the integrand of (7.10) is  $\sec^4 d(\theta, w)$ , so the Voronoi decomposition localises the identity exactly, and  $\text{vol}(V_c(W)) \geq 8$  would follow from

$$\int_{\Omega_w} \sec^4 d(\theta, w) d\theta \geq \frac{16}{\pi^2} \sigma(\Omega_w) \quad (7.14)$$

at every  $w$ , that is from the mean of  $\sec^4$  over each cell being at least  $16/\pi^2 = 1.621138\dots$ . At the root configuration every cell gives exactly that, so (7.14) is sharp. It is also false. A direction may carry nine contacts at  $60^\circ$ : writing  $w' = \frac{1}{2}w + \frac{\sqrt{3}}{2}n$  with  $n$  a unit vector orthogonal to  $w$ , the condition  $\langle w', w'' \rangle \leq \frac{1}{2}$  becomes  $\langle n', n'' \rangle \leq \frac{1}{3}$ , and nine unit vectors of  $\mathbb{R}^3$  with pairwise inner products at most  $\frac{1}{3}$  exist, by the case of nine points in Schütte and van der Waerden's determination of the optimal spherical codes of small size [SW51]. The cell of  $w$  is then smaller than any cell of the root configuration, and the mean of  $\sec^4$  over it is  $1.5732\dots$ ; moving the nine contacts out to  $61^\circ$ , so that every inner product is strictly below  $\frac{1}{2}$ , still gives  $1.5954\dots$ . Enlarging the configuration only shrinks  $\Omega_w$  and lowers the mean, so every completion violates (7.14) there.

The Delaunay decomposition does not localise at all. Inside a Delaunay cell the nearest contact direction need not be a vertex of that cell, so  $\int_T \sec^4 \delta$  is not a functional of  $T$ ; on a 22-point configuration there are directions whose nearest contact lies outside the cell containing them and beats every vertex of that cell by more than  $0.1$  in cosine. Even the functional it would have to be, the distance to the cell's own vertices, fails the corresponding inequality. For the regular spherical simplex of edge  $60^\circ$ , parametrised by the barycentric coordinates  $\lambda$  of the cone it spans, the identities  $|x|^2 = \frac{1}{2}(1+q)$  with  $q = \sum_i \lambda_i^2$  and  $\langle x, v_i \rangle = \frac{1}{2}(1+\lambda_i)$  turn the mean of  $\sec^4$  into

$$\frac{4 \mathbb{E}[(1+M)^{-4}]}{\mathbb{E}[(1+q)^{-2}]} = \frac{4/5}{0.518254\dots} = 1.543643\dots, \quad \mathbb{E}[(1+M)^{-4}] = \frac{1}{5}, \quad (7.15)$$

$M$  being the largest coordinate and both expectations taken over a uniformly distributed  $\lambda$ . The value falls short of  $16/\pi^2$  by  $0.077495\dots$ , which is  $4.78\%$ . That simplex has circumradius  $37.7612\dots^\circ$ , so Proposition 7.38 below does not exclude it from a saturated configuration, and it does occur as a Delaunay cell of a contact configuration; at edge  $62^\circ$ , where every inner product is strictly below  $\frac{1}{2}$ , the mean is still  $1.5955\dots$ . It occurs nowhere in the root system, whose  $60^\circ$  graph has clique number three, and that is the point of the example: the locally densest arrangement is not one the extremal configuration contains.

So the deficit that Proposition 7.18 shows is needed is not additive over cells. Cells worse than round exist in both decompositions, and what makes the root configuration extremal is the way its cells fit together, not the shape of any one of them. All of this is checked by `local_cell_obstruction.py`, whose integrals over  $S^3$  are Monte Carlo and are labelled as such; the margins above are percentages, not last digits.

*Remark 7.20* (What the estimate leaves). Two contact counts survive Corollary 7.17, and they are of different kinds. At  $m = 24$  the configuration is a full contact shell and the question is the one Conjecture 1.4 asks, whether two or more of the twenty-four directions

can leave the roots; the estimate returns 7.7989... there, a shortfall of 0.2011. At  $m = 23$  the configuration need not be a deviation of the root system at all, so Conjecture 1.4 does not reach it and Corollary 7.17 stops one short; the estimate returns 7.9167..., a shortfall of 0.0833. Section 7.5 disposes of the first count and reduces the second to the configurations with bounded cell, and Sections 7.11 and 7.12 prove the bound for those.

**Proposition 7.21** (Every cell reaches out to  $37.7612\dots^\circ$ ). *Let  $W$  be a contact configuration with bounded cell. Then every vertex of every spherical Voronoi cell  $\Omega_w$  is at angular distance at least  $R_0 = \arccos \sqrt{5/8} = 37.7612\dots^\circ$  from  $w$ , and in particular the covering radius of  $W$  on  $S^3$  is at least  $R_0$ .*

*Proof.* A vertex  $V$  of  $\Omega_w$  is a point of  $S^3$  at which at least four cells meet, so there are at least four points of  $W$ , one of them  $w$ , at the common distance  $R = d(V, w)$  from  $V$  and none nearer. Let  $u_1, \dots, u_4$  be unit vectors in the tangent space at  $V$  pointing along the geodesics from  $V$  to four of them. Four unit vectors in  $\mathbb{R}^3$  cannot have all six pairwise inner products below  $-\frac{1}{3}$ , since  $0 \leq |\sum_i u_i|^2 = 4 + 2 \sum_{i < j} \langle u_i, u_j \rangle$  forces  $\sum_{i < j} \langle u_i, u_j \rangle \geq -2$  and there are six terms. Take a pair with  $\langle u_i, u_j \rangle \geq -\frac{1}{3}$ . The two corresponding points of  $W$  lie at distance  $R$  from  $V$  in directions at angle  $\alpha$  with  $\cos \alpha \geq -\frac{1}{3}$ , so by the spherical law of cosines their separation  $\gamma$  satisfies

$$\cos \gamma = \cos^2 R + \sin^2 R \cos \alpha \geq \cos^2 R - \frac{1}{3} \sin^2 R.$$

Packing validity gives  $\gamma \geq 60^\circ$ , hence  $\cos \gamma \leq \frac{1}{2}$ , so  $\cos^2 R - \frac{1}{3}(1 - \cos^2 R) \leq \frac{1}{2}$  and  $\cos^2 R \leq \frac{5}{8}$ . The farthest point of  $S^3$  from  $W$  is a vertex of the cell containing it, so the covering radius is at least  $R_0$  as well.  $\blacksquare$

This is a genuine improvement on what Theorem 7.16 extracts along the way, since that argument gives up at  $r_{24} = 34.0986\dots^\circ$ . It does not help. Knowing that the survival function is positive out to  $R_0$  bounds it below only by the measure of a cap of radius  $R_0 - r$  about one deep hole, and feeding that into the integral changes the constant in the fifth decimal: 7.79899 becomes 7.79902, and even allowing one such hole per cell and treating them as disjoint gives only 7.79966. The missing 0.140 is a statement about how much of the sphere lies far from  $W$ , not about how far the farthest point is.

The overlaps the estimate throws away can be put back exactly, and it is worth doing, because the answer says how much room the method has left.

**Lemma 7.22** (Three caps do not meet). *Three directions of a contact configuration have circumradius at least  $\arccos \sqrt{2/3} = 35.2644\dots^\circ$  on their span. Consequently, if  $r < \arccos \sqrt{2/3}$ , no point of  $S^3$  lies within  $r$  of three contact directions at once.*

*Proof.* Let  $G$  be the Gram matrix of the three directions, with unit diagonal and off-diagonal entries at most  $\frac{1}{2}$ , and let  $z$  be the point of their span equidistant from all three, at angular distance  $R$ . Writing  $z = \sum a_i w_i$ , the conditions  $\langle z, w_i \rangle = \cos R$  and  $|z| = 1$  give  $Ga = (\cos R)\mathbf{1}$  and  $1 = a^\top Ga = (\cos R)^2 \mathbf{1}^\top G^{-1} \mathbf{1}$ . Now  $\mathbf{1}^\top G^{-1} \mathbf{1} = \max_{x \neq 0} (\mathbf{1}^\top x)^2 / (x^\top G x)$ , and  $x = \mathbf{1}$  already gives

$$\mathbf{1}^\top G^{-1} \mathbf{1} \geq \frac{9}{3 + 2 \sum_{i < j} g_{ij}} \geq \frac{9}{3 + 3} = \frac{3}{2},$$

so  $\cos^2 R \leq \frac{2}{3}$ . A point within  $r$  of all three would give a triple of circumradius at most  $r$ .  $\blacksquare$

**Proposition 7.23** (Putting the overlaps back). *Let  $W$  be a contact configuration with  $|W| = m$  and bounded cell, write  $\gamma_{ij}$  for the angle between  $w_i$  and  $w_j$ , and let*

$$\Lambda(r, \gamma) = 4\pi \int_{\gamma/2}^r \sin^2 t \left(1 - \tan \frac{\gamma}{2} \cot t\right) dt \quad (r \geq \frac{\gamma}{2}), \quad (7.16)$$

*the measure of the intersection of two caps of radius  $r$  whose centres are  $\gamma$  apart, and  $\Lambda(r, \gamma) = 0$  for  $r < \gamma/2$ . Then for every  $R \leq r_* := \arccos \sqrt{2/3} = 35.2644\dots^\circ$ ,*

$$\text{vol}(V_c(W)) \geq \frac{1}{4} \left[ 2\pi^2 + \int_0^R (2\pi^2 - mC(r)) d(\sec^4 r) \right] + \frac{1}{4} \sum_{i < j} \int_0^R \Lambda(r, \gamma_{ij}) d(\sec^4 r). \quad (7.17)$$

*At  $R = r_m$  the first bracket is the covering bound  $\frac{\pi m}{3} \tan^3 r_m$  of Theorem 7.16, and  $r_m < r_*$  for every  $m \geq 23$ . Below  $r_*$  the right-hand side is not an estimate at all: by Lemma 7.22 the caps of radius  $r$  meet only in pairs there, so  $\sigma(\{\delta \leq r\}) = mC(r) - \sum_{i < j} \Lambda(r, \gamma_{ij})$  exactly, and (7.17) is the true value of the integral truncated at  $R$ , that is the volume of  $V_c(W)$  inside the ball of radius  $\sec R$ ; the largest admissible  $R$  is  $r_*$ , where the ball has radius  $\sqrt{3}/2$ .*

*Proof.* For  $\Lambda$ , split the lens by the hyperplane bisecting  $w_i$  and  $w_j$ ; the half on the  $w_i$  side is the part of the cap about  $w_j$  lying beyond that hyperplane, which is at angular distance  $\gamma/2$  from  $w_j$ . Parametrise the cap about  $w_j$  by the distance  $t$  and a direction on the 2-sphere it meets at distance  $t$ , of measure  $4\pi \sin^2 t dt$ ; the coordinate of that direction along the bisector normal is uniform on  $[-1, 1]$ , and the half-space condition reads  $s \geq \cot t \tan(\gamma/2)$ , which is vacuous for  $t \leq \gamma/2$  and otherwise excludes a fraction  $\frac{1}{2}(1 + \cot t \tan \frac{\gamma}{2})$ . Doubling gives (7.16). The rest is the layer-cake step in the proof of Theorem 7.16, run with the exact survival function on  $[0, R]$  and with the integrand discarded above  $R$ : writing  $\sec^4 \min(\delta, R) = 1 + \int_0^R \mathbf{1}[\delta > r] d(\sec^4 r)$  and integrating over  $S^3$  gives the bracket, and the survival function is  $2\pi^2 - mC(r) + \sum_{i < j} \Lambda(r, \gamma_{ij})$  on  $[0, R]$  by Lemma 7.22. ■

At the configuration obtained by deleting one root, where  $m = 23$  and the 88 close pairs are all at exactly  $60^\circ$ , the right-hand side of (7.17) is

$$7.997885\dots \text{ at } R = r_{23}, \quad 8.034340\dots \text{ at } R = r_*, \quad (7.18)$$

against  $7.916728\dots$  for the plain estimate and  $\frac{25}{3}$  for the true volume. The first of these is the number one obtains by stopping the integration where the covering estimate stops, at the radius  $r_{23} = 34.6106\dots^\circ$  at which the caps would exactly exhaust the sphere if they were disjoint, and it falls short of 8 by  $0.002115\dots$ . The second carries the same exact identity to its natural limit  $r_*$ , the smallest circumradius of three contact directions, and it passes 8 with  $0.034340\dots$  to spare. The pairwise estimate is therefore not blocked at the deletion configuration; what it asks of an arbitrary configuration of 23 directions is worked out in Proposition 7.41, and Section 7.11 takes the matter further. At  $m = 24$  the same computation gives  $7.858738\dots$  at  $R = r_{24}$  and  $7.907070\dots$  at  $R = r_*$ , against the true value 8; there the truncation discards the deep holes of the root system, which sit at  $45^\circ$ .

*Remark 7.24* (What the constant is and is not). Two things should be said about  $7.798989\dots$  so that it is not mistaken for more than it is.

As a density statement it is weak. A per-cell bound  $v$  gives packing density at most  $\text{vol}(B^4)/v$ , so  $7.798989\dots$  gives  $0.632749\dots$ , against  $\pi^2/16 = 0.616850\dots$  for  $D_4$ ; the linear programming bound recalled in Section 21 is stronger as a density bound. What the covering estimate has that a density bound does not is that it is a statement about one cell at a time, which is what Conjecture 1.4 asks for, and that is why it settles the range  $m \leq 22$  outright.

The estimate itself is crude, and deliberately so: the only fact about the configuration it uses is that  $m$  caps cannot cover more than  $m$  times a cap's worth of the sphere. Proposition 7.23 says how much of that crudeness is repairable from pair data alone, and the answer is almost all of it at  $m = 23$  and rather little at  $m = 24$ . What neither version reaches is the tail: both stop at  $r_m$ , while the survival function stays positive out to the covering radius,  $45^\circ$  at the root configuration. Going further means a lower bound on the covering radius better than  $r_m$ , and Proposition 7.21 shows what that is worth, which is the fifth decimal. The rest of the tail is a statement about how much of the sphere lies far from  $W$ , and it is the same statement as the rigidity of the kissing arrangement.

Everything in this subsection, the cap-area formula, the radial identity, the layer-cake step, the closed form, the whole of Table 2 including its monotonicity, and the absence of any violation of (7.12) at the root configuration, at its subsets and at random configurations, is checked by `covering_bound.py` in the supplement, and Lemma 7.22 together with (7.16) and the two values above by `second_order_estimate.py`.

**7.5. Twenty-four contacts, and what is left.** At the top of the range the problem changes character completely, because a contact configuration of the maximal size is rigid.

**Theorem 7.25** (Twenty-four contacts). *Let  $W$  be a contact configuration with  $|W| = 24$ . Then  $W$  is the image of the 24 normalised roots of  $D_4$  under an orthogonal map, and consequently  $V_c(W)$  is a regular 24-cell and  $\text{vol}(V_c(W)) = 8$ .*

The theorem was first proved by de Laat, Leijenhorst and de Muinck Keizer [LLM24, Thm. 5.3], by a semidefinite computation. It is proved here in full. The proof has two halves. The first is an inequality: a contact configuration has at most 24 elements, and one with exactly 24 has all its pairwise inner products in  $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$ . That is Theorem 7.30 below, and it is a consequence of an explicit positive kernel on the two-element subsets of  $S^3$ . The kernel was found by the authors of [LLM24] and deposited with their paper [LLM24d]; what is proved here is that any kernel with its properties bounds the size of a contact configuration (Lemma 7.27), that a kernel built from positive definite blocks in the way theirs is built is positive (Lemma 7.28), that the deposited data have every property required, which is a finite computation carried out in Section 7.5.3, and that at the value 24 the bound forces the inner products. No statement of [LLM24] is used; what is taken from that work is the certificate itself, which is data, and the construction of the kernels in which it is expressed, which is a definition. The second half is classical, and is the following lemma.

**Lemma 7.26** (From the inner products to the root system). *Let  $W$  be a set of 24 unit vectors in  $\mathbb{R}^4$  whose pairwise inner products lie in  $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$ . Then  $\sqrt{2}W$  is the set of roots of a copy of the  $D_4$  root lattice, and  $W$  is the image of the normalised roots of  $D_4$  under an orthogonal map.*

*Proof.* The vectors of  $\sqrt{2}W$  have norm 2 and integer inner products, so they generate an integral lattice  $L \subset \mathbb{R}^4$  of rank at most 4, and they are among the vectors of norm 2 of  $L$ . The vectors of norm 2 of an integral lattice form a root system, simply laced, since two of them have inner product in  $\{0, \pm 1, \pm 2\}$  and the reflection in one of them preserves the lattice [CS88, Ch. 4], and the lattice they generate is a root lattice, an orthogonal sum of lattices of the types  $A_n$ ,  $D_n$  and  $E_n$  by the classification of simply laced root systems [Hum90, Ch. 2]; it has a basis of roots, its simple roots. Since  $\sqrt{2}W$  generates  $L$ , the lattice  $L$  is such a root lattice, of rank  $r \leq 4$ , and  $|\sqrt{2}W| = 24$  is at most the number of roots of  $L$ . The root lattices of rank at most 4 and their numbers of roots are  $A_1$ ,  $A_2$ ,  $A_3$ ,  $A_4$  with 2, 6,

12, 20,  $D_4$  with 24, and the orthogonal sums  $A_1^2$ ,  $A_1A_2$ ,  $A_1^3$ ,  $A_1A_3$ ,  $A_2^2$ ,  $A_1^2A_2$ ,  $A_1^4$  with 4, 8, 6, 14, 12, 10, 8. Only  $D_4$  reaches 24, so  $L$  is isometric to the  $D_4$  lattice and  $\sqrt{2}W$  is its full set of roots. The  $D_4$  root system is unique up to isometry, which gives the last statement.

The finite content of this argument is checked, without appeal to the classification, by `root_lattices_rank4.py` and again by Lean's kernel in `D4RootLattices.lean`, which also exhibits in every lattice with 24 roots four of them with the Cartan matrix of  $D_4$  as Gram matrix, so that the lattice contains a copy of the  $D_4$  lattice of its own determinant and is that lattice: a lattice generated by norm-2 vectors has a basis of such vectors, whose Gram matrix has 2 on the diagonal and  $-1$ , 0 or 1 off it; the script enumerates every positive definite matrix of that shape of order 1 to 4, there are 1, 3, 23 and 393 of them, counts the integer solutions of  $x^T G x = 2$  for each, exactly and within a rigorous box, and finds the largest counts 2, 6, 12 and 24, the count 24 occurring only at determinant 4 with each root having 8 neighbours at inner product 1, 6 at 0, 8 at  $-1$  and one at  $-2$ , which is  $D_4$ ; the next count at rank 4 is 20, at determinant 5, which is  $A_4$ . ■

**7.5.1. The bound of the second level, and its equality case.** For  $t \geq 0$  let  $\mathcal{I}_t$  be the set of subsets of  $S^3$  of size at most  $t$  whose pairwise inner products are at most  $\frac{1}{2}$ ; the empty set belongs to  $\mathcal{I}_t$ , and a contact configuration is an element of  $\mathcal{I}_t$  for  $t$  large enough. Every subset of an element of  $\mathcal{I}_t$  is again one. For a symmetric real function  $K$  on  $\mathcal{I}_2 \times \mathcal{I}_2$  and a set  $Q \in \mathcal{I}_4$  put

$$A_2K(Q) = \sum_{\substack{J_1, J_2 \in \mathcal{I}_2 \\ J_1 \cup J_2 = Q}} K(J_1, J_2), \quad (7.19)$$

the sum running over ordered pairs, so that  $A_2K(\emptyset) = K(\emptyset, \emptyset)$ . Call  $K$  *positive* if for every finite family  $J_1, \dots, J_m$  of elements of  $\mathcal{I}_2$  the matrix  $(K(J_i, J_j))_{i,j}$  is positive semidefinite.<sup>12</sup> This is the second level of the hierarchy that de Laat and Vallentin [dLV15] set up for packing problems in discrete geometry, and the bound it gives is the following; we include the short proof because the equality case, which is what the theorem needs, is read off from it.<sup>13</sup>

**Lemma 7.27** (The bound of the second level). *Let  $K$  be positive, and suppose that  $A_2K(\{x\}) \leq -1$  for every  $x \in S^3$  and that  $A_2K(Q) \leq 0$  for every  $Q \in \mathcal{I}_4$  with  $|Q| \geq 2$ . Then every contact configuration  $C$  satisfies*

$$|C| \leq K(\emptyset, \emptyset).$$

*If  $|C| = K(\emptyset, \emptyset)$ , then  $A_2K(\{x\}) = -1$  for every  $x \in C$  and  $A_2K(Q) = 0$  for every  $Q \subseteq C$  with  $2 \leq |Q| \leq 4$ .*

*Proof.* The set  $C$  is finite, since its points are at pairwise angular distance at least  $\pi/3$  on a compact sphere, so we may argue with finite sums. Let  $\mathcal{J}$  be the family of subsets of  $C$

<sup>12</sup>The hierarchy of [dLV15] works with continuous positive kernels on the compact space  $\mathcal{I}_2$ , because it is stated for an optimisation problem whose feasible set has to be closed. For the bound on one given code only the finite Gram matrices above enter, so continuity is not needed and is not assumed here; the certificate kernel is in fact a polynomial in inner products and is continuous.

<sup>13</sup>The bound is the generalisation to four points of the Delsarte linear programming bound and of the three-point bound of Bachoc and Vallentin [BV08]: with  $\mathcal{I}_1$  in place of  $\mathcal{I}_2$  the same argument gives  $|C| \leq K(\emptyset, \emptyset)$  for a positive kernel on the one-element subsets of  $S^3$ , which is the Delsarte bound in the form of [DGS77], and the equality argument is the one that identifies the extremal codes there, for instance in [CK07]. The only thing that changes with the level is the set of subsets over which the pairs are grouped.

of size at most 2; it lies in  $\mathcal{I}_2$ . Positivity of  $K$  on this family, with every coefficient equal to 1, gives

$$0 \leq \sum_{J_1, J_2 \in \mathcal{I}} K(J_1, J_2) = \sum_{\substack{Q \subseteq C \\ |Q| \leq 4}} A_2 K(Q),$$

the second expression being the first with the pairs grouped by their union  $Q$ , which is a subset of  $C$  of size at most 4 and hence an element of  $\mathcal{I}_4$ . The term  $Q = \emptyset$  is  $K(\emptyset, \emptyset)$ . The  $|C|$  terms with  $|Q| = 1$  are each at most  $-1$ , and every term with  $|Q| \geq 2$  is at most 0. Hence  $0 \leq K(\emptyset, \emptyset) - |C|$ , which is the bound. If  $|C| = K(\emptyset, \emptyset)$  the chain

$$0 \leq \sum_Q A_2 K(Q) = K(\emptyset, \emptyset) + \sum_{|Q|=1} A_2 K(Q) + \sum_{|Q| \geq 2} A_2 K(Q) \leq K(\emptyset, \emptyset) - |C| + 0 = 0$$

collapses, so the two sums take the values  $-|C|$  and 0 that bound them; since each singleton term is at most  $-1$  and each remaining term at most 0, every singleton term equals  $-1$  and every remaining term equals 0.<sup>14</sup> ■

The kernels to which the lemma is applied have a particular form, and their positivity is a matter of that form alone.

**Lemma 7.28** (Gram kernels are positive). *Let  $\Lambda$  be a finite set, let  $H$  be a real or complex inner product space, and for each  $\lambda \in \Lambda$  let  $\psi_{\lambda,1}, \dots, \psi_{\lambda,n_\lambda}$  be maps from  $\mathcal{I}_2$  to  $H$ . Put*

$$Z_\lambda(J_1, J_2)_{ab} = \langle \psi_{\lambda,a}(J_1), \psi_{\lambda,b}(J_2) \rangle,$$

*the inner product being conjugate linear in its first argument when  $H$  is complex, and let  $X_\lambda$  be a real symmetric positive semidefinite matrix of order  $n_\lambda$  for each  $\lambda$ . Then the function*

$$K(J_1, J_2) = \sum_{\lambda \in \Lambda} \sum_{a,b=1}^{n_\lambda} (X_\lambda)_{ab} Z_\lambda(J_1, J_2)_{ab}$$

*satisfies  $K(J_2, J_1) = \overline{K(J_1, J_2)}$ , and its real part is a positive kernel on  $\mathcal{I}_2$ ; when  $K$  is real valued,  $K$  itself is one.*<sup>15</sup>

*Proof.* Write  $X_\lambda = R_\lambda^\top R_\lambda$  with  $R_\lambda$  real, which is possible for a real symmetric positive semidefinite matrix. For  $J_1, \dots, J_m \in \mathcal{I}_2$  and real numbers  $c_1, \dots, c_m$ ,

$$\begin{aligned} \sum_{i,j} c_i c_j K(J_i, J_j) &= \sum_{\lambda} \sum_c \left\langle \sum_{i,a} c_i (R_\lambda)_{ca} \psi_{\lambda,a}(J_i), \sum_{j,b} c_j (R_\lambda)_{cb} \psi_{\lambda,b}(J_j) \right\rangle \\ &= \sum_{\lambda, c} \left\| \sum_{i,a} c_i (R_\lambda)_{ca} \psi_{\lambda,a}(J_i) \right\|^2 \geq 0, \end{aligned}$$

<sup>14</sup>This is the whole of the argument by which an exact bound identifies a configuration, and it is worth seeing how little it uses. It does not use that  $C$  is maximal in any sense other than  $|C| = K(\emptyset, \emptyset)$ , it does not use that the kernel is optimal, and it does not use the structure of  $K$  beyond positivity and the two families of inequalities. What it needs, and what a bound of 24.10 can never give, is that the number  $K(\emptyset, \emptyset)$  is attained by a code; that is why the certificate has to have objective value exactly 24 and not merely below 25, and why its rounding to an exact rational point with objective exactly 24 is the decisive step of its construction.

<sup>15</sup>Nothing about the maps  $\psi_{\lambda,a}$  is used except that they take values in an inner product space, so the positivity of the certificate kernel does not depend on the representation theory that produces the zonal matrices, only on the zonal matrices being Gram matrices, which is their definition (7.21). The representation theory is what makes the entries of  $Z_\lambda$  polynomials in the inner products and what makes every invariant positive kernel expressible in this form; the second of these properties, the completeness of the family, is what an optimiser needs and is not needed here at all, since only one kernel is ever used and it is given.

using  $(X_\lambda)_{ab} = \sum_c (R_\lambda)_{ca} (R_\lambda)_{cb}$  and the sesquilinearity of the inner product. The symmetry  $K(J_2, J_1) = \overline{K(J_1, J_2)}$  follows from  $\langle v, w \rangle = \overline{\langle w, v \rangle}$  and the symmetry of  $X_\lambda$ . The displayed quantity is real and nonnegative, so it equals its real part, which is  $\sum_{i,j} c_i c_j \operatorname{Re} K(J_i, J_j)$ ; that is positivity of  $\operatorname{Re} K$ , and of  $K$  when the two agree.  $\blacksquare$

**7.5.2. The certificate kernel.** The kernel that settles twenty-four points is of the form in Lemma 7.28, with the index set  $\Lambda$  the sixty signatures  $\lambda = (\lambda_1, \lambda_2)$ ,  $\lambda_1 \geq \lambda_2 \geq 0$ ,  $\lambda_1 + \lambda_2 \leq 14$ , of irreducible representations of  $O(4)$ , and with the maps  $\psi_{\lambda,a}$  the functions that [LLM24, §2–3] attach to a subset  $J$  of  $S^3$  of size at most 2: an irreducible representation  $\rho_\lambda$  of  $GL(2, \mathbb{C})$  on a space  $W_\lambda$  carrying a Hermitian inner product, a fixed vector  $w_k \in W_\lambda$  for each admissible index  $a = (i, j, k)$ , a section  $s(J) \in O(4)$  carrying a reference configuration  $\epsilon$  onto  $J$ , a fixed  $2 \times 4$  complex matrix  $\omega$ , and a scalar  $\xi_{\lambda,a}(J)$  which is a polynomial in the inner product of the two points of  $J$  times powers of  $\sqrt{2 \pm 2\langle x, y \rangle}$ , and then

$$\psi_{\lambda,a}(J): \gamma \mapsto \xi_{\lambda,a}(J) \rho_\lambda(\omega \gamma s(J) \epsilon) w_k, \quad \gamma \in O(4), \quad (7.20)$$

an element of the space  $H = L^2(O(4), W_\lambda)$  of square integrable  $W_\lambda$ -valued functions on  $O(4)$  with the inner product  $\langle f_1, f_2 \rangle = \int_{O(4)} \langle f_1(\gamma), f_2(\gamma) \rangle d\gamma$  against Haar probability measure. The *zonal matrices* are the Gram matrices

$$\begin{aligned} Z_\lambda(J_1, J_2)_{ab} &= \langle \psi_{\lambda,a}(J_1), \psi_{\lambda,b}(J_2) \rangle \\ &= \xi_{\lambda,a}(J_1) \xi_{\lambda,b}(J_2) \int_{O(4)} \langle \rho_\lambda(\omega \gamma s(J_1) \epsilon) w_{k_1}, \rho_\lambda(\omega \gamma s(J_2) \epsilon) w_{k_2} \rangle d\gamma, \end{aligned} \quad (7.21)$$

which are invariant under the diagonal action of  $O(4)$  on  $(J_1, J_2)$  because Haar measure is, and whose entries are polynomials in the inner products among the points of  $J_1 \cup J_2$ ; the scalars  $\xi$  are chosen so that this is so, and the  $\sqrt{2 \pm 2u}$  they carry are what absorb the square roots that the section  $s(J)$  introduces. This is the construction of [LLM24], which extends the one of [dLV15] and of the equiangular lines computations that preceded it, and it is recomputed from scratch in Section 7.5.3. The *certificate kernel* is

$$K(J_1, J_2) = \sum_{\lambda \in \Lambda} \sum_{a,b} (X_\lambda)_{ab} Z_\lambda(J_1, J_2)_{ab}, \quad (7.22)$$

where the  $X_\lambda$  are the sixty rational matrices of the deposited data [LLM24d]. Its values are rational polynomials in the inner products, so it is real valued, and it is  $O(4)$ -invariant. For  $Q \in \mathcal{I}_4$  the quantity  $A_2 K(Q)$  of (7.19) is therefore a polynomial in the inner products among the points of  $Q$ ; call it  $p_1, p_2(u), p_3(u_1, u_2, u_3)$  and  $p_4(u_1, \dots, u_6)$  according to  $|Q| = 1, 2, 3, 4$ , the  $u$  being the inner products of the pairs of  $Q$  in a fixed order. Since a set of at most four points of  $\mathbb{R}^4$  is determined up to an orthogonal map by its Gram matrix, these polynomials determine  $A_2 K$  on all of  $\mathcal{I}_4$ .<sup>16</sup>

**Proposition 7.29** (What the verification establishes). *The following statements about the deposited data hold, each by an exact computation described in Section 7.5.3.*

- (i) *Each of the sixty matrices  $X_\lambda$ , and each of the sixty-seven further matrices  $M$  that enter the sums of squares of (iii), is a real symmetric positive definite matrix with rational entries.*

<sup>16</sup>Four points of  $\mathbb{R}^4$  with a given Gram matrix  $G$  are the columns of a  $4 \times 4$  matrix  $P$  with  $P^\top P = G$ , and any two such matrices differ by an orthogonal factor on the left, since  $P_2 = P_1 U$  with  $U$  orthogonal whenever  $P_1^\top P_1 = P_2^\top P_2$ ; for fewer than four points the same holds after adding zero columns. This is why the constraints of the second level are polynomial identities in one, three and six inner products and why the domain of each is described by the positivity of a Gram matrix. It is also why the level stops at four: five points of  $\mathbb{R}^4$  have Gram matrices constrained by a determinant condition, and the third level would need it.

- (ii)  $K(\emptyset, \emptyset) = 24$ .
- (iii) *There are polynomials  $\sigma_1, \dots, \sigma_4$ , each a sum of terms  $w \cdot \langle M, mm^\top \rangle$  with  $M$  one of the matrices of (i),  $m$  a vector of monomials in the inner products of  $Q$ , and  $w$  a weight which is a nonnegative constant, or  $(u+1)(\frac{1}{2}-u)$  for an inner product  $u$  of  $Q$ , or the determinant of the Gram matrix of  $Q$ , or an elementary symmetric function of the family of the  $(u+1)(\frac{1}{2}-u)$  over the inner products of  $Q$ , or, for  $|Q| = 4$ , an elementary symmetric function of the four  $3 \times 3$  principal minors of the Gram matrix of  $Q$ , such that the four identities*

$$p_1 + 1 + \sigma_1 = 0, \quad p_2 + \sigma_2 = 0, \quad p_3 + \sigma_3 = 0, \quad p_4 + \sigma_4 = 0$$

*hold identically as polynomials in the inner products.*

- (iv) *The polynomial  $\sigma_2$  has degree 16, and its zeros in  $[-1, \frac{1}{2}]$  are exactly  $-1, -\frac{1}{2}, 0$  and  $\frac{1}{2}$ , of multiplicities 1, 2, 2 and 1.*

Every weight in (iii) is nonnegative at every  $Q \in \mathcal{I}_4$ : the inner products of  $Q$  lie in  $[-1, \frac{1}{2}]$ , so each  $(u+1)(\frac{1}{2}-u)$  is nonnegative; the Gram matrix of  $Q$  is positive semidefinite, so its determinant and its principal minors are nonnegative; and an elementary symmetric function of nonnegative quantities is nonnegative. Each  $\langle M, mm^\top \rangle = m^\top M m$  is nonnegative because  $M$  is positive definite. Hence  $\sigma_k(Q) \geq 0$  on  $\mathcal{I}_4$  for  $k = 1, \dots, 4$ , and (iii) says

$$A_2 K(\{x\}) = -1 - \sigma_1 \leq -1, \quad A_2 K(Q) = -\sigma_{|Q|}(Q) \leq 0 \quad (2 \leq |Q| \leq 4), \quad (7.23)$$

for every  $x \in S^3$  and every  $Q \in \mathcal{I}_4$  with  $|Q| \geq 2$ .

**Theorem 7.30** (Twenty-four points). *Let  $C \subset S^3$  be a set whose pairwise inner products are at most  $\frac{1}{2}$ . Then  $|C| \leq 24$ . If  $|C| = 24$ , then every pairwise inner product of  $C$  lies in  $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$ .*

*Proof.* By Proposition 7.29(i) and Lemma 7.28, the certificate kernel (7.22) is positive; it is real valued, so no real part has to be taken. By (7.23) it satisfies the hypotheses of Lemma 7.27, and by Proposition 7.29(ii) that lemma gives  $|C| \leq K(\emptyset, \emptyset) = 24$ . If  $|C| = 24$  the equality case of the lemma gives  $A_2 K(\{x, y\}) = 0$  for every pair of distinct points  $x, y \in C$ , that is  $\sigma_2(\langle x, y \rangle) = 0$  by (7.23), and Proposition 7.29(iv) places  $\langle x, y \rangle$  in  $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$ . ■

The first assertion is Musin's theorem [Mus08] that the kissing number in four dimensions is 24, obtained here a second time from a different certificate. It is the second assertion that the present paper needs, and it is what the earlier bound 24.10 of Bachoc and Vallentin [BV08], at the first level of the same hierarchy, and Musin's argument, which does not identify the configuration, do not give: it is the equality case, and the equality case is available only because the bound is exactly 24 and not merely below 25.

*Proof of Theorem 7.25.* A contact configuration is a subset of  $S^3$  whose pairwise inner products are at most  $\frac{1}{2}$ , that is a spherical code of minimal angle at least  $\pi/3$ . By Theorem 7.30 the pairwise inner products of its 24 points lie in  $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$ , and Lemma 7.26 makes it a copy of the root system. The cell of the root configuration is the 24-cell of volume 8 by Lemma 2.3, and volume is invariant under orthogonal maps. ■

7.5.3. *The certificate, checked.* What remains is Proposition 7.29, which is a statement about a finite set of rational numbers, and we are exact about where those numbers come from and what was done with them. The certificate is the discovery of de Laat, Leijenhorst and de Muinck Keizer [LLM24]: they computed the second level of the hierarchy of [dLV15]

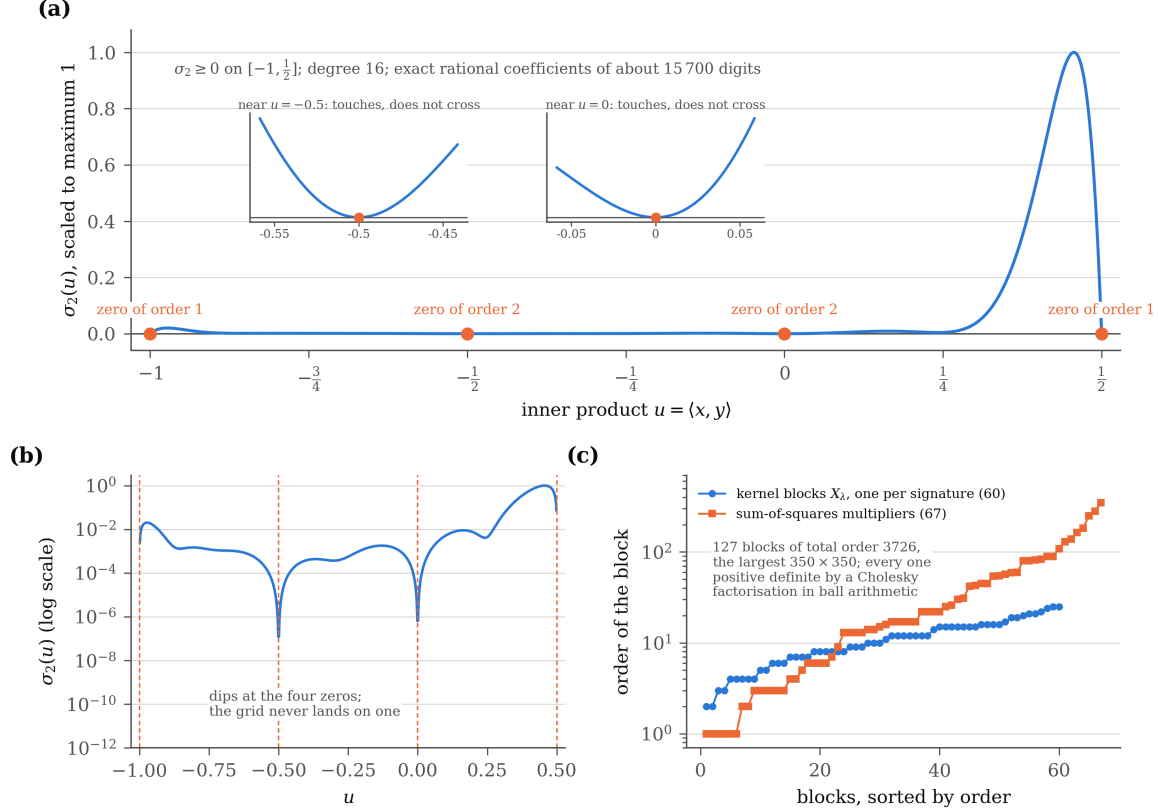


FIGURE 12. The certificate, as Proposition 7.29 reads it. (a) The two-point polynomial  $\sigma_2$  on  $[-1, \frac{1}{2}]$ , evaluated exactly from the deposited rational coefficients and scaled by its maximum: it is nonnegative, it vanishes to first order at  $-1$  and  $\frac{1}{2}$  and to second order at  $-\frac{1}{2}$  and  $0$ , and nowhere else; the insets show the two double zeros, where the curve touches the axis without crossing it. (b) The same on a logarithmic scale. (c) The orders of the 127 positive definite blocks, the sixty kernel blocks  $X_\lambda$  and the sixty-seven sum-of-squares multipliers; the largest is  $350 \times 350$ .

for spherical codes in  $S^3$  with maximal inner product  $\frac{1}{2}$ , reduced by the symmetry of the problem to the form (7.22) with truncation degrees 14 and 16, solved it to forty digits in 256-bit floating-point arithmetic by the clustered low-rank solver of [LdL24], about two weeks on eight cores with 128 gigabytes of memory, rounded the numerical optimum to an exact rational feasible point with objective value exactly 24, and deposited that point, with their own verification code, at 4TU.ResearchData [LLM24d], under the MIT licence. The archive is `LasserreSphericalCodes.zip`, MD5 02acd5270f7b3fa799abdeb5291706fd, 234 megabytes in 331 files under `proofs/4_24`; recording the checksum fixes which bytes the statements below refer to, and the licence permits the files to be redistributed, which the supplementary archive of this paper [BBMB26] does, with the licence notice, so that every input to Proposition 7.29 travels with the paper.<sup>17</sup> Their paper is, at the time of writing, a preprint; nothing below depends on it.

<sup>17</sup>The point of redistributing rather than pointing is that a proof which depends on a file should not depend on a server. The MIT licence asks only that its notice accompany the copy, and it does. The checksum is recorded so that the copy in the supplementary archive, the copy at 4TU.ResearchData and the

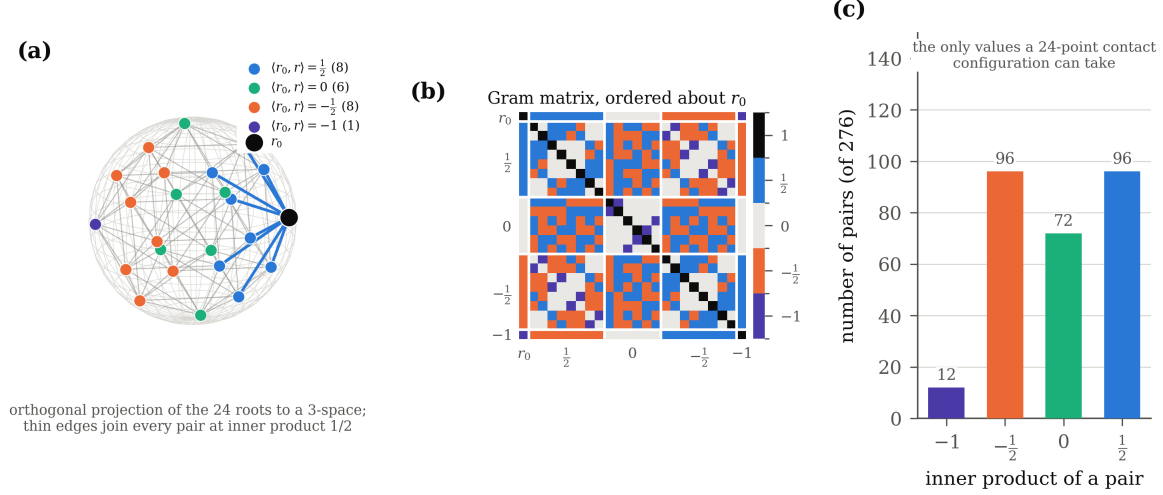


FIGURE 13. What Theorem 7.30 forces. (a) The twenty-four normalised roots of  $D_4$ , projected orthogonally to a generic 3-space so that no two coincide, with every pair at inner product  $\frac{1}{2}$  joined; one root  $r_0$  is singled out and the others are coloured by their inner product with it, eight at  $\frac{1}{2}$ , six at 0, eight at  $-\frac{1}{2}$  and the antipode at  $-1$ . (b) The Gram matrix of the twenty-four directions, rows and columns ordered by these classes; it takes no value outside  $\{1, \frac{1}{2}, 0, -\frac{1}{2}, -1\}$ . (c) The 276 pairs by inner product: 96, 72, 96 and 12. The theorem says that every contact configuration of size 24 has a Gram matrix with the values of (b), and Lemma 7.26 that it then is (a) up to an orthogonal map.

We have verified Proposition 7.29 from the deposited data in code of our own that shares nothing with theirs (`11m24_certificate_check.py`, Python with the exact rational arithmetic of FLINT and the ball arithmetic of Arb [Joh17], the programs of the directory `zonal`, and `D4InnerProducts.lean`). The verification has seven steps, and we describe them in the order in which they are run. The data are read and their format checked: 127 positive semidefinite blocks of total dimension 3726, the largest  $350 \times 350$ , sixty of them indexed by the signatures  $\lambda$  of  $O(4)$  with  $|\lambda| \leq 14$  and each of those with exactly as many rows as the paper's description of the admissible index tuples prescribes, the rest the multipliers of the sums of squares for the constraints on two, three and four points; every one of the 127 blocks is positive definite, by a Cholesky factorisation in ball arithmetic at 256 bits in which every pivot is a ball inside the positive reals (the least about  $1.4 \times 10^{-15}$ ), and the 81 blocks of size at most 16 also by an exact  $LDL^T$  over the rationals; each of the 125 prefactors of the sum-of-squares terms is a nonnegative multiple of one of the weights that describe the domain, so that every such term is nonnegative on it; the objective, the entry of the block  $\lambda = (0, 0)$  that is the value of the bound, is exactly 24; and the two-point polynomial  $p_2$ , computed exactly from the data, has degree 16 and vanishes on  $[-1, \frac{1}{2}]$  at  $-1, -\frac{1}{2}, 0, \frac{1}{2}$  and nowhere else, with multiplicities 1, 2, 2, 1, by a Sturm sequence over the rationals, the last step being checked by Lean's kernel as well.

The two steps that the memory had put out of reach are done here as well. Step 3 is the construction of the zonal matrices  $Z_\lambda$ , the polynomials in the six inner products of four

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points through which the kernel is expressed; step 5 is the check, which uses them, that the four constraint polynomials  $A_2 K(Q) + (\text{sum of squares})$  are identically equal to their right-hand sides. The authors' implementation of step 3 takes about three days and 128 gigabytes. That memory is not intrinsic to it. What step 3 wants, for each signature  $\lambda$  and each admissible pair of indices  $k_1, k_2$ , is

$$P(S)_{k_1, k_2} = \int_{O(4)} \rho_{0, k_1}(\omega \gamma \varepsilon) \rho_{0, k_2}(\omega \gamma S) d\gamma,$$

and the integral is taken monomial by monomial in the sixteen entries of  $\gamma$ . Their code expands the product of the two representation polynomials first and holds the expansion in a table indexed by the exponent matrix of  $\gamma$ , with a polynomial in  $S$  against each index; that table is the 128 gigabytes. It need never be formed. The answer is homogeneous of degree  $|\lambda| \leq 14$  in the seven free entries of  $S$ , so it has at most  $\binom{20}{6} = 38\,760$  coefficients whatever  $\lambda$  is, and each of the two factors of the integrand is a product of powers of single entries of a  $2 \times 2$  matrix; so their monomials can be walked in triples, each triple added straight into the coefficient of  $P(S)$  it belongs to, with nothing held but the factors themselves. Two facts keep the walk short. The integral vanishes unless every row sum and every column sum of the exponent matrix is even, which leaves  $7.85 \times 10^9$  of the  $2.39 \times 10^{11}$  triples; and it is unchanged by permuting rows or columns, so its values can be stored against the canonical form of the matrix, of which at most two and a half million occur in any one of the 490 entries. Carried out this way, in exact rational arithmetic, the whole of step 3 takes about two hours on two cores and never needs more than 400 megabytes.

Step 5 then follows. The four constraint polynomials are assembled from the zonal matrices and from the certificate's own blocks and factored sum-of-squares data, the sum (7.19) being taken over ordered pairs, which gives it 3, 9, 12 and 6 terms for  $|Q| = 1, 2, 3, 4$ , an unordered pair  $J_1 \neq J_2$  entering with the factor 2 that its two orders supply and a pair  $J_1 = J_2$  once; and all four are identically zero: the one-point constraint, the two-point constraint in one variable, the three-point constraint in three variables and the four-point constraint in six, that last one assembled out of pieces carrying up to seventy-five thousand coefficients of several thousand digits. The two sides of that last constraint end on the same 53 572 monomials, and on each one the coefficient from the sum-of-squares blocks is the exact negative of the coefficient from the zonal matrices, both about 15 700 digits long, so the constraint is zero term by term and not by any cancellation of rounding (Figure 14). None of this shares code with the authors' Julia package; the inputs are the deposited data and the description of the construction in their paper. Three checks that do not depend on the outcome fix the construction itself. The integration routine reproduces the seven values of the test distributed with their own integration code, and satisfies seventy further identities that follow from the rows and the columns of an orthogonal matrix being unit vectors. For the signatures  $\lambda = (k, 0)$  and two single points at inner product  $u$  the matrix  $Z_\lambda$  has to be a multiple of the zonal spherical harmonic kernel of  $S^3$ ; it is, exactly, with ratio  $8^k / (k+1)^2$  for every  $k \leq 14$ , the denominator being the dimension of the space of harmonics of degree  $k$  on  $S^3$ . And  $Z_\lambda$  has to be a positive semidefinite kernel on the subsets of size at most two of any set of points, which nothing in the computation forces; on random four-point sets the least eigenvalue is at the precision of the arithmetic for every signature, including those with  $\lambda_2 > 0$ , which the comparison with the Gegenbauer polynomials does not reach.

The seven steps establish Proposition 7.29 item by item. Step 2 is (i): every block is positive definite, by a Cholesky factorisation in ball arithmetic whose pivots are balls inside the positive reals, and the small ones also by an exact  $LDL^T$  over the rationals. Step 6

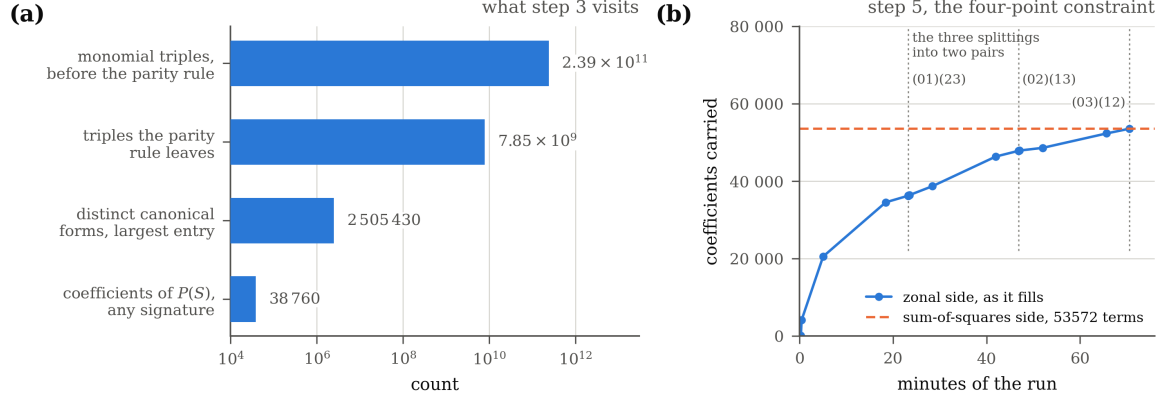


FIGURE 14. The two steps of the verification of the certificate that its authors' own code [LLM24d] leaves out, as they were carried out here. (a) Why step 3 fits in a few hundred megabytes. The integral over  $O(4)$  is taken one monomial triple at a time; the parity rule on the exponent matrix discards all but about a thirtieth of the triples, the invariance under permuting rows and columns collapses those onto at most 2 505 430 canonical forms in the largest of the 490 entries, and the answer never has more than  $\binom{20}{6} = 38\,760$  coefficients whatever the signature. The first two counts are the counters in the kernel; the third is the size of its memo table. (b) Step 5 for the four-point constraint, the only one large enough to be difficult. The curve is the zonal side of the constraint filling up, one signature at a time, through the three ways of splitting four points into two pairs; it stops at 53 572 coefficients, which is where the sum-of-squares side stops as well. The two sides sit on the same monomials and each coefficient of one is the exact negative of the other, so the difference is zero at every one of the 53 572 places.

is (ii). Steps 3, 4 and 5 are (iii): the zonal matrices are built from the definition (7.21), the prefactors are checked to be nonnegative multiples of the domain weights, and the four identities are checked to hold with every coefficient exactly zero. Step 7 is (iv), by a Sturm sequence over the rationals, and the last statement is checked by Lean's kernel as well. Step 1 is the reading of the data. What the computation therefore establishes is exactly the list of Proposition 7.29; Lemmas 7.27 and 7.28 are what turn that list into Theorem 7.30, and they are proved above. Nothing in the argument is read from [LLM24] rather than run or proved here: the authors of the certificate proved the same bound and the same equality case in their Sections 2 to 5, and a reader who wants a second account of Lemma 7.27 will find it there, but the present proof does not cite it.

Everything after Theorem 7.30, the passage from the four inner products to the root system, is Lemma 7.26, proved above by a classical argument and checked by an exact enumeration, in Python and in Lean. Every statement below that invokes Theorem 7.25 therefore rests on Proposition 7.29, that is on a finite exact computation reproducible from the supplementary archive, and on nothing outside this paper. All the other results in this paper are independent even of that.

**Corollary 7.31** (The multi-direction conjecture). *Conjecture 1.4 holds. Indeed a packing configuration at the all-contact corner with a full active set, twenty-four neighbours at distance exactly 2, is a contact configuration of size 24, so by Theorem 7.25 it is a copy of the*

root system; if the  $24 - m$  undeviated directions are the roots  $u_i$ ,  $i \notin D$ , the orthogonal map fixing them is the identity, so every tilt  $\varphi_j$  vanishes and  $\text{vol}(V_c) = 8$  with equality.

**7.6. A volume identity exact at the root system.** The truncated estimates of Section 7.4 and Section 7.11 cut the cell at a radius below  $\sqrt{2}$ . At  $m = 24$  no such truncation can be sharp: the 24-cell has circumradius  $\sqrt{2}$  (Lemma 2.3), so a ball of smaller radius removes volume from it, and the lower bound falls strictly below the true value 8 at the root configuration. This subsection truncates at  $\sqrt{2}$  itself and keeps three terms of inclusion and exclusion. The resulting lower bound is exact at the root configuration, and it converts the statement  $\text{vol}(V_c) \geq 8$  for 24 contacts into one inequality between the pair and triple angles of the contact directions, of the same form as the inequality proved for 23 contacts in Section 7.12. Remark 7.36 records what the three-point relaxation of that inequality does and does not deliver.

Throughout,  $B = \{x \in \mathbb{R}^4 : |x| \leq \sqrt{2}\}$ , and for a unit vector  $w$  we write  $K(w) = \{x \in B : \langle x, w \rangle > 1\}$ . By orthogonal invariance the numbers

$$\begin{aligned} c &= \text{vol } K(w), \\ \omega_2(u) &= \text{vol}(K(w) \cap K(w')), \\ \omega_3(u, v, t) &= \text{vol}(K(w) \cap K(w') \cap K(w'')) \end{aligned}$$

depend only on the inner products  $u = \langle w, w' \rangle$  and, for  $\omega_3$ , on

$$(u, v, t) = (\langle w, w' \rangle, \langle w, w'' \rangle, \langle w', w'' \rangle).$$

**Lemma 7.32** (Third-order bound). *Let  $w_1, \dots, w_m$  be unit vectors in  $\mathbb{R}^4$ , write  $u_{ij} = \langle w_i, w_j \rangle$  and  $V = \{x : \langle x, w_i \rangle \leq 1 \text{ for all } i\}$ . Then*

$$\text{vol}(V) \geq 2\pi^2 - mc + \sum_{i < j} \omega_2(u_{ij}) - \sum_{i < j < k} \omega_3(u_{ij}, u_{ik}, u_{jk}), \quad (7.24)$$

with equality if and only if  $V \setminus B$  and the set of points of  $B$  lying in at least four of the sets  $K(w_i)$  both have measure zero.

*Proof.* First  $\text{vol}(V) \geq \text{vol}(V \cap B) = \text{vol}(B) - \text{vol}(\bigcup_i K(w_i))$ , since  $B \setminus V = \bigcup_i K(w_i)$ , and  $\text{vol}(B) = \frac{1}{2}\pi^2(\sqrt{2})^4 = 2\pi^2$ . For a point lying in exactly  $n \geq 1$  of the sets, the identity  $\sum_{j=0}^3 (-1)^j \binom{n}{j} = -\binom{n-1}{3}$  gives

$$\binom{n}{1} - \binom{n}{2} + \binom{n}{3} = 1 + \binom{n-1}{3} \geq 1,$$

with equality exactly when  $n \leq 3$ . So the indicator of the union is bounded above, pointwise, by the sum of the indicators of the sets, less the sum over pairs, plus the sum over triples, with equality off the points lying in four or more of the sets. Integration over  $B$  gives (7.24), and the two inequalities used are equalities exactly under the stated conditions. ■

**Lemma 7.33** (The cap and the pair term). *The cap volume is  $c = \frac{1}{2}\pi^2 - \frac{4}{3}\pi$ . For  $-1 \leq u \leq 0$  one has  $\omega_2(u) = 0$ , and  $\omega_3(u, v, t) = 0$  unless  $u, v$  and  $t$  are all positive. For  $0 < u < 1$ , with  $\gamma = \arccos u$ ,*

$$\begin{aligned} \omega_2(u) &= 2\pi \int_{\gamma/2}^{\pi/4} \left(1 - \frac{1}{2} \sec^2 \theta\right)^2 d\theta = 2\pi \left(F\left(\frac{\pi}{4}\right) - F\left(\frac{\gamma}{2}\right)\right), \\ F(\theta) &= \theta - \frac{3}{4} \tan \theta + \frac{1}{12} \tan^3 \theta. \end{aligned} \quad (7.25)$$

In particular  $\omega_2(\frac{1}{2}) = \pi(9\pi + 26\sqrt{3} - 72)/54 = 0.0760762770\dots$

*Proof.* Write  $x = (y, z)$  with  $y = \langle x, w \rangle$  and  $z$  in the orthogonal complement of  $w$ . Then  $K(w)$  is the set  $1 < y \leq \sqrt{2}$ ,  $|z|^2 \leq 2 - y^2$ , and

$$\begin{aligned} c &= \int_1^{\sqrt{2}} \frac{4}{3} \pi (2 - y^2)^{3/2} dy = \frac{16}{3} \pi \int_{\pi/4}^{\pi/2} \cos^4 \phi d\phi \\ &= \frac{16}{3} \pi \left( \frac{3\pi}{32} - \frac{1}{4} \right), \end{aligned}$$

by the substitution  $y = \sqrt{2} \sin \phi$ . If  $x \in K(w) \cap K(w')$  then  $\langle x, w + w' \rangle > 2$  and  $|w + w'|^2 = 2 + 2u$ , so  $|x|^2 > 2/(1 + u)$ ; for  $u \leq 0$  this exceeds 2 and the intersection is empty, and the same argument applied to any two of three directions gives the statement on  $\omega_3$ . For  $0 < u < 1$  decompose  $x = y + z$  with  $y$  in the plane  $P$  spanned by  $w, w'$  and  $z \perp P$ . The conditions are  $\langle y, w \rangle > 1$ ,  $\langle y, w' \rangle > 1$  and  $|z|^2 \leq 2 - |y|^2$ , so  $\omega_2(u)$  is the integral of  $\pi(2 - |y|^2)_+$  over the region of  $P$  cut out by the first two. In polar coordinates on  $P$  with  $w$  at angle 0 and  $w'$  at angle  $\gamma$ , the ray at angle  $\theta \in (\gamma - \frac{\pi}{2}, \frac{\pi}{2})$  meets that region in  $\rho > \max(\sec \theta, \sec(\gamma - \theta))$ , and the reflection  $\theta \mapsto \gamma - \theta$  exchanges the two terms, so it suffices to take  $\theta \geq \gamma/2$ , where the maximum is  $\sec \theta$ , and to double. Since

$$\int_{\sec \theta}^{\sqrt{2}} \pi(2 - \rho^2) \rho d\rho = \pi \left( 1 - \frac{1}{2} \sec^2 \theta \right)^2$$

for  $\sec \theta \leq \sqrt{2}$ , while for  $\theta > \pi/4$  the ray does not meet the region inside  $B$ , (7.25) follows, and  $\gamma/2 < \pi/4$  because  $u > 0$ . Differentiation gives  $F'(\theta) = 1 - \sec^2 \theta + \frac{1}{4} \sec^4 \theta = (1 - \frac{1}{2} \sec^2 \theta)^2$ . At  $\gamma = \pi/3$ ,  $F(\frac{\pi}{4}) = \frac{\pi}{4} - \frac{2}{3}$  and  $F(\frac{\pi}{6}) = \frac{\pi}{6} - \frac{13}{18\sqrt{3}}$ . ■

**Proposition 7.34** (Exactness at the root system). *For the 24 normalised roots of  $D_4$ , (7.24) holds with equality, and*

$$8 = 32\pi - 10\pi^2 + 96\omega_2(\tfrac{1}{2}) - 96\omega_3(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}). \quad (7.26)$$

Consequently  $\omega_3(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}) = \frac{1}{16}\pi^2 - \pi + \frac{13\sqrt{3}}{27}\pi - \frac{1}{12} = 0.0118567031\dots$

*Proof.* The cell is the 24-cell, of volume 8, whose vertices all have norm  $\sqrt{2}$  (Lemma 2.3); so it lies in  $B$ . By Lemma 7.33 a point of  $K(w_i) \cap K(w_j)$  forces  $u_{ij} > 0$ , and the only positive inner product between distinct normalised roots is  $\frac{1}{2}$ . Suppose a point of  $B$  lies in four of the sets. The four corresponding roots, scaled to norm  $\sqrt{2}$ , then have Gram matrix  $I + J$ , with 2 on the diagonal and 1 off it, whose determinant is 5. They are therefore linearly independent and span a sublattice  $L$  of full rank of the  $D_4$  lattice, whose determinant is 4; but  $\det L = 4[D_4 : L]^2$ , and 5 is not of that form. So every point of  $B$  lies in at most three of the sets, and Lemma 7.32 gives equality. In the sum over pairs only the pairs at inner product  $\frac{1}{2}$  contribute; each root has eight such neighbours ( $e_1 + e_2$  has  $e_1 \pm e_3$ ,  $e_1 \pm e_4$ ,  $e_2 \pm e_3$ ,  $e_2 \pm e_4$ ), giving 96 pairs. In the sum over triples only triangles of roots pairwise at  $\frac{1}{2}$  contribute; each such pair lies in exactly three of them (for  $e_1 + e_2$  and  $e_1 + e_3$  the third root is  $e_1 + e_4$ ,  $e_1 - e_4$  or  $e_2 + e_3$ ), giving  $96 \cdot 3/3 = 96$  triangles. With  $2\pi^2 - 24c = 32\pi - 10\pi^2$  from Lemma 7.33 this is (7.26), and the value of  $\omega_3$  follows from the value of  $\omega_2(\frac{1}{2})$ . ■

The value of  $\omega_3(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$  obtained this way agrees with a direct quadrature of the triple intersection to  $1 \times 10^{-7}$ , and the cap, pair and triple volumes agree with Monte Carlo estimates in  $\mathbb{R}^4$  within their standard errors (`m24_exact_reduction.py`, Appendix F).

**Corollary 7.35** (The 24-contact case as an inequality between angles). *For a contact configuration  $W = \{w_1, \dots, w_{24}\}$  put*

$$E(W) = \sum_{i < j} \omega_2(u_{ij}) - \sum_{i < j < k} \omega_3(u_{ij}, u_{ik}, u_{jk}).$$

*Then  $\text{vol}(V_c(W)) \geq 32\pi - 10\pi^2 + E(W)$ . The root configuration has  $E = \tau$ , where*

$$\tau = 8 - 32\pi + 10\pi^2 = 6.1650790960\dots,$$

*and every  $W$  with  $E(W) \geq \tau$  has  $\text{vol}(V_c(W)) \geq 8$ .*

*Proof.* Lemma 7.32 with  $m = 24$ , and Proposition 7.34. ■

*Remark 7.36* (The three-point relaxation of Corollary 7.35). The inequality  $E(W) \geq \tau$  for all contact configurations of 24 directions would give the volume statement of Theorem 7.25 at the three-point level, without the four-point kernel of Theorem 7.30, and it has the form treated in Section 7.12. Splitting each pair over the 22 triples that contain it,  $E(W) = \sum_{i < j < k} h(u_{ij}, u_{ik}, u_{jk})$  with  $h(u, v, t) = \frac{1}{22}(\omega_2(u) + \omega_2(v) + \omega_2(t)) - \omega_3(u, v, t)$ . In the notation of Section 7.12, a function  $f = \sum_k f_k G_k$  with  $f_k \geq 0$  for  $k \geq 1$  and positive semidefinite matrices  $F_k$  satisfying

$$\begin{aligned} h(u, v, t) &\geq \frac{1}{22}(f(u) + f(v) + f(t)) + F(u, v, t) \\ &\quad + \frac{1}{22}(F(1, u, u) + F(1, v, v) + F(1, t, t)) \end{aligned}$$

on every admissible triple give  $E(W) \geq \frac{1}{2}(576f_0 - 24f(1)) - 4F(1, 1, 1)$  for every  $W$  of size 24. Because the root configuration attains  $\tau$ , a certificate reaching  $\tau$  must be sharp, and exact rounding needs it to be attained at a finite degree.

The dual of that problem is a probability measure  $p$  on admissible triples subject to the two-point conditions  $1 + \frac{23}{3} \mathbb{E}_p[P_k(u) + P_k(v) + P_k(t)] \geq 0$  for  $1 \leq k \leq d$  and, writing  $\Sigma_k = S_k(1, u, u) + S_k(1, v, v) + S_k(1, t, t)$ , the three-point conditions

$$24 S_k(1, 1, 1) + 552 \mathbb{E}_p[\Sigma_k] + 12144 \mathbb{E}_p[S_k] \succeq 0$$

for  $0 \leq k \leq d$ , in the matrices  $S_k$  of [BV08]. Every such  $p$  satisfies  $2024 \mathbb{E}_p[h] \geq V_d$ , where  $V_d$  is the value of the best degree- $d$  certificate, so a single feasible  $p$  with  $2024 \mathbb{E}_p[h] < \tau$  rules out every degree- $d$  certificate at once. Restricting  $p$  to a finite set of triples only shrinks the dual feasible set, so such a  $p$  is a proof and not an estimate.

At degree 6 the computation points to one, and we state what it shows and no more. `m24_primal_sdp.py` solves the dual on a grid of admissible triples with every constraint held with a margin  $\varepsilon = 10^{-6}$ , and rechecks the returned measure against the constraints without the margin: the measure deposited in the supplement has value 5.9424 on 2170 triples, and a rerun with a later solver gives 5.9851 on 522 triples, with least eigenvalue  $10^{-6}$  in the moment matrices and least two-point slack  $7.7 \times 10^{-3}$ ; both values lie below  $\tau = 6.1650791$  by more than 0.18. That is strong evidence that no degree-6 certificate of this form exists. It is not a proof in the sense of Section 7.12: a proof would round the measure to rationals, check the moment matrices by an exact  $LDL^\top$ , and bound  $\omega_3$  rigorously in the value, and the supplement does not carry such a check. Nothing in this paper depends on the conclusion.

At degrees 8 through 20, on a grid of 1264 triples, the dual minimum is 6.0395, 6.0700, 6.0879, 6.1060, 6.1256, 6.1416 and 6.1516, short of  $\tau$  by 0.126, 0.095, 0.077, 0.059, 0.039, 0.023 and 0.014. These are double-precision values, but the direction of the discretisation is favourable: restricting the measure to a grid only shrinks the dual feasible set and so only raises the minimum, and refining the grid by a factor of 2.6 at degrees 6 and 8 moves the

value by less than 0.005. At degree 20 the shortfall is still  $1.4 \times 10^{-2}$ , where a certificate that is to be rounded exactly would need it to be zero.

Four further computations say that the shortfall is structural rather than numerical. First, the same two-point and three-point conditions, read as conditions on a code of  $N$  points instead of 24, remain feasible up to  $N = 26.61$  at degree 4,  $N = 24.77$  at degree 6 and  $N = 24.24$  at degree 8: the relaxation does not know that 24 is the largest number of contacts available. Since  $E(W) \geq \tau$  holds with equality at the root configuration and, by Theorem 7.25, only there, a certificate for it is a certificate of uniqueness, and a relaxation that does not see maximality does not see uniqueness. Second, the minimising  $p$  is not a perturbation of the root configuration: it carries about a third of its pair mass at  $u = \frac{1}{2}$  and spreads the rest continuously over  $(-1, 0)$ , where the root configuration has atoms at  $-1$ ,  $-\frac{1}{2}$  and  $0$  alone. Requiring the two-point slacks at  $k = 2$  and  $k = 4$ , which vanish exactly at the root configuration, to exceed  $\varepsilon$  raises the degree-8 minimum only from 6.0395 to 6.0489 at  $\varepsilon = 0.1$ , and is infeasible at  $\varepsilon = 0.3$ ; so the shortfall is not sharpness at the root configuration, and treating a neighbourhood of it separately does not remove it. Third, the largest  $s$  for which the moment matrices admit  $\succeq sI$  falls from 0.75 at degree 4 to  $1.7 \times 10^{-2}$  at degree 6 and to about  $10^{-4}$  at degrees 8 and above: from degree 8 on the dual has no usable interior. That is a property of the feasible set, not of the arithmetic, and it is what makes interior-point solvers degrade there. Fourth, the root configuration never becomes a local minimum of the relaxation. Minimising the directional derivative of the objective over the directions the degree- $d$  conditions leave open at the root configuration gives  $-1.5705 \times 10^{-4}$  at degree 6 and  $-1.0951 \times 10^{-4}$ ,  $-8.7210 \times 10^{-5}$ ,  $-7.8417 \times 10^{-5}$  and  $-7.6223 \times 10^{-5}$  at degrees 8, 10, 12 and 14, each computed on the support of the direction with the closed form of  $\omega_3$  rather than a table, and each agreeing to four figures with the value the table gives. Fitting  $L + Cq^d$  to the five returns  $L = 7.25 \times 10^{-5}$  with  $q = 0.654$  and residuals below  $9 \times 10^{-7}$ , and dropping any one of the five moves  $L$  only within  $[7.03, 7.43] \times 10^{-5}$ . The improvement available at the root configuration therefore does not tend to zero with the degree, where a certificate reaching  $\tau$  would need it to be zero. The sweep is `m24_descent_sweep.py`.

That the three-point level is the wrong level is not a conclusion of ours alone. Cohn, de Laat and Leijenhorst [CdLL24] settle a family of spherical codes by three-point bounds with exact rounding, and record that the one member of it they cannot settle is the 24 points of  $R$  in  $\mathbb{R}^4$ , for which, in their words, three-point bounds do not seem to be sharp. Two of the three are authors of [LLM24], and the four-point computation there is what they went to instead.

Extended precision therefore does not decide this question in the affirmative, and Theorem 7.25 is proved from the four-point kernel of Theorem 7.30. What Corollary 7.35 adds is an exact reduction of its volume consequence to a single inequality between pair and triple angles; what the computations above add is the location of the obstruction. Closing the 24-contact case at the three-point level calls not for a longer computation at higher precision, but for a method that certifies the uniqueness of the root configuration, which is what the second level of the hierarchy supplies and the first does not.

*Remark 7.37* (How far the elementary constraints reach). They reach a definite distance and stop, and we record where. The strongest of them is a second moment. For 24 unit vectors in  $\mathbb{R}^4$  the Gram matrix  $G = (u_{ij})$  is positive semidefinite of rank at most 4 with trace 24, so Cauchy–Schwarz applied to its nonzero eigenvalues gives  $\|G\|_F^2 \geq 24^2/4 = 144$ ,

that is

$$\sum_{i < j} u_{ij}^2 \geq 60, \quad (7.27)$$

with equality exactly when the configuration is a tight frame. The root system attains it, its frame operator being  $6I$ : the 96 pairs at  $\frac{1}{2}$  contribute 24, and the 96 at  $-\frac{1}{2}$  and the 12 at  $-1$  contribute 36.

Write  $P$  for the set of pairs with  $u_{ij} > 0$ , the only ones on which  $\omega_2$  and  $\omega_3$  are supported. Since  $u_{ij} \leq \frac{1}{2}$  there, each such pair contributes at most  $\frac{1}{4}$  to the second moment, so a bound  $\sum_P u_{ij}^2 \geq 24$  would force  $|P| \geq 96$ , exactly the count of the root system. That bound does not follow from (7.27) alone: it needs the pairs outside  $P$  to carry at most 36, which is the root system's own value and is not available in advance.

Even granting it the route stops short. The ratio  $\omega_2(u)/u^2$  increases on  $(0, \frac{1}{2}]$ , from 0.0487 at  $u = 0.05$  to 0.3043 at  $u = \frac{1}{2}$ , so for a fixed second moment  $\sum_P \omega_2(u_{ij})$  is smallest when the mass is spread over as many pairs as possible at as small a value as possible. A budget  $\sum_P u_{ij}^2 = 24$  spread over all 276 pairs, at  $u = 0.2949$ , gives  $\sum_P \omega_2 = 5.1357$ , short of  $\tau = 6.1651$  by 1.0294, with the triple term still to be subtracted; the same budget concentrated on 96 pairs at  $\frac{1}{2}$  gives the root system's 7.3033. The second moment fixes how much inner-product mass a configuration carries but not where that mass sits, and the difference between the two placements exceeds the entire margin to be proved.

What is missing here and in Remark 7.36 is the same thing: a reason for the positive inner products to concentrate at  $\frac{1}{2}$  instead of spreading. That reason is the uniqueness of the root system among 24-point configurations with  $u_{ij} \leq \frac{1}{2}$ , and a treatment of  $m = 24$  has to supply it. The equality case of the second-level bound, Theorem 7.30, supplies it, which is why the argument goes through that kernel.

The extendability of a configuration turns out to be exactly a statement about the reach of its own cell, which is worth recording because it converts the remaining case into a question about one polytope.

**Proposition 7.38** (Extendability and the circumradius). *Let  $W$  be a contact configuration with bounded cell and write*

$$g(W) = \min_{\theta \in S^3} \max_{w \in W} \langle \theta, w \rangle = \cos(\text{covering radius of } W).$$

*The following are equivalent:*

- (i) *some  $v \in S^3$  has  $\langle v, w \rangle \leq \frac{1}{2}$  for all  $w \in W$ , that is  $W \cup \{v\}$  is again a contact configuration;*
- (ii)  *$V_c(W)$  contains a point of norm 2, that is its circumradius is at least 2;*
- (iii)  *$g(W) \leq \frac{1}{2}$ , that is the covering radius of  $W$  on  $S^3$  is at least  $60^\circ$ .*

*We call  $W$  saturated when these fail, so that  $g(W) > \frac{1}{2}$ .*

*Proof.* For a unit vector  $v$ , the condition  $\langle v, w \rangle \leq \frac{1}{2}$  for all  $w$  says exactly  $\langle 2v, w \rangle \leq 1$  for all  $w$ , that is  $2v \in V_c(W)$ ; and  $|2v| = 2$ . So (i) and (ii) are the same statement, since  $V_c(W)$  is convex and contains the origin, so it meets the sphere of radius 2 if and only if its circumradius is at least 2. For (iii), the circumradius is  $\max_\theta \rho(\theta) = \max_\theta 1/\cos \delta(\theta)$  by Lemma 7.15, which is  $1/g(W)$ ; and  $1/g(W) \geq 2$  is  $g(W) \leq \frac{1}{2}$ .  $\blacksquare$

**Corollary 7.39** (The reduction to twenty-three contacts). *Every contact configuration with bounded cell satisfies  $\text{vol}(V_c) \geq 8$ , with equality only for the root configuration, except possibly for a configuration of exactly 23 directions whose cell has circumradius less than 2.*

*Proof.* By Corollary 7.17 the bound holds when  $|W| \leq 22$ , with strict inequality. By Theorem 7.25 it holds with equality when  $|W| = 24$ . Let  $|W| = 23$ . If the circumradius of  $V_c(W)$  is at least 2 then by Proposition 7.38 there is a  $v$  with  $W' = W \cup \{v\}$  a contact configuration of size 24, and  $V_c(W') \subseteq V_c(W)$ , so  $\text{vol}(V_c(W)) \geq \text{vol}(V_c(W')) = 8$  by Theorem 7.25; the inequality is strict because  $V_c(W)$  contains a point of norm 2 while the 24-cell has circumradius  $\sqrt{2}$ . The only case not covered is  $|W| = 23$  with circumradius below 2. ■

The remaining case is therefore a single, sharply posed statement, and it is no longer about deviations at all.

**Theorem 7.40** (The 23-point case). *Every contact configuration of 23 directions with bounded cell has  $\text{vol}(V_c) > 8$ . In particular a contact configuration of 23 directions whose cell has circumradius less than 2, if one exists, has cell volume above 8.*

The proof occupies the rest of this section and is of a different kind from everything before it. Section 7.7 to Section 7.10 approach the statement through the configuration: what a saturated set of 23 directions would have to look like, how much of a root system it can contain, and how the codes of 23 points behave as their minimal angle is pushed to  $60^\circ$ ; none of that is the proof, and all of it shapes the one that Section 7.11 and Section 7.12 give. There the statement is turned into an inequality between the pair angles of any 23 contact directions (Theorem 7.69), and the inequality is established by a semidefinite certificate that reads the triples of directions, verified in exact and interval arithmetic (Theorem 7.73). Two remarks first on the size of what is asked. The configuration obtained by deleting one root from  $D_4$  has cell the 24-cell together with the pyramid of Lemma 13.1, of volume  $\frac{25}{3} = 8.333\dots$ , and its circumradius is exactly 2: the apex  $2u_0$  is the point that the deleted root would occupy. So that configuration sits exactly on the boundary of Proposition 7.38(ii), and is extendable by the single direction  $u_0$ . And the covering estimate of Theorem 7.16 gives  $7.916728\dots$  at  $m = 23$ , short of 8 by  $0.083272\dots$ , against a true value of  $\frac{25}{3}$  at the configuration just described; so the estimate has to be improved by about a fifth of its slack there, not by all of it, which is a different situation from  $m = 24$ , where the true value is 8 and no slack exists at all. Figure 15(a) plots the range, and Figure 15(b) records what settles each part of it.

**7.7. The saturated case, written out.** Everything Theorem 7.40 asserts can be written down explicitly, and it is worth doing so, because what it concerns is a single polytope with twenty-three facets and no deviation parameters at all. Fix a contact configuration  $W = \{w_1, \dots, w_{23}\}$  with bounded cell and circumradius below 2, and write  $s_{ij} = \langle w_i, w_j \rangle$ . The same statement takes three exact forms.

*Radially.* By Lemma 7.15 the quantity to bound is  $\frac{1}{4} \int_{S^3} \sec^4 \delta$ , so the target is

$$\int_{S^3} \sec^4(\delta(\theta)) d\theta \geq 32. \quad (7.28)$$

Saturation says  $\delta < 60^\circ$  everywhere, and Proposition 7.21 says  $\max \delta \geq 37.7612\dots^\circ$ , so the distance function is confined to a band of width about  $22^\circ$ .

*On the boundary.* By Lemma 7.10 the same target is  $\sum_i \text{vol}_3(F_i) \geq 32$ , an average of  $\frac{32}{23} = 1.391304\dots$  per facet. That number is larger than the  $\frac{4}{3}$  of a facet of the root configuration, and larger than the  $16\sqrt{2} - \frac{64}{3} = 1.294084\dots$  of the antiprism facet of Remark 7.12. So a counterexample would be a configuration whose facets, each containing the ball of radius  $1/\sqrt{3}$  by Lemma 7.11, average less than  $\frac{32}{23}$ ; and every facet shape that occurs at twenty-four contacts already averages less than that. This is the exact sense in which the remaining

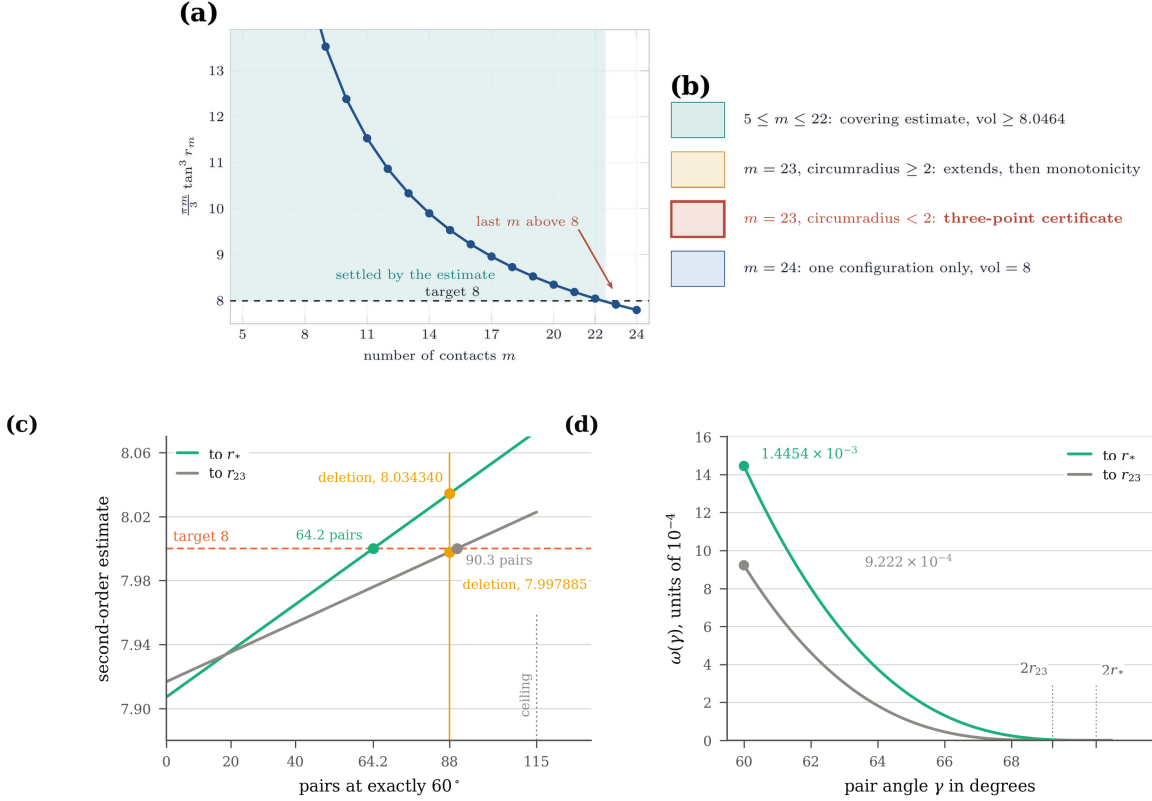


FIGURE 15. (a) The covering bound (7.12) at every admissible contact count  $m$ , against the target 8. It clears the target for all  $m \leq 22$  and falls below at  $m = 23$ . (b) How each part of the range of contact counts is settled: the estimate below 22 (Corollary 7.17), extension followed by monotonicity at 23 when the circumradius reaches 2 (Proposition 7.38), and the classification at 24 (Theorem 7.25). The third row is settled by the pair inequality of Theorem 7.69 and its certificate, Theorem 7.73. (c) The pairwise estimate at  $m = 23$ , for the two integration limits  $r_{23}$  and  $r_*$ : the right-hand side of (7.17) as a function of the number of pairs at exactly  $60^\circ$ , with the bracket alone at 0 pairs, the deletion configuration at 88 on both lines, the crossings of the target at 90.3 and at 64.2 pairs, and the elementary ceiling of Proposition 7.41 at 115. (d) The per-pair weight  $\omega(\gamma)$  for each integration limit:  $0.000922\dots$  and  $0.001445\dots$  at  $60^\circ$ , vanishing at  $2r_{23} = 69.2213\dots^\circ$  and at  $2r_* = 70.5288\dots^\circ$ .

case is not a perturbation of the root configuration: it must have larger facets than the root configuration has, not smaller.

*At the vertices.* If  $I$  indexes four active directions at a vertex  $z_I$  of the cell, then  $\langle z_I, w_i \rangle = 1$  for  $i \in I$ , so writing  $z_I = \sum_{i \in I} a_i w_i$  gives  $G_I a = \mathbf{1}$  and

$$|z_I|^2 = \mathbf{1}^\top G_I^{-1} \mathbf{1}, \quad G_I = (s_{ij})_{i,j \in I}. \quad (7.29)$$

Circumradius below 2 therefore says  $\mathbf{1}^\top G_I^{-1} \mathbf{1} < 4$  at every vertex. Two cases of (7.29) are worth recording. Four pairwise orthogonal active directions give  $G_I = \text{Id}$  and  $|z_I|^2 = 4$  exactly, so a saturated configuration has no vertex at which four active directions are

mutually orthogonal. And four directions with all pairwise inner products equal to  $g$  give  $|z_I|^2 = 4/(1+3g)$ , which is at least 4 precisely when  $g \leq 0$ ; so at every vertex of a saturated configuration the active directions must be positively correlated on average.

**Proposition 7.41** (What the pairwise estimate would need). *For  $m = 23$  write*

$$\omega(\gamma) = \frac{1}{4} \int_0^{r_*} \Lambda(r, \gamma) d(\sec^4 r)$$

*for the contribution a single pair at angle  $\gamma$  makes to the right-hand side of (7.17) at  $R = r_*$ . Then  $\omega$  is strictly decreasing on  $[60^\circ, 2r_*]$  and vanishes from  $2r_* = 70.5288\dots^\circ$  on, with  $\omega(60^\circ) = 0.001445408\dots$ , and a contact configuration of 23 directions with bounded cell satisfies  $\text{vol}(V_c) \geq 8$  as soon as*

$$\sum_{i < j} \omega(\gamma_{ij}) \geq 0.092855570\dots, \quad (7.30)$$

*which holds in particular once it has 65 pairs at exactly  $60^\circ$ . The deletion configuration has 88, and no configuration of 23 directions has more than 115.*

*Proof.* That  $\omega$  is decreasing and eventually zero is immediate from (7.16), since  $\Lambda(r, \gamma)$  is decreasing in  $\gamma$  and vanishes for  $r \leq \gamma/2$ . Inequality (7.30) is (7.17) at  $R = r_*$  with the value 7.907144430... of the bracket subtracted from 8. Numerically  $7.907144430 + 65 \times 0.001445409 = 8.001096\dots$ , while 64 pairs give 7.999651..., which is the sense in which 65 suffice and 64 do not; the crossing is at 64.24 pairs.

For the last assertion, fix  $w \in W$  and let  $S$  be the set of  $w_i$  with  $\langle w, w_i \rangle = \frac{1}{2}$ . Write  $w_i = \frac{1}{2}w + \frac{\sqrt{3}}{2}u_i$  with  $u_i$  a unit vector of  $w^\perp$ . Then  $\langle w_i, w_j \rangle = \frac{1}{4} + \frac{3}{4}\langle u_i, u_j \rangle$ , so the contact condition  $\langle w_i, w_j \rangle \leq \frac{1}{2}$  reads  $\langle u_i, u_j \rangle \leq \frac{1}{3}$ : the  $u_i$  are unit vectors of  $\mathbb{R}^3$  at pairwise angle at least  $\arccos \frac{1}{3} = 70.5287\dots^\circ$ , and the caps of angular radius  $35.2643\dots^\circ$  about them are disjoint on  $S^2$ . Each such cap has area  $2\pi(1 - \cos 35.2643\dots^\circ) = 1.152986\dots$  against the  $4\pi$  of  $S^2$ , so  $|S| \leq 10$ . Every vertex of the graph of pairs at  $60^\circ$  therefore has degree at most 10, and a graph on 23 vertices of maximum degree 10 has at most  $\lfloor 230/2 \rfloor = 115$  edges. ■

*Remark 7.42* (The one route that is not blocked). With the integration stopped at  $r_{23}$ , as in (7.18), the estimate misses 8 at the deletion configuration by three pairs, and that is the number often quoted as closing off the pairwise route. Carried to  $r_*$  it does not miss: the deletion has 88 pairs against the 64.24 that (7.30) asks for, a surplus of 0.034, and the weight falls away quickly once a pair opens up, with  $\omega(62^\circ)/\omega(60^\circ) = 0.55504\dots$  and  $\omega(66^\circ)/\omega(60^\circ) = 0.09077\dots$ . So what would settle the local bound at twenty-three contacts, for every configuration and not only for the saturated ones, is a lower bound on the weighted number of pairs at or near  $60^\circ$  that every contact configuration of 23 directions satisfies. Proposition 7.44 says how much of it the pair angles supply on their own, and Section 7.11 sets the statement out in full. Figure 15(c) and (d) set the numbers down, for both integration limits. The script is `pair_budget.py` for the limit  $r_{23}$  and `three_point_reduction.py` for  $r_*$ .

Two further facts about the pair angles are worth setting down, because together they say how far a bound of this kind can be pushed and where it stops. The first is a lower bound on the total overlap of the contact caps that a saturated configuration is forced to have.

**Proposition 7.43** (Covering multiplicity). *Let  $W$  be a saturated contact configuration with  $|W| = m$ . Then*

$$\sum_{i < j} \Lambda(\frac{\pi}{3}, \gamma_{ij}) \geq \frac{1}{2} \left( \frac{(m C(\pi/3))^2}{2\pi^2} - m C(\pi/3) \right), \quad (7.31)$$

where  $C(r) = \pi(2r - \sin 2r)$  is the measure of a cap of radius  $r$ . At  $m = 23$  the right-hand side is  $155.172054\dots$ , against  $162.566121\dots$  at the deletion configuration, so the inequality is within five per cent of binding there.

*Proof.* Write  $N(\theta)$  for the number of  $w \in W$  with  $\langle \theta, w \rangle > \frac{1}{2}$ . Then  $\int_{S^3} N = m C(\pi/3)$ , and  $\int_{S^3} N(N-1)$  counts ordered pairs of caps containing  $\theta$ , so  $\int_{S^3} N^2 = m C(\pi/3) + 2 \sum_{i < j} \Lambda(\frac{\pi}{3}, \gamma_{ij})$ . Cauchy–Schwarz gives  $\int N^2 \geq (\int N)^2 / 2\pi^2$ , and rearranging is (7.31). ■

The second fact is negative, and it is the sharpest thing we can say about the route. Everything the estimate of Proposition 7.23 sees of a configuration is the multiset of its pair angles. One can ask how far that multiset alone can carry it, by relaxing the multiset to an arbitrary measure subject to every inequality of the same character that a genuine configuration satisfies.

**Proposition 7.44** (What the pair angles alone can give). *Let  $\mu$  range over the nonnegative measures on  $[60^\circ, 180^\circ]$  of total mass  $\binom{23}{2} = 253$  subject to*

$$\int \Lambda(t, \gamma) d\mu \geq 23 C(t) - 2\pi^2 \quad \text{for all } t, \quad \int G_k(\cos \gamma) d\mu \geq -\frac{23}{2} \quad \text{for } k \geq 1,$$

where  $G_k$  are the Gegenbauer polynomials of  $S^3$  normalised by  $G_k(1) = 1$ . The pair-angle multiset of every contact configuration of 23 directions gives such a  $\mu$ . The minimum of  $\int \omega d\mu$  over this family, for the weight  $\omega$  of Proposition 7.41, is  $0.07380\dots$ , four fifths of the  $0.092855\dots$  that (7.30) requires; for the weight integrated only to  $r_{23}$  it is  $0.04157\dots$  against  $0.083272\dots$ , one half.

*Proof.* Feasibility is Bonferroni’s inequality  $\sigma(\bigcup_i C_i(t)) \geq \sum_i \sigma(C_i(t)) - \sum_{i < j} \Lambda(t, \gamma_{ij})$  together with  $\sigma(\bigcup_i C_i(t)) \leq 2\pi^2$  for the first family, and the positive definiteness of  $G_k$  on  $S^3$ , which gives  $\sum_{i,j} G_k(\langle w_i, w_j \rangle) \geq 0$  and hence  $\sum_{i < j} G_k(\cos \gamma_{ij}) \geq -\frac{23}{2}$ , for the second. The values are the optima of finite linear programmes: in `covering_multiplicity.py` on a grid of 601 angles with 24 radii between  $r_{23}$  and  $45^\circ$  and 12 degrees, for the weight integrated to  $r_{23}$ , and in `three_point_reduction.py` on a grid of 1201 angles with 60 radii between  $r_{23}$  and  $90^\circ$  and 24 degrees, for both weights. Adding radii or degrees does not raise either optimum, and the same measure attains both: 103.7 pairs at  $62.3^\circ$ , 90.0 at  $103.0^\circ$ , 39.2 at  $122.6^\circ$  and 20.1 at  $154.1^\circ$ . ■

*Remark 7.45* (Where a proof would have to look). Proposition 7.44 is a stronger statement than the one (7.18) makes, and a more useful one. It is not that the deletion configuration falls short, which it does not once the integration is carried to  $r_*$ : it is that nothing which reads only the pair angles gets past four fifths of the way, whatever constraints of that type are imposed and however they are combined. The minimising measure is instructive. It gives every direction nine neighbours at  $62.3^\circ$  and nearly eight at  $103^\circ$ , which is a perfectly good local picture at any one direction, and the two constraints available to a pair-angle relaxation, Bonferroni and positive definiteness, cannot see that the twenty-three such pictures do not fit together on one sphere. Saturation itself is invisible at that level too, since the covering radius is not a function of the pair angles, which is why Proposition 7.43 does not help either. A proof of the bound at twenty-three contacts has to read more of the

configuration than its pair angles record; the classification of Theorem 7.60 is one such thing, and the three-point form of Section 7.11 is another.

One family of candidates can be ruled out completely, and it is the family one would look at first. A configuration of a prime number of directions with any symmetry at all transitive on them is a single orbit of a rotation of that order, and in dimension four such an orbit carries a single parameter.

**Proposition 7.46** (No symmetric candidate). *No contact configuration of 23 directions in  $\mathbb{R}^4$  admits a group of symmetries transitive on its directions.*

*Proof.* Let  $W$  be such a configuration and  $G \leq O(4)$  transitive on it. Then 23 divides  $|G|$ , so  $G$  has an element  $g$  of order 23 by Cauchy's theorem. Since  $\det g$  is a 23rd root of unity in  $\{\pm 1\}$  it is 1, so  $g \in SO(4)$  and  $g$  is a rotation through angles  $t_1 = 2\pi a/23$  and  $t_2 = 2\pi b/23$  in two orthogonal planes  $P_1, P_2$ , with  $(a, b) \neq (0, 0)$ . The cell is bounded, so  $W$  spans  $\mathbb{R}^4$  and  $g$  cannot fix every member of  $W$ ; the orbit of any  $w_0 \in W$  under  $\langle g \rangle$  therefore has 23 elements and is all of  $W$ .

Write  $w_0 = p + q$  with  $p \in P_1$ ,  $q \in P_2$  and  $c = |p|^2$ , so  $|q|^2 = 1 - c$  and  $c \in [0, 1]$ . Then  $w_k = g^k w_0$  and

$$\langle w_0, w_k \rangle = \langle p, g^k p \rangle + \langle q, g^k q \rangle = c \cos(kt_1) + (1 - c) \cos(kt_2),$$

which does not depend on the directions of  $p$  and  $q$  inside their planes and is affine in  $c$ . Transitivity makes every pairwise inner product one of these, so the contact condition is the system of 22 affine inequalities

$$c \cos(kt_1) + (1 - c) \cos(kt_2) \leq \frac{1}{2}, \quad k = 1, \dots, 22,$$

in the single variable  $c \in [0, 1]$ . Each of the 528 pairs  $(a, b)$  is therefore decided by a one-variable linear programme. None of the 528 is feasible. The quantity

$$\delta(a, b) = \min_{c \in [0, 1]} \max_{1 \leq k \leq 22} \left( c \cos(kt_1) + (1 - c) \cos(kt_2) - \frac{1}{2} \right)$$

is the least violation available to the type  $(a, b)$ , and it is positive exactly when that type is infeasible. Its minimum over the 528 types is  $0.0740002839\dots$ , attained at  $(a, b) = (8, 17)$ . The maximum of affine functions is convex, so each  $\delta(a, b)$  is found exactly by evaluating at the endpoints and at the crossings of pairs of the 22 functions; the computation is carried out at 60 digits in `saturation_search.py`, and a margin of 0.074 in an inner product is not a rounding effect.  $\blacksquare$

*Remark 7.47* (What that removes). Proposition 7.46 is not the proof of Theorem 7.40, and it is worth being exact about what it does. A saturated configuration, if one exists, has no symmetry acting transitively on its directions: it is not a single orbit of anything. That removes every construction of the kind one reaches for first, and it is consistent with the rest of what is known about the case, since the configurations that do exist at 23 directions are the deletions, whose symmetry group is the stabiliser of a root in the Weyl group and has four orbits on the remaining directions, of sizes 1, 8, 6, 8. The search of `saturation_search.py` looks for the rest: it minimises  $\max_I |z_I|$  directly, from perturbed deletions, from random starts and from partially completed root systems, and over 202 starts it reached 12 contact configurations, all with the inner-product multiset of a deletion and all with  $g$  exactly  $\frac{1}{2}$ . That is exploration, and we report it as such.

The one configuration that comes close to the description is the deletion of a root, and it is not merely on the wrong side of the circumradius condition: it is rigid, so a saturated configuration cannot be reached by bending it.

**Proposition 7.48** (The deletion configurations are rigid). *Let  $W = \mathbb{R} \setminus \{u_0\}$  be the contact configuration obtained by deleting one root. Every first-order motion of  $W$  through contact configurations is an infinitesimal rotation.*

*Proof.* Scale the roots to  $a_i$  with  $|a_i|^2 = 2$ , so that the contact condition reads  $\langle a_i, a_j \rangle \leq 1$  and every quantity below is an integer. Write  $i \sim j$  when  $\langle a_i, a_j \rangle = 1$ ; deleting the root  $a_0$  leaves 23 directions and a set  $T$  of 88 unordered pairs  $\{i, j\}$  with  $i \sim j$ . A first-order motion is a tuple  $d = (d_1, \dots, d_{23})$  with

$$\langle d_i, a_i \rangle = 0 \quad \text{for every } i, \quad \langle d_i, a_j \rangle + \langle a_i, d_j \rangle \leq 0 \quad \text{for every } \{i, j\} \in T, \quad (7.32)$$

the first condition keeping each direction on its sphere and the second keeping each tight pair from closing. The infinitesimal rotations  $d_i = Aa_i$  with  $A$  antisymmetric satisfy both with equality and span a space of dimension 6.

Let  $c_i = \langle a_i, a_0 \rangle$  record the position of  $a_i$  relative to the deleted root; the values  $-2, -1, 0, 1$  occur 1, 8, 6, 8 times. Put

$$y_{ij} = \begin{cases} 1, & (c_i, c_j) \in \{(-1, -1), (-1, 1)\}, \\ 2, & (c_i, c_j) \in \{(-2, -1), (-1, 0), (0, 1)\}, \\ 3, & (c_i, c_j) = (1, 1), \end{cases} \quad \mu_i = \begin{cases} -6, & c_i = -1, \\ -8, & \text{otherwise,} \end{cases}$$

the first for  $\{i, j\} \in T$  with the pair written so that  $c_i \leq c_j$ , the second for each  $i$ . Those six are the only types that occur among the pairs of  $T$ , and a direct integer computation gives the equilibrium relation

$$\sum_{j: j \sim i} y_{ij} a_j + \mu_i a_i = 0 \quad \text{for every } i. \quad (7.33)$$

Pair (7.33) with  $d_i$  and sum over  $i$ . The diagonal terms contribute  $\mu_i \langle d_i, a_i \rangle = 0$ , and each unordered pair of  $T$  is met twice, contributing  $y_{ij}(\langle d_i, a_j \rangle + \langle a_i, d_j \rangle)$ , so

$$\sum_{\{i, j\} \in T} y_{ij} (\langle d_i, a_j \rangle + \langle a_i, d_j \rangle) = 0.$$

Every weight is positive and every bracket is nonpositive by (7.32), so every bracket vanishes: a first-order motion holds all 88 pairs at equality. The linear system consisting of those 88 equalities together with the 23 conditions  $\langle d_i, a_i \rangle = 0$  has, over  $\mathbb{Q}$ , a solution space of dimension exactly 6, and the rotations already occupy 6 dimensions of it, so the two coincide.  $\blacksquare$

Since the deletion configuration is a tensegrity framework, 23 bars from the centre and 88 struts, infinitesimal rigidity is rigidity in the sense of Roth and Whiteley [RW81, Thm. 5.7]: no continuous deformation through contact configurations moves it except by rotation. A saturated 23-point configuration therefore cannot be a small perturbation of a deletion, which is the same conclusion the facet average reached above, now from the contact graph, not from the volume. Both are checked by `rigidity23.py` in the supplement, the stress and the dimension count in exact integer arithmetic.

The same argument, with nothing deleted, places the root system itself.

**Corollary 7.49** (The root system is rigid). *Every first-order motion of  $\mathbb{R}$  through contact configurations of 24 directions is an infinitesimal rotation, and  $\mathbb{R}$  is rigid: a contact configuration of 24 directions sufficiently close to  $\mathbb{R}$  is a rotation of it.*

The first half of this is not new. Cohn, Jiao, Kumar and Torquato [CJKT11, Prop. 3.3] prove that the  $D_4$  root system is infinitesimally jammed, through the  $A_2$  subsystems and

not through a stress. We give the proof again because the stress, and not the jamming alone, is what Theorem 7.51 needs, and because the same stress handles the deletion of Proposition 7.48. Neither their argument nor ours carries a radius as it stands; supplying one is what Theorem 7.51 does.

*Proof.* The set  $T$  is now the set of the 96 pairs at  $\langle a_i, a_j \rangle = 1$ , the edges of the 24-cell, and every direction lies in 8 of them (Figure 22(a), drawn on page 83). The eight roots at  $60^\circ$  from a root  $a_i$  sum to  $4a_i$ , so the constant stress, weight 1 on every pair of  $T$  and  $\mu_i = -4$ , satisfies (7.33); pairing it with a first-order motion as before forces all 96 brackets to vanish, and the system of those 96 equalities with the 24 conditions  $\langle d_i, a_i \rangle = 0$  has, over  $\mathbb{Q}$ , a solution space of dimension 6, which the rotations fill. Rigidity follows from infinitesimal rigidity by [RW81, Thm. 5.7], the pairs at inner products 0,  $-1$  and  $-2$  staying strictly slack under a small motion. The count and the dimension are checked by `rigidity24.py`, and the equilibrium relation, the sum of the eight neighbours of every root being four times the root, by Lean's kernel in `D4Stress.lean`. ■

Corollary 7.49 is the local half of Theorem 7.25: the root system is isolated among contact configurations of 24 directions, and that much is elementary. What the semidefinite kernel of Theorem 7.30 supplies, and what no argument in a neighbourhood of  $R$  can, is the global half, that there is no such configuration at any distance from  $R$ ; it is the one ingredient of the general theorems that rests on a certificate rather than on a closed argument, and we have not found a way to derive it from anything local.

The rigidity of Corollary 7.49 comes with no radius. Roth and Whiteley produce a neighbourhood but not a measure of it, and a statement of isolation with no number in it cannot be combined with anything quantitative. It can be made explicit, and cheaply: the stress argument that proves Corollary 7.49 proves rather more if the quadratic terms are carried along instead of discarded, and what the carrying needs is one constant, the smallest nonzero singular value of the linear map that the argument linearises.

Write  $\mathcal{T} = \{\tau \in (\mathbb{R}^4)^{24} : \langle \tau_i, a_i \rangle = 0\}$  for the tangent space, a subspace of dimension 72, and  $\Lambda : \mathcal{T} \rightarrow \mathbb{R}^{96}$  for the *rigidity operator*

$$(\Lambda\tau)_{ij} = \frac{1}{\sqrt{2}}(\langle a_i, \tau_j \rangle + \langle a_j, \tau_i \rangle), \quad \{i, j\} \in T, \quad (7.34)$$

where  $T$  is, as in the proof of Corollary 7.49, the set of the 96 pairs at  $\langle a_i, a_j \rangle = 1$  and the factor normalises the roots. The first-order motions of Corollary 7.49 are the  $\tau$  with  $\Lambda\tau \leq 0$ , and that corollary says  $\ker \Lambda$  is the 6-dimensional space of infinitesimal rotations.

**Lemma 7.50** (Spectrum of the rigidity operator). *The singular values of  $\Lambda$  are 0, 1,  $\sqrt{5/2}$ ,  $\sqrt{3}$ , 2, with multiplicities 6, 29, 8, 21, 8. In particular the smallest nonzero singular value is exactly 1, so*

$$\|\Lambda\tau\|_2 \geq \|\tau\|_2 \quad \text{for every } \tau \in \mathcal{T} \text{ orthogonal to } \ker \Lambda. \quad (7.35)$$

*Proof.* What has to be established is a finite exact statement about one integral matrix, and it is established by computation, carried out twice on independent code paths: in integer arithmetic in Python and again inside Lean's kernel. What follows sets out what is computed and why it determines the spectrum.

Let  $\Lambda' = \sqrt{2}\Lambda$ , whose coefficients  $\langle a_i, \cdot \rangle$  are integral, and let  $P$  be the orthogonal projection onto  $\mathcal{T}$ , with blocks  $I - \frac{1}{2}a_i a_i^\top$ , which is rational. Then  $N = P\Lambda'^\top \Lambda' P$  has  $4N$  integral, and the nonzero eigenvalues of  $N$  are twice the squares of the nonzero singular values of  $\Lambda$ . A direct computation in integer arithmetic gives the matrix identity

$$4N(4N - 8I)(4N - 20I)(4N - 24I)(4N - 32I) = 0,$$

so every eigenvalue of  $N$  lies in  $\{0, 2, 5, 6, 8\}$ , and the rank of  $4N - cI$  over a field of positive characteristic, which bounds the rank over  $\mathbb{Q}$  from below, is 66, 67, 88, 75, 88 for  $c = 0, 8, 20, 24, 32$ ; the five multiplicities that result sum to 96, so each bound is met and the multiplicities are 30, 29, 8, 21, 8. The kernel of  $N$  contains the 24 radial directions, on which  $P$  vanishes, together with  $\ker \Lambda$ , and  $30 = 24 + 6$  accounts for it exactly. Dividing by two and taking square roots gives the singular values. The identity and the ranks are computed by `rigidity_spectrum.py` by fraction-free elimination over  $\mathbb{Z}$ , with no floating point at any step; the 96 tight pairs it starts from, and the neighbour sum of Corollary 7.49, are checked again by Lean's kernel in `D4Stress.lean`.  $\blacksquare$

**Theorem 7.51** (A radius for the rigidity). *Let  $W = \{w_1, \dots, w_{24}\} \subset S^3$  be a contact configuration and put*

$$d(W) = \min_{R \in O(4)} \min_{\pi} \left( \sum_{i=1}^{24} |w_{\pi(i)} - Ra_i/\sqrt{2}|^2 \right)^{1/2},$$

*the minimum over orthogonal maps and over bijections  $\pi$ . Then either  $d(W) = 0$ , and  $W$  is an orthogonal image of the normalised root system, or*

$$d(W) \geq \frac{2}{\sqrt{577}} = 0.0832611\dots$$

*Proof.* Both minima are over compact sets and are attained; relabel and rotate so that they are attained at  $\pi = \text{id}$  and  $R = I$ , write  $u_i = a_i/\sqrt{2}$ , and put  $\varepsilon_i = w_i - u_i$ ,  $e_i = |\varepsilon_i|$  and  $\|\varepsilon\|^2 = \sum_i e_i^2 = d(W)^2$ .

Since  $|w_i| = |u_i| = 1$ , expanding  $|u_i + \varepsilon_i|^2 = 1$  gives

$$\rho_i := \langle u_i, \varepsilon_i \rangle = -\frac{1}{2}e_i^2, \quad (7.36)$$

so the radial part of each displacement is quadratically small. Put  $\tau_i = \varepsilon_i - \rho_i u_i$ , so that  $\tau \in \mathcal{T}$  and  $|\tau_i|^2 = e_i^2 - \rho_i^2$ .

Minimality in  $R$  fixes the gauge: differentiating  $R \mapsto \sum_i |w_i - Ru_i|^2$  along  $R = e^{sA}$  with  $A$  antisymmetric and setting  $s = 0$  gives  $\sum_i \langle \varepsilon_i, Au_i \rangle = 0$ , and since  $\langle u_i, Au_i \rangle = 0$  the same holds for  $\tau$ . So  $\tau$  is orthogonal to every infinitesimal rotation, hence to  $\ker \Lambda$  by Corollary 7.49, and (7.35) applies.

Now the contact condition, kept to second order. For a pair  $\{i, j\} \in T$ , expanding  $\langle w_i, w_j \rangle \leq \frac{1}{2}$  and using  $\langle u_i, \varepsilon_j \rangle = \langle u_i, \tau_j \rangle + \frac{1}{2}\rho_j$ ,

$$g_{ij} := (\Lambda\tau)_{ij} \leq -\frac{1}{2}(\rho_i + \rho_j) - \langle \varepsilon_i, \varepsilon_j \rangle \leq \frac{1}{4}(e_i^2 + e_j^2) + e_i e_j$$

by (7.36) and Cauchy–Schwarz. Every index lies in exactly eight pairs of  $T$ , so  $\sum_T (e_i^2 + e_j^2) = 8\|\varepsilon\|^2$  and  $\sum_T e_i e_j \leq \frac{1}{2} \sum_T (e_i^2 + e_j^2) = 4\|\varepsilon\|^2$ , and therefore  $\sum_T g_{ij}^+ \leq 6\|\varepsilon\|^2$  for the positive parts. The constant stress of Corollary 7.49 gives  $\sum_T g_{ij} = 0$  exactly, so the negative parts sum to as much as the positive ones and

$$\|\Lambda\tau\|_2 \leq \|\Lambda\tau\|_1 = 2 \sum_T g_{ij}^+ \leq 12\|\varepsilon\|^2.$$

With (7.35) this reads  $\|\tau\|_2 \leq 12\|\varepsilon\|^2$ . Finally  $\sum_i \rho_i^2 = \frac{1}{4} \sum_i e_i^4 \leq \frac{1}{4}\|\varepsilon\|^4$  by (7.36), so

$$\|\varepsilon\|^2 = \|\tau\|_2^2 + \sum_i \rho_i^2 \leq 144\|\varepsilon\|^4 + \frac{1}{4}\|\varepsilon\|^4 = \frac{577}{4}\|\varepsilon\|^4,$$

and  $\|\varepsilon\|$  is either 0 or at least  $2/\sqrt{577}$ .  $\blacksquare$

**Corollary 7.52** (The twenty-four-contact case near the root system). *A contact configuration  $W$  with  $|W| = 24$  and  $d(W) < 2/\sqrt{577}$  is an orthogonal image of the normalised roots, so  $V_c(W)$  is a regular 24-cell and  $\text{vol}(V_c(W)) = 8$ . Nothing in this uses a certificate.*

*Proof.* Theorem 7.51 gives  $d(W) = 0$ , and the volume is then Lemma 2.3 together with the invariance of volume under orthogonal maps.  $\blacksquare$

We state plainly where the line between the certificate and the closed arguments of this paper now falls. The global classification at 24 contacts is Theorem 7.30, proved from the verified kernel. What this section supplies is the quantitative statement that the classification by itself does not give: an explicit isolation radius about the normalised root system,  $2/\sqrt{577}$  in the root-sum-of-squares displacement and about 0.017 per direction, within which Theorem 7.25 is proved by hand and the kernel is not used at all. Beyond that radius the kernel is doing the work, and it is doing it on the whole of the rest of a 66-dimensional quotient,  $(S^3)^{24}$  modulo  $O(4)$ , which is by far the larger part. The constant  $577/4$  is not optimal and a sharper reading of the same inequalities would enlarge the ball, but no argument of this kind reaches a configuration that is nowhere near  $R$ , which is the whole of the remaining difficulty. Figure 16(a) and (b) draw the neighbour sum that makes the stress work, Figure 16(c) the spectrum that supplies the constant, and Figure 16(d) the two sides of the final inequality.

Beyond that, what we have is negative and numerical. Searches started from each of the twenty-four deletions, from greedy random packings and from annealed random starts produced no contact configuration of 23 directions with  $g(W) > \frac{1}{2}$ ; the largest value found was  $\frac{1}{2}$  itself, attained exactly at the deletions, and the random starts reached no 23-point contact configuration at all. That is exploration and not evidence of nonexistence, and we report it as such; the script is `extendability.py`. What a proof has to supply is clear from Proposition 7.18: the covering estimate already uses the areas of the spherical Voronoi cells to the full, so the missing  $0.083272\dots$  has to come from their shape, and that is what the truncated volume of Section 7.11 reads off, pair by pair, from the lenses in which neighbouring caps overlap.

**7.8. How much of a root system a configuration can keep.** Proposition 7.48 says that a saturated configuration of 23 directions is not a bent deletion. The following makes that quantitative and turns it into a statement with no smallness in it: such a configuration can share at most 20 of its directions with any root system, so at least three of them are new. The argument is elementary and uses only the coordinates.

Take  $R = \{(e_i \pm e_j)/\sqrt{2} : 1 \leq i < j \leq 4\}$  and let  $S \subseteq R$ . Write

$$A(S) = \{v \in S^3 : \langle v, r \rangle \leq \tfrac{1}{2} \text{ for every } r \in S\} \quad (7.37)$$

for the directions that may be added to  $S$ , so that a contact configuration  $W$  with  $W \cap R = S$  satisfies  $W \subseteq S \cup A(S)$ . Call  $\{i, j\}$  the *support* of the four roots  $(\pm e_i \pm e_j)/\sqrt{2}$ , and call a support *filled* for  $S$  when all four of its roots lie in  $S$ . The six supports pair off into three couples of complementary index pairs,

$$\{1, 2\} \mid \{3, 4\}, \quad \{1, 3\} \mid \{2, 4\}, \quad \{1, 4\} \mid \{2, 3\}, \quad (7.38)$$

and these three couples are what carry the whole argument. Throughout,  $t = 1/\sqrt{2}$  and  $v = (x_1, x_2, x_3, x_4)$ .

**Lemma 7.53** (A filled support). *If  $\{i, j\}$  is filled for  $S$ , then every  $v \in A(S)$  satisfies  $|x_i| + |x_j| \leq t$ , and hence  $x_i^2 + x_j^2 \leq \frac{1}{2}$ .*

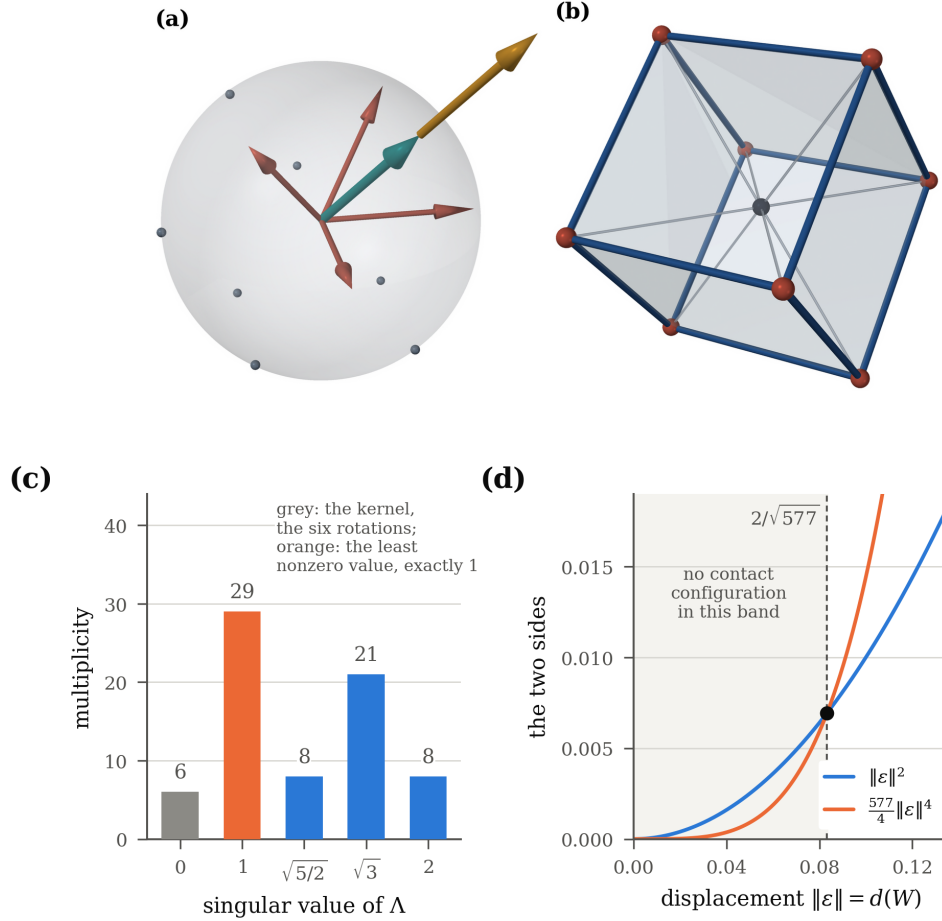


FIGURE 16. (a) The neighbour sum in dimension three, where it fits in one picture: a contact direction of the face-centred cubic lattice (teal), the four contact directions at  $60^\circ$  from it (clay), and their sum (ochre), which is twice the direction itself. (b) The same identity in dimension four, drawn where it lives. The eight roots at inner product 1 from a root  $\alpha$  fill the hyperplane  $\langle \alpha, x \rangle = 1$ , a copy of  $\mathbb{R}^3$ , as the vertices of a cube centred at  $\alpha/2$  (dark); eight times that centre is  $4\alpha$ . (c) The singular values of the rigidity operator  $\Lambda$  on the 72-dimensional tangent space, with multiplicities (Lemma 7.50). The kernel is the six infinitesimal rotations, and the smallest nonzero value is the constant that turns the first-order argument into a radius. (d) The two sides of the inequality that closes Theorem 7.51. Feasibility forces  $\|\varepsilon\|^2 \leq \frac{577}{4}\|\varepsilon\|^4$ , so the displacement is either zero or at least  $2/\sqrt{577}$ : the shaded band holds no contact configuration of 24 directions.

*Proof.* The four conditions  $\langle v, (\pm e_i \pm e_j)/\sqrt{2} \rangle \leq \frac{1}{2}$  read  $\pm x_i \pm x_j \leq t$  over the four sign patterns, and the pattern matching the signs of  $x_i$  and  $x_j$  gives  $|x_i| + |x_j| \leq t$ . Then  $x_i^2 + x_j^2 \leq (|x_i| + |x_j|)^2 \leq t^2 = \frac{1}{2}$ .  $\blacksquare$

**Lemma 7.54** (A filled couple). *If both supports of one of the couples (7.38) are filled for  $S$ , then  $A(S) = R \setminus S$ .*

*Proof.* Let  $\{i, j\}$  and  $\{k, l\}$  be the couple and let  $v \in A(S)$ . By Lemma 7.53,  $x_i^2 + x_j^2 \leq \frac{1}{2}$  and  $x_k^2 + x_l^2 \leq \frac{1}{2}$ , and the two sides add to  $|v|^2 = 1$ , so both are equalities. Equality in  $x_i^2 + x_j^2 \leq (|x_i| + |x_j|)^2$  forces  $x_i x_j = 0$ , and equality in  $(|x_i| + |x_j|)^2 \leq t^2$  forces  $|x_i| + |x_j| = t$ ; so one of  $x_i, x_j$  is zero and the other is  $\pm t$ , and the same holds for  $x_k, x_l$ . Therefore  $v = \pm t e_p \pm t e_q$  with  $p \in \{i, j\}$  and  $q \in \{k, l\}$ , which is a root. A root in  $A(S)$  cannot lie in  $S$ , since  $\langle v, v \rangle = 1 > \frac{1}{2}$ ; and every root outside  $S$  does lie in  $A(S)$ , because distinct roots have inner product at most  $\frac{1}{2}$ .  $\blacksquare$

**Proposition 7.55** (Twenty-two roots). *If a contact configuration  $W$  satisfies  $|W \cap R| \geq 22$  for some root system  $R$ , then  $W \subseteq R$ .*

*Proof.* Put  $S = W \cap R$ . At most two roots are missing, and they lie in at most two of the six supports, so at least one of the three couples has both supports filled. By Lemma 7.54,  $A(S) = R \setminus S$ , and  $W \subseteq S \cup A(S) = R$ .  $\blacksquare$

At 21 the conclusion changes, and exactly one family of exceptions appears.

**Proposition 7.56** (Twenty-one roots). *If a contact configuration  $W$  satisfies  $|W \cap R| = 21$  for some root system  $R$ , then  $W \subseteq R$  or  $|W| \leq 22$ .*

*Proof.* Let  $S = W \cap R$  and let  $\rho_1, \rho_2, \rho_3$  be the missing roots. If their supports occupy at most two of the six index pairs, some couple is filled, so  $A(S) = \{\rho_1, \rho_2, \rho_3\}$  by Lemma 7.54 and  $W \subseteq R$ . So assume the three supports are distinct and meet each couple of (7.38) once. Choosing one index pair from each couple gives eight possibilities, and inspection of the list shows that four of them are *stars*, three pairs sharing an index, as with  $\{1, 2\}, \{1, 3\}, \{1, 4\}$ , and four are *triangles*, the three pairs inside a triple of indices, as with  $\{1, 2\}, \{1, 3\}, \{2, 3\}$ . Permuting the coordinates reduces to those two.

Let  $v \in A(S)$  and suppose  $v$  is not a root. For each  $m$  the set  $S \cup \{\rho_m\}$  misses only two roots, so  $A(S \cup \{\rho_m\})$  consists of roots by Lemma 7.54, and therefore does not contain  $v$ . Since  $v \in A(S)$ , the only condition that can fail at  $S \cup \{\rho_m\}$  is the one at  $\rho_m$  itself, so

$$\langle v, \rho_m \rangle > \frac{1}{2}, \quad m = 1, 2, 3. \quad (7.39)$$

Suppose first that the supports form the star  $\{1, 2\}, \{1, 3\}, \{1, 4\}$ . Sign changes in the coordinates 2, 3, 4 are symmetries of  $R$ , so we may take  $\rho_m = (\varepsilon_m e_1 + e_m)/\sqrt{2}$  with  $\varepsilon_m \in \{\pm 1\}$  for  $m = 2, 3, 4$ . The supports  $\{2, 3\}, \{2, 4\}$  and  $\{3, 4\}$  are filled, so  $|x_m| \leq t$  for  $m = 2, 3, 4$  by Lemma 7.53, and (7.39) reads  $\varepsilon_m x_1 + x_m > t$ . If  $\varepsilon_m = 1$  this gives  $x_1 > t - x_m \geq 0$ , and if  $\varepsilon_m = -1$  it gives  $x_1 < x_m - t \leq 0$ ; the two cannot both occur, so the three signs agree, and after a sign change in the first coordinate  $\rho_m = (e_1 + e_m)/\sqrt{2}$  for  $m = 2, 3, 4$ . Those three roots are pairwise at  $60^\circ$ .

Suppose instead that the supports form the triangle  $\{1, 2\}, \{1, 3\}, \{2, 3\}$ , and write  $\rho_1 = (a e_1 + b e_2)/\sqrt{2}$ ,  $\rho_2 = (c e_1 + d e_3)/\sqrt{2}$ ,  $\rho_3 = (f e_2 + g e_3)/\sqrt{2}$  with signs  $a, \dots, g$ . The supports  $\{1, 4\}, \{2, 4\}, \{3, 4\}$  are filled, so  $|x_m| \leq t$  for  $m = 1, 2, 3$ . Adding the three inequalities (7.39) gives

$$(a + c)x_1 + (b + f)x_2 + (d + g)x_3 > 3t,$$

and the left side is at most  $2tk$ , where  $k$  counts the nonzero terms among  $a + c, b + f, d + g$ . Hence  $k = 3$ , that is  $a = c, b = f$  and  $d = g$ , and  $\rho_1, \rho_2, \rho_3$  are again pairwise at  $60^\circ$ . Changing the signs of the coordinates 1, 2, 3 as needed makes the triple  $\{(e_1 + e_2), (e_1 + e_3), (e_2 + e_3)\}/\sqrt{2}$ , and the orthogonal map with matrix  $\frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$ , which permutes  $R$ , carries it to  $\{(e_1 + e_2), (e_1 + e_3), (e_1 - e_4)\}/\sqrt{2}$ ; one more sign change in the fourth coordinate returns the star.

So in every case we may take  $\rho_m = (e_1 + e_m)/\sqrt{2}$  for  $m = 2, 3, 4$ , and it remains to read off  $A(S)$  for that  $S$ . The supports  $\{2, 3\}$ ,  $\{2, 4\}$ ,  $\{3, 4\}$  are filled, so  $|x_2| + |x_3|$ ,  $|x_2| + |x_4|$  and  $|x_3| + |x_4|$  are all at most  $t$ . The maximum of  $p^2 + q^2 + r^2$  over  $\{p, q, r \geq 0 : p + q \leq t, p + r \leq t, q + r \leq t\}$  is attained at a vertex of that polytope, and its vertices are the origin, the three points  $te_k$  and  $\frac{t}{2}(1, 1, 1)$ , with values 0,  $t^2$  and  $\frac{3}{4}t^2$ . Hence

$$x_2^2 + x_3^2 + x_4^2 \leq t^2 = \frac{1}{2}, \quad \text{so} \quad x_1^2 \geq \frac{1}{2}. \quad (7.40)$$

Each support  $\{1, m\}$  keeps three of its four roots, all but  $(e_1 + e_m)/\sqrt{2}$ , and these give  $x_1 - x_m \leq t$  and  $|x_m| - x_1 \leq t$ . If  $x_1 \leq -t$  the second forces  $x_m = 0$  for  $m = 2, 3, 4$ , hence  $v = -e_1$ , which the first contradicts at  $m = 2$ . So  $x_1 \geq t$ , and then  $x_m \geq x_1 - t \geq 0$  for  $m = 2, 3, 4$ : on  $A(S)$  the first coordinate is at least  $t$  and the other three are nonnegative. Consequently, for  $v, v' \in A(S)$ ,

$$\langle v, v' \rangle = x_1x'_1 + x_2x'_2 + x_3x'_3 + x_4x'_4 \geq x_1x'_1 \geq t^2 = \frac{1}{2}, \quad (7.41)$$

with equality only if  $x_1 = x'_1 = t$ . That makes (7.40) an equality, so  $(x_2, x_3, x_4)$  is one of the vertices  $te_k$ , and  $v \in \{\rho_1, \rho_2, \rho_3\}$ ; the same holds for  $v'$ .

Now  $W \setminus S$  lies in  $A(S)$  and its elements are pairwise at  $60^\circ$  or more. By (7.41) any two of them are missing roots, so either every element of  $W \setminus S$  is a root, in which case  $W \subseteq R$ , or  $W \setminus S$  has a single element and  $|W| = 22$ . ■

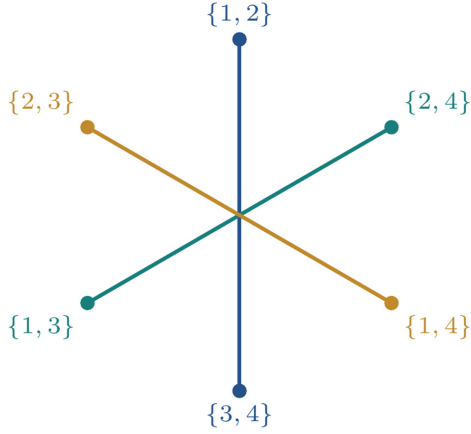
**Corollary 7.57** (A saturated configuration is far from every root system). *A contact configuration of 23 directions whose cell has circumradius below 2 meets every root system in at most 20 directions.*

*Proof.* Let  $|W| = 23$  and suppose  $|W \cap R| \geq 21$ . By Proposition 7.55 and Proposition 7.56, either  $|W| \leq 22$ , which is false, or  $W \subseteq R$  and  $W$  is  $R$  with one root  $u_0$  removed. In the second case the cell of  $W$  contains the point  $2u_0$ , by Lemma 13.1, so its circumradius is 2. ■

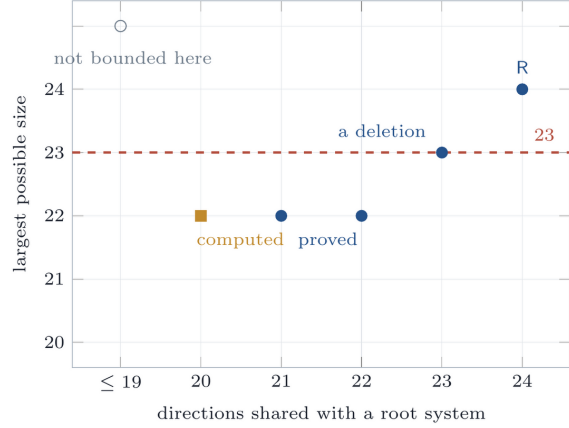
Figure 17 collects the two ingredients. The bound of Corollary 7.57 is the first statement we have that separates a hypothetical saturated configuration from the root system by a fixed amount, not by an unspecified small one: three of its twenty-three directions lie outside every copy of  $R$ , so it is not reachable from a deletion by exchanging one or two directions.

*Remark 7.58* (The next rung, and what it costs). The same reduction can be tried at twenty roots. Of the 10626 ways to remove four roots, only four orbits under the symmetry group of  $R$  leave a cell of circumradius above 2, and in each of those the removed set is a triple pairwise at  $60^\circ$  together with one further root. For each of the four we minimised the largest of the three pairwise inner products over triples of directions in  $A(S)$ , requiring each direction to stay an angle  $\delta$  away from every root, so that the removed roots themselves are excluded. At  $\delta = 0$  the value is  $\frac{1}{2}$  in all four, and the minimiser is three of the removed roots; at  $\delta = 2^\circ$  the smallest of the four values is already 0.538256..., at  $\delta = 5^\circ$  it is 0.592985..., at  $\delta = 10^\circ$  it is 0.676602..., and it increases from there. So the doorways at twenty roots hold two directions but not three, which would push Corollary 7.57 to nineteen. We record this as a computation and not as a proof: it is a search over a family of local minima, not an exhaustive argument, and the case analysis that would make it one is four times the length of Proposition 7.56. The script is `root_meet.py`, with `root_deletions_exact.py` carrying out the same enumeration for one, two and three removed roots in exact integer arithmetic as a check on Proposition 7.55 and Proposition 7.56.

(a)



(b)



each chord joins a support to its complement;  
two removed roots meet at most two supports,  
so one whole chord stays filled

FIGURE 17. (a) The six supports of the roots, joined in the three couples of complementary index pairs. Removing two roots touches at most two supports and leaves a whole couple filled, which is Lemma 7.54; removing three can meet each couple once, and the eight ways of doing that are four stars and four triangles. (b) The largest size a contact configuration can have, against the number of directions it shares with a root system. The value 23 occurs only inside the root system, where the circumradius is exactly 2, so the saturated case is pushed to twenty directions or fewer.

**7.9. The remaining case as a question about codes.** A contact configuration of  $m$  directions is the same object as a spherical code of  $m$  points on  $S^3$  with minimal angle at least  $60^\circ$ , and the question is how much room a code of that size has. Write  $\theta(m)$  for the largest minimal angle available to  $m$  points on  $S^3$ . Musin's theorem [Mus08] gives  $\theta(25) < 60^\circ$ , and Theorem 7.25 gives  $\theta(24) = 60^\circ$ , since a code of 24 points at more than  $60^\circ$  would be a contact configuration of that size and hence a copy of the root system, whose minimal angle is exactly  $60^\circ$ . The value at  $m = 23$  is what decides the shape of the set of configurations that Theorem 7.40 is about.

**Proposition 7.59** (Slack, and its absence). *If  $\theta(m) > 60^\circ$ , the contact configurations of  $m$  directions form a set with nonempty interior in  $(S^3)^m$ . If  $\theta(m) = 60^\circ$ , that set has empty interior, and every contact configuration of  $m$  directions attains  $\theta(m)$ .*

*Proof.* The contact configurations of size  $m$  are exactly the  $m$ -point codes of minimal angle at least  $60^\circ$ , and the minimal angle is a continuous function on  $(S^3)^m$ . If  $\theta(m) > 60^\circ$ , a code attaining  $\theta(m)$  has a neighbourhood on which the minimal angle stays above  $60^\circ$ , and every configuration in it is a contact configuration. If instead  $\theta(m) = 60^\circ$ , then a contact configuration  $W$  has minimal angle exactly  $60^\circ$ , so some pair  $w_i, w_j$  is at  $60^\circ$ ; rotating  $w_i$  towards  $w_j$  through any positive angle, however small, gives a configuration at distance as small as one likes from  $W$  whose minimal angle is below  $60^\circ$ , and which is therefore not a contact configuration. So  $W$  is not an interior point. ■

At  $m \leq 22$  the first alternative is the one that holds, and the configurations at those sizes fill out an open set; that is why Corollary 7.17 has to hold uniformly over a family of positive dimension, and it does. At  $m = 23$  every computation we know of, our own in Remark 7.62 and the published tables [Slo], returns  $\theta(23) = 60^\circ$ , and if that is the true value the second alternative takes over: the configurations that Theorem 7.40 is about are extremal codes, not a family with interior. That is the structural difference between 23 and every size below it, and a set with empty interior can of course turn out to have no saturated member at all. We have no proof that  $\theta(23) = 60^\circ$ ; what Section 7.10 shows is how the codes of 23 points behave as their minimal angle is pushed towards  $60^\circ$  from below.

**Theorem 7.60** (The 23-point case from a uniqueness statement). *Suppose every set of 23 unit vectors in  $\mathbb{R}^4$  with pairwise inner products at most  $\frac{1}{2}$  is the image under an orthogonal map of  $R \setminus \{u\}$  for some root  $u$ . Then no saturated 23-point contact configuration exists, and the case that Theorem 7.40 covers is empty.*

*Proof.* Let  $W$  be a contact configuration with  $|W| = 23$ , and take an orthogonal  $O$  and a root  $u$  with  $W = O(R \setminus \{u\})$ . Every  $w \in W$  is  $Or$  for a root  $r \neq u$ , so  $\langle Ou, w \rangle = \langle u, r \rangle \leq \frac{1}{2}$ , and  $W \cup \{Ou\}$  is again a contact configuration. By Proposition 7.38 the circumradius of  $V_c(W)$  is then at least 2, so  $W$  is not saturated, and Corollary 7.39 then needs nothing beyond Theorem 7.25.  $\blacksquare$

*Remark 7.61* (Why a uniqueness statement and not a bound). The hypothesis of Theorem 7.60 is what Theorem 7.25 proves at 24 points, asked at 23, and one difference between the two carries the whole of the difficulty. At 24 the code has maximal size, so the semi-definite relaxation is tight there and complementary slackness forces every inner product into  $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$ , which is Theorem 7.30; the classification is then combinatorial. At 23 nothing is tight. No upper bound on the size of a code in terms of its minimal angle can be of any use here, whatever the method that produces it: such a bound is at least 24 at  $60^\circ$ , because a code of 24 points at that angle exists, so it cannot exclude 23 points at the same angle. What is wanted is a classification at one fixed size below the maximum, which is a different kind of statement and, as far as we know, has not been attempted in this dimension.

*Remark 7.62* (What the search returns). We looked for a 23-point code of minimal angle above  $60^\circ$ , and for a contact configuration of 23 directions whose Gram matrix is unlike that of a deletion, by Riesz continuation: minimise  $\sum_{i < j} |w_i - w_j|^{-s}$  over the sphere with  $s$  running through 4, 16, 64, 256, 1024, then reduce the largest inner product directly. Calibration comes first. Over 120 starts the same procedure returns 0.5000071... at  $m = 24$ , where the true value is  $\frac{1}{2}$ ; 0.4980170... at  $m = 22$ , a minimal angle of  $60.1311\dots^\circ$ , against the  $60.1398863^\circ$  of the published tables [Slo]; and, over 60 starts, 0.5374065... at  $m = 25$ , where no contact configuration exists. So the procedure locates the root system, sees the slack at 22 and sees the obstruction at 25, and one can read how close to  $\frac{1}{2}$  it gets when  $\frac{1}{2}$  is the answer. At  $m = 23$ , over 1500 random starts, the smallest largest inner product it reached was 0.5000078..., which is nearer to  $\frac{1}{2}$  than the calibration run at  $m = 24$  managed; no start went below  $\frac{1}{2}$ ; and of the 32 outcomes that came within  $5 \times 10^{-3}$  of feasibility, all 32 had the inner-product multiset of a deletion, namely  $-1$  eleven times,  $-\frac{1}{2}$  eighty-eight times, 0 sixty-six times and  $\frac{1}{2}$  eighty-eight times, which is an invariant of the orbit under  $O(4)$ . The tables [Slo] record  $60.0000000^\circ$  at 23 points, the same value as at 24, and  $60.1398863^\circ$  at 22. None of this proves the hypothesis of Theorem 7.60, and we do not offer it as a proof; what it does is say which statement to aim at. The script is `spherical_code_23.py`.

One finite family can be searched completely instead of sampled, and it is the natural place for a code that is not a subset of a root system to hide. The 600-cell has 120 vertices, the unit icosians, and its inner products are  $0, \pm\frac{1}{2}, \pm\cos 36^\circ, \pm\cos 72^\circ$  and  $\pm 1$ ; two vertices are closer than  $60^\circ$  exactly when they are joined by an edge, at  $36^\circ$ . Its subsets with all inner products at most  $\frac{1}{2}$  are therefore the independent sets of the edge graph, which is 12-regular, and the largest of them have 24 elements and are the twenty-five inscribed copies of  $R$ , five through each vertex. The question is whether a 23-element independent set can avoid all twenty-five.

**Proposition 7.63** (Codes among the vertices of the 600-cell). *Let  $I \subset S^3$  be the vertex set of the regular 600-cell. A subset of  $I$  whose pairwise inner products are all at most  $\frac{1}{2}$  has at most 24 elements; the subsets with 24 elements are the twenty-five inscribed copies of  $R$ ; and every subset with 23 elements is one of those copies with a single vertex removed. In particular no saturated contact configuration of 23 directions lies in  $I$ .*

*Proof.* Write  $\phi = \frac{1}{2}(1 + \sqrt{5})$ . Doubling the coordinates puts every vertex of the 600-cell in  $\mathbb{Z}[\phi]^4$ , with each coordinate one of  $0, \pm 1, \pm 2, \pm\phi, \pm(\phi - 1)$ , and  $4\langle u, v \rangle$  is then computed exactly in  $\mathbb{Z}[\phi]$ ; the nine values that occur are  $0, \pm 2, \pm 4, \pm 2\phi$  and  $\pm(2\phi - 2)$ , of which only  $2\phi$ , that is  $\langle u, v \rangle = \cos 36^\circ$ , exceeds 2. So the sets in question are exactly the independent sets of the graph  $\Gamma$  on  $I$  whose edges join vertices at  $36^\circ$ , and every vertex of  $\Gamma$  has degree 12, its vertex figure being an icosahedron. The symmetry group of the 600-cell is transitive on  $I$ , so it suffices to enumerate the independent sets through one fixed vertex  $v_0$ , and this is a finite search which we carry out to the end: a depth-first enumeration over bitsets, with the standard pruning that an independent set meets every clique of  $\Gamma$  at most once, so that a greedy partition of the remaining candidates into cliques bounds what can still be added.

The search returns no independent set of size 25 through  $v_0$ ; exactly five of size 24, each of which has all its pairwise inner products in  $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$  and is therefore a copy of  $R$  (by Theorem 7.25, or directly, since the 24-point contact configurations inside  $I$  are checked one by one); and exactly 115 of size 23, every one of which is contained in one of the five copies of  $R$  through  $v_0$ . Since a copy of  $R$  through  $v_0$  has 23 other vertices and  $5 \times 23 = 115$ , the sets of size 23 through  $v_0$  are precisely those five copies with one of their other vertices removed, and nothing else. Summing over the 120 vertices,  $115 \times 120/23 = 600 = 25 \times 24$  independent sets of size 23 in all, one for each pair consisting of an inscribed copy of  $R$  and one of its vertices, and  $5 \times 120/24 = 25$  inscribed copies.

A set of 23 vertices of one copy of  $R$  is a deletion, and is extended by the vertex it lacks, so by Proposition 7.38 its cell has circumradius 2 and it is not saturated.

The enumeration is carried out three times, independently. Once in Python, with the vertices in  $\mathbb{Z}[\phi]$  and the graph and the search as described; once in C, on the graph the Python script writes out; and once in Lean 4, where the vertex list, the nine inner-product values, the degree 12, the twenty-five cells, their inner products and the fact that five pass through each vertex are all checked by the kernel from the  $\mathbb{Z}[\phi]$  coordinates, and the three counts 0, 5 and 115 are established by the same depth-first search compiled to native code and run by `native_decide`. The scripts are `cell600_exact.py` and `cell600_enum.c`, and the Lean development is described in Section 23. ■

*Remark 7.64* (What the 600-cell does and does not decide). Proposition 7.63 closes the family one reaches for first (Figure 20(a), drawn on page 79). The unit icosians form the largest finite subgroup of the unit quaternions, the binary icosahedral group of order 120; they contain the twenty-four Hurwitz units, which are  $R$ , in twenty-five different ways; and a subset of them with all inner products at most  $\frac{1}{2}$  is the first thing to try when looking

for a code built on the golden ratio rather than on  $\sqrt{2}$ . The proposition says that no such subset of 23 points escapes a copy of  $R$ . It says nothing about a code whose coordinates lie outside  $\mathbb{Q}(\sqrt{5})$ , and nothing about the continuous families that Section 7.10 finds once the minimal angle is allowed to drop below  $60^\circ$ , none of which has the inner-product spectrum of a subset of the 600-cell. The random sampling of maximal independent sets that preceded the enumeration, two million of them drawn by greedy growth along random orders of the vertices, had already found sizes 10 to 22 and 24 and never 23; the enumeration replaces that observation by a statement. The same enumeration on the 48 unit quaternions of the binary octahedral group, the vertices of two 24-cells in dual position (Figure 18(b)), in which two directions are closer than  $60^\circ$  exactly when they lie in different copies and are  $45^\circ$  apart, returns the two copies of  $R$  and their forty-eight deletions and nothing else; that one is a second of computation, in  $\mathbb{Z}[\sqrt{2}]$ , and is `octahedral148_exact.py`.

**7.10. The largest inradius as the minimal angle rises.** The remaining case can be put as the value of one function at one point, and the function can be computed away from that point. For  $\delta \geq 0$  let

$$K(\delta) = \{W = (w_1, \dots, w_{23}) \in (S^3)^{23} : \langle w_i, w_j \rangle \leq \tfrac{1}{2} + \delta \text{ for all } i \neq j\},$$

the configurations of 23 directions whose minimal angle is at least  $\arccos(\tfrac{1}{2} + \delta)$ , and let

$$h(\delta) = \max_{W \in K(\delta)} g(W), \quad g(W) = \min_{\theta \in S^3} \max_i \langle \theta, w_i \rangle,$$

so that  $g$  is the function of Proposition 7.38, the cosine of the covering radius, and the inradius of  $\text{conv}(W)$  whenever that hull contains the origin. The contact configurations are  $K(0)$ , and by Proposition 7.38 a member of  $K(0)$  is saturated exactly when  $g > \tfrac{1}{2}$ .

**Lemma 7.65** (The limit from above). *The function  $h$  is nondecreasing on  $[0, \infty)$ ,  $h(0) \geq \tfrac{1}{2}$ , and  $h(\delta) \rightarrow h(0)$  as  $\delta \downarrow 0$ . No saturated contact configuration of 23 directions exists if and only if  $h(0) = \tfrac{1}{2}$ ; and this holds as soon as  $h(\delta) \leq \tfrac{1}{2} + C\delta^\kappa$  on some interval  $(0, \delta_0)$ , for any constants  $C$  and  $\kappa > 0$ .*

*Proof.* The map  $(\theta, W) \mapsto \max_i \langle \theta, w_i \rangle$  is continuous, and  $|\max_i \langle \theta, w_i \rangle - \max_i \langle \theta, w'_i \rangle| \leq \max_i |w_i - w'_i|$ , so  $g$  is continuous on  $(S^3)^{23}$ , indeed 1-Lipschitz in the maximum norm. Each  $K(\delta)$  is a closed subset of the compact space  $(S^3)^{23}$ , so the maximum defining  $h(\delta)$  is attained; the sets increase with  $\delta$ , so  $h$  is nondecreasing; and  $K(0)$  contains the deletions of a root, at which  $g = \tfrac{1}{2}$ , so  $h(0) \geq \tfrac{1}{2}$ . For the limit take  $\delta_n \downarrow 0$  and  $W_n \in K(\delta_n)$  with  $g(W_n) = h(\delta_n)$ . By compactness a subsequence converges to some  $W$ , and  $W \in K(\delta_m)$  for every  $m$ , since  $K(\delta_m)$  is closed and contains  $W_n$  for all  $n \geq m$ ; hence  $W \in \bigcap_m K(\delta_m) = K(0)$ . By continuity  $g(W) = \lim h(\delta_n)$ , and  $g(W) \leq h(0)$  because  $W \in K(0)$ , while  $h(\delta_n) \geq h(0)$  by monotonicity; so the limit is  $h(0)$ , along every such sequence. The equivalence is the definition of saturation, and the last statement follows by letting  $\delta \downarrow 0$  in the bound. ■

So the hypothesis of Theorem 7.60 is the statement  $h(0) = \tfrac{1}{2}$ , and  $h$  can be looked at for  $\delta > 0$ , where the constraint set has interior and a local optimiser has room to move. We did this with the trust-region sequential linear programme described in Appendix F, which maximises the smallest facet offset of  $\text{conv}(W)$  over tangent moves of bounded size subject to the linearised contact constraints, accepting a move only when the true inradius, read from a fresh convex hull, improves without any constraint being violated. For each  $\delta$  on a decreasing schedule the starts are the twenty best configurations of the previous level, restored to the new constraint, together with a batch of fresh random starts; three passes were run, with different seeds and with 40, 60 and 80 fresh starts per level, and Table 3

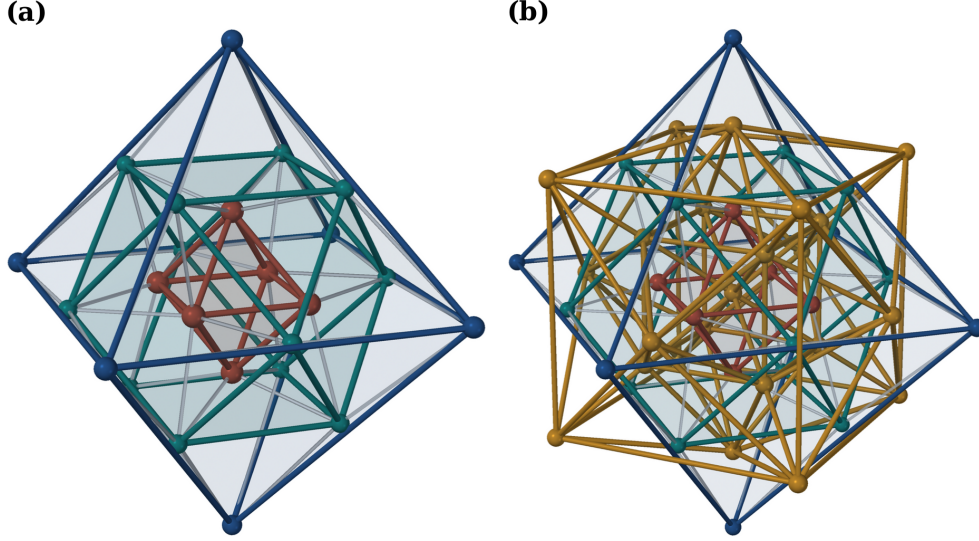


FIGURE 18. (a) The 24 contact directions of  $D_4$  seen from a deep hole, radius proportional to angle: an octahedron at  $45^\circ$  (clay), a cuboctahedron at  $90^\circ$  (teal), an octahedron at  $135^\circ$  (blue), and the 96 pairs at  $60^\circ$  as edges. (b) The same, with the dual 24-cell added (ochre): the 48 directions of the binary octahedral group, a cube, an octahedron and a cube about the hole, which is now itself a direction.

records the largest inradius found at each level over the three. Every number in the table is a lower bound for  $h(\delta)$  obtained by local search, and nothing in this subsection is a proof.

Three things stand out. First, for  $\delta \geq 0.009$ , that is for minimal angles up to  $59.40^\circ$ , the best configurations have inradius between 0.65 and 0.70: their covering radius is between  $46^\circ$  and  $49^\circ$ , so twenty-three caps of radius  $49^\circ$  about their directions cover the sphere, where the deletion of a root needs caps of radius  $60^\circ$  and the root system itself, with twenty-four directions, needs  $45^\circ$ . These configurations are not perturbed deletions in any sense. Their inner-product multisets have no pair at  $-1$  at all, where a deletion has eleven antipodal pairs, and none within  $10^{-3}$  of  $-\frac{1}{2}$  or of 0; between three and fifty-three of their pairs sit at the imposed bound  $\frac{1}{2} + \delta$ , and fewer than three per cent of their inner products lie within  $10^{-3}$  of a value taken by the 600-cell, so none lies in the 600-cell either. Second, between  $\delta = 0.009$  and  $\delta = 0.008$  they vanish. At every level from 0.008 down the best inradius is  $\frac{1}{2} + O(\delta)$ , with  $h(\delta) - \frac{1}{2}$  between  $1.1\delta$  and  $2\delta$  across the whole range down to  $\delta = 5 \times 10^{-5}$ , and the configurations attaining it are deletions of a root with their 88 tight pairs opened by amounts of order  $\delta$ : every one of them keeps the eleven antipodal pairs of a deletion, and as  $\delta$  decreases their inner-product multisets settle onto the deletion's, which is reached to  $10^{-3}$  by  $\delta = 3 \times 10^{-4}$ . The transition is not gradual: a configuration of the first kind, restored to the constraint at  $\delta = 0.008$  by the feasibility programme, does not reach it, because its minimal angle sits at a local maximum below  $59.47^\circ$  from which no local move raises it. Third, at  $\delta = 0$  every feasible endpoint of every pass is a deletion of a root, with the inner-product multiset  $-1$  eleven times,  $-\frac{1}{2}$  eighty-eight times, 0 sixty-six times,  $\frac{1}{2}$  eighty-eight times, and  $g = \frac{1}{2}$  to fifteen digits. Figure 19 draws the table.

*Configurations with a symmetry.* A configuration invariant under a nontrivial element of  $O(4)$  is a union of orbits of that element, and inside such a class the search has far fewer

TABLE 3. The largest inradius  $g$  found at each level of the continuation, over three passes. The third column is the minimal angle  $\arccos(\frac{1}{2} + \delta)$  imposed, the fifth the covering radius  $\arccos g$  of the best configuration, and the last two the number of feasible endpoints across the three passes and how many of them are deletions of a root. At  $\delta = 0$  every endpoint is a deletion and  $g = \frac{1}{2}$  exactly.

| $\delta$ | min. angle | $h(\delta) \geq$ | cov. radius | endpoints | deletions |
|----------|------------|------------------|-------------|-----------|-----------|
| 0.05     | 56.633°    | 0.676201         | 47.453°     | 180       | 0         |
| 0.03     | 57.995°    | 0.695650         | 45.921°     | 239       | 0         |
| 0.02     | 58.668°    | 0.686522         | 46.645°     | 159       | 0         |
| 0.015    | 59.003°    | 0.673135         | 47.691°     | 113       | 0         |
| 0.01     | 59.336°    | 0.657060         | 48.924°     | 104       | 0         |
| 0.009    | 59.403°    | 0.653786         | 49.172°     | 34        | 0         |
| 0.008    | 59.469°    | 0.508771         | 59.418°     | 2         | 0         |
| 0.007    | 59.536°    | 0.508968         | 59.405°     | 11        | 0         |
| 0.005    | 59.669°    | 0.505913         | 59.608°     | 15        | 0         |
| 0.003    | 59.801°    | 0.504484         | 59.703°     | 22        | 0         |
| 0.002    | 59.868°    | 0.502914         | 59.807°     | 26        | 0         |
| 0.0015   | 59.901°    | 0.502552         | 59.831°     | 34        | 0         |
| 0.001    | 59.934°    | 0.501392         | 59.908°     | 40        | 0         |
| 0.0007   | 59.954°    | 0.501123         | 59.926°     | 42        | 0         |
| 0.0005   | 59.967°    | 0.500878         | 59.942°     | 46        | 0         |
| 0.0003   | 59.980°    | 0.500480         | 59.968°     | 52        | 0         |
| 0.0002   | 59.987°    | 0.500378         | 59.975°     | 56        | 0         |
| 0.0001   | 59.993°    | 0.500202         | 59.987°     | 56        | 0         |
| 0.00005  | 59.997°    | 0.500081         | 59.995°     | 63        | 0         |
| 0        | 60.000°    | 0.500000         | 60.000°     | 51        | 51        |

parameters and can be run from enough starts to cover it. Proposition 7.46 disposes of the transitive case. The script `symmetric_search.py` treats every rotation of order  $n \leq 12$ , through angles  $2\pi a/n$  and  $2\pi b/n$  in two orthogonal planes with  $\gcd(a, b, n) = 1$ , together with the two improper involutions, a reflection in a hyperplane and  $\text{diag}(1, -1, -1, -1)$ ; for each it lists every way of writing 23 as a sum of orbit sizes, the generic orbit having  $n$  points and a point of either plane an orbit of  $n/\gcd(a, n)$  or  $n/\gcd(b, n)$ , with at most six points on either plane since they lie on a great circle. That gives 73 orbit structures, with between 5 and 35 parameters each, and in each the continuation of Table 3 is run from sixteen random starts on the orbit representatives, through  $\delta = 0.05, 0.02, 0.01, 0.005, 0.002, 0$ . Eleven of the seventy-three structures reached a feasible endpoint at  $\delta = 0$ , sixty-nine endpoints in all, and every one of them is a deletion of a root: the structures are those with a rotation of order 2, 3 or 4 fixing a plane pointwise, a reflection with five fixed points, and  $\text{diag}(1, -1, -1, -1)$ , which are symmetries that a deletion has, its symmetry group being the stabiliser of the deleted root in the symmetry group of  $R$ . The remaining sixty-two structures never reached the constraint at  $\delta = 0$  from any start. At  $\delta = 0.01$  the best symmetric configuration has inradius 0.6599, in line with Table 3. No symmetric configuration crossed into  $K(0)$  except the deletions.

*Remark 7.66* (What this says about the remaining case). The picture is the one the classification of Theorem 7.60 would present if it holds, and it says something about where a proof of it would have to work. By Lemma 7.65 it would be enough to show  $h(\delta) \leq \frac{1}{2} + C\delta$  on

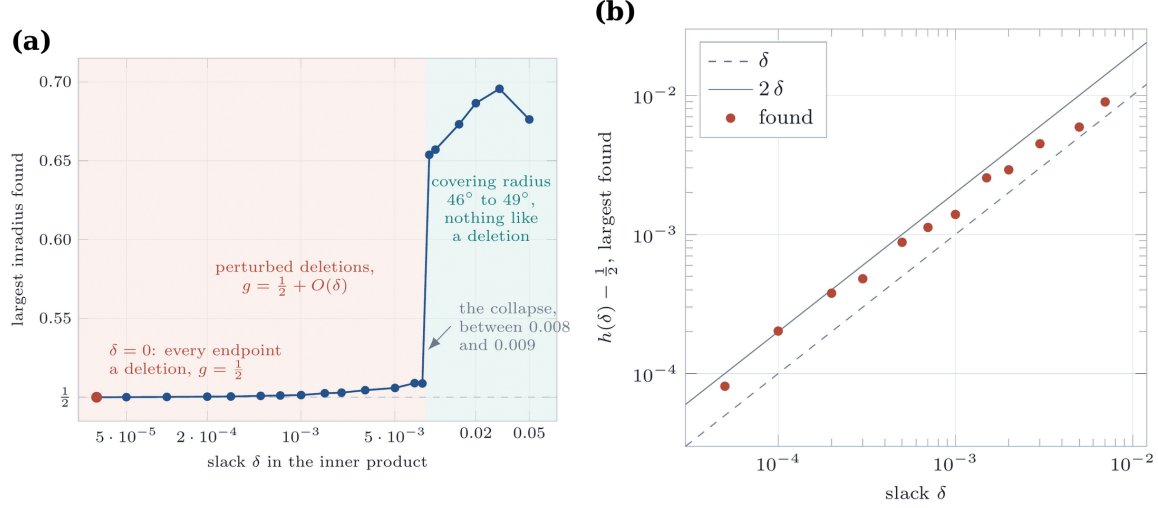


FIGURE 19. (a) The continuation of Table 3: the largest inradius found, against  $\delta$  on a logarithmic axis. The plateau near  $\frac{2}{3}$  runs down to  $\delta = 0.009$  and is carried by configurations far from any deletion; from  $\delta = 0.008$  down it collapses to  $\frac{1}{2} + O(\delta)$ . (b) The same table below the collapse:  $h(\delta) - \frac{1}{2}$  against  $\delta$  on logarithmic axes, with the lines  $\delta$  and  $2\delta$  for comparison. The approach to  $\frac{1}{2}$  is linear in  $\delta$ , which is what a perturbed deletion gives.

some interval  $(0, \delta_0)$ , and the data put the constant near 2 for small  $\delta$ ; but the same data put  $h(0.009)$  above 0.65, so no such bound holds with a uniform constant beyond  $\delta_0 \approx 0.008$ , and an argument would have to work inside a window of about half a degree in the minimal angle. That is the quantitative form of what Remark 7.61 said: the classification at 23 points is a statement about codes of minimal angle exactly  $60^\circ$ , and codes at  $59.3^\circ$  already behave in a completely different way. It also says what a counterexample, if there is one, would have to look like: a saturated configuration is a member of  $K(0)$  with  $g > \frac{1}{2}$ , hence a member of every  $K(\delta)$  with  $h(\delta) \geq g$ , and it would have to have escaped a search that, at every level below 0.009, found nothing but deletions. We found no such configuration, in three passes of the continuation and in 300 direct maximisations of  $g$  over  $K(0)$  from the three kinds of start described in Appendix F: 94 of those reached a feasible endpoint, every one of them a deletion with  $g = \frac{1}{2}$ , and the 100 random starts, brought down through the same schedule of slacks, never reached the constraint at  $\delta = 0$  at all, which is the collapse seen from the other side. That is exploration, and it is reported as such.

**7.11. The remaining case as a strict inequality.** Everything in this section so far has approached Theorem 7.40 through the configuration: what a saturated set of 23 directions would have to look like, and why none has been found. This subsection approaches it through the cell, and arrives at a statement of a different character: an inequality between the pair angles of the configuration which, if it holds for every contact configuration of 23 directions, proves Theorem 7.40 outright, which the one configuration we can write down satisfies with room to spare, and which Section 7.12 then establishes. The tools are the ones already in hand, the covering bound of Theorem 7.16 and the exact overlaps of Proposition 7.23; what is new is the observation of where the integration ought to stop, and what stopping there buys.

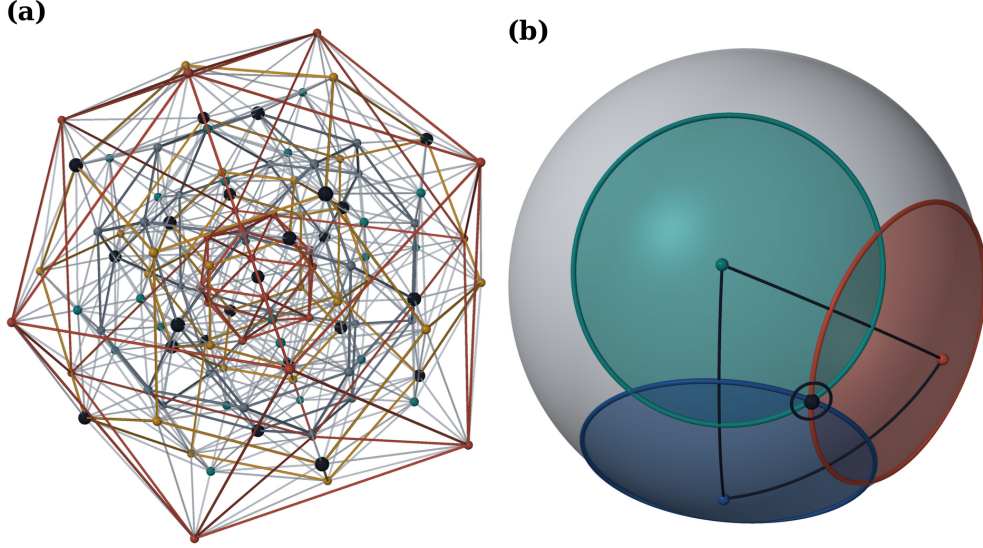


FIGURE 20. (a) The 600-cell seen from one vertex, its 119 other vertices in shells at  $36^\circ$  to  $144^\circ$  with 708 of the 720 edges at  $36^\circ$ ; the dark points are an inscribed 24-cell whose far vertex is the one at infinity, that is, a deletion. (b) The cap lemma on a great 2-sphere: three contact directions  $60^\circ$  apart carry caps of radius  $r_* = 35.26^\circ$  that meet in exactly one point, the circumcentre of their triangle (ringed); any smaller radius separates them.

**Lemma 7.67** (Contact directions in a cap). *If  $k$  contact directions lie in a closed cap of angular radius  $r$  on  $S^3$ , then  $\sin^2 r \geq \frac{k-1}{2k}$ . In particular two contact directions need a cap of radius at least  $30^\circ$ , three need  $r_* = \arcsin(1/\sqrt{3}) = 35.2644\dots^\circ$ , and four need  $r_4 = \arcsin \sqrt{3/8} = 37.7612\dots^\circ$ .*

*Proof.* A point  $\theta$  of the cap about  $c$  satisfies  $|\theta - (\cos r)c|^2 = 1 - 2\cos r \langle \theta, c \rangle + \cos^2 r \leq \sin^2 r$ , so the cap lies in the Euclidean ball of radius  $\sin r$  about  $(\cos r)c$ . For points  $x_1, \dots, x_k$  of that ball with centroid  $q$ ,

$$\begin{aligned} \sum_{i < j} |x_i - x_j|^2 &= k \sum_i |x_i - q|^2 \\ &\leq k \sum_i |x_i - (\cos r)c|^2 \leq k^2 \sin^2 r, \end{aligned}$$

the first step being the standard identity and the second the fact that the centroid minimises the sum of squared distances. Contact directions have  $|x_i - x_j|^2 = 2 - 2\langle x_i, x_j \rangle \geq 1$ , so  $\binom{k}{2} \leq k^2 \sin^2 r$ .  $\blacksquare$

The case  $k = 3$  is drawn in Figure 20(b), and the lens in which the caps of two contact directions overlap, whose measure is the  $\Lambda(r, \gamma)$  of (7.43) below, in Figure 21(a). The cases  $k = 3$  and  $k = 4$  are Lemma 7.22 and Proposition 7.21 again, since  $\arccos \sqrt{2/3} = \arcsin \sqrt{1/3}$  and  $\arccos \sqrt{5/8} = \arcsin \sqrt{3/8}$ ; the lemma puts them on one footing and is the reason the two radii  $r_*$  and  $r_4$  govern what follows.

**Proposition 7.68** (The cell inside a ball). *Let  $W$  be a contact configuration of  $m$  directions with bounded cell, let  $U(r)$  be the measure of the set of points of  $S^3$  farther than  $r$  from every*

direction of  $W$ , and let  $R \geq 1$ . Then

$$\text{vol}(V_c(W) \cap B(R)) = \frac{1}{4} \left[ 2\pi^2 + \int_0^{\arccos(1/R)} U(r) d(\sec^4 r) \right] \leq \text{vol}(V_c(W)), \quad (7.42)$$

and for  $r < r_4$

$$U(r) = 2\pi^2 - mC(r) + \sum_{i < j} \Lambda(r, \gamma_{ij}) - \sum_{i < j < k} \Lambda_3(r; \gamma_{ij}, \gamma_{ik}, \gamma_{jk}), \quad (7.43)$$

where  $\Lambda_3$  is the measure of the intersection of three caps of radius  $r$  about directions at the given pairwise angles, and the last sum is absent for  $r < r_*$ . Consequently, with

$$A_4 = \frac{1}{4} \left[ 2\pi^2 + \int_0^{r_4} (2\pi^2 - 23C(r)) d(\sec^4 r) \right] = 7.647558\dots, \\ \omega_4(\gamma) = \frac{1}{4} \int_0^{r_4} \Lambda(r, \gamma) d(\sec^4 r),$$

and  $\omega_3$  the same integral of  $\Lambda_3$ , every contact configuration of 23 directions with bounded cell satisfies

$$\text{vol}(V_c) \geq \text{vol}(V_c \cap B(\sqrt{8/5})) = A_4 + \sum_{i < j} \omega_4(\gamma_{ij}) - \sum_{i < j < k} \omega_3(\gamma_{ij}, \gamma_{ik}, \gamma_{jk}). \quad (7.44)$$

*Proof.* The radial function of  $V_c(W)$  in the direction  $\theta$  is  $\sec \delta(\theta)$  by Lemma 7.15, so that of  $V_c(W) \cap B(R)$  is  $\min(\sec \delta, R)$  and the volume is  $\frac{1}{4} \int_{S^3} \min(\sec \delta, R)^4$ . Writing  $\rho = \arccos(1/R)$ ,

$$\min(\sec \delta, R)^4 = \sec^4 \min(\delta, \rho) = 1 + \int_0^\rho \mathbf{1}[\delta(\theta) > r] d(\sec^4 r),$$

and integrating over  $\theta$  gives (7.42), the inequality because  $V_c(W) \cap B(R) \subseteq V_c(W)$ . For (7.43),  $U(r)$  is  $2\pi^2$  less the measure of the union of the closed caps of radius  $r$  about the directions of  $W$ , and by Lemma 7.67 no point lies in four of those caps when  $r < r_4$ , nor in three when  $r < r_*$ ; the indicator of a union of sets no four of which meet is the sum of the indicators of the sets, less that of the pairwise intersections, plus that of the triple intersections, exactly. Integrating (7.43) against  $d(\sec^4 r)$  over  $[0, r_4]$  gives (7.44), and  $\sec r_4 = \sqrt{8/5}$ .  $\blacksquare$

The three-dimensional analogue of all this can be drawn, and Figure 21(b) and Figure 22(b) draw it: the Voronoi cell of the face-centred cubic lattice, whose contact directions are the twelve vertices of a cuboctahedron, cut by the ball of radius  $\sqrt{3/2}$ , and the caps of radius  $r_*$  about those twelve directions, which overlap in pairs and meet three at a time only at the eight triangle circumcentres; the uncovered measure there is exactly the two-term inclusion and exclusion, as it is on  $S^3$  below  $r_*$ .

At the deletion of a root the three-point bound (7.44) is 8.140848...: the 88 tight pairs contribute  $88\omega_4(60^\circ) = 88 \times 0.00565947\dots = 0.498033\dots$ , the 84 triangles of tight pairs subtract  $84\omega_3(60^\circ, 60^\circ, 60^\circ) = 84 \times 5.65 \times 10^{-5} = 0.00474\dots$ , and no other pair or triple contributes, since  $\Lambda(r, \gamma) = 0$  for  $\gamma \geq 2r_4 = 75.52^\circ$ . A direct quadrature of  $\min(\sec \delta, \sqrt{8/5})^4$  over  $S^3$  returns 8.14068, and the same quadrature at the full root system, with 24 directions, returns 7.96841: there the truncation discards the deep holes at  $45^\circ$  and the bound falls below the true value 8, which says that at 24 contacts the three-point information below  $r_4$  is not enough, as it need not be. At 23 contacts it is enough at the one configuration known, with 0.14 to spare, and the pair-only bound (7.17) at  $R = r_*$  is enough there too, with 0.034. That is the content of the following statement, which we regard as the right form of the remaining case.

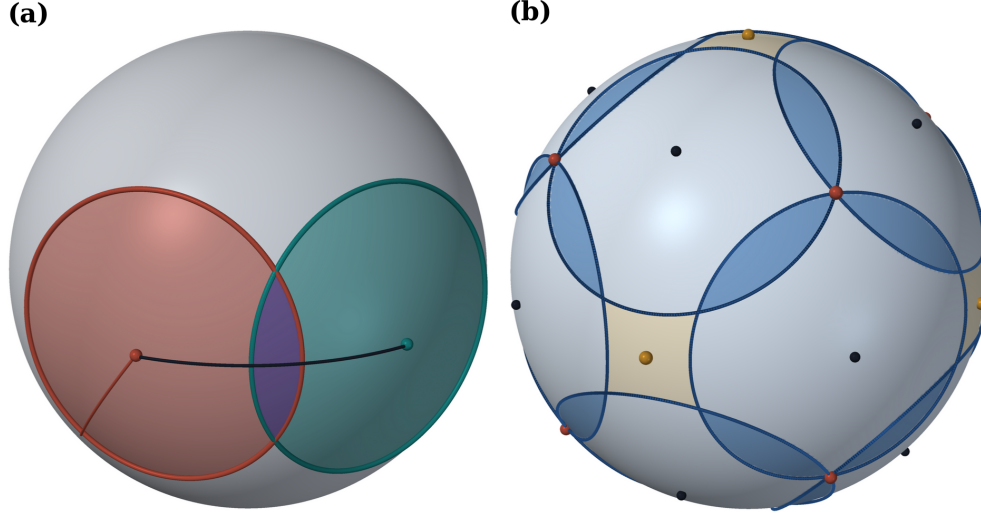


FIGURE 21. (a) Two contact directions at angle  $\gamma = 60^\circ$  and their caps of radius  $r = r_*$ . The lens in which the caps overlap (violet) has measure  $\Lambda(r, \gamma)$ , and  $\omega(\gamma)$  integrates it against  $d(\sec^4 r)$ ; the dark arc is  $\gamma$  and the clay arc is  $r$ . (b) The caps of radius  $r_*$  about the twelve contact directions of the same lattice, the sphere shaded by how many caps cover it: one (pale blue), two in the lenses, three at the eight triangle circumcentres (clay), and none in the six square holes (ochre).

**Theorem 7.69** (The remaining case as an inequality between pair angles). *Suppose that every contact configuration of 23 directions with bounded cell satisfies*

$$\sum_{i < j} \omega(\gamma_{ij}) \geq 8 - A_* = 0.092855570 \dots, \quad (7.45)$$

where

$$A_* = \frac{1}{4} \left[ 2\pi^2 + \int_0^{r_*} (2\pi^2 - 23 C(r)) d(\sec^4 r) \right] = 7.907144430 \dots$$

and  $\omega$  is the weight of Proposition 7.41; equivalently, that the cell of every such configuration has volume at least 8 inside the ball of radius  $\sqrt{3}/2$ . Then every such configuration has  $\text{vol}(V_c) \geq 8$ , with strict inequality wherever (7.45) is strict, and Theorem 7.40 follows. The same conclusion follows if every such configuration satisfies the three-point inequality  $\sum_{i < j} \omega_4(\gamma_{ij}) - \sum_{i < j < k} \omega_3 \geq 8 - A_4 = 0.352442 \dots$

*Proof.* By Proposition 7.23 at  $R = r_*$ , or by (7.44), the hypothesis gives  $\text{vol}(V_c) \geq \text{vol}(V_c \cap B(\sqrt{3}/2)) \geq 8$  for every contact configuration of 23 directions with bounded cell, strictly when the pair sum exceeds  $8 - A_*$ . A configuration whose cell has circumradius below 2 has bounded cell, so Theorem 7.40 is the bound just proved once (7.45) is known to hold strictly, which is what Theorem 7.73 supplies. ■

**Remark 7.70** (What kind of statement this is). Three things distinguish (7.45) from the classification of Theorem 7.60, and they are the reason for stating it.

First, it is strict where the classification is not. The classification asks for a property of codes of minimal angle exactly  $60^\circ$ , and the deletion of a root sits on the boundary of that property, with  $g = \frac{1}{2}$ ; Remark 7.61 explains why no relaxation can see a boundary. Inequality (7.45) holds at the deletion with 0.034 to spare, and at every configuration

reached in Section 7.10 with more: the least value of  $\text{vol}(V_c \cap B(\sqrt{3/2}))$  found over the configurations of Table 3 and over a direct minimisation of that quantity on  $K(\delta)$  is 8.028, at  $\delta = 0.05$ , and the least value of  $\text{vol}(V_c \cap B(\sqrt{8/5}))$  is 8.129 (`truncated_volume.py`). A statement with slack is one that a relaxation can certify, and Section 7.12 does.

Second, it needs no classification and no saturation. It is the  $m = 23$  case of the 24-cell conjecture itself, in the form  $\text{vol}(V_c) \geq 8$  for every configuration of 23 contacts, saturated or not; what Section 7.5 contributes is only that this case is the last one, the others being settled by Corollary 7.17 and by Theorem 7.25.

Third, it is the kind of statement a semidefinite relaxation can certify, and the two-point relaxation already delivers four fifths of it. Proposition 7.44 minimises the left-hand side of (7.45) over every pair-angle distribution that satisfies the Bonferroni inequality at every radius and is positive definite in every degree, and finds  $0.07380\dots$  against  $0.09285\dots$ . The measure that attains the minimum gives each direction nine neighbours at  $62.3^\circ$  and nearly eight at  $103^\circ$ , a local picture that is consistent at any one direction and cannot be consistent at all twenty-three at once; telling the two apart is exactly what three-point information is for. The semidefinite bounds of Bachoc and Vallentin [BV08], which carry the distribution of triples of directions and which de Laat, Leijenhorst and de Muinck Keizer [LLM24] sharpened, at the second level, to an exact classification at 24 points (Theorem 7.30), are the instrument built for that, and the objective (7.45) is linear in the pair distribution, as such bounds require. A relaxation can fall short of a true inequality; what makes this one a good candidate is that the target it has to reach is fixed,  $0.092855\dots$ , that the pair angles alone reach  $0.0738$ , and that the gap is not a boundary phenomenon but a quantity with room on either side of it. The next subsection carries the computation out, and the relaxation does reach the target.

**7.12. The inequality between pair angles holds.** The inequality (7.45) is a lower bound for a linear functional of the pair distribution of a spherical code of fixed size and fixed minimal angle, and the method built for statements of that shape is the semidefinite programming bound of Bachoc and Vallentin [BV08], which reads the distribution of triples of directions. We set it up in the form we need, prove the one lemma that turns a feasible point of the programme into an inequality for every configuration, and then describe the certificate and its verification, which is carried out in exact rational arithmetic where the objects are rational and in interval arithmetic where they are not. Throughout,  $N = 23$ ,  $u_{ij} = \langle w_i, w_j \rangle = \cos \gamma_{ij}$ , and  $\omega$  is the weight of Proposition 7.41, regarded as the function of the inner product  $u = \cos \gamma$ : increasing on  $[\frac{1}{3}, \frac{1}{2}]$  from 0 to  $0.00144540\dots$  and identically 0 below  $\frac{1}{3}$ , since  $\Lambda(r, \gamma) = 0$  for  $\gamma \geq 2r_*$ .

*Positive definite kernels on the three-sphere.* For  $k \geq 0$  let  $G_k$  be the Gegenbauer polynomial of  $S^3$  normalised by  $G_k(1) = 1$ , that is  $G_k(\cos \theta) = \sin((k+1)\theta)/((k+1)\sin \theta)$ , the Chebyshev polynomial of the second kind divided by  $k+1$ . By Schoenberg's theorem [Sch42] every finite  $W \subset S^3$  satisfies  $\sum_{x,y \in W} G_k(\langle x, y \rangle) \geq 0$ , which is the linear programming bound of Delsarte, Goethals and Seidel [DGS77] in its primal form. The three-point matrices are built from the Legendre polynomials  $P_k$ , which play the same role on  $S^2$ . Fix a degree  $d$  and, for  $0 \leq k \leq d$ , let  $Y_k(u, v, t)$  be the square matrix of order  $d - k + 1$  with entries

$$\begin{aligned} Y_k(u, v, t)_{ij} &= T_i(u) T_j(v) ((1 - u^2)(1 - v^2))^{k/2} \\ &\quad \times P_k\left(\frac{t - uv}{\sqrt{(1 - u^2)(1 - v^2)}}\right), \end{aligned} \tag{7.46}$$

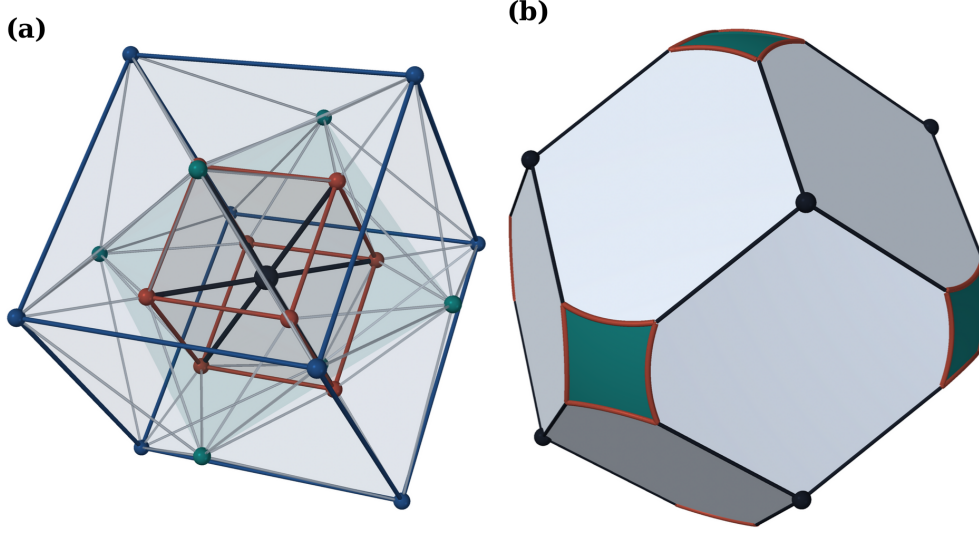


FIGURE 22. (a) The contact graph of the root system from one root (dark, at the centre), the other twenty-three at radius proportional to angle: a cube at  $60^\circ$  (clay), an octahedron at  $90^\circ$  (teal), a cube at  $120^\circ$  (blue), and the 88 tight pairs among them; the eight at the centre are drawn dark, and they sum to four times it. (b) The truncated cell in dimension three: the Voronoi cell of the face-centred cubic lattice at inradius 1, a rhombic dodecahedron, cut by the ball of radius  $(3/2)^{1/2}$ . The ball passes through the eight three-valent vertices and replaces the six four-valent corners by spherical squares (teal).

for  $0 \leq i, j \leq d - k$ , where  $T_i$  is the Chebyshev polynomial of the first kind; since  $P_k$  has the parity of  $k$ , each entry is a polynomial in  $u, v, t$ . Bachoc and Vallentin write  $u^i v^j$  in place of  $T_i(u)T_j(v)$ ; the two families differ by a fixed congruence of the matrices and are interchangeable in everything below, and the Chebyshev basis is the better conditioned one on  $[-1, \frac{1}{2}]$ . Let  $S_k$  be the average of  $Y_k$  over the six permutations of  $(u, v, t)$ .

**Lemma 7.71** (Positivity of the three-point matrices). *For every finite  $W \subset S^3$  and every  $k$ , the matrix  $\sum_{(x,y,z) \in W^3} S_k(\langle x, y \rangle, \langle x, z \rangle, \langle y, z \rangle)$  is positive semidefinite.*

*Proof.* Fix  $e \in S^3$  and write  $x^\perp = x - \langle e, x \rangle e$  for  $x \in S^3$ . With  $u = \langle e, x \rangle$ ,  $v = \langle e, y \rangle$ ,  $t = \langle x, y \rangle$  one has  $|x^\perp|^2 = 1 - u^2$  and  $\langle x^\perp, y^\perp \rangle = t - uv$ , so the scalar factor of (7.46) is  $Q_k(x, y) = |x^\perp|^k |y^\perp|^k P_k(\langle x^\perp, y^\perp \rangle / (|x^\perp| |y^\perp|))$ , read as 0 for  $k \geq 1$  and as 1 for  $k = 0$  when one of the projections vanishes. The kernel  $P_k(\langle \xi, \eta \rangle)$  is positive definite on the unit sphere of  $e^\perp$ , a copy of  $S^2$ , by [Sch42], and multiplying a positive definite kernel by  $a(x)a(y)$  keeps it positive definite; so  $Q_k$  is positive definite on  $S^3$ , and for any functions  $a_0, \dots, a_{d-k}$  the matrix with entries  $\sum_{x,y \in W} a_i(x)a_j(y)Q_k(x, y)$  is positive semidefinite, because  $c^\top M c = \sum_{x,y} b(x)Q_k(x, y)b(y)$  with  $b = \sum_i c_i a_i$ . Taking  $a_i(x) = T_i(\langle e, x \rangle)$  gives  $\sum_{x,y \in W} Y_k(\langle e, x \rangle, \langle e, y \rangle, \langle x, y \rangle) \geq 0$  for every  $e$ ; summing over  $e \in W$  gives the same for the sum over  $W^3$  of  $Y_k$ , and that sum is unchanged by permuting the three arguments, so it equals the sum of  $S_k$ . ■

*From a feasible point to an inequality.*

**Lemma 7.72** (The certificate lemma). *Let  $f = \sum_{k=0}^d f_k G_k$  with  $f_k \geq 0$  for  $k \geq 1$ , let  $F(u, v, t) = \sum_{k=0}^d \langle F_k, S_k(u, v, t) \rangle$  with each  $F_k$  positive semidefinite of order  $d - k + 1$ , and suppose that*

$$\omega(u) + \omega(v) + \omega(t) - f(u) - f(v) - f(t) - F(u, v, t) - \frac{F(1, u, u) + F(1, v, v) + F(1, t, t)}{N - 2} \geq 0 \quad (7.47)$$

for every admissible triple, that is every  $(u, v, t) \in [-1, \frac{1}{2}]^3$  with  $1 + 2uv t - u^2 - v^2 - t^2 \geq 0$ . Then every contact configuration of  $N = 23$  directions satisfies

$$\sum_{i < j} \omega(u_{ij}) \geq B := \frac{N}{2}(N f_0 - f(1)) - \frac{N F(1, 1, 1)}{6(N - 2)}. \quad (7.48)$$

*Proof.* The inner products of three distinct directions of a contact configuration form an admissible triple, since their Gram matrix is positive semidefinite with unit diagonal and its determinant is  $1 + 2uv t - u^2 - v^2 - t^2$ . Sum (7.47) over the  $\binom{N}{3}$  triples  $i < j < k$ . Each pair lies in  $N - 2$  triples, so the  $\omega$  terms give  $(N - 2) \sum_{i < j} \omega(u_{ij})$  and the  $f$  terms give  $(N - 2) \sum_{i < j} f(u_{ij}) = \frac{N-2}{2} (\sum_{i,j} f(u_{ij}) - N f(1)) \geq \frac{N-2}{2} (N^2 f_0 - N f(1))$  by Schoenberg's theorem, since  $G_0 = 1$ . The triple sum of  $F$  and the pair sum of  $F(1, u, u)$  are read off from the sum of  $F$  over all  $N^3$  ordered triples of  $W$ , which is nonnegative by Lemma 7.71: the  $N$  triples with all three entries equal contribute  $N F(1, 1, 1)$ ; the  $3N(N - 1)$  with exactly two equal contribute  $F(1, u_{ij}, u_{ij})$  each, by the symmetry of  $F$ , that is  $6 \sum_{i < j} F(1, u_{ij}, u_{ij})$ ; and the  $N(N - 1)(N - 2)$  with all entries distinct contribute  $6 \sum_{i < j < k} F(u_{ij}, u_{ik}, u_{jk})$ . Hence  $\sum_{i < j < k} F \geq - \sum_{i < j} F(1, u_{ij}, u_{ij}) - \frac{N}{6} F(1, 1, 1)$ , while the last term of (7.47) summed over triples is exactly  $\sum_{i < j} F(1, u_{ij}, u_{ij})$ . Putting the three parts together,  $(N - 2) \sum_{i < j} \omega(u_{ij}) \geq \frac{N-2}{2} (N^2 f_0 - N f(1)) - \frac{N}{6} F(1, 1, 1)$ , which is (7.48).  $\blacksquare$

The lemma uses the minimal angle only through the domain of (7.47), and the number 23 only through  $N$ . Its content is that the pair sum on the left of (7.45) is bounded below by the value of a semidefinite programme in the variables  $f_k$  and  $F_k$ , and that a feasible point of that programme with value above  $8 - A_*$  is a proof of Theorem 7.69's hypothesis. Such a point exists at degree 8.

**Theorem 7.73** (The pair inequality). *There are numbers  $f_0, \dots, f_8$  and symmetric matrices  $F_0, \dots, F_8$ , of orders  $9, 8, \dots, 1$ , listed in the supplementary material, with  $f_k > 0$  for  $k \geq 1$ , every  $F_k$  positive definite, satisfying (7.47) on the admissible domain, and with*

$$B = 0.0929000002 \dots > 8 - A_* = 0.0928555702 \dots$$

Consequently every contact configuration of 23 directions satisfies (7.45) with  $0.0000444$  to spare, every such configuration with bounded cell has  $\text{vol}(V_c) \geq \text{vol}(V_c \cap B(\sqrt{3}/2)) = A_* + \sum_{i < j} \omega(\gamma_{ij}) \geq A_* + B > 8$ , and Theorem 7.40 holds.

The proof is the verification of the three properties of the certificate, and we describe it in the detail that a reader who wants to repeat it would need; the scripts named are in Appendix F.

*Proof. The certificate.* The numbers are double-precision floating-point numbers, which are dyadic rationals, and they are taken as the exact rationals they denote; nothing is rounded. All computations below in which only these rationals and the integers enter are done in exact rational arithmetic.

*Positivity.* The nine matrices are symmetric and positive definite: an exact  $LDL^T$  elimination over  $\mathbb{Q}$  returns positive pivots throughout, and  $f_1, \dots, f_8$  are positive. The bound

$B$  is computed exactly from (7.48); here  $F(1, 1, 1)$  is the sum of the entries of  $F_0$ , because every entry of  $Y_k(1, 1, 1)$  vanishes for  $k \geq 1$  and  $T_i(1) = 1$ . These two steps, with the certificate written into the source as exact rationals, are also carried out by the Lean kernel (`lean/D4Certificate.lean`).

*The constants.*  $A_*$  is elementary. With  $S(r) = \sec^4 r$  one has  $\int r dS = r \sec^4 r - \tan r - \frac{1}{3} \tan^3 r$  and  $\int \sin 2r dS = \frac{8}{3} \tan^3 r$ , and  $\sec^2 r_* = \frac{3}{2}$ ,  $\tan r_* = 1/\sqrt{2}$ , so that

$$A_* = \frac{9\pi^2}{8} - \frac{207\pi}{8} r_* + \frac{253\pi}{12\sqrt{2}}, \quad r_* = \arctan \frac{1}{\sqrt{2}}, \quad (7.49)$$

and  $8 - A_*$  lies in  $[0.092855570293, 0.092855570294]$  by interval arithmetic at forty digits. The weight  $\omega$  is elementary too. The lens measure of Proposition 7.23 integrates to

$$\Lambda(r, \gamma) = 4\pi \left[ \frac{r - \gamma/2}{2} - \frac{\sin 2r - \sin \gamma}{4} - \frac{\tan(\gamma/2)}{2} (\sin^2 r - \sin^2(\gamma/2)) \right],$$

and  $\omega(\gamma) = \int_{\gamma/2}^{r_*} \Lambda(r, \gamma) \tan r \sec^4 r dr$  is a combination of the four primitives  $\int \tan r \sec^4 r dr = \frac{1}{4} \sec^4 r$ ,  $\int r \tan r \sec^4 r dr = \frac{1}{4} r \sec^4 r - \frac{1}{4} (\tan r + \frac{1}{3} \tan^3 r)$ ,  $\int \sin 2r \tan r \sec^4 r dr = \frac{2}{3} \tan^3 r$  and  $\int \sin^2 r \tan r \sec^4 r dr = \frac{1}{4} \tan^4 r$ , with coefficients depending on  $\gamma$  alone. The closed form agrees with the quadrature of Proposition 7.41 to fifteen digits at every angle tested, and it is differentiated symbolically to give  $d\omega/du$  and  $d^2\omega/du^2$  as functions of  $\gamma$ , with  $du = -\sin \gamma d\gamma$ . On  $(60^\circ, 2r_*)$  the function  $\omega$  is analytic, and it is decreasing in  $\gamma$ , since  $\Lambda(r, \gamma)$  is and the range of integration shrinks; so  $\omega$  is increasing in  $u$  on  $[\frac{1}{3}, \frac{1}{2}]$ .

*Tables.* At 12001 angles  $\gamma_i$  equally spaced from  $2r_*$  down to  $60^\circ$ , interval arithmetic gives an upper bound  $u_i$  of  $\cos \gamma_i$ , a lower bound of  $\omega(\gamma_i)$  and an enclosure of  $\omega'(u)$  at  $u = \cos \gamma_i$ ; consecutive  $u_i$  differ by at most  $1.44 \times 10^{-5}$ , and  $|\omega''(u)| \leq 0.289712$  on the whole range, by interval evaluation of the second derivative on 3000 subintervals. For  $u \geq u_i$  the monotonicity gives  $\omega(u) \geq \omega(\gamma_i)$ , and the mean value theorem gives  $|\omega'(u) - \omega'(\cos \gamma_i)| \leq 0.289712(u - \cos \gamma_i)$ .

*The inequality (7.47).* The polynomial  $P(u, v, t) = f(u) + f(v) + f(t) + F(u, v, t) + (F(1, u, u) + F(1, v, v) + F(1, t, t))/21$  is expanded exactly, from the recurrences for  $T_i$ ,  $P_k$  and  $G_k$ , into 449 monomials of total degree at most 16 with rational coefficients; it is symmetric in  $u, v, t$ , so (7.47) need only be verified on the ordered part  $-1 \leq u \leq v \leq t \leq \frac{1}{2}$  of the domain, and it reads  $Q := 1000(\omega(u) + \omega(v) + \omega(t)) - P \geq 0$  after the scaling by 1000 under which the certificate was computed. The verification is a branch and bound over boxes, in floating-point interval arithmetic with every operation rounded outward. On a box with centre  $c$  and half-widths  $\rho_1, \rho_2, \rho_3$  whose  $u, v$  and  $t$  ranges lie inside  $(\frac{1}{3}, \frac{1}{2}]$ , Taylor's theorem gives

$$Q \geq Q(c) - \sum_v |\partial_v Q(c)| \rho_v - \frac{1}{2} \sum_{v,w} \left( \sup_{\text{box}} |\partial_v \partial_w Q| \right) \rho_v \rho_w,$$

where  $Q(c)$  and  $\partial_v Q(c)$  are bounded through the tables and the exact polynomial evaluated at the point  $c$ , and the second derivatives through the bound on  $\omega''$  and the interval evaluation of the second derivatives of  $P$  over the box. A variable whose range is not inside  $(\frac{1}{3}, \frac{1}{2}]$  has its  $\omega$  term replaced by 0, which is a lower bound, with derivatives 0. A box on which the right-hand side is nonnegative is verified; a box on which  $1 + 2uvt - u^2 - v^2 - t^2$  is negative throughout is outside the domain and is discarded; any other box is bisected along the direction that contributes most to the Taylor remainder. Starting from  $[-1, \frac{1}{2}]^3$ , the process ends after 44 levels with 313 780 boxes verified and 13 517 discarded, in under ten minutes (Figure 24(a)), and no box reaching the minimum width  $10^{-6}$  undecided. The same branch

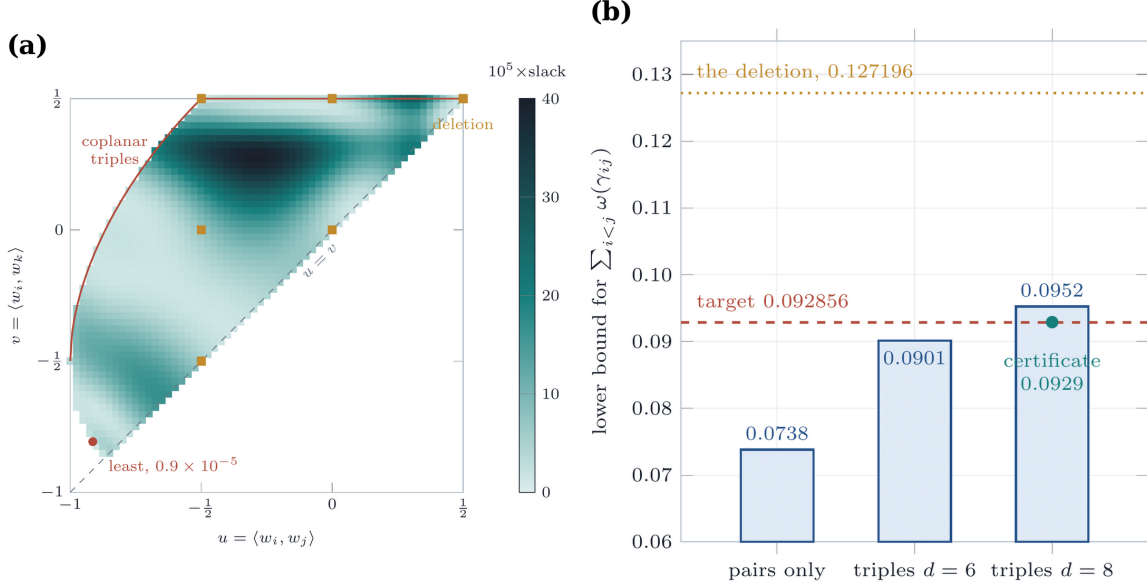


FIGURE 23. (a) The certificate of Theorem 7.73: the slack of the condition (7.47), the amount by which its left-hand side exceeds 0, on the slice of the admissible domain in which one pair of the triple is at  $60^\circ$ . It is positive throughout, least near a coplanar triple with one nearly antipodal pair, and largest, about  $4 \times 10^{-4}$ , in the interior. The six squares are the triples that occur in a deletion of a root. (b) The values of the relaxations against the target  $8 - A_*$ : the pair angles alone (Proposition 7.44), the three-point relaxation at degrees 6 and 8, and the certificate, at which the bound is fixed at 0.0929 and the least slack is maximised.

and bound is then carried out a second time, independently and in exact arithmetic, by a Lean program (`lean/certificate`): it expands  $P$  from the certificate on its own, multiplies it by 315 so that every coefficient becomes an integer over a power of two, and runs the subdivision with the same Taylor form in dyadic arithmetic, integers and exponents only, with no rounding at any point, taking from outside Lean nothing but the tables of bounds for  $\omega$ ,  $\omega'$  and  $\omega''$ ; it processes 421 881 boxes, fewer than the Python run because its nested evaluation of the polynomials encloses more tightly than a sum of monomials, subdivides down to the width  $2^{-20}$  where it has to, and returns true, which Lean records as a theorem under `native_decide`. Hence (7.47) holds on the admissible domain, Lemma 7.72 gives (7.45) for every contact configuration of 23 directions, and Theorem 7.69 gives the rest. ■

*Remark 7.74* (How the certificate was found, and how tight it is). The programme of Lemma 7.72, with (7.47) imposed on a sample of admissible triples and  $B$  maximised (Figure 23), is solved by an interior-point method (`three_point_sdp.py`); the sample is refined four times by adding the triples at which the current solution violates (7.47) most, on a check over about 1.15 million further triples, and the value reported is reduced by  $\binom{23}{3}/21$  times the largest violation that remains, so that it is a bound over the whole check. At degree 6 the value is 0.09011, that is 97.0 per cent of  $8 - A_*$ ; at degree 8 it is 0.09523, or 102.6 per cent. The pair angles alone, in Proposition 7.44, gave 79.5 per cent; the triples supply the rest. The certificate of Theorem 7.73 is then obtained by fixing  $B$  at 0.0929 and maximising instead the least slack of (7.47) over the sample, with  $F_k - 10^{-7}I$  and  $f_k - 10^{-7}$

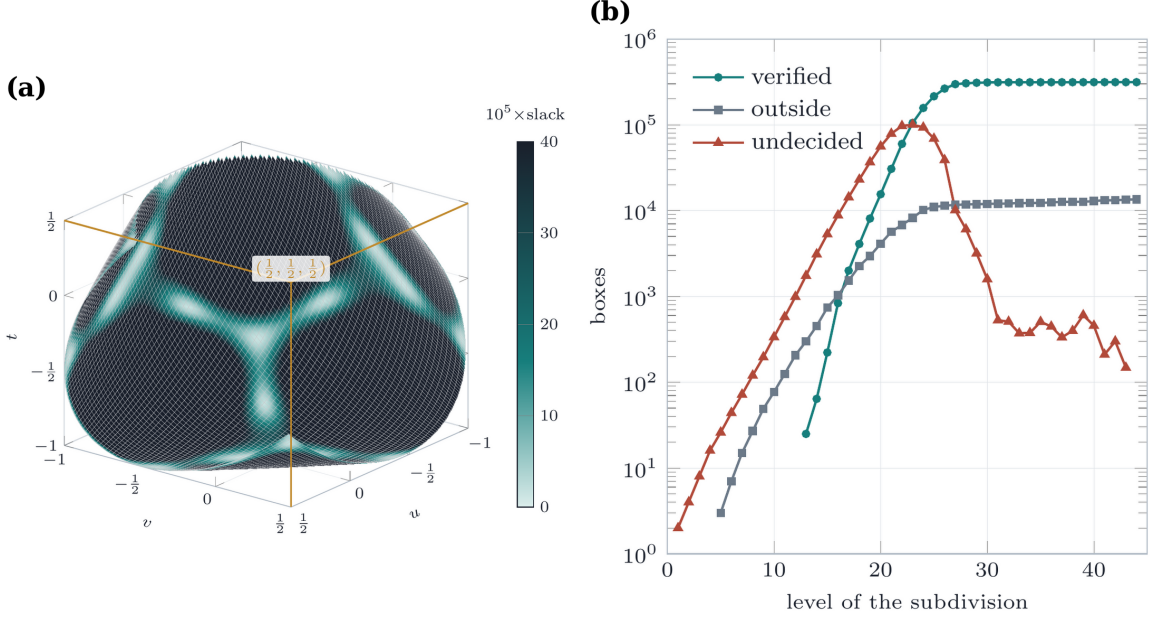


FIGURE 24. (a) The domain of the certificate condition: the admissible region of Lemma 7.72 in three dimensions, where the triples  $(u, v, t)$  of pairwise inner products of three contact directions fill the part of the cube  $[-1, \frac{1}{2}]^3$  enclosed by the surface of coplanar triples. The surface is coloured by the slack of (7.47), which is least along the curves where two of the three directions are nearly antipodal. (b) The branch and bound of the proof of Theorem 7.73, level by level: the boxes live, the boxes discarded on the enclosure, and the width reached.

constrained to be positive semidefinite so that the solution survives being read as exact rationals: the least slack is  $1.35 \times 10^{-5}$  on the final sample of 21 648 triples and  $1.30 \times 10^{-5}$  on the check, against  $\omega(60^\circ) = 0.00145$ . Two features of the solution are worth recording. The two-point part is almost inert,  $f_0 = 0.000927$  and  $f_1, \dots, f_8$  below  $3 \times 10^{-8}$ , so it is the matrices  $F_k$  that carry the argument. And the inequality is far from tight at the deletion of a root, where the pair sum is 0.1272 against  $B = 0.0929$ ; the relaxation had room, which is the situation Remark 7.70 asked for, and it is why a relaxation could succeed here where, by Remark 7.61, none could have classified the codes. Finally, the checker itself was tested on a certificate it should reject: adding  $10^{-5}$  to  $f_0$ , which raises  $P$  by  $3 \times 10^{-5}$  and so exceeds the slack, produces a certificate for which the same programme exhibits the admissible triple  $(-\frac{1}{16}, -\frac{1}{16}, -\frac{1}{16})$  at which (7.47) fails, in under two minutes.

*Remark 7.75* (What is verified where). The proof of Theorem 7.73 rests on three kinds of computation, and the reader should know which is which. The exact rational steps, positivity of the matrices and the value of  $B$ , are verified by the Lean kernel from the same numbers, and the expansion of  $P$  is an exact rational computation in Python whose result is checked against an independent floating-point evaluation of the same expressions at random triples, agreeing to  $10^{-13}$ . The constants  $A_*$ ,  $\omega$ ,  $\omega'$  and  $\omega''$  are evaluated from closed forms, (7.49) among them, in the interval arithmetic of `mpmath`. The branch and bound is run twice: in Python, in an interval arithmetic on IEEE doubles with outward rounding by one unit in the last place after every operation, and in Lean, in exact dyadic arithmetic,

where the expansion of  $P$ , its derivatives, every enclosure and every bisection are integer computations and the only data taken from outside are the tables of bounds for  $\omega$  and its two derivatives, so that no floating-point operation enters the Lean run at all. The Lean theorem states that the program returns true and is settled by `native_decide`, which runs the compiled program and therefore trusts the compiler as well as the kernel; what remains outside any proof assistant is the evaluation of the closed forms at the table points and the mathematics that makes the program a proof, the Taylor form and the monotonicity of  $\omega$ , which are written out above. The Taylor form is applied only on boxes inside the open interval where  $\omega$  is analytic.

## 8. THE DEVIATION DOMAIN AND ITS FUNDAMENTAL TRIANGLE

The identities of Section 7 hold for any number of deviating directions. From here to Section 13 we return to the case of one, which is where the exact structure lives, and set up the domain that case is played out on.

**8.1. The two remaining parameter spaces.** Theorem 7.1 makes clear that the domain covered by Theorem 1.1 enlarges in two structurally different directions:

- *More directions at once* ( $m \geq 2$ ): the parameter space is now a choice of  $m$  pairwise-compatible directions on  $S^3$ , together with an assignment of each to its own 3-dimensional tangent deviation. This is the subject of Section 15 onward.
- *One direction, but off the root-aligned family* ( $m = 1$ ,  $\eta$  not root-aligned): the parameter space is the full unit 2-sphere of deviation directions orthogonal to the undeveloped root, here reduced (Section 8.2) to a triangle by two exact symmetries. This is the subject of Section 9 onward.

**8.2. The fundamental domain of deviation directions.** Fix a root  $\alpha_0 \in \mathbb{R}$  and let  $u_0 = \alpha_0/\sqrt{2}$  be the corresponding contact direction. The tangent space orthogonal to  $u_0$  is 3-dimensional, and a deviation direction is any unit vector  $\eta$  in it; the deviating cap direction is then  $u_1(\varphi, \eta) = \cos \varphi u_0 + \sin \varphi \eta$ .

**Lemma 8.1** (Two exact symmetries). *Two elements of  $\text{Stab}(u_0) \leq W(D_4)$  act on the space of deviation directions  $\eta$ : a symmetry  $S$  of order 2 and a symmetry  $H$  (a signed permutation realising a Hadamard-type transformation) also of order 2, generating a group of order 4 whose fundamental domain on the unit circle of directions orthogonal to two further fixed root directions  $v_1, w_1$  (themselves canonically determined by  $u_0$ ) is a single triangle with vertices  $v_1, w_1$ , and  $v_2 = Sw_1$ .*

This reduces the classification of *every* deviation direction to the two boundary arcs of this triangle, the arc from  $v_1$  to  $w_1$  and the arc from  $w_1$  to  $v_2$ , parametrised respectively as

$$\eta_{v_1 w_1}(s) = \cos s v_1 + \sin s w_1, \quad \eta_{w_1 v_2}(s) = \cos s w_1 + \sin s v_2, \quad s \in [0, \frac{\pi}{4}].$$

(Working with  $s \in [0, \pi/4]$  on each arc, rather than a full quarter-turn, comes from a further symmetry fixing the arc's own midpoint, proved directly for each arc in Sections 9 and 11.) Every deviation direction on the interior of the triangle is covered by interpolating between the two arcs. Section 9 through Section 12 treat the two arcs in full, which is where the exact breakpoint structure and the exact defect polynomials live; the interior is not reachable by that route, and one might hope that some soft convexity argument would drag it along once the boundary was settled. It does not: Section 8.6 tests the natural candidate and finds it false. The interior is settled instead by the global argument of Section 13, which does not

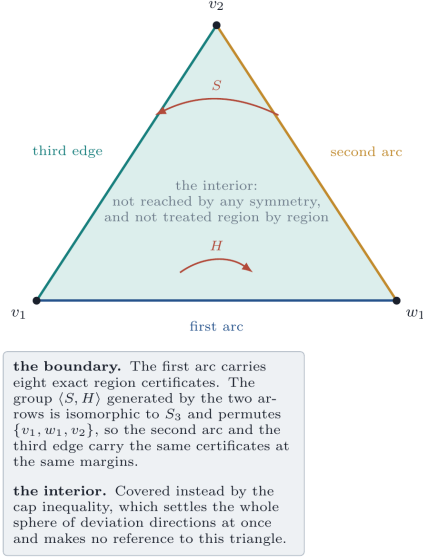


FIGURE 25. The fundamental domain of Lemma 8.1: one triangle with vertices  $v_1, w_1, v_2$ , obtained from the full sphere of deviation directions by the two order-2 symmetries  $S$  and  $H$ . The three edges carry the certificates of Sections 10 and 12, through the identifications of Propositions 8.3 and 8.5; the interior is settled separately, in Section 13, by an argument that does not use the triangle at all.

use the triangle at all. Figure 25 shows the triangle and which argument covers which part of it.

The immediate target of Section 9 through Section 12 is therefore the following statement, which Corollary 8.6 will establish with explicit margins and which Theorem 13.7 will later subsume:  $\Delta(\varphi, s) \geq 0$  for every  $\varphi$  in the packing-valid range and every  $s \in [0, \pi/4]$  on both boundary arcs  $\eta_{v_1 w_1}$  and  $\eta_{w_1 v_2}$ .

**8.3. Where the worst direction sits.** Lemma 8.1 reduces the classification to the triangle's two boundary arcs as a matter of the symmetry group's own fundamental domain, and says nothing about where in the triangle the defect is actually smallest. That is a separate question, and the answer shapes the rest of this section, so we record what a direct search finds.

Minimising  $\Delta(\varphi, \eta)$  numerically over the entire transverse 2-sphere, at each of many sampled tilts, and comparing the result against direct evaluation along three symmetry-distinguished one-parameter families, gives the following picture at chamber  $\mathfrak{k}_1$ . The three families are the six root directions orthogonal to  $u_0$ , the two coordinate-axis directions orthogonal to  $u_0$  (the vertex  $v_1$  family), and the half-integer-sign family exemplified by  $e_* = (-1, 1, -1)/2$  (the vertex  $w_1$  family). One of the three matches the global minimum for almost the whole range of tilts: the coordinate family on  $\varphi \in (0, 0.16] \cup [1.26, \pi/2)$ , the root family on  $\varphi \in [0.18, 1.09]$ . The exception is the window  $\varphi \in (1.09, 1.26)$ , where the minimiser lies strictly below all three.

A finer search in that window locates it. At ten tilts spanning and bracketing the window, from  $\varphi = 1.05$  to  $\varphi = 1.30$ , a Nelder–Mead search with 30 random restarts per tilt over the full 2-sphere returns a minimiser one of whose coordinates  $(a, b, c)$  relative to the frame

$(v_1, w_1, v_2)$  vanishes to at least five decimal places, values of order  $10^{-5}$  against coordinates of order 1. To the resolution of the search, the minimiser is therefore on one of the three great circles  $a = 0, b = 0, c = 0$  through pairs of  $\{v_1, w_1, v_2\}$ , and never at an interior point with all three coordinates bounded away from zero. For the five tilts  $\varphi \in \{1.15, 1.18, 1.20, 1.22, 1.24\}$  it sits on the circle  $b = 0$  through  $v_1$  and  $v_2$ , which is the third edge of the triangle rather than either of the two arcs; at the window's ends it sits on  $a = 0$  or  $c = 0$ .

That is a numerical finding, not a proof, and the theorem of Section 13 does not depend on it. Its use here is to say which part of the boundary the certificates of the next four sections have to reach, and the answer includes the third edge. The next subsection supplies the exact symmetry that puts the third edge inside the first arc's certificate.

**8.4. The third edge, closed by an exact symmetry.** The preceding paragraph leaves the  $v_1$ - $v_2$  edge as the one part of the fundamental triangle's boundary carrying no certificate of any kind. We resolve this here: not by a new certificate computation, but by exhibiting the exact isometry the preceding paragraph asks for and verifying it directly.

**Lemma 8.2** (The third edge is the image of the first arc). *Let  $S = \text{diag}(1, 1, 1, -1)$  in the coordinates of Lemma 8.1 (so  $u_0 = (1, 1, 0, 0)/\sqrt{2}$ ,  $v_1 = (1, -1, 0, 0)/\sqrt{2}$ ,  $w_1 = (0, 0, 1, 1)/\sqrt{2}$ ,  $v_2 = (0, 0, 1, -1)/\sqrt{2}$ ). Then, exactly:*

$$Su_0 = u_0, \quad Sv_1 = v_1, \quad Sv_2 = w_1, \quad S^2 = \text{Id}.$$

Moreover  $S$  permutes the 24 roots of  $\mathbf{R}$  bijectively (a root  $\pm e_i \pm e_j$  maps to another root of the same index pair, with the sign at index 4 flipped when  $i = 4$  or  $j = 4$  and left unchanged otherwise), so  $S$  is an automorphism of the root system fixing  $u_0$ .

*Proof.* Immediate:  $S$  negates only the fourth coordinate, and  $u_0, v_1$  both have a zero fourth coordinate, while  $v_2$  and  $w_1$  differ exactly in the sign of that coordinate.  $S^2 = \text{diag}(1, 1, 1, 1) = \text{Id}$  since  $(-1)^2 = 1$ . The root-permutation claim is a finite check on all 24 roots; both this and the four displayed identities were verified by exact computer algebra (Appendix F), with no floating point at any stage. ■

**Proposition 8.3** (The  $v_1$ - $v_2$  edge is certified for free). *For every  $\varphi$  in the packing-valid range and every  $s$ ,*

$$\Delta(\varphi, \eta_{v_1 v_2}(s)) = \Delta(\varphi, \eta_{v_1 w_1}(s)),$$

where  $\eta_{v_1 v_2}(s) = \cos s v_1 + \sin s v_2$  is the third edge and  $\eta_{v_1 w_1}$  is the first arc's own parametrisation (Lemma 8.1). Consequently, by Theorems 9.1 and 10.1,  $\Delta \geq 0$  on the entire  $v_1$ - $v_2$  edge for every  $s \in [0, \pi/4]$ , with exactly the same margin structure already recorded for the first arc (seven of the eight regions exact and zero-margin, one with the same explicit thin margin arising from the elliptic obstruction noted there): no new certificate is computed for this edge, because none is needed.

*Proof.* Since  $S$  fixes  $u_0$  linearly,  $S$  carries the deviated direction  $u_1(\varphi, \eta) = \cos \varphi u_0 + \sin \varphi \eta$  to  $u_1(\varphi, S\eta)$  for every  $\varphi, \eta$ . Since  $S$  is also an automorphism of  $\mathbf{R}$  fixing  $u_0$  (Lemma 8.2),  $S$  carries the entire packing configuration at deviation  $\eta$  (the 23 undeviated roots, permuted among themselves, together with the one deviating root  $u_1(\varphi, \eta)$ ) to a congruent configuration at deviation  $S\eta$ : the two Voronoi cells are related by the isometry  $S$ , hence have identical volume, hence identical defect. Applying this with  $\eta = \eta_{v_1 v_2}(s)$  and using  $Sv_1 = v_1$ ,  $Sv_2 = w_1$  from Lemma 8.2 (so, by linearity,  $S\eta_{v_1 v_2}(s) = \cos s v_1 + \sin s w_1 = \eta_{v_1 w_1}(s)$  exactly, for every  $s$ , not merely the sampled values) gives the displayed identity. ■

The third edge, which Section 8.3 identified as where the worst direction sits for part of the transition window, therefore carries the first arc's certificate without any further

computation. The isometry that does it is  $S$ , the same symmetry already used in Section 11 to reduce the second arc's range to  $[0, \pi/4]$ , applied here in the direction that connects the first arc to the third edge.

**8.5. The full symmetry group of the fundamental triangle, and its consequence for the second arc.** Lemma 8.2 used the single symmetry  $S$ . Taken together with the earlier symmetry  $H$  of Lemma 8.1, it generates a group large enough to settle the second arc as well.

**Lemma 8.4** (The fundamental triangle's full symmetry group). *The group generated by  $S = \text{diag}(1, 1, 1, -1)$  and the Hadamard matrix  $H = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$  has order exactly 6, every element fixes  $u_0$ , and every element is an automorphism of  $\mathbf{R}$ . The induced action on  $\{v_1, w_1, v_2\}$  realises all six permutations of this three-element set: the group is (isomorphic to) the symmetric group  $S_3$ , acting on  $\{v_1, w_1, v_2\}$  exactly as its full permutation group.*

*Proof.* Direct enumeration: the six words in  $S, H$  up to the relations  $S^2 = H^2 = \text{Id}$  produce six distinct  $4 \times 4$  matrices, each verified exactly (symbolic  $\mathbb{Q}(\sqrt{2})$  arithmetic, no floating point) to fix  $u_0$ , to permute the 24 roots of  $\mathbf{R}$  bijectively, and to induce a distinct one of the six permutations of  $\{v_1, w_1, v_2\}$  on evaluation. In particular  $M := SH$  has order 3 and acts as the cycle  $v_1 \mapsto v_2 \mapsto w_1 \mapsto v_1$ . ■

**Proposition 8.5** (The second arc is the image of the first). *For every  $\varphi$  in the packing-valid range and every  $s$ ,*

$$\Delta(\varphi, \eta_{w_1 v_2}(s)) = \Delta(\varphi, \eta_{v_1 w_1}(s)).$$

*Proof.* By Lemma 8.4,  $M = SH$  fixes  $u_0$  and is an automorphism of  $\mathbf{R}$ , so (exactly as in the proof of Proposition 8.3) it carries the packing configuration at any deviation  $\eta$  to a congruent one at deviation  $M\eta$ , giving  $\Delta(\varphi, \eta) = \Delta(\varphi, M\eta)$  for every  $\varphi, \eta$ . Since  $Mw_1 = v_1$  and  $Mv_2 = w_1$  (Lemma 8.4), linearity gives  $M\eta_{w_1 v_2}(s) = \cos s v_1 + \sin s w_1 = \eta_{v_1 w_1}(s)$  exactly, for every  $s$ . Substituting  $\eta = \eta_{w_1 v_2}(s)$  gives the displayed identity. ■

Both identities were checked, independently of the algebra above, by direct numerical evaluation of the volume-defect function itself (halfspace intersection and convex-hull volume, the same method used elsewhere in this paper to sanity-check symbolic identities): a fine active-facet-pattern scan across  $s \in \{0.02, \dots, 0.78\}$ , spanning every combinatorial sub-part of the second arc including the B-bounded band, found the first and second arcs' breakpoints coinciding exactly (to grid resolution) at every sampled  $s$ , and the defect values agreeing to floating-point tolerance throughout, including precisely at curve B's own predicted crossing, where Theorem 11.1 records vertex  $b$  as "a vertex that plays no role at all on the first arc." That description remains correct in the sense meant there (vertex  $b$  itself, as a labelled root, has no counterpart among the vertices Theorem 9.1 names on the first arc); Proposition 8.5 shows the *curve* it produces nonetheless has an exact counterpart on the first arc, reached under  $M$  by a differently-labelled vertex, the two arcs' independent derivations found the same underlying function through different combinatorial routes, without the connection between them being noticed at the time.

**Corollary 8.6** (Positivity on the whole boundary).  *$\Delta(\varphi, s) \geq 0$  for every  $\varphi$  in the packing-valid range and every  $s \in [0, \pi/4]$  on both boundary arcs  $\eta_{v_1 w_1}$  and  $\eta_{w_1 v_2}$ , with the same margin structure as Theorems 9.1 and 10.1 (seven of the eight regions exact and zero-margin, one with an explicit, quantified thin margin arising from the elliptic obstruction recorded there).*

*Proof.* By Propositions 8.3 and 8.5, both boundary arcs (and the third edge, though the conjecture as stated does not require it) are identical, as functions of  $(\varphi, s)$ , to the first arc. Theorems 9.1 and 10.1 already establish  $\Delta \geq 0$  throughout the first arc's own domain, with the stated margin structure. Substituting the identities gives the same bound, with the same margins, on both named arcs. ■

This settles the sign on the whole boundary of the fundamental triangle at the rigour standard already established for the first arc alone, and it does so with the first arc's margin structure intact. No further certificate on the second arc's region types is needed: the second-arc computation of Section 12, including the B-bounded sub-part of Section 12.4, becomes redundant once Proposition 8.5 is available. We have left that account in place because it is what led us to look for the symmetry.

The symmetry argument stops at the boundary. No element of the group of Lemma 8.4, and none of any larger group acting here, moves a generic interior point onto an edge, so the triangle's two-dimensional interior is not reachable this way. It is reached in Section 13 by an argument that abandons the triangle altogether.

**8.6. The interior: a falsified shortcut, and what the interior's own structure looks like.** Since the boundary is now fully settled, it is natural to ask whether some soft structural argument drags the interior along for free without requiring a two-parameter closed-form derivation. We tried the most natural candidate, report that it fails, and record what direct computation shows about the interior's own structure instead.

#### 8.6.1. A candidate reduction, tested and falsified.

*Candidate argument (false, recorded for the record).* If  $\Delta(\varphi, \cdot)$  were concave, for every fixed  $\varphi$ , along every geodesic arc of the transverse 2-sphere lying inside the closed fundamental triangle, then for any interior point  $Q$ , picking a geodesic through  $Q$  with both endpoints on the triangle's boundary would give  $\Delta(\varphi, Q) \geq \min(\Delta(\varphi, \text{endpoint}_1), \Delta(\varphi, \text{endpoint}_2)) \geq 0$ , the last inequality by Corollary 8.6 together with Proposition 8.3 (which covers the third edge). This would settle the whole triangle with no further derivation.

This argument is false. Testing it directly, geodesics from each vertex to random points on the opposite edge, and fully generic interior-to-interior chords, sampled at 25 values of  $\varphi$  each, finds small positive discrete second differences (apparent local convexity, not concavity) at several points, concentrated toward  $\varphi$  near  $\pi/2$ . Rather than report this as an ambiguous grid artifact, we ran it down: fixing one such point and shrinking the step size geometrically across eight orders of magnitude ( $h = 0.08$  down to  $h \approx 3 \times 10^{-4}$ ) gives  $d^2(h)/h^2$  converging cleanly to  $+0.3056$ , constant to four significant figures over the last five halvings, the signature of a nonzero positive second derivative at that point, not discretisation noise (which would shrink faster than  $h^2$  and show no stable limit) and not a combinatorial breakpoint nearby (the local vertex count is confirmed constant at 37 throughout a window around the point). The candidate argument is dead:  $\Delta(\varphi, \cdot)$  is not concave along every interior geodesic, so the interior cannot be reduced to the boundary this way, at any  $\varphi$  in the domain (the effect was not confined to  $\varphi$  near  $\pi/2$ , only larger there). The finding is negative, and we record it as such: it rules out the whole class of arguments that would have let the boundary drag the interior along.

**8.6.2. The interior's own combinatorial structure.** Direct evaluation of the number of active vertices of the local Voronoi region, scanned over a fine grid of the closed triangle at a fixed  $\varphi = 1.45$  (in the region-top band, where both boundary arcs have the simple 32-vertex type

throughout), shows that the boundary's combinatorial simplicity is special to the boundary: it does *not* extend to the interior. A grid of 861 points finds the 32-vertex type only along a band adjacent to the edges (25% of the sampled points), a 34-vertex type in a second band just inside that (17%), a 37-vertex type filling most of the remaining interior (53%, the plurality type), and a further 40-vertex type occupying a smaller region near the triangle's centre (5%). The interior is therefore combinatorially richer than either boundary arc: it is not one region but at least four, and we have not attempted to locate the exact curves separating them or to derive a volume formula on any one of them; the map above only establishes that such curves exist and roughly where.

This partition is not arbitrary: it visibly respects the  $S_3$  symmetry of Lemma 8.4. The map is invariant under the reflection fixing  $v_1$  and exchanging  $w_1, v_2$  (i.e. the symmetry  $S$  of Lemma 8.2), exactly as it must be, since  $S$  is an exact isometry of the whole configuration and therefore permutes combinatorial type among congruent points. By the same argument, the full 6-element group of Lemma 8.4 acts on the interior, not only on the boundary it was originally used for: every one of its elements is an automorphism of  $\mathbb{R}$  fixing  $u_0$ , hence carries any interior deviation direction to a congruent configuration at the image direction, with identical defect and identical combinatorial type. This is a real, if modest, scope reduction for any future attempt at the interior: it suffices, in principle, to derive the combinatorial map and the volume formula on a single fundamental domain of the triangle under this  $S_3$  action (one-sixth of the triangle, e.g. the wedge from  $v_1$  to the midpoint of arc  $v_1w_1$  to the triangle's centre) and extend by the group, instead of treating the interior as one undifferentiated two-parameter domain six times the necessary size.

Taken together, these two findings are what closed off the regional approach to the interior. The natural soft reduction is false, not merely unproved, and the interior's own combinatorial structure is strictly richer than the boundary's, so a region-by-region derivation there would have meant at least four new volume formulas before any certificate could be attempted. That is the point at which we stopped refining the triangle and looked for an argument that does not see it, which is the subject of Section 13.

## 9. BREAKPOINT CLASSIFICATION ON THE FIRST ARC

**9.1. Setup.** Fix the arc  $\eta_{v_1w_1}(s) = \cos s v_1 + \sin s w_1$ ,  $s \in [0, \pi/4]$ , of Lemma 8.1, and write  $u_1(\varphi, s) = \cos \varphi u_0 + \sin \varphi \eta_{v_1w_1}(s)$  for the deviating cap direction. As  $\varphi$  increases from 0 at fixed  $s$ , the active-facet pattern of the local Voronoi region changes at finitely many exact values of  $\varphi$ , the arc's *breakpoints*, each the locus where some further root  $\beta \in \mathbb{R}$  achieves  $\langle \beta, u_1(\varphi, s) \rangle = 1$  simultaneously with the cap constraint.

**Theorem 9.1** (Complete breakpoint classification, first arc). *Exactly four exact curves account for every active-facet-set transition of  $u_1(\varphi, s)$  across the entire domain  $s \in [0, \pi/4]$ :*

- (1) the universal curve  $\varphi = \pi/3$ , arising from the 8-fold degenerate vertex  $2u_0$  and independent of  $s$ ;
- (2) a curve  $W$  from the sign vertex  $w = (1, 1, 1, 1)/\sqrt{2}$ , satisfying  $\sin s = \tan(\varphi/2)$ ;
- (3) a curve  $A$ , the image of  $W$  under the symmetry  $H$  of Lemma 8.1, from the axis vertex  $a = \sqrt{2}e_1$ , satisfying  $\cos s = \tan(\varphi/2)$ ;
- (4) a curve  $Y$  from the sign vertex  $y = (1, -1, 1, 1)/\sqrt{2}$ , satisfying  $\sin \varphi (\sin s + \cos s) = 1$ .

Each curve is derived by an exact symbolic solve of the corresponding tangency condition after the Weierstrass substitution  $u = \tan(\varphi/2)$  reduces it to a low-degree polynomial equation in  $u$  (linear, in every one of the four cases here); completeness, that no further breakpoint curve exists anywhere on the domain, is confirmed by a dense numerical scan

(over 150 values of  $s$  and, at each, a refined bisection search over  $\varphi$ ) finding at most four breakpoints at any sampled  $s$  and matching every one, to a tolerance far smaller than the curves' own separation, against the four curves above.

**9.2. The eight combinatorial regions.** Sorting the four curves by  $\varphi$ -value at a given  $s$  produces the domain's combinatorial atlas: eight distinct region types, established by an exact vertex-pattern comparison (not merely a curve-crossing count) at representative points of each candidate region and cross-checked against a dense grid scan with no unexplained region boundary. We label them I, ..., VIII, purely as bookkeeping: region I ("region top") is the topmost band, bounded below by curve A throughout the entire  $s$ -range and above by  $\varphi = \pi/2$ , the only region needing no internal crossover point, since A and  $\varphi = \pi/2$  never meet inside the domain; regions II through VII subdivide the remaining domain according to which pair of the curves W, A, Y (and the universal curve) locally bound it; and region VIII is the bottommost band. Section 10 gives a complete positivity certificate on every one of these eight regions.

## 10. CERTIFICATES ON THE FIRST ARC

**10.1. Method: exact remapping versus box-covering.** Two complementary exact techniques are used across the eight regions of Theorem 9.1, chosen region by region according to whether the region's boundary curve is itself rational in Weierstrass coordinates. Exact zero-margin certificates. When a region's lower (or upper) boundary curve is rational in  $u = \tan(\varphi/2)$ ,  $v = \tan(s/2)$ , as is the universal curve and the W, A curves of Theorem 9.1, a further exact substitution maps the curved region onto an exact unit square, after which the volume-defect numerator and denominator become two explicit polynomials with coefficients in  $\mathbb{Z}[\sqrt{2}]$  (or, where a crossover point of the region's own two bounding curves is used to rescale the domain,  $\mathbb{Z}[\sqrt{5}]$ ). Converting both to the Bernstein basis over the unit square by exact arithmetic and checking the sign of every coefficient either proves  $\Delta_k > 0$  on the entire open region with *no excluded margin*, or reports precisely which coefficients fail to cooperate.

Box-covering certificates. When the region's own boundary is not rational (an elliptic obstruction on several regions, discussed below), the region is instead covered by many small rectangular boxes in the plain Weierstrass coordinates, with corners chosen numerically to sit safely inside the true domain; the same exact Bernstein sign certificate is then run independently on each box, at the cost of a thin, explicitly quantified excluded margin near the true boundary.

**10.2. A correction to the region count.** An initial grid scan (at  $60 \times 90$  resolution) reported *nine* combinatorially distinct types in the reduced domain  $s \in (0, \pi/4)$ ,  $\varphi \in (0, \pi/2)$ . Re-running the same scan at much higher resolution ( $150 \times 200$ ), with an explicit 0.01-radian margin excluded around each of the four breakpoint curves (to rule out grid points landing on, or numerically near, a curve, which can spuriously register as an extra type owing to floating-point near-degeneracy in the underlying half-space-intersection routine), finds only *eight* distinct types. Tracing the exact pairwise intersections of the four breakpoint curves within  $s \in (0, \pi/4)$  confirms exactly three crossings, the Y-curve (a relabelling of one of the curves counted among W, A, Y in Theorem 9.1) meets the universal line at  $s_1 = 0.169918$ , the W-curve meets Y at  $s_2 = \arctan(\frac{1}{2})$ , and W meets the universal line at  $s_3 = \arcsin(1/\sqrt{3})$ , and one region initially split by this crossover ("region 4", bounded above first by W then by Y) is confirmed, by an independent direct convex-hull volume computation at representative points on each side of the crossover agreeing with a single symbolic volume formula

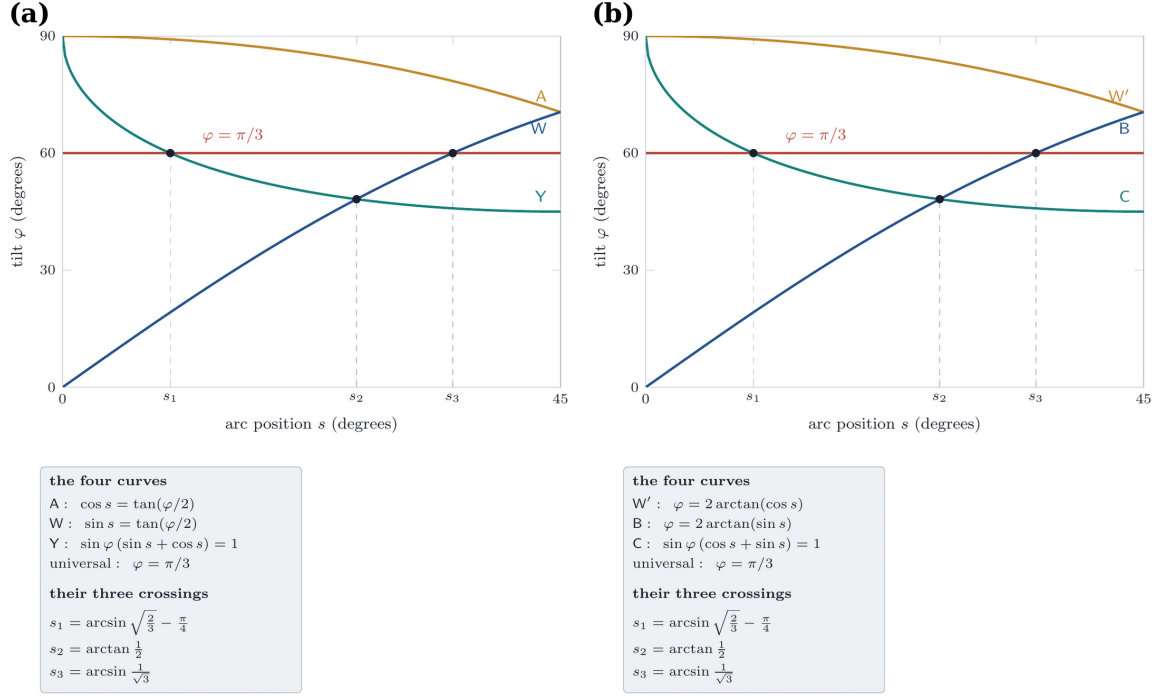


FIGURE 26. (a) The four breakpoint curves of the first arc, drawn to scale from the closed forms of Theorem 9.1: the universal curve  $\varphi = \pi/3$  and the curves A, W, Y. They meet in exactly three points, at the abscissae  $s_1$ ,  $s_2$ ,  $s_3$  listed at the right, and the eight combinatorial regions of Section 10.2 are the cells of the resulting arrangement. The curves A and W meet only at  $s = \pi/4$ , where the reflection symmetry of the arc identifies them. (b) The four breakpoint curves of the second arc, drawn to scale from the closed forms of Theorem 11.1. Comparing with Figure 26(a) shows what Proposition 8.5 later proves: the three equations here are exactly those of A, W and Y on the first arc, and the arrangement, crossings included, is the same one, even though the curves come from different vertices of  $\mathcal{V}$ . The band immediately below region top splits into three sub-parts at the crossings  $s_1$ ,  $s_2$ ,  $s_3$ .

to twelve decimal places, to be *one* combinatorial region throughout, not two, accounting for the discrepancy. It is nonetheless certified in two computational pieces (Section 10.3), since only one piece's own boundary happens to be rational.

### 10.3. The eight regions, one by one.

**Theorem 10.1** (Certificates, first arc). *Every one of the eight regions of Section 10.2 carries a positivity certificate for  $\Delta$ : two regions (region top and region 4's first piece) carry an exact, zero-margin Bernstein certificate; the remaining regions carry a box-covering certificate with an excluded margin tightened, across successive refinements, to well under one percent of the region's own area. In every case the certificate's numerator and denominator Bernstein coefficients are of a single, consistent sign throughout the region, so  $\Delta$  itself has a single, consistent sign there, matching a direct numerical spot-check at an interior point.*

We record all eight in turn, each with its representative-point vertex count (moving vertices, from the deviating cap, plus fixed vertices, confirmed unchanged across several sample

points spanning the region), its exact fixed-volume contribution (the boundary simplices not involving the deviating cap, which sum to an exact rational constant independent of  $(\varphi, s)$ ), and its certificate method.

Region top (region I). Bounded below by the Hadamard-image curve  $A$  ( $\cos s = \tan(\varphi/2)$ ) and above by  $\varphi = \pi/2$ , for the *entire* range  $s \in (0, \pi/4)$ , the only region needing no internal crossover point at all, since its two bounding curves never meet inside the reduced domain. Both boundary values of  $\varphi$  are simple in Weierstrass coordinates, giving the simplest possible remapping onto a unit square, with resulting numerator and denominator of degree at most 12 in each variable (Lemma 3.3) and all-one-sign Bernstein coefficients throughout: an exact, zero-margin certificate.

Region 2. Bounded by  $\pi/3 < \varphi <$  the Y-curve ( $\sin \varphi(\sin s + \cos s) = 1$ ) for  $s \in (0, s_1)$ , the second region to receive a full vertex/volume treatment historically. A generic point has 29 vertices (24 fixed +5 moving); of 24 fixed points, 23 belong to the same axis/sign-vertex families used throughout, and the 24th is the universal breakpoint's own eightfold-degenerate fixed point. The fixed-only boundary-simplex sum is exactly  $29/4$ . Box-covering certificate.

Region 4 (two-piece). Bounded above, for all  $s \in (0, \pi/4)$ , by the lower envelope of the W- and Y-curves, crossing exactly at  $s_2 = \arctan(\frac{1}{2})$  (equivalently  $\varphi_2 = \arccos(\frac{2}{3})$ , verified by exact symbolic reduction of both defining equations to 0 under this substitution):  $\varphi_{\max}(s) = 2 \arctan(\sin s)$  for  $s < s_2$  (piece 4a) and  $\varphi_{\max}(s) = \arcsin(1/(\sin s + \cos s))$  for  $s > s_2$  (piece 4b). Piece 4a's own ceiling curve is itself rational in Weierstrass coordinates ( $u = 2v/(1+v^2)$ ), so a further exact substitution ( $u = w \cdot 2v/(1+v^2)$ ,  $v = (\sqrt{5} - 2)v_2$ , mapping the crossover point itself to a rational corner) gives an exact unit-square remapping over  $\mathbb{Z}[\sqrt{5}]$ , with numerator and denominator of degree (10, 18) and (10, 20) respectively, an exact, zero-margin certificate, the second (after region top) of the two exact certificates on this arc. Piece 4b's ceiling is an elliptic curve (not rationally parametrisable): the plain Weierstrass-mapped numerator and denominator, degree (10, 10) with plain rational coefficients, are certified by box-covering instead, using only numerically-chosen safe box corners against the (non-algebraic, numerically-located) ceiling.

Region 6. Bounded below by the universal line and above by the W-curve, for  $s \in (s_3, \pi/4)$ . A representative point has 36 vertices (22 fixed +14 moving), the first region whose scale exposed two accuracy issues in the underlying pipeline (documented and fixed generically, not specific to this region): a 5-facet degenerate vertex, resolved by trying every 3-subset of active facets and keeping the algebraically simplest representation; and `sympy`'s default Bareiss determinant algorithm hanging indefinitely on some of this region's symbolic  $4 \times 4$  simplex matrices despite modest entry size, resolved by forcing `method='berkowitz'` throughout (every determinant then computes in well under a second). Box-covering certificate.

Region A. Bounded below by W, then (past  $s_1$ ) by Y, and above by the universal line, for  $s \in (0, s_2)$ , the first region whose own interior contains an internal breakpoint of its floor curve while remaining a single combinatorial type throughout (confirmed, not assumed, by an independent convex-hull check on each side of  $s_1$  agreeing to twelve decimal places). A representative point has 32 vertices (23 fixed +9 moving); the fixed-only sum is exactly  $65/12$ . Box-covering certificate.

Region D. A “lens” region, bounded above by the universal line and below by Y then W, pinching to zero width at *both* ends ( $s_1$  and  $s_2$ ), the first region to do so, since  $\theta_Y(s_1) = \theta_W(s_3) = \pi/3$  exactly by the very definitions of  $s_1, s_3$  is not directly relevant here, but the analogous coincidence at this region's own two ends is. A representative point has 37

vertices (22 fixed +15 moving, the largest moving-vertex count of any region so far); the fixed-only sum is exactly  $29/6$ . Box-covering certificate.

Region E. Bounded below by Y throughout and above by W then the universal line, pinching to zero width only at its left edge  $s_2$ . A representative point has 37 vertices (21 fixed +16 moving, the most moving vertices of any region in this atlas); the fixed-only sum is exactly  $17/4$ . The floor curve is the same non-rational Y-curve responsible for region 4b's elliptic obstruction. Box-covering certificate.

Region C. Bounded above by A throughout and below by Y, then the universal line, then W, the only region whose own floor switches identity *twice*, yet confirmed, by direct polytope facet computation at representative points in all three floor sub-ranges, to remain a single combinatorial type throughout: the floor's three curves meet smoothly at  $s_1$  and  $s_3$ , so (unlike regions A, D, E) this region does not pinch to zero width at either internal crossover, narrowing only near the domain's own right edge  $s = \pi/4$ , where floor and ceiling coincide by the underlying symmetry. A representative point has 34 vertices (23 fixed +11 moving); the fixed-only sum is exactly  $11/2$ . Box-covering certificate.

**10.4. What this does and does not establish.** Combined, the eight certificates of Theorem 10.1 establish  $\Delta > 0$  on the entire first arc's domain, except for the thin, explicitly quantified excluded margins recorded above (none exceeding a few tenths of one percent of their region's area, and none arising on the two exact regions). On its own this settles the sign along the first arc away from two short windows; the windows are closed in Section 8.5, and the whole sphere of directions is settled by a different route in Section 13.

## 11. BREAKPOINT CLASSIFICATION ON THE SECOND ARC

**11.1. Setup.** The second boundary arc of Lemma 8.1 is  $\eta_{w_1 v_2}(s) = \cos s w_1 + \sin s v_2$ ,  $s \in [0, \pi/4]$ , reduced to this range by an exact symmetry  $S$  that swaps  $w_1 \leftrightarrow v_2$  and fixes  $u_0, v_1$  (verified directly at the level of coordinates). We classify its breakpoints completely.

**Theorem 11.1** (Complete breakpoint classification, second arc). *Exactly four exact curves account for every active-facet-set transition of  $u_1(\varphi, s) = \cos \varphi u_0 + \sin \varphi \eta_{w_1 v_2}(s)$  across the entire domain  $s \in [0, \pi/4]$ :*

- (1) *the universal curve  $\varphi = \pi/3$ , inherited unchanged from Theorem 9.1 (it is proved once, for every choice of deviation direction, and simply restricts to this arc);*
- (2) *a curve W',  $\varphi = 2 \arctan(\cos s)$ , from the same sign vertex  $w = (1, 1, 1, 1)/\sqrt{2}$  that gave curve W on the first arc, but with a different functional dependence on  $s$ , since the same vertex plays a structurally different role on the two arcs;*
- (3) *a curve B (newly identified here),  $\varphi = 2 \arctan(\sin s)$ , from the sign vertex  $b = (1, 1, 1, -1)/\sqrt{2}$ , a vertex that plays no role at all on the first arc;*
- (4) *a curve C (newly identified here),  $\sin \varphi (\cos s + \sin s) = 1$ , from the one-axis vertex  $c = \sqrt{2} e_3$ , the same functional form as curve Y on the first arc, but arising here from a different vertex.*

*The two vertices responsible for the first arc's remaining curves,  $y = (1, -1, 1, 1)/\sqrt{2}$  and  $a = \sqrt{2} e_1$ , are confirmed here to touch nowhere in this arc's open interior.*

As on the first arc, each curve is derived by an exact symbolic solve after the Weierstrass substitution, and completeness is confirmed by a dense scan (150 values of  $s$ , 900-point  $\varphi$ -grid with bisection refinement) finding at most four breakpoints at any sampled  $s$ , every one matching one of the four curves above to a tolerance of  $2 \times 10^{-4}$ , far smaller than the curves' own separation, with zero unexplained transitions across the entire scan.

**11.2. The combinatorial atlas.** Sorting the four curves by  $\varphi$ -value locates three exact crossing points on  $s \in (0, \pi/4)$ :

$$\begin{aligned} s_1 &= 0.169918\dots \quad (\text{C} = \text{universal}), \\ s_2 &= \arctan(\tfrac{1}{2}) = 0.463648\dots \quad (\text{B} = \text{C}), \\ s_3 &= \arcsin(\tfrac{1}{\sqrt{3}}) = 0.615480\dots \quad (\text{B} = \text{universal}), \end{aligned}$$

each located both by an exact symbolic solve and confirmed numerically to eleven significant figures. These three crossings divide the domain into four  $s$ -intervals, within each of which the four curves have a fixed relative order, giving a coarse combinatorial atlas of approximately seven region types (a scope estimate, established at the same resolution as the first arc's own atlas but not yet independently re-derived by the finer vertex-pattern method that corrected the first arc's own region count from an initial nine down to eight); we treat seven as a solid working count, not a fully hardened one. The arrangement is drawn in Figure 26(b).

**11.3. Region top.** The topmost region, bounded above by  $\varphi = \pi/2$  and below by curve  $W'$ , spanning the entire  $s \in (0, \pi/4)$  domain with no internal splitting, is the natural first target for a certificate, since its lower boundary is itself rational in Weierstrass coordinates (the same functional form,  $u = (1 - v^2)/(1 + v^2)$ , as the first arc's own curve A). Section 12 gives a complete, exact, zero-margin certificate for this region.

**11.4. The band immediately below region top.** The next region down, bounded above by  $W'$  and below, depending on  $s$ , by whichever of curves C, the universal curve, or B is locally uppermost, is combinatorially a single connected region across each of the three  $s$ -sub-intervals determined by the crossings  $s_1, s_3$  above, but the identity of its *lower* boundary curve changes between them:

$$\text{lower boundary} = \begin{cases} \text{C} & s \in (0, s_1), \\ \text{universal } (\varphi = \pi/3) & s \in (s_1, s_3), \\ \text{B} & s \in (s_3, \pi/4). \end{cases}$$

This is confirmed directly: combinatorial-pattern stability is checked at eight sample points spanning the middle sub-interval  $s \in (s_1, s_3)$  and found identical at every one (34 vertices, the same active-facet pattern throughout), confirming this middle sub-part is a single connected combinatorial region on its own. Section 12 gives a direct certificate on this middle sub-part, the widest of the three by  $s$ -range. The two remaining sub-parts are handled not by a direct computation but by the identification of Proposition 8.5, which carries the first arc's certificates onto the whole of this one.

## 12. CERTIFICATES ON THE SECOND ARC

### 12.1. Region top: an exact, zero-margin certificate.

**Theorem 12.1** (Certificate, second arc, region top). *On the region bounded above by  $\varphi = \pi/2$  and below by curve  $W'$  (Theorem 11.1), for the entire domain  $s \in (0, \pi/4)$ , the volume-defect function  $\Delta$  admits an exact, zero-margin Bernstein-basis positivity certificate: after the same Weierstrass-plus-crossover remapping used on the first arc's own region top, the resulting numerator has 286 Bernstein coefficients (degree (12, 21)) and the denominator 231 (degree (10, 20)), every one of the 286 numerator coefficients strictly negative and every one of the 231 denominator coefficients strictly negative, exactly by  $\mathbb{Z}[\sqrt{2}]$  arithmetic, no floating point at any stage.*

Since numerator and denominator are of the same constant sign throughout, their ratio is strictly positive on the entire open region, with *no* excluded margin, the same standard as the strongest certificates on the first arc. The vertex and volume stage underlying this certificate finds 32 vertices (8 moving under the deviating cap, 24 fixed) at a representative point, confirmed unchanged across five sample points spanning nearly the full  $s$ -range; the exact fixed-volume contribution is  $77/12$ , and the closed-form moving volume is checked against a direct convex-hull volume computation to eleven significant figures at the representative point before any symbolic remapping is attempted. An independent sanity check evaluates the volume-defect quantity numerically at 55 points scattered across the full  $(s, \varphi)$ -domain: all 55 strictly positive, with zero exceptions, fully consistent with the certificate.

**12.2. The band below region top: a box-covering attempt.** For the middle sub-part of the band described in Section 11.4, bounded above by  $W'$  and below by the universal curve  $\varphi = \pi/3$ , for  $s$  in the sub-interval  $(s_1, s_3) = (0.169918, 0.615480)$ , an exact zero-margin certificate of the type used in Section 12.1 is not attempted, because the lower boundary here is the constant curve  $\varphi = \pi/3$ : mapping between it and the curved upper boundary  $W'$  would introduce  $\sqrt{3}$  into the arithmetic on top of the  $\sqrt{2}$  already present, a complication this paper's exact-sign machinery (built for  $\mathbb{Q}(\sqrt{2})$ ) does not handle. We instead attempt the same box-covering fallback already used for the harder regions of the first arc.

**Theorem 12.2** (Certificate, second arc, band below region top). *On the middle sub-part described above, the volume-defect numerator and denominator, mapped to plain Weierstrass coordinates  $(u, v)$  (degree (12, 12) and (10, 11) respectively, exact  $\mathbb{Z}[\sqrt{2}]$  coefficients throughout, each mapping step verified against a 200-digit numerical evaluation of the original trigonometric expression), are certified on all 40 boxes covering the sub-interval: every one of the 169 Bernstein coefficients of  $N$ , and every one of the 169 Bernstein coefficients of  $D$ , on every box, is non-negative (in fact, on this run, strictly positive with none exactly zero). By the same same-sign argument used throughout this paper, this proves  $\Delta > 0$  on the entire covered region, modulo the excluded margin recorded below.*

This is a second region-by-region result on the second arc, on top of region top (Theorem 12.1): the excluded margin is a strip of about 1.7% of the sub-interval's own  $s$ -range at its two ends (from the safety factor used when choosing box corners near  $s_1$  and  $s_3$ ), together with a fixed 0.1–0.2% safety strip against each box's own numerically-located floor ( $\varphi = \pi/3$ ) and ceiling ( $W'$ ), wider than the tightest margins achieved elsewhere in this paper (Section 10's own regions have been refined to under 0.2%), reflecting that this certificate was run once, at a moderate resolution, and not iteratively tightened; we expect, by direct analogy with the earlier region-4b tightening reported in Section 10, that this margin is similarly reducible at the cost of runtime, but we have not done so here and do not claim a tighter margin than the one actually certified.

**12.3. The band's C-bounded sub-part.** The first of the two remaining sub-parts of the band described in Section 11.4, bounded above by  $W'$  and below by curve  $C$  (Theorem 11.1), for  $s \in (0, s_1) = (0, 0.169918)$ , is combinatorially stable (34 vertices, 11 moving under the deviating cap and 23 fixed, identical at eight sample points spanning the sub-interval) and admits a certificate by exactly the same box-covering method used for the sub-part of Theorem 12.2, adapted only by replacing the constant lower boundary  $\varphi = \pi/3$  with the curve  $C$  (numerically evaluated, rather than mapped exactly, when placing each box's lower corner, box-covering does not require the boundary to be rational in the working coordinates).

**Theorem 12.3** (Certificate, second arc, band C-sub-part). *On the sub-part described above, the exact fixed-volume contribution is  $17/3$ ; the moving-volume numerator and denominator, mapped to plain Weierstrass coordinates, have degree  $(12, 12)$  and  $(10, 11)$  respectively (the same degrees as Theorem 12.2's own middle sub-part, exact  $\mathbb{Z}[\sqrt{2}]$  coefficients throughout, each mapping step verified against a 200-digit numerical evaluation of the original trigonometric expression). All 40 boxes covering the sub-interval are certified: every one of the 169 Bernstein coefficients of  $N$ , and every one of the 169 Bernstein coefficients of  $D$ , on every box, is strictly positive, with none exactly zero.  $\Delta > 0$  on the entire covered sub-part, modulo the excluded margin recorded below.*

The excluded margin here is wider than Theorem 12.2's own: approximately 4% of the sub-interval's own  $s$ -range at its two ends (a wider safety factor was used near  $s = 0$  and near  $s_1$ , where curve C's numerical evaluation is closer to degenerate), together with the same 0.1–0.2% safety strip against each box's own floor and ceiling. As with Theorem 12.2, we expect this margin is reducible with a finer grid but have not pursued that here, and report the certificate exactly at the resolution actually run.

**12.4. The band's B-bounded sub-part.** One sub-part of the band, bounded above by  $W'$  and below by curve B for  $s \in (s_3, \pi/4)$ , does not yield to the same method. Its combinatorial data behave exactly as elsewhere: the representative point has 34 vertices, 11 moving and 23 fixed, and the pattern is identical at eight sample points spanning the sub-interval, with fixed volume contribution 6. The moving-volume step is where it stops. One of the 74 moving boundary simplices has a symbolic determinant whose exact simplification over the relevant algebraic extension grows without settling, and we did not obtain a closed form for the volume there. The certificate for this sub-part therefore does not come from the direct computation.

It does not need to. Section 8.5 shows that the second arc, this sub-part included, is the first arc's own defect function in disguise, so the certificates of Section 10 already cover it. The three direct certificates recorded above remain useful as an independent check of that identification on part of the domain, computed by a route that shares nothing with it.

**12.5. Coverage of the boundary.** Figure 27(a) shows the octant, and Figure 27(b) the certificate carried by each region of the first arc.

Read directly, the region-by-region computations of this section settle three of the second arc's units and leave the rest open. Read through Proposition 8.5, they are not needed at all: the second arc and the third edge both carry the first arc's certificates, at the first arc's margins. The interior of the octant is a separate matter, settled by a different argument in Section 13.

### 13. POSITIVITY FOR EVERY DEVIATION DIRECTION

The certificates of Sections 10 and 12 are exact, but they are local: each one governs a single combinatorial region of a single one-dimensional arc. This section proves the positivity of  $\Delta(\varphi, \eta)$  at every point of the full two-sphere of deviation directions and at every tilt, by an argument of a different kind. The defect is first rewritten as the deficiency of a cap cut from one fixed polytope; that polytope is then enclosed in a cross-polytope, for which the extremal cap can be determined outright.

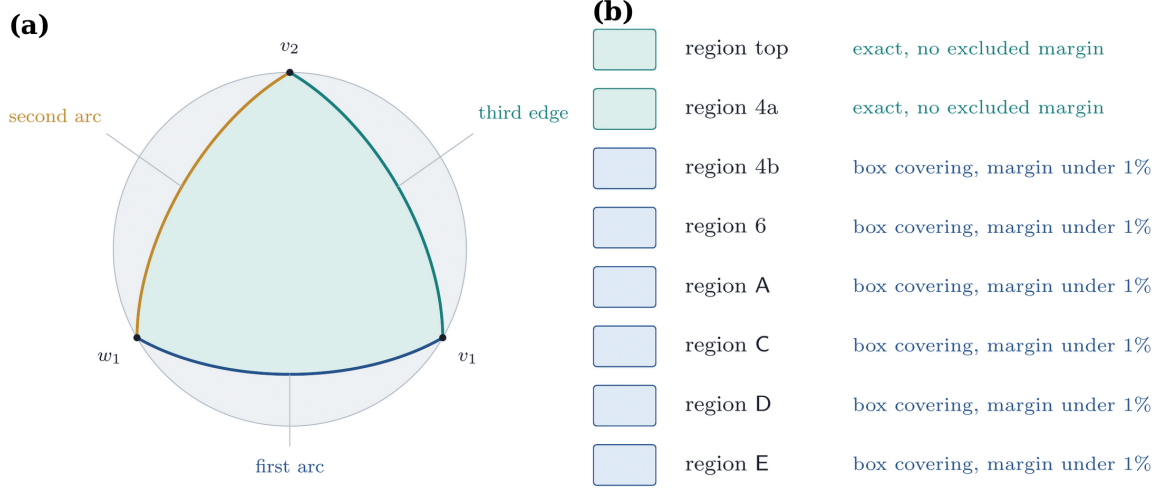


FIGURE 27. (a) The transverse sphere of deviation directions and the spherical octant left by the two symmetries of Lemma 8.1. The vertices  $v_1, w_1, v_2$  are pairwise orthogonal, so the octant's three edges are quarter great circles, and  $\langle S, H \rangle \cong S_3$  fixes  $u_0$  and permutes them; a certificate on the first arc is therefore a certificate on the whole boundary of the octant. (b) The eight combinatorial regions of the first arc and the kind of certificate each carries. Two are exact with no excluded margin; the other six are certified by box covering, with the excluded margin in each case a few tenths of one percent of the region's own area. Through the identifications of Propositions 8.3 and 8.5, these eight certificates cover the whole boundary of the octant.

**13.1. The defect as a cap deficiency.** Throughout,  $u_0 = \alpha_0/\sqrt{2}$  is the undeviated contact direction,  $u_1 = \cos \varphi u_0 + \sin \varphi \eta$  the deviated one, and

$$Q = \bigcap_{i \neq 0} \{x \in \mathbb{R}^4 : \langle x, u_i \rangle \leq 1\}$$

the intersection of the half-spaces of the twenty-three contact directions that stay at roots. The local cell at the all-contact corner is  $Q \cap \{\langle x, u_1 \rangle \leq 1\}$ , so

$$\Delta(\varphi, \eta) = \text{vol}(Q \cap \{\langle x, u_1 \rangle \leq 1\}) - 8. \quad (13.1)$$

Write  $p = \langle x, u_0 \rangle$ , and let  $N$  denote the eight roots making an angle of  $60^\circ$  with  $\alpha_0$ , so that  $\langle u_0, u_i \rangle = \frac{1}{2}$  for  $i \in N$ .

**Lemma 13.1** (Structure of  $Q$ ). *Let  $F \subset u_0^\perp$  be the octahedral facet of  $\mathcal{V}$  at  $u_0$ , translated to the origin, so that the facet is  $u_0 + F$ . Then  $F$  is a regular octahedron whose six vertices are  $\pm v_1, \pm v_2, \pm v_3$ , where  $v_1, v_2, v_3$  are three pairwise orthogonal contact directions orthogonal to  $u_0$ ; in particular  $(u_0, v_1, v_2, v_3)$  is an orthonormal basis of  $\mathbb{R}^4$ ,  $\text{vol}_3(F) = \frac{4}{3}$ , and*

$$F = \left\{ y \in u_0^\perp : \sum_i |\langle y, v_i \rangle| \leq 1 \right\}.$$

Moreover  $Q \cap \{p \geq 1\}$  is the pyramid  $\text{Pyr}$  with apex  $2u_0$  and base  $u_0 + F$ , of volume  $\frac{1}{3}$ , while  $Q \cap \{p \leq 1\} = \mathcal{V}$ . Hence  $\text{vol}(Q) = 8 + \frac{1}{3} = \frac{25}{3}$ .

*Proof.* Six roots are orthogonal to  $\alpha_0$ , and they fall into three orthogonal pairs  $\pm \alpha_{i_1}, \pm \alpha_{i_2}, \pm \alpha_{i_3}$ ; put  $v_j = \alpha_{i_j}/\sqrt{2}$ . Each  $u_0 \pm v_j$  satisfies  $\langle u_0 \pm v_j, u_i \rangle \leq 1$  for every root

and meets six of those constraints with equality, so the six points  $u_0 \pm v_j$  are vertices of  $\mathcal{V}$  lying on the facet at  $u_0$ ; since a facet of the 24-cell is a regular octahedron with six vertices, these are all of them. A regular octahedron with vertices  $\pm v_j$  in an orthonormal frame is  $\{y : \sum_j |\langle y, v_j \rangle| \leq 1\}$  and has volume  $\frac{2^3}{3!} \cdot 2 = \frac{4}{3}$ .

For  $i \in N$  set  $m_i = u_i - \frac{1}{2}u_0$ , so  $m_i \perp u_0$  and  $|m_i| = \frac{\sqrt{3}}{2}$ . Writing  $x = pu_0 + y$  with  $y \perp u_0$ , the constraint  $\langle x, u_i \rangle \leq 1$  reads  $\langle y, m_i \rangle \leq 1 - \frac{p}{2}$ . At  $p = 1$  these eight constraints cut out exactly  $F$ , so at height  $p$  they cut out  $(2 - p)F$ ; that is,

$$\bigcap_{i \in N} \{\langle x, u_i \rangle \leq 1\} = \{pu_0 + y : y \in (2 - p)F\}. \quad (13.2)$$

Let  $1 \leq p \leq 2$  and  $y \in (2 - p)F \subseteq F$ . The other fifteen constraints are then slack: for the root  $-\alpha_0$  one has  $\langle x, -u_0 \rangle = -p \leq -1$ ; for a root  $\alpha_j$  orthogonal to  $\alpha_0$ ,  $\langle x, u_j \rangle = \langle y, \pm v_k \rangle \leq 1$  because  $\max_{y \in F} \langle y, v_k \rangle = 1$ ; and for a root  $\alpha_j$  with  $\langle u_0, u_j \rangle = -\frac{1}{2}$ , necessarily  $u_j = -u_i$  for some  $i \in N$ , whence  $\langle x, u_j \rangle = -\frac{p}{2} + \langle y, -m_i \rangle \leq -\frac{1}{2} + \frac{1}{2} = 0$ . So on  $\{p \geq 1\}$  the set  $Q$  agrees with Equation (13.2), which there is the cone over  $u_0 + F$  with apex  $2u_0$ , of volume  $\int_1^2 \text{vol}_3((2 - p)F) dp = \frac{4}{3} \int_0^1 \lambda^3 d\lambda = \frac{1}{3}$ . Finally  $\mathcal{V} \subseteq \{p \leq 1\}$ , and below that hyperplane  $Q$  and  $\mathcal{V}$  are cut out by the same constraints apart from the one at  $\alpha_0$ , which is inactive there.  $\blacksquare$

**Proposition 13.2** (Cap form of the defect). *For a unit vector  $u$  put  $\text{Cap}(u) = \{x \in Q : \langle x, u \rangle \geq 1\}$ . Then*

$$\Delta(\varphi, \eta) = \frac{1}{3} - \text{vol}(\text{Cap}(u_1)).$$

*Proof.* By Lemma 13.1,  $\text{vol}(Q) = \frac{25}{3}$ , and Equation (13.1) gives  $\Delta = \text{vol}(Q) - \text{vol}(\text{Cap}(u_1)) - 8$ .  $\blacksquare$

The deviation parameters have now left the body and entered the cutting half-space:  $Q$  is one fixed polytope with 25 vertices, and the question is how much of it a half-space at unit distance from the origin can cut off. At  $u_1 = u_0$  the answer is  $\frac{1}{3}$ , the pyramid.

**13.2. A cross-polytope around  $Q$ .** Let  $(u_0, v_1, v_2, v_3)$  be the orthonormal frame of Lemma 13.1 and write  $x_0 = \langle x, u_0 \rangle$ ,  $x_j = \langle x, v_j \rangle$  for the coordinates in that frame.

**Lemma 13.3** (Enclosure).  *$Q \subseteq B$ , where*

$$B = \{x \in \mathbb{R}^4 : |x_0| + |x_1| + |x_2| + |x_3| \leq 2\}$$

*is the cross-polytope of radius 2 in the frame  $(u_0, v_1, v_2, v_3)$ .*

*Proof.* Sixteen roots make an angle of  $60^\circ$  or  $120^\circ$  with  $\alpha_0$ , and they come in eight antipodal pairs. If  $\alpha_i$  is one of the eight at  $60^\circ$  then  $u_i = \frac{1}{2}u_0 + \frac{\sqrt{3}}{2}n_i$  with  $n_i$  a unit vector orthogonal to  $u_0$ ; since the eight vectors  $n_i$  are the outer normals of the eight faces of the regular octahedron  $F$ , they must be  $(\pm v_1 \pm v_2 \pm v_3)/\sqrt{3}$ , one for each choice of signs. Hence

$$\langle x, u_i \rangle = \frac{1}{2}(x_0 \pm x_1 \pm x_2 \pm x_3),$$

and the eight constraints  $\langle x, u_i \rangle \leq 1$  say exactly that  $x_0 \pm x_1 \pm x_2 \pm x_3 \leq 2$  for all eight sign patterns, that is,  $x_0 + |x_1| + |x_2| + |x_3| \leq 2$ . The eight roots at  $120^\circ$  are the negatives of these, and give  $-x_0 + |x_1| + |x_2| + |x_3| \leq 2$ . All sixteen roots are distinct from  $\alpha_0$ , so all sixteen constraints are among those defining  $Q$ .  $\blacksquare$

The enclosure is efficient where it matters:  $\text{vol}(B) = \frac{32}{3}$  against  $\text{vol}(Q) = \frac{25}{3}$ , and, more to the point, the half-space  $\{x_0 \geq 1\}$  cuts from  $B$  the cone over the octahedron  $\{\sum_{j \geq 1} |x_j| \leq 1\}$  with apex  $2u_0$ , of volume exactly  $\frac{1}{3}$ . On the critical direction the enclosure is therefore not lossy at all. Figure 28(a) shows the three bodies in section.

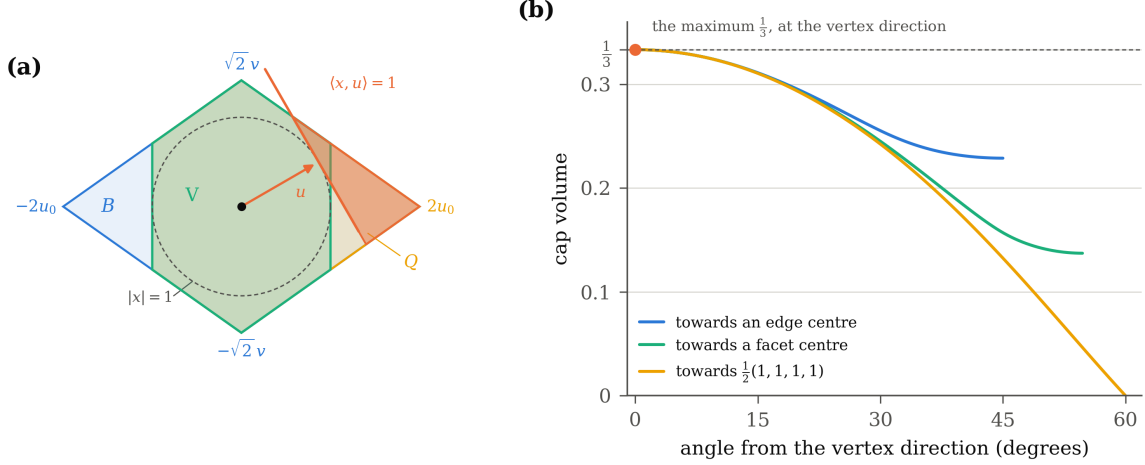


FIGURE 28. (a) The section of the reference cell  $\mathcal{V}$ , of  $Q$ , and of the cross-polytope  $B$  by the plane spanned by the base direction  $u_0$  and one transverse direction. Coordinates are exact: the section of  $\mathcal{V}$  is a hexagon, that of  $Q$  the same hexagon with its right edge replaced by the two edges meeting at the apex  $2u_0$ , and that of  $B$  a rhombus through the same apex and the same two vertices  $\pm\sqrt{2}v$ . The dashed circle is the unit sphere, on which every contact hyperplane is tangent, and the shaded triangle is the section of the cap cut by a half-space at unit distance whose normal is tilted  $30^\circ$  from  $u_0$ . (b) The volume of the cap cut from the cross-polytope by a half-space at unit distance, along three great circles leaving a vertex direction: towards an edge centre, towards a facet centre, and towards  $\frac{1}{2}(1, 1, 1, 1)$ , where the half-space meets  $B$  only in its four positive vertices and the cap volume falls to zero. The maximum  $\frac{1}{3}$  of Theorem 13.6 is attained at the vertex direction, marked.

**13.3. The extremal cap of the cross-polytope.** We now determine the largest volume that a half-space at unit distance from the centre can cut from  $B$ . Write  $B_1^4 = \{\|x\|_1 \leq 1\}$ , so  $B = 2B_1^4$ , and let  $e_0, e_1, e_2, e_3$  be the frame vectors.

**Lemma 13.4** (Cap formula). *Let  $u$  be a unit vector with coordinates  $u_k$  in the frame, and set  $a_k = u_k^2$ , so that  $a_0 + a_1 + a_2 + a_3 = 1$ . Define*

$$g(a) = \begin{cases} a(2\sqrt{a} - 1)^4, & a \geq \frac{1}{4}, \\ 0, & a \leq \frac{1}{4}. \end{cases}$$

Then

$$\text{vol}(B \cap \{\langle x, u \rangle \geq 1\}) = \frac{1}{3} g[a_0, a_1, a_2, a_3],$$

where  $g[\cdot]$  denotes the third-order divided difference of  $g$  at the four nodes.<sup>18</sup>

<sup>18</sup>We use nothing about divided differences beyond the following, so we record it here instead of sending the reader elsewhere. For a function  $h$  and pairwise distinct nodes  $y_0, \dots, y_n$ , the divided difference is defined by  $h[y_0] = h(y_0)$  and

$$h[y_0, \dots, y_n] = \frac{h[y_1, \dots, y_n] - h[y_0, \dots, y_{n-1}]}{y_n - y_0},$$

and unwinding the recursion gives the closed form  $h[y_0, \dots, y_n] = \sum_i h(y_i) / \prod_{j \neq i} (y_i - y_j)$ , from which it is visibly a symmetric function of the nodes. Three consequences are used below. First, if  $h$  is  $C^n$  then  $h[y_0, \dots, y_n] = h^{(n)}(\xi)/n!$  for some  $\xi$  in the smallest interval containing the nodes; in particular the divided

*Proof.* Both sides are unchanged by a signed permutation of the coordinates, so we may assume  $u_k \geq 0$  for all  $k$ . Split  $B$  into the sixteen simplices  $S_\sigma = \text{conv}\{0, 2\sigma_0 e_0, \dots, 2\sigma_3 e_3\}$ ,  $\sigma \in \{\pm 1\}^4$ , each of volume  $\frac{2^4}{4!} = \frac{2}{3}$ . For a 4-simplex  $\Delta$  with vertices  $p_0, \dots, p_4$  at which a linear form  $f$  takes distinct values  $f_0, \dots, f_4$ , one has

$$\text{vol}(\Delta \cap \{f \geq s\}) = \text{vol}(\Delta) \sum_{i=0}^4 \frac{(f_i - s)_+^4}{\prod_{j \neq i} (f_i - f_j)}. \quad (13.3)$$

<sup>19</sup> On  $S_\sigma$  the form  $\langle \cdot, u \rangle$  takes the value 0 at the origin and  $2\sigma_k u_k$  at  $2\sigma_k e_k$ . With  $s = 1$  the vertex 0 contributes nothing, and a vertex  $2\sigma_k e_k$  contributes only when  $\sigma_k = +1$  and  $2u_k > 1$ . Summing the contribution of the  $k$ -th vertex over the eight sign choices of the remaining coordinates gives

$$\sum_{\sigma_l = \pm 1, l \neq k} \prod_{l \neq k} \frac{1}{2u_k - 2\sigma_l u_l} = \prod_{l \neq k} \left( \frac{1}{2u_k - 2u_l} + \frac{1}{2u_k + 2u_l} \right) = \prod_{l \neq k} \frac{u_k}{u_k^2 - u_l^2},$$

so that

$$\text{vol}(B \cap \{\langle x, u \rangle \geq 1\}) = \frac{2}{3} \sum_k \frac{(2u_k - 1)_+^4}{2u_k} \prod_{l \neq k} \frac{u_k}{u_k^2 - u_l^2} = \frac{1}{3} \sum_k \frac{(2u_k - 1)_+^4 u_k^2}{\prod_{l \neq k} (u_k^2 - u_l^2)}.$$

The  $k$ -th summand is  $g(a_k)/\prod_{l \neq k} (a_k - a_l)$ , and the sum is the divided difference. Coincident nodes are handled by continuity, both sides being continuous in  $u$ .  $\blacksquare$

The shape of the answer is not an accident. A divided difference of a truncated power is the classical representation of a B-spline, and the volume of a slice of a simplex is exactly such a quantity; the identification goes back to Curry and Schoenberg [CS66], and it is what makes the next lemma available.

Two properties of  $g$  drive everything that follows. Writing  $t = \sqrt{a}$  for  $a \geq \frac{1}{4}$ , repeated differentiation gives

$$\begin{aligned} g' &= (2t - 1)^3(6t - 1), & g'' &= \frac{6(2t - 1)^2(4t - 1)}{t}, \\ g''' &= 96 - \frac{60}{t} + \frac{3}{t^3}, \end{aligned} \quad (13.4)$$

$$\text{and } g'''' = \frac{3(20t^2 - 3)}{2t^5}.$$

difference extends continuously to coincident nodes, with  $h[y, \dots, y] = h^{(n)}(y)/n!$ , so all the formulas below may be read at coincident nodes by continuity. Second, if  $h^{(n-1)}$  is nondecreasing then  $h[y_0, \dots, y_n] \geq 0$  for all nodes, immediate by the mean-value form where  $h$  is  $C^n$  and in general by uniform approximation, which is the case we need, our  $g$  being  $C^3$  with  $g'''$  nondecreasing but not  $C^4$  at  $a = \frac{1}{4}$ . Third, adding a node differentiates:  $\partial h[y_0, \dots, y_n]/\partial y_i = h[y_0, \dots, y_n, y_i]$ . These three facts turn a statement about the fourth derivative of one variable into a monotonicity statement in four.

<sup>19</sup>Equation (13.3) is the only fact about simplices the proof uses, and it has a short derivation worth recording. Put  $\Delta = \{\lambda \in \mathbb{R}^5 : \lambda_i \geq 0, \sum \lambda_i = 1\}$  in barycentric coordinates, so that  $f = \sum_i f_i \lambda_i$  and the uniform probability measure on  $\Delta$  pushes forward under  $f$  to the distribution of  $\sum_i f_i \lambda_i$  for  $\lambda$  uniform on the simplex. The coordinates  $\lambda_i$  are the normalised spacings of four independent uniform points on an interval, and the density of  $\sum f_i \lambda_i$  is  $4!$  times the third-order divided difference of  $t \mapsto (t - x)_+^3/3!$  in the nodes  $f_i$ , which is the Curry–Schoenberg B-spline with those knots [CS66]. Integrating that density from  $s$  upwards raises the truncated power by one degree and produces exactly the displayed sum. Equivalently, and with no probability at all: both sides of Equation (13.3) are, as functions of  $s$ , piecewise polynomials of degree 4 that vanish for  $s \geq \max_i f_i$  and agree with  $\text{vol}(\Delta)$  for  $s \leq \min_i f_i$ , with matching derivatives at every breakpoint, so they agree identically. For  $d$ -simplices the exponent 4 is replaced by  $d$  throughout.

**Lemma 13.5** (Monotonicity).  *$g$  is of class  $C^3$  on  $[0, \infty)$ , its third derivative is nondecreasing, and consequently  $g[a_0, a_1, a_2, a_3]$  is nondecreasing in each node.*

*Proof.* All three expressions in Equation (13.4) vanish at  $t = \frac{1}{2}$ , so  $g, g', g'', g'''$  all vanish at  $a = \frac{1}{4}$  and match the identically zero branch; hence  $g \in C^3$ . For  $t \geq \frac{1}{2}$  one has  $20t^2 - 3 \geq 2 > 0$ , so  $g''' > 0$  there, while  $g''' \equiv 0$  on  $[0, \frac{1}{4}]$ ; therefore  $g'''$  is nondecreasing on  $[0, \infty)$ . A function whose  $(n-1)$ st derivative is nondecreasing has all  $n$ th-order divided differences nonnegative, so  $g[y_0, \dots, y_4] \geq 0$  for any five nodes.<sup>20</sup> Since

$$\frac{\partial}{\partial a_i} g[a_0, a_1, a_2, a_3] = g[a_0, a_1, a_2, a_3, a_i],$$

the divided difference is nondecreasing in each node. ■

**Theorem 13.6** (Extremal cap of the cross-polytope). *For every unit vector  $u \in \mathbb{R}^4$ ,*

$$\text{vol}(B \cap \{\langle x, u \rangle \geq 1\}) \leq \frac{1}{3},$$

*with equality if and only if  $u = \pm e_k$  for some  $k$ .*

Figure 28(b) shows the cap volume along three great circles leaving a vertex direction, computed from the enumerated cell and independent of the argument below.

*Proof.* By Lemma 13.4 the claim is that  $F(a) := g[a_0, a_1, a_2, a_3] \leq 1$  on the simplex  $a_k \geq 0$ ,  $\sum_k a_k = 1$ , with equality only at the vertices. Let  $S = \{k : a_k > \frac{1}{4}\}$ ; since the nodes sum to 1,  $|S| \leq 3$ . We may relabel so that  $a_0 \geq a_1 \geq a_2 \geq a_3$ .

*Case  $|S| = 0$ .* All four nodes lie in  $[0, \frac{1}{4}]$ , where  $g$  vanishes identically together with all its derivatives, so  $F(a) = 0$ .

*Case  $|S| = 1$ .* Only  $a_0$  exceeds  $\frac{1}{4}$ , and again  $g$  vanishes to infinite order at the other three nodes, so the Newton form of the divided difference collapses to a single term:

$$F(a) = \frac{g(a_0)}{(a_0 - a_1)(a_0 - a_2)(a_0 - a_3)}.$$

Write  $c = \sqrt{a_0} \in (\frac{1}{2}, 1]$ . The nodes  $a_1, a_2, a_3$  range over the polytope  $P_c = \{\beta \in [0, \frac{1}{4}]^3 : \beta_1 + \beta_2 + \beta_3 = 1 - c^2\}$ , and  $\beta \mapsto \sum_l \log(a_0 - \beta_l)$  is concave, so  $\prod_l (a_0 - \beta_l)$  attains its minimum over  $P_c$  at a vertex of  $P_c$ , that is, at a point where at most one coordinate lies strictly between 0 and  $\frac{1}{4}$ . Three ranges of  $c$  occur.

If  $c \geq \frac{\sqrt{3}}{2}$  the extreme point is  $(1 - c^2, 0, 0)$  and the required inequality  $g(a_0) \leq \prod_l (a_0 - \beta_l)$  reads  $(2c - 1)^4 \leq c^2(2c^2 - 1)$ , that is,

$$(1 - c)(14c^3 - 18c^2 + 7c - 1) \geq 0.$$

The cubic has a single real root, at  $c = 0.7410\dots < \frac{\sqrt{3}}{2}$ , and is positive beyond it, so the product is nonnegative on  $[\frac{\sqrt{3}}{2}, 1]$  and vanishes only at  $c = 1$ .

<sup>20</sup>The stronger conclusion one would like here is that  $g[a_0, a_1, a_2, a_3]$  is Schur-convex in  $(a_0, \dots, a_3)$ , which would reduce Theorem 13.6 to evaluating  $F$  at the majorisation maximum  $(1, 0, 0, 0)$  in one step; see [MOA11, Ch. 3] for the criterion. It is not available. Schur-convexity of a symmetric divided difference of this form would follow from  $g^{(5)} \geq 0$ , and  $g^{(5)}(t^2) = \frac{45}{4}t^{-7}(1 - 4t^2)$  is negative at every  $t > \frac{1}{2}$ , which is the whole interior of the support. The one place the fifth derivative is positive is the point mass at  $a = \frac{1}{4}$ , where  $g'''$  jumps from 0 to 96. Monotonicity in each node separately, which needs only  $g''' \geq 0$ , is therefore the strongest tool of this kind that the function actually supports, and it is why the proof below splits into cases instead of reducing to a single extremal configuration.

If  $\frac{1}{\sqrt{2}} \leq c \leq \frac{\sqrt{3}}{2}$  the extreme point is  $(\frac{1}{4}, \frac{3}{4} - c^2, 0)$  and the inequality reads  $(2c - 1)^4 \leq (c^2 - \frac{1}{4})(2c^2 - \frac{3}{4})$ , that is,  $\frac{1}{16}(2c - 1)q(c) \geq 0$  with

$$q(c) = -112c^3 + 200c^2 - 102c + 13.$$

Here  $q'(c) = -2(168c^2 - 200c + 51)$  has its roots at  $\frac{50 \pm \sqrt{358}}{84}$ , of which only  $\frac{50 + \sqrt{358}}{84} = 0.8205 \dots$  lies in  $[\frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}]$ . So  $q$  rises and then falls on that interval and attains its minimum at an endpoint, where  $q(\frac{1}{\sqrt{2}}) = 113 - 79\sqrt{2}$  and  $q(\frac{\sqrt{3}}{2}) = 163 - 93\sqrt{3}$ . Both are positive, since  $113^2 = 12769 > 12482 = 2 \cdot 79^2$  and  $163^2 = 26569 > 25947 = 3 \cdot 93^2$ .

If  $\frac{1}{2} \leq c \leq \frac{1}{\sqrt{2}}$  the extreme point is  $(\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - c^2)$ , the required inequality becomes  $16(c - \frac{1}{2})^4 c^2 \leq 2(c - \frac{1}{2})^3(c + \frac{1}{2})^3$ , that is,  $8(c - \frac{1}{2})c^2 \leq (c + \frac{1}{2})^3$ , that is,  $r(c) \geq 0$ , where

$$r(c) = 8\left((c + \frac{1}{2})^3 - 8(c - \frac{1}{2})c^2\right) = -56c^3 + 44c^2 + 6c + 1.$$

Again  $r'(c) = -168c^2 + 88c + 6$  has a single root,  $\frac{22 + \sqrt{736}}{84} = 0.5849 \dots$ , in  $[\frac{1}{2}, \frac{1}{\sqrt{2}}]$ , so  $r$  attains its minimum on that interval at an endpoint;  $r(\frac{1}{2}) = 8$  and  $r(\frac{1}{\sqrt{2}}) = 23 - 11\sqrt{2} > 0$ , the latter because  $23^2 = 529 > 242 = 2 \cdot 11^2$ .

In all three ranges  $F(a) \leq 1$ , with equality only when  $c = 1$ , that is  $a = (1, 0, 0, 0)$  and  $u = \pm e_0$ .

*Case  $|S| \geq 2$ .* Here  $a_0 \geq a_1 \geq \frac{1}{4}$ . By Lemma 13.5,  $F$  is nondecreasing in each node, so on any box  $\prod_k [l_k, h_k]$  one has  $F(a) \leq F(h)$ . The region

$$R = \left\{a : a_0 \geq a_1 \geq a_2 \geq a_3 \geq 0, \sum_k a_k = 1, a_1 \geq \frac{1}{4}\right\}$$

is covered by 303 boxes, obtained by bisecting a uniform initial grid of 216 cubes of side  $\frac{1}{8}$  and refining only where needed, and on each box the value of  $F$  at the upper corner is a rational number at most 1; the largest of the 303 corner values is  $0.99755 \dots$ . The nodes at a corner are taken at rational points  $t_k$  with  $t_k^2$  at least the corner coordinate, and ties are broken upward, both of which only increase the bound. The computation is exact rational arithmetic throughout, and is reproduced by `cap_inequality_certificate.py` in the supplement. Since  $\sup_R F = 0.76583 \dots$ , attained at  $(\frac{3}{4}, \frac{1}{4}, 0, 0)$ , the certificate has a margin of about a quarter and does not depend on the fineness of the grid beyond what is stated.  $\blacksquare$

### 13.4. The single-deviation theorem.

**Theorem 13.7** (Direction-of-deviation positivity). *For every unit deviation direction  $\eta$  orthogonal to  $u_0$  and every tilt  $\varphi \in [0, \pi]$ ,*

$$\Delta(\varphi, \eta) \geq 0,$$

*with equality if and only if  $\varphi = 0$ .*

*Proof.* By Lemma 13.3,  $\text{Cap}(u_1) \subseteq B \cap \{\langle x, u_1 \rangle \geq 1\}$ , so Theorem 13.6 gives  $\text{vol}(\text{Cap}(u_1)) \leq \frac{1}{3}$ , and Proposition 13.2 turns this into  $\Delta \geq 0$ .

For the equality case, suppose  $\Delta(\varphi, \eta) = 0$ , so  $\text{vol}(\text{Cap}(u_1)) = \frac{1}{3}$ . Then the cap of  $B$  also has volume  $\frac{1}{3}$ , so  $u_1 = \pm e_k$  by Theorem 13.6. If  $u_1 = -u_0$  then  $\text{Cap}(u_1) = \{x \in Q : \langle x, u_0 \rangle \leq -1\}$ , which is empty because  $-a_0$  is a root and  $Q$  therefore lies in  $\{\langle x, u_0 \rangle \geq -1\}$ ; if  $u_1 = \pm v_j$  then  $\pm v_j$  is itself one of the contact directions defining  $Q$ , so again the cap is empty. In each of these cases  $\Delta = \frac{1}{3}$ , not 0. The remaining possibility is  $u_1 = u_0$ , that is  $\varphi = 0$ , where the cap is the pyramid of Lemma 13.1 and  $\Delta = 0$ .  $\blacksquare$

Theorem 1.1 can now be proved in the form stated in the introduction.

**Corollary 13.8** (Single-deviation cell bound). *Let  $c$  be a centre of a unit-ball packing of  $\mathbb{R}^4$  at which at most one active neighbour direction fails to be a root direction of  $R$ , with nothing assumed about that neighbour's deviation direction or about the size of its deviation. Then  $\text{vol}(V_c) \geq 8$ , with equality if and only if the active neighbours of  $c$  form a copy of  $\sqrt{2}R$  centred at  $c$ .*

*Proof.* By Sections 2.4 and 2.6 the volume is minimised at the all-contact corner, and by Theorem 7.1 the deficit there is  $\Delta(\varphi, \eta)$  for the appropriate parameters, which is nonnegative by Theorem 13.7. If the active set is a proper subset of  $R \setminus \{\alpha_0\}$ , the cell contains  $Q \cap \{\langle x, u_1 \rangle \leq 1\}$  and the same bound applies a fortiori. Equality forces  $\varphi = 0$  and a full active set.  $\blacksquare$

Two features of the argument deserve comment. First, it is insensitive to which combinatorial cell of Section 3 the parameters fall in: all 176 of them are covered at once, because the cap inequality is a statement about a single polytope and makes no reference to the combinatorial type of the perturbed cell. Second, the enclosure  $Q \subseteq B$  discards a seventh of the volume of  $Q$  and yet loses nothing at the critical direction, which is what makes the estimate sharp; the same argument applied to the 24-cell in place of  $Q$ , or to a smaller enclosing body, would not have this property.

**13.5. A quantitative form near the base direction.** Theorem 13.7 gives a sign and no more. For small and moderate tilts one can say how large the defect is, by replacing  $Q$  with a cone in place of a cross-polytope. The eight roots at  $60^\circ$  alone cut out

$$K = \{x : x_0 + |x_1| + |x_2| + |x_3| \leq 2\},$$

a cone with apex  $2u_0$ , and  $Q \subseteq B \subseteq K$ . The cap of a cone is a union of simplices, so it has a closed form.

**Proposition 13.9** (Cone bound). *Let  $\eta$  be a unit vector orthogonal to  $u_0$ , with coordinates  $\eta_j = \langle \eta, v_j \rangle$  in the frame of Lemma 13.1, and write  $c = \cos \varphi$ ,  $\sigma = \sin \varphi$ . If  $\sigma \max_j |\eta_j| < c$ , then*

$$\Delta(\varphi, \eta) \geq \frac{1}{3} - \frac{c^2 (2c - 1)_+^4}{3 \prod_{j=1}^3 (c^2 - \sigma^2 \eta_j^2)}. \quad (13.5)$$

*The right-hand side equals  $\varphi^2/3 + O(\varphi^4)$  as  $\varphi \rightarrow 0$ , and it is strictly positive for every such  $\eta$  if and only if  $c > c_0$ , where  $c_0 = 0.7410861\dots$  is the unique real root of  $14c^3 - 18c^2 + 7c - 1$ ; that is, for  $\varphi < 42.17598\dots^\circ$ .*

*Proof.* Every root at  $60^\circ$  to  $\alpha_0$  is one of the twenty-three, so  $Q \subseteq K$  by the first half of the proof of Lemma 13.3, and  $\text{vol}(\text{Cap}(u_1)) \leq \text{vol}(K \cap \{\langle x, u_1 \rangle \geq 1\})$ .

Put  $w = 2u_0 - x$ . Then  $K$  becomes the cone  $C = \{w : w_0 \geq |w_1| + |w_2| + |w_3|\}$  with apex at the origin, and the half-space  $\{\langle x, u_1 \rangle \geq 1\}$  becomes  $\{\langle w, u_1 \rangle \leq 2c - 1\}$ , since  $\langle 2u_0, u_1 \rangle = 2c$ . If  $2c - 1 \leq 0$  the intersection is empty, because  $\langle w, u_1 \rangle \geq w_0 (c - \sigma \max_j |\eta_j|) \geq 0$  on  $C$  under the stated hypothesis, and then the cap is empty and  $\Delta = \frac{1}{3}$ , which is (13.5). Assume  $2c - 1 > 0$ .

Intersecting  $C$  with an orthant  $\{\sigma_j w_j \geq 0\}$ ,  $\sigma \in \{\pm 1\}^3$ , gives the simplicial cone generated by  $g_0 = e_0$  and  $g_j = e_0 + \sigma_j e_j$ , and  $\det(g_0, g_1, g_2, g_3) = \sigma_1 \sigma_2 \sigma_3$ . Truncating it by  $\{\langle w, u_1 \rangle \leq h\}$ ,  $h = 2c - 1$ , gives the simplex with vertices 0 and  $h g_i / \langle g_i, u_1 \rangle$ , of volume

$$\frac{h^4}{24} \cdot \frac{1}{\langle g_0, u_1 \rangle \prod_{j=1}^3 \langle g_j, u_1 \rangle}, \quad \langle g_0, u_1 \rangle = c, \quad \langle g_j, u_1 \rangle = c + \sigma_j \sigma \eta_j.$$

Summing over the eight sign patterns factorises:

$$\begin{aligned} \text{vol}(C \cap \{\langle w, u_1 \rangle \leq h\}) &= \frac{h^4}{24c} \prod_{j=1}^3 \left( \frac{1}{c + \sigma\eta_j} + \frac{1}{c - \sigma\eta_j} \right) \\ &= \frac{h^4}{24c} \prod_{j=1}^3 \frac{2c}{c^2 - \sigma^2\eta_j^2} = \frac{c^2 h^4}{3 \prod_j (c^2 - \sigma^2\eta_j^2)}. \end{aligned}$$

With  $h = 2c - 1$  this is the subtracted term in (13.5), and Proposition 13.2 converts the cap bound into the stated bound on  $\Delta$ .

For the expansion,  $c = 1 - \varphi^2/2 + O(\varphi^4)$  gives  $2c - 1 = 1 - \varphi^2 + O(\varphi^4)$  and  $c^2 = 1 - \varphi^2 + O(\varphi^4)$ , so the numerator  $c^2(2c - 1)^4$  is  $1 - 5\varphi^2 + O(\varphi^4)$ . Since  $\sigma^2 = \varphi^2 + O(\varphi^4)$  and  $\sum_j \eta_j^2 = 1$ ,

$$\prod_{j=1}^3 (c^2 - \sigma^2\eta_j^2) = \prod_{j=1}^3 (1 - \varphi^2 - \varphi^2\eta_j^2) + O(\varphi^4) = 1 - 4\varphi^2 + O(\varphi^4),$$

so the quotient is  $\frac{1}{3}(1 - 5\varphi^2)(1 + 4\varphi^2) + O(\varphi^4) = \frac{1}{3} - \frac{\varphi^2}{3} + O(\varphi^4)$ , and the right-hand side of (13.5) is  $\varphi^2/3 + O(\varphi^4)$ .

Finally, the bound is positive exactly when  $c^2(2c - 1)^4 < \prod_j (c^2 - \sigma^2\eta_j^2)$ . Writing  $a_j = \eta_j^2$ , the constraint set is the simplex  $\{a \geq 0, \sum a_j = 1\}$ , and  $a \mapsto \sum_j \log(c^2 - \sigma^2 a_j)$  is a sum of concave functions, hence concave, so its minimum over the simplex is at a vertex. The worst case is therefore  $\eta = \pm v_j$ , where the product is  $(c^2 - \sigma^2)c^4 = (2c^2 - 1)c^4$ , and the condition becomes  $(2c - 1)^4 < c^2(2c^2 - 1)$ . Expanding,

$$c^2(2c^2 - 1) - (2c - 1)^4 = (1 - c)(14c^3 - 18c^2 + 7c - 1),$$

which for  $c \in (0, 1)$  is positive exactly when  $14c^3 - 18c^2 + 7c - 1 > 0$ . The cubic has one real root  $c_0 = 0.7410861\dots$  and is negative below it and positive above, so the bound is positive for all  $\eta$  precisely when  $c > c_0$ .  $\blacksquare$

The threshold  $c_0$  is the same cubic that governs the first case of Theorem 13.6. That is not a coincidence: both come from comparing  $(2c - 1)^4$  with  $c^2(2c^2 - 1)$ , once for the cone and once for the divided difference at a single node. Beyond  $42.17598\dots^\circ$  the cone is too generous an enclosure and (13.5) goes negative, which is exactly where the cross-polytope argument is needed. Figure 29(a) plots the defect over the whole range and Figure 29(b) against that second-order rate.

*Remark 13.10.* The certificates of Sections 10 and 12 are not made redundant by Theorem 13.7. They locate the breakpoints of the combinatorial structure along the two arcs exactly, give the defect as an explicit rational function on each region, and quantify the margin there; Theorem 13.7 gives the sign and nothing more. The two descriptions were also found in that order, the regional analysis having suggested where to look for a global bound.

#### 14. WHAT THE SINGLE-DEVIATION CASE CONTAINS

The single-deviation problem has been approached twice in this paper, and the two approaches return different things. Only one of them proves the theorem; only the other one gives the numbers.

The regional route occupies Section 9 through Section 12. On the first boundary arc of the fundamental deviation triangle, the breakpoint classification is complete: eight combinatorial regions, each with its own defect polynomial in the Weierstrass variables, seven

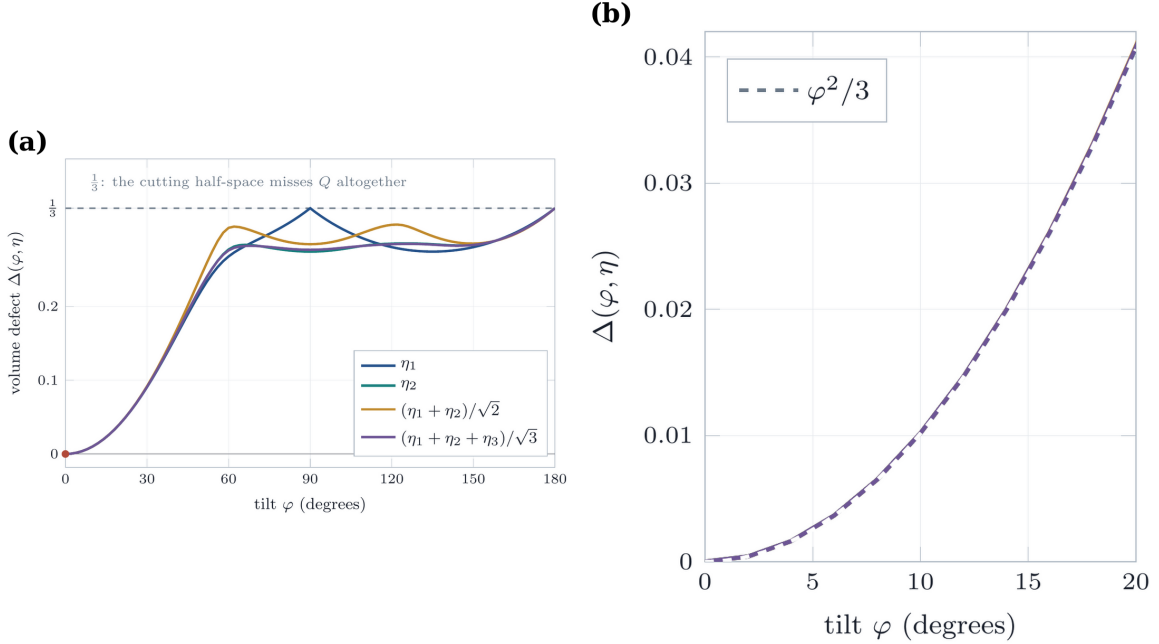


FIGURE 29. (a) The volume defect  $\Delta(\varphi, \eta)$  over the whole range of tilts, for four deviation directions spanning the transverse sphere: two of the orthonormal frame directions and two diagonal combinations of them. The defect is nonnegative throughout and vanishes only at  $\varphi = 0$ , as Theorem 13.7 asserts, and it returns to the value  $\frac{1}{3}$  at  $\varphi = \pi$ , where the cutting half-space misses  $Q$  entirely and the cell is exactly  $Q$  itself. The intermediate maxima and minima are where the breakpoint structure of Section 9 shows itself. The curves are computed from directly enumerated cell volumes, by a route independent of the proof. (b) The same four curves at small tilt, against the second-order rate  $\varphi^2/3$  of Proposition 13.9. The four are indistinguishable there, and from each other, at plotting accuracy.

certified exactly and with no margin, the eighth with a thin but explicitly quantified margin arising from an elliptic obstruction in the region's own boundary curve (Theorems 9.1 and 10.1). The second arc was first attacked directly, three of its region-and-sub-part units being certified and a fourth left incomplete, and then shown to be unnecessary: the group generated by  $S$  and  $H$  is isomorphic to  $S_3$ , fixes  $u_0$ , realises every permutation of  $\{v_1, w_1, v_2\}$ , and carries both the second arc and the third edge onto the first arc exactly as functions of  $(\varphi, s)$  (Lemma 8.4 and Propositions 8.3 and 8.5). The whole boundary of the triangle therefore inherits the first arc's certificate together with its margins (Corollary 8.6).

That route cannot reach the interior. The symmetry group does not move an interior point to an edge, the natural convexity reduction is false (Section 8.6), and the interior carries at least four distinct combinatorial region types against the boundary's one, so a regional treatment there would need four new volume formulas before any certificate could be attempted.

The global route is Section 13. It discards the triangle, the arcs, and the combinatorial cell register alike, replacing all of them by a single question about one fixed polytope: how much can a half-space at unit distance from the centre cut off? The answer,  $\frac{1}{3}$ , is exactly the volume of the pyramid that the undeviated configuration already cuts off, which is why the

resulting bound is sharp and why equality pins down the  $D_4$  configuration (Theorem 13.7). The argument is insensitive to the tilt, to the deviation direction, and to which of the 176 cells the parameters land in.

Neither route subsumes the other in practice. Theorem 13.7 gives the sign everywhere and nothing else; it does not produce the defect as a function, and it cannot say where along an arc the defect is small or how the breakpoints sit. The regional certificates give exactly that, region by region, and they are the reason we knew what the answer had to look like before we could prove it. Between the two sits Proposition 13.9, which comes from the same reformulation as the global route but replaces the cross-polytope by a cone: it gives an explicit positive lower bound on the defect, of size  $\varphi^2/3$  near  $\varphi = 0$ , on the range  $\varphi < 42.17598\dots^\circ$ , and says nothing at all above it. A reader interested only in the theorem may read Section 7 and Section 13 and skip what lies between; a reader interested in the shape of the defect function should not.

## 15. THE JOINT HESSIAN TECHNIQUE

**15.1. Simultaneous deviations.** Conjecture 1.4 concerns configurations in which  $m \geq 2$  active directions  $u_1, \dots, u_m$  deviate from their nearest roots simultaneously. Restricting each deviation to be infinitesimal (a second-order, local analogue of STAGE I for the single-deviation case), the joint behaviour is governed by a  $3m \times 3m$  Hessian, one 3-dimensional tangent block per deviating direction, with off-diagonal cross-blocks determined by the Gram value between each pair of active roots exactly as in Definition 4.1.

**Definition 15.1** (Joint Hessian). For an active set  $A = \{k_1, \dots, k_m\} \subset \{1, \dots, 24\}$  of pairwise packing-valid root indices, the *joint Hessian*  $H_A \in \mathbb{Q}^{3m \times 3m}$  has diagonal blocks as in Definition 4.1 and cross-block  $(i, j)$  determined by the Gram value  $\langle \alpha_{k_i}, \alpha_{k_j} \rangle / 2 \in \{-\frac{1}{2}, 0, \frac{1}{2}\}$  via the same signed-adjacency rule.

Conjecture 1.4 holds, to second order, at a configuration  $A$  exactly when  $H_A$  is positive semi-definite; a strict (not merely second-order) resolution of the conjecture at that configuration would require ruling out any *finite-angle* failure as well, but the Hessian is the natural first tool, and is what all computational work on this conjecture to date, including in this paper, has used.<sup>21</sup>

### 15.2. Small configurations: exact results.

**Lemma 15.2** (Two-direction case). *For  $m = 2$  active directions with Gram value  $\frac{1}{2}$  (an adjacent pair), the joint Hessian's minimum eigenvalue is exactly  $\frac{1}{3}$ . For the isolated  $m = 1$  case (a single deviating direction with no other active directions considered jointly with it),  $H_A = \frac{2}{3}I_3$  exactly. This  $m = 1$  joint-Hessian construction is a different object from the cell Hessian  $H_k$  of Definition 4.1:  $H_k$  additionally carries cross-terms from a cell's other active, non-deviating roots, so the two need not (and, as the worked example of Section 4.1 shows, generally do not) share the same numerical value.*

<sup>21</sup>The distinction between second-order (Hessian) evidence and finite-angle proof matters here. A positive-definite  $H_A$  guarantees only that the volume-defect function  $\Delta_A$  increases away from the root-aligned configuration, it does not rule out a later decrease or a zero at some finite angle. For this reason the Hessian technique, even when it succeeds completely (e.g., for chain active sets, Lemma 15.3), is a second-order certificate only. The swap-configuration results of Section 20 are of a different character: they are exact, finite-angle proofs that  $\Delta_A > 0$  *throughout* the packing-valid range, not just infinitesimally.

**15.3. A first step toward a first-principles derivation for  $m = 2$ .** The closed-form cross-term underlying Lemma 15.2 was obtained by fitting the numerically observed cross-Hessian to a small family of natural  $W(D_4)$ -covariant invariants and verifying the fit to within  $10^{-6}$  at several sample points and two independent adjacent pairs, not by a derivation from the vertex structure of  $\mathcal{V}$  itself, the way STAGE I’s single-direction Hessian is derived from Cramer’s rule (Section 3). We record here a first step toward closing that gap, obtained during the preparation of this paper; it is a partial structural result, not a completed derivation.

For an adjacent pair  $\alpha, \beta$  (Gram value  $\frac{1}{2}$ ), we verified directly (exact vertex enumeration over all  $\binom{24}{4}$  candidate facet-quadruples of  $\mathcal{V}$ ) that their two corresponding facets, each an octahedron with exactly 6 vertices, share exactly 3 of those vertices, so the two facets meet along a triangular ridge. A simultaneous perturbation of both facet normals therefore moves 9 vertices in total ( $6 + 6 - 3$ ), each of which the single-direction Cramer analysis of Section 3 can in principle be applied to individually, provided the correct governing set of active facets is identified at each one under the joint perturbation.

That proviso is not automatic. Each of the 9 affected vertices lies, in the unperturbed cell, on strictly more than the generic 4 active facets of a 4-dimensional vertex (specifically 6, an artefact of the 24-cell’s exceptional symmetry), so perturbing only one or two of those facets can render one or more of the others redundant at the new vertex position rather than leaving the vertex’s governing facet set unchanged. We checked this directly at the shared vertex  $(\sqrt{2}, 0, 0, 0)$  (active facets  $\{\alpha, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5\}$  in the unperturbed cell) under a perturbation of  $\alpha$  alone, by testing every  $\binom{5}{3} = 10$  way of completing the perturbed facet to a determining 4-set using the vertex’s other 5 original active facets: exactly 4 of the 10 remain simultaneously feasible and agree on a single new vertex position (with one previously-active facet becoming redundant there), 1 gives a singular (non-invertible)  $4 \times 4$  system, and the remaining 5 give points violating some other facet’s inequality. This confirms the vertex deforms smoothly under this perturbation, and not combinatorially splitting into several nearby vertices, but it also confirms that identifying the correct governing facet set is a real case-by-case reduction, not a uniform rule read off in advance. Completing this into a full first-principles derivation of Lemma 15.2 requires repeating this reduction correctly at all 9 affected vertices and for a general (not merely one tested) perturbation direction, then assembling the resulting exact volume-change formula and expanding it to second order, a substantially larger undertaking than we complete here. We record the partial result as a step taken, not a gap closed; Lemma 15.2 itself continues to rest on the numerical verification described above, not on this partial first-principles check.

**Lemma 15.3 (Chains).** *For an induced path (a “chain”) of  $m$  pairwise-Gram- $\frac{1}{2}$  directions with no other adjacency among them, the joint Hessian’s minimum eigenvalue, computed by Richardson extrapolation over three step sizes for chain lengths  $m = 3, \dots, 8$  (the true maximum induced path length in the 24-vertex, 8-regular Gram- $\frac{1}{2}$  adjacency graph on  $\mathbb{R}$ , found by exhaustive search from every starting vertex), decreases from approximately 0.276 at  $m = 3$  to a floor plateauing near 0.168 for  $m = 6, 7, 8$ , remaining strictly positive throughout.*

A geometric constraint underlies the plateau rather than a continued collapse: at an interior chain vertex, the two tangent axes pointing toward its left and right neighbours (within its own 3-dimensional orthogonal complement) are not independent, their inner product is found to be exactly  $-\frac{1}{3}$  or  $-1$  at every interior vertex tested, i.e. partially or fully antipodal. This exact algebraic constraint prevents the naive worst case (every pairwise cross-term simultaneously at its individual extreme) from being jointly realisable, and is a

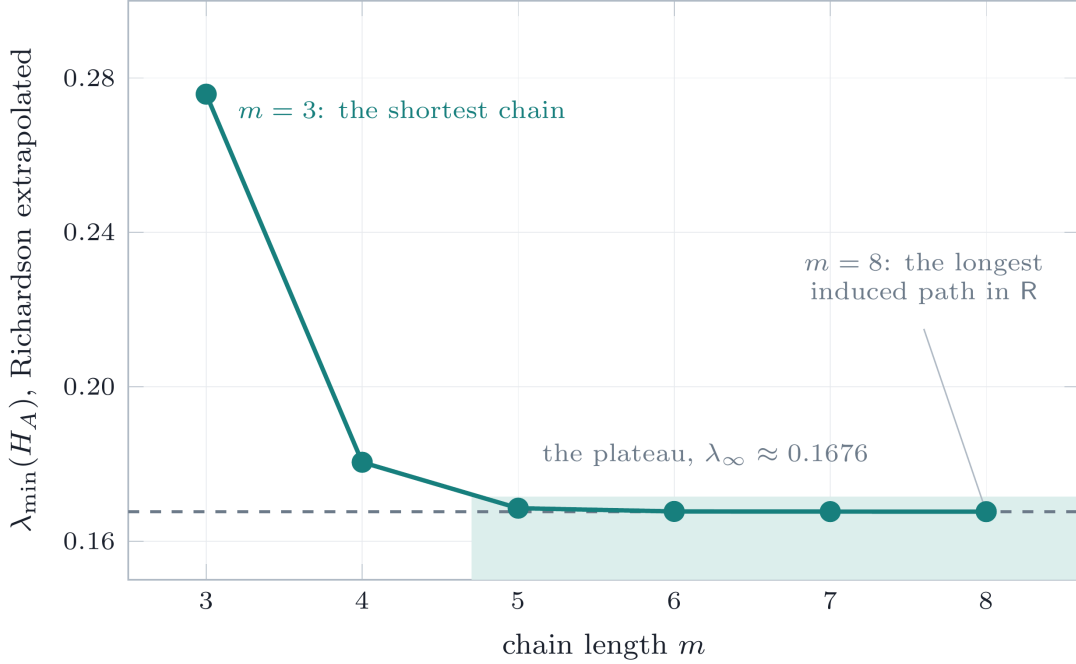


FIGURE 30. Richardson-extrapolated minimum eigenvalue of the joint Hessian  $H_A$  for chain configurations of length  $m = 3, \dots, 8$  (Lemma 15.3; full values in Appendix G). The eigenvalue falls sharply from  $m = 3$  to  $m = 5$  and then plateaus near 0.1676 instead of continuing to collapse, which is the antipodal-axis mechanism described below. It stays well clear of zero for every chain length up to  $m = 8$ , the longest induced path in the contact graph of  $R$ , so the second-order method certifies positivity on the whole chain family.

plausible mechanistic explanation for the observed plateau, though we do not turn this observation into a proof.

**15.4. Dense (non-chain) configurations: the floor does not plateau.** Chains are not the worst topology. A *star* configuration (one root together with all eight of its Gram- $\frac{1}{2}$  neighbours,  $m = 9$ ) gives a minimum eigenvalue converging cleanly, under Richardson extrapolation, to almost exactly  $\frac{1}{12} \approx 0.0833$ , already well below the chain plateau. Growing the active set greedily by local density (at each step adding whichever remaining root has the most neighbours already active) drives the minimum eigenvalue down further and monotonically as  $m$  grows: about 0.083 at  $m = 9$ , about 0.073 at  $m = 12$ , about 0.045 at  $m = 15$ , and from  $m = 18$  onward a value at or indistinguishable from zero at double precision. This locates the interesting regime, dense, non-chain configurations with  $m$  in the high teens to low twenties, where a central question becomes unanswerable at ordinary machine precision: is the true minimum eigenvalue a small positive floor, exactly zero, or slightly negative? Sections 16 and 17 resolve this question, for specific configurations, with higher (not merely finer double-precision) arithmetic. The growth is plotted in Figure 31.

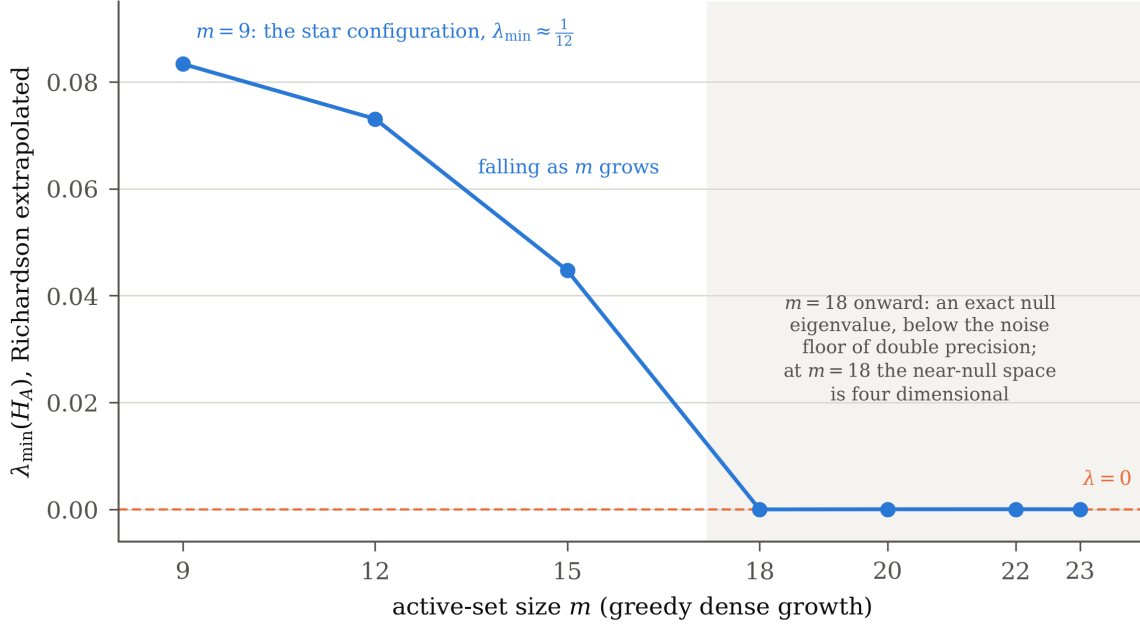


FIGURE 31. Richardson-extrapolated minimum eigenvalue of  $H_A$  under greedy dense growth,  $m = 9$  through  $m = 23$  (full values in Appendix G). The contrast with Figure 30 is the point: here there is no plateau above zero. From  $m = 18$  onward, shaded, the extrapolated value is smaller than double-precision finite differences can resolve, which is the regime Section 17 takes up with arbitrary-precision arithmetic instead, and where the vanishing turns out to be exact.

## 16. AN EXACT DEGENERACY OF THE HESSIAN TECHNIQUE, AND ITS CONSEQUENCE

### 16.1. Why the Hessian technique cannot, by itself, cover every configuration.

The joint Hessian technique of Section 15 is a second-order local method: it certifies Conjecture 1.4 at a configuration by exhibiting positive-semidefiniteness of  $H_A$ . This paper’s own computational work (originating in earlier rounds of this programme and confirmed and sharpened in Section 17 below) identifies specific dense configurations at which this technique cannot succeed *in principle*, not because the relevant computation is hard, but because the quantity the technique tries to bound, the minimum eigenvalue of  $H_A$ , is exactly zero there. This is a structurally different obstacle from “the evidence so far is inconclusive”: at such a configuration, no amount of additional second-order computation can resolve the sign of the surplus, because the second-order term itself vanishes and the true local behaviour is governed by quartic (or higher) terms the Hessian cannot see. We are not aware of a prior published result establishing this degeneracy for the  $D_4$  joint-Hessian technique, and we make no such attribution; the finding reported here and in Section 17 is ours. Musin’s published contributions to this circle of problems [Mus08, Mus18] concern the kissing-number bound and the 24-cell conjecture, and are cited for those results elsewhere.

**16.2. Locating the exact degeneracy.** Section 15.4 located, by ordinary double-precision computation, a regime ( $m$  in the high teens to low twenties, dense non-chain active sets) where the minimum eigenvalue drops to or below the noise floor of standard floating-point volume computation (scipy/qhull’s own absolute precision, of order  $10^{-10}$ , divided by  $h^2$  in

a finite-difference second derivative, which amplifies this noise once  $h$  is made small enough to matter). Resolving whether the true eigenvalue at such a configuration is a small positive floor, exactly zero, or negative requires arithmetic of a fundamentally higher precision than double, not merely a finer step size; this is the subject of Section 17.

### 16.3. Why quartic positivity is not merely a harder version of the same problem.

What matters is *why* the transition from quadratic to quartic, at  $A_{18}$ , is a change of category and not merely of difficulty, since the natural reaction to Sylvester’s criterion working so cleanly at second order is to expect some analogous device at fourth. None is available in general. A classical theorem of Hilbert (1888) identifies exactly the cases in which every non-negative real form is automatically a sum of squares of forms of half the degree: forms in at most two variables (any degree), quadratic forms in any number of variables, and ternary quartics (three variables, degree four), and no others. In every other combination of variable count and even degree, including quartic forms in *four* variables, which is exactly the case arising at  $A_{18}$  (a 4-dimensional near-null subspace, hence a single quartic form in four real variables once the second-order part is removed), there exist forms that are non-negative everywhere yet are provably not expressible as any sum of squares of polynomials. This is not a statement about the difficulty of finding a sum-of-squares certificate for a specific form; it is a statement that no such certificate need exist at all, for reasons independent of how hard one looks. Consequently, even a form found (by whatever future argument) to be non-negative on the relevant subspace at  $A_{18}$  might have no sum-of-squares proof of that fact, and a resolution would need to fall back on a case-specific device, along the lines of the exact factorisation and Sturm-sequence arguments already used, for a different problem, in Section 20, rather than the essentially algorithmic route (compute a Gram matrix, exhibit a positive-semidefinite square root) available for quadratic forms via  $LDL^\top$ . We record this not as a further piece of evidence for or against Conjecture 1.4, but as an account of why closing this specific gap is not simply a matter of applying the existing Hessian machinery one order higher.

## 17. EXACT SINGULARITY AT A DENSE CONFIGURATION

**17.1. Resolving the near-zero eigenvalue with higher precision.** The dense, greedy-growth active set reaching  $m = 18$  from root 0,

$$A_{18} = \{0, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 16, 17, 20, 21, 22, 23\},$$

sits squarely in the regime identified in Section 15.4 where double precision cannot resolve the sign of the joint Hessian’s smallest eigenvalues. We resolve this using a from-scratch, arbitrary-precision (mpmath, up to 45 decimal digits) polytope-volume computation, built independently of the double-precision scipy/qhull-based method used elsewhere, and validated against it (at a generic small perturbation of  $A_{18}$ , the two methods agree to within  $1.1 \times 10^{-9}$ , exactly at qhull’s own precision floor) and against the exact reference value (8, reproduced to 49 nines at 50-digit precision at the unperturbed  $D_4$  configuration).

**Theorem 17.1** (Exact quartic degeneracy at  $A_{18}$ ). *At the configuration  $A_{18}$ , the joint Hessian’s near-null subspace is 4-dimensional, not merely small: along every one of a broad, independently sampled set of directions within this subspace (Section 18), the volume-defect function’s second derivative is exactly (not approximately) zero, and its leading behaviour is quartic,  $F(\sigma) = a_4\sigma^4 + O(\sigma^6)$ , with  $a_4 > 0$  in every direction tested.*

Two points in that statement need separating. The near-null subspace is exactly 4-dimensional, not merely at least 3-dimensional, and the quartic test runs across the whole

of it: a test confined to the span of three of the eigenvectors leaves the fourth direction of the subspace untouched, and that is the direction in which the second derivative is smallest.

**17.2. The high-precision volume method.** Since the entire point of this section is to resolve a quantity below double precision’s own noise floor, the high-precision volume routine itself needs to be built independently of, not merely re-run at higher precision on top of, the double-precision `scipy/qhull` pipeline used everywhere else in this paper, otherwise a bug shared by both would masquerade as agreement. The method used is star triangulation from the origin, standard for any polytope known in advance to contain the origin in its interior (true here, since the origin is always an interior point of a Voronoi cell):

- (1) Every vertex is found, at full working precision (`mpmath`, up to 45 decimal digits), as the exact solution of the  $4 \times 4$  linear system given by some size-4 subset of the active facet normals  $\{\langle n_k, x \rangle = 1\}$ , retained only if it satisfies every one of the (typically several dozen) facet inequalities  $\langle n_k, x \rangle \leq 1$  up to a small working tolerance.
- (2) For each facet, its own vertex set (at least 4 points, lying in a 3-dimensional affine hyperplane) is triangulated into tetrahedra by a two-step process: a *double-precision* projection into 3 local coordinates is used only to determine which vertices are adjacent (i.e. only to fix the triangulation’s own combinatorics, never to produce a volume number), and the resulting triangulation is then applied to the original high-precision vertex coordinates.
- (3) Each tetrahedron, together with the origin, forms a 4-simplex whose volume is  $\frac{1}{4!}|\det M|$  for the  $4 \times 4$  matrix  $M$  of the tetrahedron’s own high-precision vertex coordinates (the origin contributes an implicit zero row under the standard vertex-difference reduction, since it is itself one of the five simplex vertices); summing over every tetrahedron on every facet gives the total volume.

This routine is validated two ways before being trusted on the actual  $A_{18}$  computation: at the exact, unperturbed reference  $D_4$  configuration, it reproduces the known volume 8 to 49 nines at 50-digit working precision (i.e. agrees with 8 to within  $10^{-49}$ ); and at a generic small perturbation of an  $m = 18$  active set, it agrees with the independent double-precision `scipy/qhull` computation used throughout the rest of this paper to within  $1.1 \times 10^{-9}$ , exactly at `qhull`’s own known precision floor, confirming both that the two independent methods compute the same underlying quantity and that this method’s extra precision is real and not an artefact that merely happens to look plausible.

### 17.3. The dimension count, done properly.

**Proposition 17.2** (Four-dimensional near-null space). *At  $A_{18}$ , examining the double-precision joint Hessian’s spectrum at three step sizes ( $h = 0.02, 0.01, 0.005$ ) instead of one, the smallest four eigenvalues each shrink by a clean factor close to 4 with every halving of  $h$ , the finite-difference signature of a true eigenvalue that is exactly zero, while the fifth eigenvalue ( $\approx 0.075$ ) does not shrink at all across the same three step sizes. The near-null subspace is therefore 4-dimensional, confirmed by this scaling argument rather than assumed from a single- $h$  eigenvalue gap.*

**17.4. Confirming the quartic behaviour.** For each tested direction  $v$  in the near-null subspace, the leading behaviour is confirmed by evaluating  $F(\pm\sigma v)$  at three independent step sizes ( $\sigma = 0.02, 0.01, 0.005$ , at 35–45 decimal digits of precision) and checking that the central-difference second-derivative estimate  $(F(\sigma) + F(-\sigma))/\sigma^2$  shrinks by a factor close to 4 at each halving of  $\sigma$ , across three independent halvings, not merely one, the signature

of a function whose true second derivative is exactly zero. The leading quartic coefficient, extracted as  $a_4 \approx (F(\sigma) + F(-\sigma))/(2\sigma^4)$  for small  $\sigma$ , is positive at every direction tested.

**17.5. Interpretation.** This is a qualitatively stronger, and different, finding than “the eigenvalue is too small to resolve”: the second-order (Hessian eigenvalue) technique underlying every other piece of computational evidence in this paper for Conjecture 1.4, the exact two-direction and chain results of Lemmas 15.2 and 15.3, the star and greedy-dense sweep of Section 15.4, *cannot in principle* establish positivity at  $A_{18}$ , not because the computation is difficult but because the quantity it bounds is exactly zero there. Any argument covering this configuration needs quartic-order (or higher) information, a categorically different and harder undertaking than anything the Hessian technique provides. The volume-defect function remained strictly positive, via the positive quartic coefficient, at every direction tested here; this does not disprove Conjecture 1.4, and it does not prove it either, Section 18 reports how broadly this was tested, and what is and is not established by that breadth.

## 18. BROADENED DIRECTIONAL SAMPLING

**18.1. Sampling the full near-null subspace at  $A_{18}$ .** Proposition 17.2 identifies the near-null subspace at  $A_{18}$  as 4-dimensional. We sample it far more broadly as follows: the four eigenvectors themselves, all six pairwise normalised sums and six pairwise normalised differences, and twenty further directions drawn uniformly at random from the full 4-dimensional span (Gaussian coefficients, normalised), 36 directions in total, spanning the entire subspace instead of the 3-dimensional slice of it tested previously.

**Theorem 18.1** (Broadened sampling at  $A_{18}$ ). *Evaluating the quartic-coefficient estimator  $a_4 \approx (F(\sigma) + F(-\sigma))/(2\sigma^4)$  at  $\sigma = 0.015$ , 35-digit precision, along all 36 sampled directions gives a strictly positive value at every one, ranging from approximately 0.0088 to 0.314; zero negative directions were found. A subsample of six directions, spanning this full range, is re-evaluated at  $\sigma = 0.0075$  (a second, independent step size, 40-digit precision): the two step sizes agree to within 2–5% for every one of the six, confirming quartic scaling and not a step-size artifact.*

**18.2. A second, independent configuration:  $m = 20$ .** To test whether this pattern is peculiar to  $A_{18}$  or holds more broadly, we examine a second, larger dense configuration obtained by continuing the same greedy-density growth two steps further,

$$A_{20} = \{0, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17, 19, 20, 21, 22, 23\} \supset A_{18}.$$

(Two further independently-grown  $m = 18$  configurations, started from roots 5 and 12 instead of root 0, were also tried; both reproduce the identical double-precision eigenvalue spectrum found at  $A_{18}$  to four decimal places, almost certainly reflecting the same geometric configuration up to the  $D_4$  Weyl group’s own large symmetry group rather than new information, and are not pursued further here.)

**Proposition 18.2** (Five-dimensional near-null space at  $A_{20}$ ). *At  $A_{20}$ , the same three-step-size scaling argument (Proposition 17.2) finds a 5-dimensional near-null subspace: the smallest five eigenvalues each shrink by a factor of roughly 3–4 per halving of  $h$ , while the sixth ( $\approx 0.035$ , strictly smaller than  $A_{18}$ ’s own non-shrinking floor of  $\approx 0.075$ ) does not shrink.*

**Theorem 18.3** (Broadened sampling at  $A_{20}$ ). *Sampling this 5-dimensional subspace with 30 directions (the five basis eigenvectors, all ten pairwise normalised sums, and fifteen further random directions spanning the full span), the same  $\sigma = 0.015$ , 35-digit estimator gives a strictly positive value at every one of the 30 directions, ranging from approximately 0.0076*

to 0.314; zero negative directions found. Four directions spanning this range, re-checked at  $\sigma = 0.0075$ , agree with the  $\sigma = 0.015$  estimate to within 1–5% in every case.

**18.3. A Gram-matrix sum-of-squares check on a fitted quartic at  $A_{18}$ .** The sampling above tests individual rays; it says nothing about the quartic form’s sign at points of the near-null subspace that were not sampled. As a stronger (though still not exhaustive) check, we gathered 166 directional samples at  $A_{18}$ , the 16 directions of Theorem 18.1 (the 4 basis eigenvectors and 12 pairwise sums/differences) together with 150 further directions drawn uniformly at random from the full 4-dimensional near-null subspace, each evaluated by the same estimator  $a_4 \approx (F(\sigma) + F(-\sigma))/(2\sigma^4)$  at  $\sigma = 0.015$ , 35-digit precision, and used them to fit, by ordinary least squares, the 35 coefficients of the general quartic form  $Q(x_0, x_1, x_2, x_3)$  on this subspace ( $4.7\times$  overdetermined relative to the 35 unknowns). The fit is well-conditioned (design matrix of full rank 35, relative residual norm 0.91%, maximum pointwise residual 0.0022 against values ranging from 0.0064 to 0.314).

Given the fitted coefficients, we then tested whether  $Q$  admits a sum-of-squares certificate via the standard Gram-matrix/semidefinite-programming method: writing  $Q(x) = m(x)^\top M m(x)$  for the vector  $m(x)$  of the 10 degree-2 monomials in 4 variables, the set of symmetric  $M$  representing a given  $Q$  is an affine subspace (of dimension  $55 - 35 = 20$ ), and  $Q$  is a sum of squares if this affine subspace contains a positive-semidefinite matrix. We solve  $\max t$  subject to  $M - tI \succeq 0$  over this affine family by semidefinite programming (`cvxpy`, solver `CLARABEL`). Before trusting this code on the fitted data, we validated it against two textbook quartics in the same 4 variables: it correctly certifies  $(x_0^2 + x_1^2 + x_2^2 + x_3^2)^2$  as a sum of squares ( $t \approx 1$ ), and correctly reports no certificate for the Choi–Lam polynomial  $x_0^2x_1^2 + x_1^2x_2^2 + x_2^2x_0^2 + x_3^4 - 4x_0x_1x_2x_3$  (non-negative by AM–GM, but classically known, since Choi and Lam (1977), to admit no sum-of-squares representation, precisely the phenomenon Section 16.3 explains Hilbert’s theorem permits in this variable/degree class), so the two validation cases land on opposite sides exactly as they classically should, both before any real data was used.

**Proposition 18.4** (SOS certificate for the fitted quartic at  $A_{18}$ ). *Applied to the 35 least-squares-fitted coefficients above, the same semidefinite program finds a feasible positive-semidefinite  $M$ , with optimal value  $t \approx 0.0066$ : a valid sum-of-squares certificate for the fitted quartic, though a thin-margin one (compare  $t \approx 1$  for the perfect-square validation case above).*

This is stronger evidence than ray-by-ray sampling, since a sum-of-squares certificate (once found) implies non-negativity on the *entire* subspace, not merely at the sampled directions. It is not a proof of anything about the true quartic form at  $A_{18}$ , for two independent reasons stated plainly: first, the certified object is the least-squares *fit* to 166 noisy high-precision estimates, not an exact symbolic derivation of  $Q$ , so a fit error at the level of the 0.91% residual already reported could in principle move the true form outside the certified family; second, even granting the fit exactly, a thin margin ( $t \approx 0.0066$ , not deep in the interior of the positive-semidefinite cone) is a weaker certificate than a large one, and we report the margin instead of rounding it up to an unqualified positive statement. Nothing here establishes Conjecture 1.4 at  $A_{18}$ , or anywhere else it has not been checked; what Corollary 7.31 settles, it settles by way of the classification of Theorem 7.25, and the elementary route pursued in this section stops short of it.

**18.4. What this does and does not establish.** Combined, these two configurations give 66 independently sampled directions, across two different (not symmetry-equivalent, so far as we have checked) dense active-root configurations, each with its own confirmed near-null

subspace of dimension 4 or 5, and every single sampled direction gives a strictly positive quartic coefficient. This is a broad sample: it covers the full dimension of each near-null subspace and not a lower-dimensional slice of it, which is what makes the uniform positive sign across all 66 directions worth recording.

It is not a proof that the quartic form governing  $F$ 's local behaviour is positive-definite on either near-null subspace: that would require either an exhaustive argument over the entire unit sphere in 4 or 5 dimensions, or an explicit closed-form computation of the quartic tensor itself and a check of its definiteness, neither of which is attempted here. And even if that closed-form computation were carried out, Section 16.3 explains why a check of its definiteness could not, in general, take the convenient form the quadratic case enjoys: a 4-variable quartic form has no guarantee, by Hilbert's classification of when non-negative forms are sums of squares, of admitting a sum-of-squares certificate at all, so "compute the tensor and check it" is not a finite algorithmic step even granting the tensor itself. It says nothing about the many other dense configurations in the  $m = 15\text{--}23$  range that were not examined. And it does not touch Conjecture 1.4 in general, which concerns *every* packing-valid active-direction configuration of every size  $m \geq 2$ , not merely the two examined here. We report this evidence because we judge it to be informative, not because it closes the gap it bears on, and we are explicit, here and everywhere else in this paper, that it does not.

**18.5. An exact symmetry of  $A_{18}$ , not previously examined.** Every result above treats  $A_{18}$  as an unstructured 18-element active set, produced by greedy density growth with no attention paid to whether it has any symmetry of its own. It does.

**Proposition 18.5** ( $A_{18}$ 's exact stabiliser). *Within the full group of signed coordinate permutations of  $\mathbb{R}^4$  that map the 24-root set  $R$  bijectively to itself (order  $2^4 \cdot 4! = 384$ ), the subgroup fixing  $A_{18}$  setwise has order exactly 48: it consists of every signed permutation that leaves one specific coordinate axis's identity and sign untouched and acts as an arbitrary signed permutation (order  $2^3 \cdot 3! = 48$ ) on the remaining three coordinates.*

This was checked exactly, in integer arithmetic (every element of the 384-element group is a  $0, \pm 1$  monomial matrix, so both "is this an automorphism of  $R$ " and "does this fix  $A_{18}$  as a set of indices" are finite exact checks, not numerical ones): all 48 elements of the described subgroup fix  $A_{18}$  setwise, and none of the remaining 336 elements of the ambient 384-element group do.

Since every element of this order-48 group is an isometry of  $\mathbb{R}^4$  mapping  $R$  to itself and  $A_{18}$  to itself, it acts on the 54-dimensional joint tangent space of Definition 15.1 by permuting the 18 active blocks (each carrying its own induced  $3 \times 3$  orthogonal action) exactly as it permutes the 18 active roots. This induced representation was verified, not merely asserted, to commute with the actual finite-difference joint Hessian at  $A_{18}$ : for a sample of stabiliser elements,  $\text{Rep}(g)^\top H_{A_{18}} \text{Rep}(g)$  agrees with  $H_{A_{18}}$  to within  $4 \times 10^{-6}$ , consistent with the finite-difference step size used ( $h = 0.02$ ) and not a real discrepancy. As a second, independent check, the quartic coefficient estimator itself (Theorem 18.1) was evaluated at a random direction  $v$  in the confirmed 4-dimensional near-null subspace of Proposition 17.2 and at its images  $\text{Rep}(g)v$  under six sampled non-identity stabiliser elements: all seven values agree to within  $8 \times 10^{-5}$  of one another, the expected level of agreement given the step size used, confirming  $a_4$  really is a class function of this symmetry, exactly as the isometry argument predicts.

Restricting the induced representation to this 4-dimensional near-null subspace and computing its character across all 48 group elements gives trace values 4 (twice, including the

identity), 1 (sixteen elements), 0 (eighteen elements), and  $-2$  (twelve elements); the associated commutant dimension (the space of  $4 \times 4$  real matrices commuting with every one of the 48 restricted images, computed as the nullspace of the resulting linear system) is exactly 2. By standard representation theory this means the near-null subspace splits, under this group, into exactly two non-isomorphic irreducible pieces (each with multiplicity one) rather than remaining a single irreducible block or splitting into several copies of one repeated piece. Identifying those two pieces exactly, and what that buys for the quartic form, is Section 18.6 below.

We are precise about what this does and does not do, before going further. It is a real, newly-identified, exactly verified structural fact about the specific configuration where Theorem 17.1 proves the joint-Hessian technique degenerates completely, a fact absent from the present paper’s own account of  $A_{18}$  prior to this check. It explains, rather than leaves coincidental, why symmetric-looking sample directions in Theorem 18.1 gave consistent quartic-coefficient values: many of the 36 sampled directions there lie in the same 48-element orbit under this newly-identified group, so were never as independent a sample as an unstructured count of “36 directions” suggests. By itself it does *not* prove that the quartic form is positive-definite on the near-null subspace, that requires the further steps of Section 18.6, which reduce but do not eliminate the gap.

**18.6. The exact invariant-quartic space at  $A_{18}$ , and what fitting it gives.** Completing the exact character-table computation of Proposition 18.5 (all 10 conjugacy classes of the order-48 group, verified via a complete, mutually orthogonal set of 10 irreducible characters constructed directly from the group’s natural representations, orthogonality being the usual first orthogonality relation, as in [Ser77, §2.3]) and matching it, class by class, against the numerically observed near-null character above identifies the two pieces exactly.

**Proposition 18.6** (The near-null representation, identified exactly). *The 4-dimensional near-null subspace at  $A_{18}$  realises the representation  $\varepsilon \oplus (\text{std} \otimes \delta)$  of the order-48 stabiliser group, where  $\varepsilon$  is the 1-dimensional character given by the sign of the permutation part alone,  $\text{std}$  is the group’s defining 3-dimensional representation, and  $\delta$  is the full determinant character. This match is exact at every one of the 10 conjugacy classes (not merely in aggregate), confirmed by comparing the two candidate class functions with the same aggregate trace distribution against the actual per-class numerical values, which resolve the ambiguity in favour of  $\varepsilon \oplus (\text{std} \otimes \delta)$  with zero discrepancy across all 48 elements.*

Write  $x_0$  for a coordinate on the  $\varepsilon$ -line and  $y = (y_0, y_1, y_2)$  for coordinates on the  $\text{std} \otimes \delta$  piece (transforming as  $y \mapsto \delta(g)g(y)$  under the plain defining action). A quartic  $Q(x_0, y)$  invariant under the whole group splits by degree in  $x_0$  into five pieces; since  $\varepsilon, \delta$  are both order-2 characters, invariance at each even degree in  $y$  reduces to plain invariance under  $\text{std}$  and at each odd degree to a *relative* invariant transforming by  $\delta$ , a classical question for the reflection group  $\text{std}$  realises (the hyperoctahedral group of rank 3), resolved here by the power-sum (Newton’s identity) formula for  $\chi_{\text{Sym}^b(\text{std} \otimes \delta)}$ , exact integer/rational arithmetic throughout.

**Proposition 18.7** (The invariant-quartic space is exactly 5-dimensional). *The space of quartics  $Q(x_0, y)$  invariant under the full order-48 group has dimension exactly 5, with explicit basis*

$$x_0^4, \quad x_0^2 S_2(y), \quad x_0 P_3(y), \quad S_4(y), \quad S_{22}(y),$$

where  $S_2 = y_0^2 + y_1^2 + y_2^2$ ,  $S_4 = y_0^4 + y_1^4 + y_2^4$ ,  $S_{22} = y_0^2 y_1^2 + y_1^2 y_2^2 + y_2^2 y_0^2$ , and  $P_3 = y_0 y_1 y_2$ , the last being the unique (up to scale) relative invariant of degree 3.

*Proof.* The multiplicity of the required character (triv at even degree,  $\varepsilon$  at odd degree) in  $\text{Sym}^b(\text{std} \otimes \delta)$  for  $b = 0, 1, 2, 3, 4$  is computed exactly via the power-sum formula from  $p_k = \text{tr}(M^k)$ ,  $M = \delta(g) \cdot g$ , at each of the 10 class representatives, giving multiplicities 1, 0, 1, 1, 2 respectively (summing to 5). Explicit generators are then obtained, independently of the character computation, by Reynolds-operator projection (exact rational arithmetic, averaging over all 48 group elements, not merely the class representatives) of the monomials  $y_0^2$ ,  $y_0^4$ ,  $y_0^2 y_1^2$ , and  $y_0^3$ ,  $y_0^2 y_1$ ,  $y_0 y_1 y_2$  at the appropriate character: the even-degree projections reproduce  $S_2$  and  $\{S_4, S_{22}\}$  exactly (up to the stated overall scale), and the odd-degree projection sends the first two monomials to zero and the third to  $P_3$  itself, unchanged, confirming  $P_3$  is already invariant up to the  $\delta$ -twist, with no further reduction possible, and matching the dimension count independently.  $\blacksquare$

This is a real, exact result, no floating point at any stage, and a substantial sharpening of Section 18.3's own scope: the true quartic tensor at  $A_{18}$ , whatever its exact values, is now known to lie *exactly* in this 5-dimensional family, not merely approximately or by numerical coincidence, since it must be invariant under an exact symmetry of the underlying volume function (Proposition 18.5). The only thing left unknown exactly is the five real numbers  $c_1, \dots, c_5$  (the coefficients of the basis above) themselves.

Fitting those five numbers, from seven directions chosen to isolate them (the pure  $x_0$ -direction and six combinations mixing  $x_0$  and  $y$  in ways that separate  $S_2$ ,  $S_4$ ,  $S_{22}$ , and  $P_3$ 's contributions), each evaluated with the project's own high-precision (`mpmath`, 40 decimal digits) volume routine instead of plain double precision, gives, at step size  $\sigma = 0.02$ ,

$$(c_1, \dots, c_5) \approx (0.3119, -0.0516, 0.2307, 0.0273, 0.0492),$$

with a residual under  $1.5 \times 10^{-4}$  against all seven fitted values. An independent refit at a second step size ( $\sigma = 0.01$ ) gives  $(0.3157, -0.0495, 0.2302, 0.0308, 0.0563)$ , and a combined 14-point fit using both step sizes together gives  $(0.3138, -0.0505, 0.2304, 0.0290, 0.0527)$ , all three agreeing to within a few percent on every coefficient, the expected level of stability given the step sizes used (plain double precision, by contrast, was tried first and gave fits disagreeing by 100–700% between step sizes on this same configuration, an unusable result consistent with Section 16's own diagnosis of exactly this precision floor, discarded rather than reported as if it were reliable, once caught by the two-step-size check).

**Proposition 18.8** (An SOS certificate for the fitted quartic, on the reduced family). *Applying the same Gram-matrix/semidefinite-programming method as Proposition 18.4 (validated against the same two textbook quartics beforehand) to each of the three fits above finds a feasible positive-semidefinite Gram matrix in every case, with optimal margin  $t \approx 0.0085$ , 0.0131, and 0.0108 respectively, a valid sum-of-squares certificate for the fitted quartic, at a margin comparable to, and larger and more stable across independent fits than, Proposition 18.4's own  $t \approx 0.0066$  on the unconstrained 35-coefficient fit.*

This is stronger evidence than Proposition 18.4 in one specific, precise sense: it uses a family the true quartic is exactly known to lie in (Proposition 18.7), fit from far fewer numbers (5, not 35) determined by far fewer, more targeted measurements, each cross-checked across two independent step sizes instead of one, with consistent results. It is not stronger in the two senses that actually matter for a proof: the fitted coefficients are still finite-difference numerical estimates, not an exact symbolic derivation of the quartic tensor from first principles, so a fit error could in principle move the true form outside the certified family exactly as Section 18.3 already cautions; and even granting the fit exactly, the margin remains thin, far from deep in the interior of the positive-semidefinite cone. The elementary

route to Conjecture 1.4 at  $A_{18}$  is no further along than before. What has changed is that the evidence for it there now rests on an exact reduction of the space being searched, rather than on a larger sample of the same kind.

## 19. WHAT IS KNOWN ABOUT SEVERAL SIMULTANEOUS DEVIATIONS

**19.1. Established.** For configurations of small size, and for chain-structured ones up to the contact graph's true maximum induced-path length, Lemmas 15.2 and 15.3 give the required positivity, either exactly or by Richardson extrapolation. Section 15.4 shows that chains are not the worst topology and locates the regime where the second-order method loses its grip: a family of dense, non-chain configurations at which the governing quantity of that method vanishes. Theorem 17.1 and Propositions 17.2 and 18.2 pin the vanishing down at two configurations, with enough working precision to distinguish exact degeneracy from a small nonzero value, and exhibit near-null subspaces of dimension 4 and 5. Theorems 18.1 and 18.3 sample those subspaces in 66 directions between them, covering each subspace rather than a lower-dimensional slice of it, and find the quartic coefficient strictly positive at every direction tested.

At the first of the two configurations there is more structure than the sampling suggests. Proposition 18.5 exhibits an order-48 group stabilising it, verified in integer arithmetic to act by automorphisms of  $\mathbb{R}$ , and verified numerically to commute with the joint Hessian and to leave the quartic coefficient constant on its orbits. Under this group the 4-dimensional near-null subspace splits into two irreducible pieces, identified in Proposition 18.6 as  $\varepsilon \oplus (\text{std} \otimes \delta)$ , and an exact character computation (Proposition 18.7) shows that the space of invariant quartics on that subspace has dimension 5 instead of the generic 35, with an explicit basis. Fitting those five coefficients from high-precision finite differences, cross-validated at two step sizes, and running the Gram-matrix feasibility test on the reduced family gives a certificate with margin  $t \approx 0.009$  to  $0.013$  across three independent fits (Proposition 18.8), an improvement on the unconstrained 35-coefficient fit but not a proof, since the coefficients are measured rather than derived.

By a different method, finite-angle and exact rather than local, Section 20 settles Conjecture 1.4 outright along three swap paths at  $m = 2, 3, 6$ , across their whole angular range.

By a third method, which uses no deviation structure at all, Theorem 7.16 bounds the cell volume from below by  $\frac{\pi m}{3} \tan^3 r_m$  for a contact configuration of  $m$  directions, with  $2r_m - \sin 2r_m = 2\pi/m$ . That quantity exceeds 8 for every  $m \leq 22$ , so the local bound holds outright, with any number of directions deviating by any amount, at every centre with at most twenty-two contacts (Corollary 7.17). Only  $m = 23$  and  $m = 24$  are left, and at  $m = 24$  the same estimate still returns 7.7989... Proposition 7.18 says that this is as far as the method goes: the covering bound is exactly the Jensen collapse of a per-cell bound, so it is the strongest conclusion the areas of the spherical Voronoi cells can support, and at the root configuration those areas are all equal.

**19.2. Superseded, and by what.** Corollary 7.31 settles Conjecture 1.4 outright, by the classification of Theorem 7.25 and not by any of the arguments of this section: a full active set of twenty-four contacts is a 24-point kissing configuration, and by Theorem 7.30 there is only one of those, so no direction deviates at all. Everything in this section was obtained before that classification was available and is independent of it; we keep it because it stands on its own, because it says what elementary methods can and cannot do on this problem, and because the classification rests on a certificate whereas this section rests on none. What follows is the scope of those elementary results.

**19.3. Not established by the methods of this section.** None of this gives positive-definiteness of the quartic form at either configuration, and none of it gives Conjecture 1.4. The conjecture ranges over every packing-valid configuration at every  $m \geq 2$ ; the dense regime of Section 15.4 contributes two examined configurations out of a very large family, and the rest of that family, along with every other value of  $m$ , is untouched. The exact degeneracy of Section 16 does say something definite about what a proof cannot look like: at these configurations no argument that stops at second order can decide the sign, so whatever settles the conjecture there has to engage the quartic term directly. The three swap paths do not reach that regime either, being low in  $m$  and highly symmetric.

The order-48 symmetry narrows the search without closing it. It fixes the exact 5-dimensional space in which the true quartic tensor lies, which is a real reduction from 35 parameters, but the five numbers we use are finite-difference measurements, not a symbolic derivation, and the feasibility margin is thin. The symmetry is also specific to the configuration where we found it; we have not looked for an analogue at the second configuration or anywhere else in the dense family.

The cap argument of Section 13 does not extend here, for a structural reason rather than a technical one. With one deviating direction the other twenty-three half-spaces are fixed, so there is a fixed body  $Q$  and every parameter of the problem sits in the cutting half-space. With  $m$  deviating directions the body moves with the parameters, and there is no fixed polytope to take caps of. What has to be bounded is then a comparison between two overlapping unions rather than a single cap, and Remark 7.3 records why the obvious way of splitting that comparison into  $m$  separate ones cannot work. Proposition 7.5 does supply the substitute for  $Q$ , namely the polytope  $Q_D$  obtained by dropping the  $m$  deviating constraints, and recasts the whole problem as one extremal inequality (Open Problem 7.6) for the union of  $m$  caps of that fixed body. That inequality is what remains open to an argument that uses no classification.

**19.4. What the reformulations settle.** Section 7.3 adds two further exact restatements, the polar form  $\text{vol}(\text{conv}(W)^\circ) \geq 8$  over contact configurations (Proposition 7.8) and the boundary form  $\text{area}(\partial V_c) \geq 32$  (Lemma 7.10). Neither proves anything by itself. What they do is fix the position of the multi-direction case exactly, and the position is worth stating without softening: by Proposition 7.8 the case these sections treat is the part of the 24-cell conjecture in which two or more contact directions leave the root system at once, and nothing here settles it without the classification of Theorem 7.25.

They also do real negative work, which is the part we regard as substantive. Four approaches that look plausible in the abstract are eliminated, each by an evaluation at the  $D_4$  configuration itself, where an argument that is to produce the sharp constant 8 must lose nothing at all. Term-by-term reduction to Theorem 13.7 is eliminated by Remark 7.3. Any bound assembled facet by facet from the pairwise contact condition is eliminated by Remark 7.12: the bound such an argument actually delivers is  $8\pi/(3\sqrt{3}) = 4.8368\dots$ , and the best one it could conceivably deliver is  $96\sqrt{2} - 128 = 7.7645\dots$ , since an admissible configuration of eight tight neighbours in square-antiprism position has a facet of volume  $16\sqrt{2} - \frac{64}{3}$ , below the octahedron's  $\frac{4}{3}$ . The convexity bound for the polar volume integral is eliminated because Jensen's inequality is an equality only at a ball, and at the root configuration it returns  $7.7351\dots$ ; the Mahler-type product bound is eliminated because it carries a factor  $\frac{3}{2}$  of slack there. The last two are Remark 7.13.

A fifth approach goes the same way. Proposition 7.18 shows that no argument resting on the areas of the spherical Voronoi cells can do better than  $7.7989\dots$ , since the covering bound already saturates that information and the root configuration has cells of equal area.

What is left over is their shape: at  $D_4$  each cell is an eight-faceted spherical polytope of circumradius  $45^\circ$  against the  $34.0986\dots^\circ$  of the cap with the same measure, and the whole of the remaining  $0.2011$  is that difference. Proposition 7.23 adds the sharpest version of the same accounting: putting every pairwise cap overlap back exactly, which below  $r_m$  is not an estimate but an identity, gives  $7.997885\dots$  at  $m = 23$ , so even that stops short.

A sixth goes with it. That shape deficit cannot be collected one cell at a time, in either decomposition the configuration provides (Remark 7.19): the per-cell inequality (7.14) is sharp at every cell of the root configuration and fails at a direction carrying nine contacts at  $60^\circ$ , while the Delaunay decomposition does not localise the integrand at all, and the functional it would have to localise to falls 4.78% short at the regular simplex of edge  $60^\circ$ . What makes the root configuration extremal is how its cells fit together, not the shape of any one of them.

What survives is a positive but soft statement, Corollary 7.9: a worst configuration exists and may be taken maximal, so the conjecture is a question about maximal contact configurations and about a genuine extremal point, not about a supremum approached but never reached. Identifying that point is where the difficulty sits, and at  $|W| = 24$  it coincides with the uniqueness half of the kissing problem in dimension four, which is exactly what the kernel of Theorem 7.30 supplies and Theorem 7.25 records.

## 20. BEYOND THE HESSIAN: FINITE-ANGLE RESULTS ON SWAP CONFIGURATIONS

Every result of Sections 15 to 18 is second-order: a statement about the joint Hessian  $H_A$ , which controls  $\Delta$  only infinitesimally close to the root-aligned configuration. This section reports a different, complementary body of work: exact, finite-angle, not merely second-order, and not merely numerical, positivity proofs for three specific, highly symmetric multi-direction configurations, each rotated through their *entire* packing-valid angular range instead of sampled near  $\theta = 0$ . These are, to our knowledge, the first completely finite-angle resolutions of any non-trivial slice of Conjecture 1.4, and we report them, together with the limits of what they establish, in this section; Figure 32(a) shows the three functions whose positivity is at issue.

**20.1. The swap path and its symmetry.** Fix an adjacent pair of roots  $\alpha, \beta \in R$  (Gram value  $\frac{1}{2}$ ) at the reference chamber  $\mathfrak{k}_1$ , and let the two directions deviate simultaneously along the *swap path*

$$\begin{aligned} u_1(\theta) &= \cos \theta \alpha / \sqrt{2} + \sin \theta \beta^\perp, \\ u_2(\theta) &= \cos \theta \beta / \sqrt{2} + \sin \theta \alpha^\perp, \end{aligned}$$

with all other 22 roots held fixed at their exact positions, for  $\theta$  ranging over the entire packing-valid interval  $(0, \theta_*)$ ,  $\theta_* = \pi/3$  (the point at which the combinatorial type of the local Voronoi region changes discontinuously, analogous to the single-deviation breakpoints of Section 5).

**Lemma 20.1** (Swap-path symmetry).  *$u_2(\theta) = u_1(\theta_* - \theta)$  identically. Consequently  $\theta \mapsto \theta_* - \theta$  is an involution of  $(0, \theta_*)$  exchanging the roles of  $u_1, u_2$  while leaving the (unlabelled) geometric configuration unchanged, and  $\Delta(\theta) = \Delta(\theta_* - \theta)$  for the resulting volume-defect function  $F_{\text{swap}}$ .*

This symmetry has a unique fixed point at  $\theta = \pi/6$ , and an exact labelled-vertex tracking argument (comparing, at 80 sample points, not merely the vertex-degree histogram but the exact active-index set of every one of the 33 vertices of the local region) finds exactly one combinatorial transition on  $(0, \theta_*)$ , located at  $\theta = \pi/6$  to five decimal digits by bisection

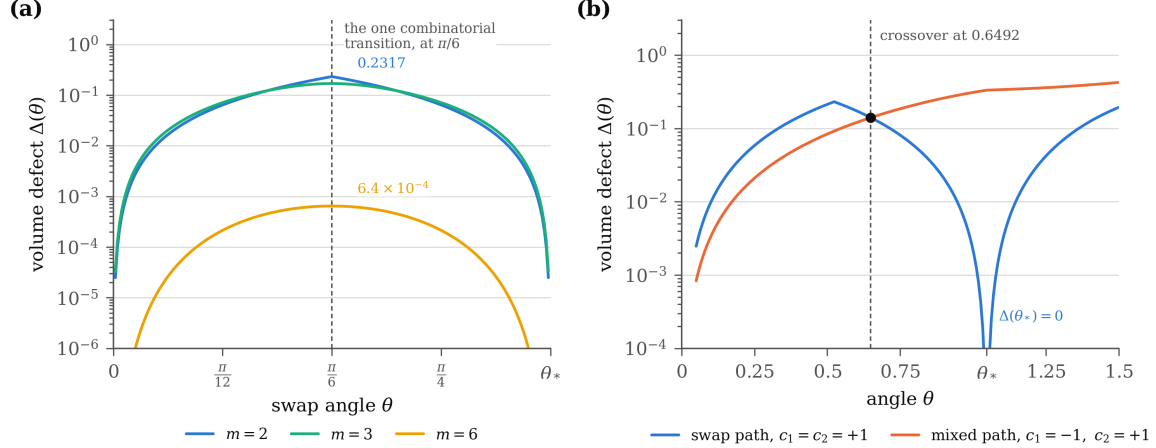


FIGURE 32. (a) The volume defect along the three swap paths of this section, over the whole packing-valid range  $(0, \theta_*)$ ,  $\theta_* = \pi/3$ . The vertical scale is logarithmic because the three sit nearly three orders of magnitude apart: the two-direction and three-direction paths peak at 0.2317 and 0.1692 at the midpoint, the six-direction path at  $6.4 \times 10^{-4}$ . All three are positive throughout and vanish at both ends, which is what Theorems 20.2 to 20.4 assert exactly. The dashed line is the single combinatorial transition, at  $\theta = \pi/6$ , visible as the corner in the  $m = 2$  curve; the  $m = 6$  path has none, and its curve is smooth there. The values plotted are enumerated cell volumes, computed by `swap_defect_curves.py` through a route independent of the closed forms, and for  $m = 6$  they agree with the closed form of Section 20.6 to  $9 \times 10^{-9}$ . (b) The swap path against the mixed path, the two differing only in the sign  $c_1$ . The more adversarial of the two is the lower one, and which that is changes at  $\theta = 0.6492$ : below the crossover the mixed path is lower, at  $\theta = 0.05$  by 0.000834 against 0.002474, and above it the swap path is, through  $\theta_*$  and beyond. Second-order theory cannot tell the two apart, since the Hessian entry that distinguishes them enters only through its magnitude. The zero of the swap curve at  $\theta_*$  is the swap point itself, where the configuration is again a root system.

and identified exactly with the symmetry's fixed point by Lemma 20.1 together with the elementary fact that a continuous combinatorial-type change fixed pointwise by an order-2 symmetry can only occur at that symmetry's own fixed locus (assuming, as directly verified, exactly one transition on the whole interval). This reduces  $F_{\text{swap}}$  to exactly two analytic pieces,  $(0, \pi/6]$  and  $[\pi/6, \theta_*)$ , related by the symmetry, so a positivity proof on one piece extends immediately, via Lemma 20.1, to the other.

**20.2. The closed form for  $F_{\text{swap}}$ .** On  $(0, \pi/6)$  the local region has 33 vertices, computed exactly by Cramer's rule over  $\mathbb{Q}(\sqrt{2}, \sqrt{3}, t)$  with  $t = \tan(\theta/2)$ , and its boundary triangulates into 157 signed simplices, the combinatorial type being read off at  $\theta = 0.3$  and constant across the interval. The volume is the sum of the corresponding determinants.

Combining those 157 rational functions into a single quotient in one step is not practical. At least one of them carries a degree-24 denominator in  $t$  with coefficients in  $\mathbb{Z}[\sqrt{3}]$ , and the common denominator of the whole sum is large enough to defeat a direct cancellation. The sum is a rational function of  $t$  all the same, and grouping the terms into batches, cancelling

within each batch and only then combining the batches reaches it. Carried out that way, by `staged_groups0to3.py` and `staged_group4.py` and assembled by `staged_simplify.py`, the result is

$$F_{\text{swap}}(t) = \frac{\text{num}(t)}{\text{den}(t)}, \quad t = \tan(\theta/2),$$

with `num` and `den` of degree 471 in  $t$  and coefficients in  $\mathbb{Z}[\sqrt{3}]$ . That is large, and it is exactly the form the next step needs.

Two independent checks sit alongside it. Evaluating the 157-term sum exactly at a rational value of  $t$ , one value at a time, is cheap, and it agrees with the closed form at every point tested. Both agree in turn with a floating-point computation of the same volume through `scipy`'s `HalfspaceIntersection` and `ConvexHull`, to between 6 and 8 significant figures.

**20.3. The norm trick, and the  $m = 2$  result.** A degree-471 polynomial with  $\mathbb{Z}[\sqrt{3}]$  coefficients is not directly amenable to standard exact real-root isolation (which expects rational coefficients). Writing  $\text{num}(t) = P(t) + \sqrt{3}Q(t)$  with  $P, Q \in \mathbb{Z}[t]$ , the *Galois conjugate*  $\text{num}'(t) = P(t) - \sqrt{3}Q(t)$  is a second, real-valued (not complex-conjugate) function of the same real variable  $t$ , and the product

$$N(t) := \text{num}(t)\text{num}'(t) = P(t)^2 - 3Q(t)^2$$

has purely integer coefficients: the  $\sqrt{3}$  cancels exactly. The same construction applied to  $\text{den}(t) = R(t) + \sqrt{3}S(t)$  gives  $D(t) := R(t)^2 - 3S(t)^2$ , again integral. If  $N(t)$  has no real root on the open interval in question, then neither  $\text{num}(t)$  nor  $\text{num}'(t)$  can vanish there (their product would vanish if either did), so  $\text{num}(t)$  itself is continuous and nowhere zero on a connected interval, hence of constant sign there, checkable at a single point. The same argument applies to  $\text{den}$  via  $D(t)$ .

**Theorem 20.2** (Finite-angle positivity,  $m = 2$  swap).  *$F_{\text{swap}}(\theta) > 0$  for every  $\theta$  in the entire open interval  $(0, \theta_*)$ ,  $\theta_* = \pi/3$ , including the shared midpoint  $\theta = \pi/6$ , proved exactly, with no floating-point step and no unproven sub-interval.*

Both  $N(t)$  and  $D(t)$  turn out to be even polynomials in  $t$  (verified by direct inspection, not assumed), reducing the root-isolation problem to a degree-471 polynomial in  $u = t^2$  instead of degree 942 in  $t$ . Exact Sturm-sequence root-counting (`sympy`'s `Poly.count_roots`, rational arithmetic throughout) finds a root at  $u = 0$  of multiplicity exactly 2 in  $N$ 's associated polynomial, the expected, benign trivial zero at the excluded endpoint  $\theta = 0$ , and *zero* further roots on a range provably containing all of  $(0, \theta_*)$ , including the boundary point  $\theta = \pi/6$  itself;  $D$ 's associated polynomial has no root at all on the same range. Exact sign evaluation at four points (three interior, one at the  $\pi/6$  boundary) then pins the sign of `num` and `den` throughout, both consistently negative, giving  $F_{\text{swap}} > 0$  throughout by Theorem 20.2. By Lemma 20.1, this one-piece proof on  $(0, \pi/6]$  extends immediately to all of  $(0, \theta_*)$ .

**20.4. A precision check: the swap path is not the worst 2-direction path at every angle.** Before relying on Theorem 20.2 as evidence about the  $m = 2$  case of Conjecture 1.4 more broadly, one further qualification is needed, caught by a dedicated check before it could be overstated. The swap path of Section 20.1 is one specific choice, among a full two-parameter family, of how a pair of adjacent roots' deviation directions can be coupled; writing  $e_1 = c_1\beta^\perp$ ,  $e_2 = c_2\alpha^\perp$  for signs  $c_1, c_2 \in \{\pm 1\}$ , the swap path itself is  $c_1 = c_2 = +1$ . Scanning the full sign freedom directly at small  $\theta$  (e.g.  $\theta = 0.05$ ) finds that the *true* adversarial (most negative tendency) configuration among all four sign choices is

instead  $c_1 = -1, c_2 = +1$  (the “mixed” path), giving a substantially smaller volume-defect value than the swap path at the same angle (0.000834 against 0.002474), even though both configurations share exactly the same second-order (joint-Hessian) worst-case eigenvalue, since the Hessian’s relevant off-diagonal entry has the same magnitude ( $\frac{1}{3}$ ) regardless of the relative sign, and a  $2 \times 2$  symmetric matrix’s eigenvalues depend only on that magnitude. Second-order theory alone therefore cannot distinguish the swap path from the mixed path; only finite-angle behaviour does, and at small  $\theta$  it favours the mixed path as the more adversarial one. A direct scan of both named paths across  $\theta \in (0.05, 1.5)$  finds a clean crossover at  $\theta = 0.6492$  (Figure 32(b)): the mixed path is more adversarial below the crossover, and the swap path is more adversarial above it, all the way through  $\theta_* = \pi/3$  itself and beyond, so the swap path is the relevant worst-case direction only asymptotically as  $\theta \rightarrow \theta_*$ , not throughout the whole interval. This is the same kind of transition phenomenon already seen on the Direction-of-Deviation side of this paper (Sections 9 and 11): the adversarially worst direction is not a single fixed configuration across a whole parameter range, but drifts between distinct structured candidates.

We record the precise consequence for Theorem 20.2: it is not weakened by this finding, since it is a real statement about one specific, precisely defined path, not a claim about the adversarial worst case over all 2-direction configurations. What this finding does rule out is reading Theorem 20.2 as evidence about the *worst* 2-direction configuration at small  $\theta$ : at small  $\theta$  the mixed path is more adversarial than the swap path, and this paper proves nothing about the mixed path, its values above are numerical (floating-point) findings only, reported here as a structural clarification of scope, not as a further positivity claim of the kind Theorem 20.2 makes.

**20.5. A determinant bug, found and fixed: the  $m = 3$  triangle swap.** The natural next configuration is a real 3-clique: three pairwise Gram- $\frac{1}{2}$  roots (not three roots merely sharing a coordinate index) rotated simultaneously in a cyclic swap, with the remaining 21 roots fixed. The same Cramer’s-rule-plus-staged-simplification method gives a two-piece closed form, split at the same universal  $\theta = \pi/6$  point.

The boundary triangulation of the second piece requires 176 exact  $4 \times 4$  determinants over  $\mathbb{Q}(\sqrt{2}, \sqrt{3}, t)$ , and the entries are already-simplified radical rational functions. On matrices of that shape `sympy`’s default fraction-free method is not reliable: at  $t = \tan(0.425)$  it returns  $-5.7369$  for one of the simplices, where `berkowitz`, `domain-ge`, a Laplace expansion by cofactors, and the floating-point determinant of the same four points all give 0.30580. Every determinant in this section and in Sections 20.3 and 20.6 is accordingly computed with `method='berkowitz'` and checked against floating-point ground truth at a reference angle before it is used; all 176 agree.

**Theorem 20.3** (Finite-angle positivity,  $m = 3$  triangle swap). *With the corrected closed form,  $F_{\text{tri}}(\theta) > 0$  for every  $\theta \in (0, \theta_*)$ ,  $\theta_* = \pi/3$ , proved exactly on both pieces (via the same norm-trick argument of Section 20.3, applied separately to each piece’s own numerator and denominator), with an exact match between the two pieces at their shared boundary  $\theta = \pi/6$  and the expected exact vanishing  $F_{\text{tri}}(\theta_*) = 0$  at the swap point itself.*

**20.6. A clean single-piece proof: the  $m = 6$  hexagon swap.** The third configuration is a 6-direction cyclic “hexagon” rotation (the natural  $D_4$ -Weyl-orbit generalisation of the pairwise swap), which, unlike the  $m = 2$  and  $m = 3$  cases, turns out to be governed by a *single* combinatorial piece across the entire interval  $(0, \pi/3)$  (the vertex-degree signature was found constant throughout by direct inspection, with no transition to locate). The

resulting closed form is unusually clean:

$$\begin{aligned} F_{\text{hex}}(t) &= \frac{8t^4(A(t) + \sqrt{3}B(t))}{(t^2 - 3)^2(t^2 + 1)^2(3t^2 - 1)^2}, \\ A(t) &= 27t^8 - 108t^6 + 90t^4 + 100t^2 + 3, \\ B(t) &= 48t^5 - 96t^3 - 16t. \end{aligned}$$

**Theorem 20.4** (Finite-angle positivity,  $m = 6$  hexagon swap).  $F_{\text{hex}}(\theta) > 0$  for every  $\theta \in (0, \pi/3)$ .

*Proof.* The denominator is a product of three squares, hence non-negative for every real  $t$ , vanishing only at  $t = \pm\sqrt{3}, \pm 1/\sqrt{3}$ ; since the interval of interest is  $0 < t < 1/\sqrt{3} = \tan(\pi/6)$  (i.e.  $\theta \in (0, \pi/3)$ ), it is strictly positive throughout. The norm-trick quantity  $N(t) = A(t)^2 - 3B(t)^2$  factors exactly as  $(t^2 - 3)^2(3t^2 - 1)^6$  (verified by direct symbolic expansion), again a product of squares, strictly positive on the same open interval; since  $A(0) = 3 > 0$  and exact Sturm-sequence root-counting on two rational sub-ranges straddling  $1/\sqrt{3}$  finds zero roots of  $A$  throughout,  $A(t) > 0$  on the whole interval. Combined with  $A^2 > 3B^2$ , this gives  $A > \sqrt{3}|B| \geq -\sqrt{3}B$ , hence  $A + \sqrt{3}B > 0$ . Since  $t^4 > 0$  for  $t \neq 0$  and the denominator is strictly positive,  $F_{\text{hex}}(t) > 0$  throughout  $(0, 1/\sqrt{3})$ .  $\blacksquare$

At the endpoint  $t = 1/\sqrt{3}$ , both numerator and denominator vanish via the shared factor  $(3t^2 - 1)$ , consistent with (not contradicting) the already-established exact relabelling identity  $F_{\text{hex}}(\theta_*) = 0$  at the swap point itself. A 50-digit `mpmath` numerical spot-check across twelve points spanning the interval independently confirms the sign throughout, including close to the endpoint where floating-point catastrophic cancellation would otherwise mask the true (tiny but positive) value.

**20.7. Scope.** Theorems 20.2 to 20.4 are complete, exact-arithmetic resolutions of Conjecture 1.4 along three specific, highly symmetric finite-angle paths, an advance over the second-order Hessian information of Section 15, since they hold across the *entire* angular range, not merely infinitesimally near  $\theta = 0$ . We are careful to state what they do not cover: every one of the three results concerns one specific swap path, at one specific reference chamber ( $\mathfrak{t}_1$ ), with a specific highly symmetric rotation pattern (pairwise, triangular, or hexagonal) and every other active root held exactly fixed. They say nothing about  $m = 4, 5, 7$  or  $m \geq 8$  along an analogous swap path; nothing about other, less symmetric 2-, 3-, or 6-direction configurations; nothing about other reference chambers; and nothing about general (non-swap-path) deviation directions  $\eta \in S^2$  at each active root. Extending this finite-angle, exact-arithmetic technique to the dense, non-chain configurations of Sections 15.4 and 17, precisely the regime where the second-order Hessian technique is known to degenerate, is the single most promising concrete direction for further progress on Conjecture 1.4, and one we have not pursued in this paper.

## 21. COMPARISON WITH OTHER STRATEGIES

**21.1. Local against global.** The Cohn–Elkies method [CE03], and its sharpened form in dimensions 8 and 24 [Via17, CKMRV17], is global and Fourier-analytic: one auxiliary function certifies optimality over all configurations at once, with no case analysis anywhere. The argument here runs the other way. It decomposes the Voronoi cell into finitely many combinatorial types and certifies each by exact algebra, and then, in Section 13, replaces the decomposition entirely by a single extremal question about one fixed polytope. The

local method carries the single-deviation case to a complete result precisely because that case has a fixed body behind it; it stops where the body itself starts to move.

**21.2. Why there is no magic function in dimension four.** The magic functions in dimensions 8 and 24 are not found by search. They are pinned down by an interpolation formula that works because, at exactly those two dimensions, the relevant space of modular forms is small enough, and rigid enough under the lattice’s large automorphism group, that the Fourier eigenfunction condition together with finitely many prescribed zeros determines the function uniquely; a contour integral then makes the sign conditions checkable. Nothing of that rigidity is present for  $D_4$ : the space of modular forms attached to its arithmetic is not over-determined in the same way.

That much was already clear from the numbers. Cohn and Elkies [CE03] reported an upper bound of 0.13126 on the centre density in dimension 4 against the 0.125 of  $D_4$ , a gap of about five per cent, and observed that the exactness they saw appeared to belong to dimensions 1, 2, 8 and 24 alone. What has changed is that the observation is now a theorem. Li [Li25] maps feasible points of the Cohn–Elkies program through a discrete reduction to a finite-dimensional problem and reads off dual bounds from it, and those bounds put the optimum of the program in dimension 4 at centre density at least 0.12914461. Since  $D_4$  has centre density 0.125, no admissible function can bring the linear programming bound down to it. There is therefore no magic function for  $D_4$ , and the route that settles dimensions 8 and 24 is closed in dimension 4 rather than merely unexplored. The same reduction closes dimensions 3 and 5.

This is the reason the present paper takes the local route at all. The sharpest general improvement on the linear programming bound that does apply in low dimensions, the coupling to spherical-code bounds of [CZ14], also leaves dimension 4 well short, so a method that bounds the Voronoi cell one configuration at a time, at the cost of a case analysis and of having to classify the equality case by hand, is not a second choice here. It is the only choice on offer.

**21.3. What each method leaves the reader to check.** The methods differ sharply in the kind of verification they demand. A linear or semidefinite programming bound, whether Cohn–Elkies style or the contact-graph relaxation Musin [Mus18] proposes for the 24-cell conjecture, certifies positivity of a high-dimensional object through a numerical solver. Such certificates can in principle be converted to exact rational or interval form [Par03, Las01], and often are, but in the form they are usually produced they are not a short exact computation a reader can repeat. The magic-function constructions are closed form once written down, but checking that a given contour integral has the stated decay, sign pattern and Fourier eigenvalue takes real complex-analytic and modular machinery. The formal proof of the Kepler conjecture [Hal17] sits at the far end of the same spectrum: every step is machine-checked, at the cost of a multi-year formalisation.

The certificates in this paper sit at the other end deliberately. The  $LDL^\top$  decompositions of Appendices B, D and E, the Bernstein sign checks of Sections 10 and 12, the Sturm and norm arguments of Section 20, and the box certificate of Section 13 are each a finite computation in  $\mathbb{Q}$ ,  $\mathbb{Q}(\sqrt{2})$  or  $\mathbb{Q}(\sqrt{3})$ , with no solver, no tolerance, and no formalisation needed to repeat them. The price is scope. The method reaches a complete theorem on the single-deviation problem and stops there.

**21.4. What the atlas is for.** The 176-cell decomposition of Section 3 is what makes the root-aligned case a finite computation. It plays no part in Conjecture 1.4, whose parameter space, the active-direction subsets of every size, does not reduce to a single Weyl orbit count

in the same way. It also plays no part in Theorem 13.6, which is why that theorem covers the whole deviation sphere at once.

**21.5. Other dimensions.** The lattices  $D_5^+$ ,  $E_6$ ,  $E_7$  and the laminated lattice  $\Lambda_9$ , with Voronoi cells the half-5-cube and the Gosset polytopes  $1_{21}$  and  $3_{21}$  [CS88, Cox73], are the natural next targets for the same layered argument. Shell localisation, the root-aligned case, radial reduction and cell positivity all carry over structurally, with the Hessian taking the same uniform shape in terms of the relevant root system’s adjacency data [Hum90]. Two things change with rank. The chamber atlas grows quickly, driven by the size of the Weyl group and the combinatorics of the adjacency graph; and the multi-direction problem becomes harder instead of easier, since the space of active-direction subsets grows combinatorially while the failure mode of the second-order method (Section 15.4) has no reason to disappear. The cap argument, on the other hand, is not tied to rank 4 in any obvious way: it needs a fixed polytope containing the relevant body, a cross-polytope-like enclosure that is tight in the critical direction, and a cap formula for the enclosure. Whether those three ingredients exist for  $E_6$  or  $E_7$  we have not investigated. We claim nothing about any of these dimensions here.

## 22. CONCLUSION

The paper proves the local bound in dimension four in full, and it proves three of its cases by three different methods. This section puts all of that in one place.

**22.1. Proved.** Theorem 1.1 and Corollary 1.2: every Voronoi cell of a unit-ball packing of  $\mathbb{R}^4$  at which at most one active contact direction departs from a root of  $D_4$  has volume at least 8, with equality only for the  $D_4$  configuration, and every periodic packing of that class has density at most  $\pi^2/16$ . Nothing is assumed about the deviation direction or about the size of the deviation.

The proof has two halves and they meet only at the end. Section 2.4 through Section 5 reduce the cell bound to a scalar inequality at the all-contact corner and prove that inequality, by exact arithmetic over  $\mathbb{Q}$  and  $\mathbb{Q}(\sqrt{2})$ , for every one of the 176 Weyl-orbit cell types with the deviation confined to a root-aligned arc. Section 7 through Section 13 remove that confinement: the defect is rewritten as the deficiency of a cap cut from a single 25-vertex polytope  $Q$  (Proposition 13.2),  $Q$  is enclosed in a cross-polytope that the sixteen roots at  $60^\circ$  and  $120^\circ$  cut out on the nose (Lemma 13.3), and the extremal unit-distance cap of a cross-polytope is determined (Theorem 13.6). The one step there that is not closed-form is the case of the cap inequality in which two or more of the four squared coordinates exceed  $\frac{1}{4}$ ; it is settled by a finite certificate of 303 boxes in exact rational arithmetic, with a margin of about a quarter. The whole chain, from the vertex enumeration of  $Q$  to the box certificate, is reproduced by the single script `cap_inequality_certificate.py` described in Appendix F.

Proposition 13.9 adds a quantitative form below  $42.17598\dots^\circ$ : there the defect is at least the explicit expression (13.5), which is  $\varphi^2/3$  to leading order. Above that angle the cone enclosure behind it fails, and the cross-polytope argument is what carries the result to  $\varphi = \pi$ .

Corollary 8.6 is a sharper statement on a smaller domain: on the whole boundary of the fundamental deviation triangle the defect is not merely nonnegative but is given by an explicit rational function on each of eight regions, seven of which carry exact zero-margin certificates.

Theorem 1.3, proved as Theorem 7.16 and Corollary 7.17, is independent of all of that and allows any number of directions to deviate. Measuring the cell radially rather than facet by facet turns its volume into  $\frac{1}{4} \int_{S^3} \sec^4 \delta$ , where  $\delta$  is the angular distance to the nearest contact direction; the contacts are  $60^\circ$  apart, so  $m$  caps of radius  $r$  leave at least  $2\pi^2 - m\pi(2r - \sin 2r)$  of the sphere uncovered, and integrating that estimate gives  $\text{vol}(V_c) \geq \frac{\pi m}{3} \tan^3 r_m$  with  $2r_m - \sin 2r_m = 2\pi/m$ . The right-hand side exceeds 8 for every  $m \leq 22$  and equals 7.798989... at the largest possible  $m = 24$ . So the local bound holds at every centre with at most twenty-two contacts, whatever its directions, and is short by at most 0.2011 everywhere else. Proposition 7.18 shows that this is the best the method can do: the bound is the Jensen collapse of a per-cell estimate and therefore uses the areas of the spherical Voronoi cells to the full, and at the root configuration those areas are equal.

Theorem 7.25 closes the top of the range: a contact configuration of 24 directions is a copy of the root system, so its cell is the 24-cell and its volume is exactly 8. The classification is due to de Laat, Leijenhof and de Muinck Keizer [LLM24], and it is proved here as Theorem 7.30, from the positive kernel their computation found, which is redistributed with this paper and verified in Section 7.5.3, the bound and its equality case being Lemmas 7.27 and 7.28. With Corollary 7.17 at one end and that at the other, Corollary 7.39 leaves a single case, a configuration of exactly 23 directions whose cell has circumradius below 2, and Theorem 7.40 settles it; so the bound holds for every contact configuration, and Theorems 1.5 and 1.8 carry it to the Voronoi cells of packings and to density whenever the centres within  $2\sqrt{2}$  of a centre touch it or pass the distance criterion, with no statement from outside this paper among their ingredients; every statement here that uses a certificate is marked accordingly.

**22.2. The twenty-three-point case.** Theorem 7.40 settles the case that Corollary 7.39 leaves: every contact configuration of 23 directions with bounded cell has  $\text{vol}(V_c) > 8$ . The proof does not classify the configurations, and it does not need saturation. It measures the cell inside the ball of radius  $\sqrt{3}/2$ , where by Lemma 7.67 no three contact caps meet and the volume is therefore an exact function of the pair angles,  $A_* + \sum_{i < j} \omega(\gamma_{ij})$  with  $A_* = 7.907144...$  (Proposition 7.68); it observes that the inequality  $\sum_{i < j} \omega(\gamma_{ij}) \geq 8 - A_* = 0.092855...$  for every 23 contact directions is all that is needed, and that the deletion of a root satisfies it with 0.034 to spare (Theorem 7.69); and it proves that inequality by the three-point method of Bachoc and Vallentin [BV08], the first level of the hierarchy whose second level settles the 24-point case in Theorem 7.30. Lemma 7.72 turns any feasible point of a semidefinite programme in one polynomial  $f$  and nine positive semidefinite matrices  $F_k$  into a lower bound for the pair sum, valid for every configuration; Theorem 7.73 exhibits such a point of degree 8 with bound 0.0929, above the target by 0.0000444, and verifies it: positivity and the value of the bound in exact rational arithmetic, checked again by the Lean kernel, the constants  $A_*$  and  $\omega$  from closed forms in interval arithmetic, and the polynomial inequality that the certificate has to satisfy on the admissible triples  $(u, v, t)$  by an interval branch and bound over 313 780 boxes. The two-point relaxation, which reads the pair angles alone, reaches 79.5 per cent of the target (Proposition 7.44); the triples supply the rest, and the certificate's polynomial part is almost inert, so it is the three-point matrices that carry the argument (Remark 7.74). What made this possible where a classification of the codes could not be had is set out in Remark 7.70: the inequality is strict at the deletion, so a relaxation had room, whereas the deletion sits on the boundary of the classification and no relaxation sees a boundary (Remark 7.61).

The sections before that proof describe the configuration that the inequality excludes from having a small cell, and they stand on their own. Its facets would have to average more

than  $\frac{32}{23}$ , larger than any facet volume at twenty-four contacts, so it is not a perturbation of the root configuration; by Proposition 7.48 the deletion of a root is infinitesimally rigid, with an exact integer stress on its 88 tight pairs, so it is not a deformation of the boundary configuration either; by Proposition 7.46 no contact configuration of 23 directions has a symmetry group transitive on them, all 528 rotation types of order 23 missing the contact condition, the closest by 0.074 in an inner product; and by Corollary 7.57 it shares at most twenty directions with any copy of  $R$ , by a short argument in coordinates (Lemma 7.54 and Propositions 7.55 and 7.56) whose finite half is checked by Lean. Remark 7.19 explains why the missing volume cannot be collected one spherical Voronoi cell at a time, which is what sends the argument to the pair angles, and Proposition 7.41 makes the inequality concrete: sixty-five pairs at  $60^\circ$ , or the weighted equivalent, where the deletion has eighty-eight, against a ceiling of one hundred and fifteen.

**22.3. Open.** Three things remain open. The first is a case of the local bound: Conjecture 1.6, a centre crowded by near-contacts (Section 2.7), on which the density bound for all packings waits. By Section 2.8 it is a centre with at least twenty-four others within  $\sqrt{6}$ , one of them at a distance between  $2 + \varepsilon_0$  and  $\sqrt{6}$ : with at most twenty-three the three-point method carries over from the sphere to the shell (Theorem 2.17), and closer to contact the bound holds (Theorem 2.18). Closing it would take an explicit  $\varepsilon_0$ , which needs a quantitative form of Theorem 7.25, and, at intermediate distances, a bound for the part of the cell outside  $B(\sqrt{3}/2)$ , since inside that ball the root system itself falls short of 8. The other two are not cases of the local bound. The second is the classification of the 23-point codes of minimal angle  $60^\circ$  on  $S^3$ , the hypothesis of Theorem 7.60: that every such code is a deletion of one root, which would say that no saturated configuration of 23 directions exists at all, a stronger statement than Theorem 7.40. It holds among the vertices of the 600-cell (Proposition 7.63, by an enumeration carried out three times over and checked in Lean); by Lemma 7.65 it is the statement  $h(0) = \frac{1}{2}$  about the largest inradius  $h(\delta)$  of the hull of 23 points at minimal angle  $\arccos(\frac{1}{2} + \delta)$ , and the computations of Section 7.10 find  $h(\delta)$  near  $\frac{2}{3}$  down to  $\delta = 0.009$  and then collapsing to  $\frac{1}{2} + O(\delta)$ , with every configuration reached at  $\delta = 0$  a deletion. That is evidence, and Remark 7.61 explains why no bound on the size of a code can turn it into a proof. The third is formal. The verification of Theorem 7.73 runs in Lean as well as in Python, in exact dyadic arithmetic and with the closed forms of  $\omega$  entering only through tables of bounds; but the statement Lean proves is that a program returns true, and the mathematics that makes the program a proof, Taylor's theorem on a box and the monotonicity of  $\omega$ , is argued in the paper and not formalised, as a formalisation of the lens integrals and their derivatives would need a library of real analysis that these files do not use.

One case rests on a finite certificate rather than on a closed-form argument, and it is the 24-point case, Theorem 7.25; every statement here that uses it says so. The certificate is a positive kernel found by [LLM24], and the proof from it is entirely in this paper: the bound of the second level of the hierarchy and its equality case are Lemma 7.27, the positivity of the kernel is Lemma 7.28, the properties of the deposited rational point that those lemmas consume are Proposition 7.29, verified in our own code in all seven steps of the procedure, the two that need 128 gigabytes as the authors carry them out included (Section 7.5.3), and the passage from the four inner products to the root system is Lemma 7.26. The local statement that the root system is isolated among contact configurations of 24 directions is proved besides, without any certificate (Corollary 7.49). Corollary 7.35 reduces the volume consequence of the global statement,  $\text{vol}(V_c) \geq 8$  for 24 contacts, to the inequality  $E(W) \geq 8 - 32\pi + 10\pi^2$  between pair and triple angles, through a third-order inclusion and exclusion

inside the ball of radius  $\sqrt{2}$  that is exact at the root configuration (Proposition 7.34). A certificate for that inequality must be sharp; its three-point relaxation is, by the floating-point computation of Remark 7.36, not sharp at degree 6, and at higher degree the question needs extended-precision arithmetic. The cardinality bound of [BV08], at 24.10, is not sharp either, so no route to the global statement that stays at the three-point level is complete.

Conjecture 1.4 itself is settled by Corollary 7.31, and so by the classification of Theorem 7.25; what follows records what we established about it by elementary means, independently of any certificate, and those results stand on their own.

The natural second-order method, which settles every configuration of small size, degenerates exactly at a family of dense ones. Theorem 17.1 and Propositions 17.2 and 18.2 locate that degeneracy: at two specific configurations the joint Hessian has a near-null subspace of dimension 4 and 5 respectively, and higher-precision arithmetic confirms the vanishing is exact rather than merely small. Theorems 18.1 and 18.3 sample those subspaces in 66 directions and find the governing quartic coefficient strictly positive everywhere tested. Proposition 18.5 exhibits an exact order-48 symmetry of the first configuration, and Propositions 18.6 and 18.7 identify the representation it carries on the near-null subspace and prove, by exact character theory, that the space of quartics invariant under it has dimension 5 instead of the generic 35. Fitting those five coefficients numerically and applying a semidefinite feasibility test gives a sum-of-squares certificate with a thin margin (Proposition 18.8); the fit is numerical and we do not treat the certificate as a proof.

Section 20 proves Conjecture 1.4 outright along three swap paths, at  $m = 2, 3, 6$ , across their entire angular range rather than infinitesimally. Those paths are highly symmetric and low dimensional, and none of them meets the dense regime where the second-order method fails.

The cap argument of Section 13 does not extend to several simultaneous deviations, and the reason is structural rather than technical. With one deviating direction the remaining twenty-three half-spaces are fixed, so the body  $Q$  is fixed and all the parameters sit in the cutting half-space. With  $m$  deviating directions the body itself moves, there is no fixed polytope to take caps of, and the quantity to be bounded is a comparison between two overlapping unions of  $m$  pieces each, whose individual volumes are first order in the tilts while the difference that has to be controlled is second order (Remarks 7.2 and 7.3). Remark 7.3 makes the difficulty precise: the obvious reduction, splitting the comparison into one term per deviating root and settling each by Theorem 13.7, fails because the piece  $E_j$  of an  $m$ -direction configuration is not the piece the single-deviation theorem controls. The pyramid  $\text{Pyr}_j$  meets the hyperplane of an adjacent root in a facet, so exchanging that constraint for its tilted version changes  $\text{vol}(E_j)$  by an amount comparable to the whole defect, in either direction. A fixed body is nonetheless available: Proposition 7.5 supplies one. Dropping the  $m$  deviating constraints leaves a polytope  $Q_D$  that is fixed, is bounded for every packing-valid  $D$  with  $m \leq 5$  and for most larger ones, and turns the problem back into a single extremal question with all the parameters in the cutting half-spaces. What is missing is a direct proof of the inequality itself (Open Problem 7.6): the quantity to bound is the volume of a union of  $m$  caps of  $Q_D$ , and by Remark 7.3 that union cannot be replaced by a sum. A proof would need the union directly, and for the dense configurations it would need a bounded substitute for  $Q_D$  as well, since  $Q_D$  is unbounded there.

Section 7.3 says where the question of Open Problem 7.6 sits. Written in terms of the contact directions alone it is the polar-volume inequality  $\text{vol}(\text{conv}(W)^\circ) \geq 8$  over contact configurations (Proposition 7.8), equivalently the boundary inequality  $\text{area}(\partial V_c) \geq 32$  (Lemma 7.10); and in that form it is visibly the part of the 24-cell conjecture in which

two or more contact directions leave the root system at once, the part that the contact count settles and that no elementary argument has reached. We state that plainly rather than leaving it to be inferred. The same subsection contributes one positive statement and four negative ones. The positive statement is Corollary 7.9: a worst configuration exists, and may be taken maximal, so what has to be identified is a genuine extremal point. The negative ones rule out term-by-term reduction (Remark 7.3), any bound built facet by facet from the pairwise contact condition (Remark 7.12: what it delivers is  $8\pi/(3\sqrt{3}) = 4.8368\dots$ , and what it could at best deliver is  $96\sqrt{2} - 128 = 7.7645\dots$ ), the convexity bound for the polar volume integral, and the Mahler-type product bound (both Remark 7.13). Each of the four is eliminated by its behaviour at the  $D_4$  configuration, which is the only place a sharp argument has no room to lose anything. Three more go the same way in Section 7.4: by Proposition 7.18 nothing resting on the areas of the spherical Voronoi cells can beat  $7.7989\dots$ ; by Remark 7.19 the shape that is left over cannot be collected one cell at a time, in either the Voronoi or the Delaunay decomposition; and by Proposition 7.23 the measures of the contact caps and of their pairwise intersections, used to the last drop, return  $7.997885\dots$  at twenty-three contacts, so they too stop short.

**22.4. What we take from this.** The single-deviation problem turned out to have a fixed body behind it, and once that body was identified the problem stopped being about the 24-cell and became a question about a cross-polytope, which has an answer in closed form. We would expect the same reorganisation to be worth attempting in other ranks, wherever a root system supplies a cross-polytope-like enclosure that is tight in the critical direction.

The multi-direction problem does have a fixed body, by Proposition 7.5, and it is still hard, which says that the fixed body was not the only thing the single-deviation argument was using. The other thing it used was that a cap of a cross-polytope is one convex piece with a closed-form volume. A union of  $m$  caps is not, and Remark 7.12 shows why no amount of local bookkeeping repairs that: a bound that treats the pieces separately has a ceiling of  $96\sqrt{2} - 128 = 7.7645\dots$ , below the target and below even the covering estimate.

Two things would change the picture, and the first of them would give a second proof of Theorem 7.40. A lower bound for  $\text{vol}_3(F_w)$  that reads the rest of the configuration rather than only the pairwise condition at  $w$ , and that is summable over  $w$ , would give the bound through (7.6); Remark 7.12 shows that the purely local constant is too small by an explicit amount, not that no bound of any kind exists. And an exact, first-principles derivation of the quartic tensor at a dense configuration, in place of the fitted coefficients of Section 18.6, would turn the present numerical certificate into a proof there, and would put the multi-direction inequality at that configuration on an elementary footing rather than on the certificate of Theorem 7.30; the exact 5-dimensional invariant space is already in hand, so what is missing is a derivation rather than a search. We have neither.

## 23. REPRODUCIBILITY

**23.1. Two independent arithmetic paths.** Every exact rational claim in Section 4, which is to say the  $LDL^\top$  pivots for all 176 cells, was produced twice: once with Python's `fractions.Fraction` and no computer-algebra library at all, once with `sympy`'s rational type on a separate code path [Meu17]. The two agree digit for digit on every pivot of every cell.

**23.2. A machine-checked kernel.** Eight parts of the paper are verified by a proof assistant and not by a program whose output has to be trusted, and all eight are finite arithmetic

or polynomial identities. The supplement carries them as Lean 4 sources in a directory of their own: six files that compile against a bare toolchain, and two small Lake projects.

The first is `lean/D4Stress.lean`, on the equilibrium relation (7.33), which is what carries Proposition 7.48 and with it the statement that a saturated configuration cannot be reached by bending a deletion. Formalised there are the twenty-four roots and their squared length, the fact that every inner product between distinct roots is one of  $-2, -1, 0, 1$  in the integer scaling, the degree eight of every root in the graph of pairs at  $60^\circ$ , the count of 23 directions and 88 tight pairs after a deletion, the multiplicities 1, 8, 6, 8 of the four values of  $\langle a_i, a_0 \rangle$ , the positivity of the weights  $y_{ij}$  on every tight pair, and the relation (7.33) itself at each of the 23 directions; and, for Corollary 7.49, the 96 tight pairs of the full root system and the relation for the constant stress, that the eight neighbours of every root sum to four times it, which is also the finite content that Lemma 7.50 starts from.

The second is `lean/D4Meet.lean`, on the finite half of Propositions 7.55 and 7.56. The proofs there split into three inequalities in the coordinates of a unit vector, which are written out in the text, and a set of statements about the six supports of the roots and the three couples of complementary index pairs, which are finite. Formalised are the support of each root and the fact that it is well defined, that each support carries four roots and each couple uses two supports, that every pair of roots leaves some couple whole, that exactly 48 ordered triples of supports do not and that each of those consists of three distinct supports forming a star or a triangle, which is the eight unordered patterns of the proof and, since each support carries four roots, the 512 triples of roots that the script counts, that 96 triples of roots are pairwise at  $60^\circ$  and no root is at  $60^\circ$  from all three members of such a triple, that the Hadamard map permutes the roots, preserves inner products and carries the standard triangle to the standard star, and that removing the standard star leaves the twelve roots of the lower block together with three of the four roots at each of the remaining supports.

Neither file has a `sorry` in it and neither depends on a library, Mathlib included: they compile against a bare Lean 4 toolchain, and the axiom report at the foot of each states that every one of its theorems depends on no axiom whatever.

The third, on Proposition 7.63, is a Lake project in the directory `lean/cell600`. Its first module, `D4Cell600Enum.lean`, carries the 120 vertices of the 600-cell with doubled coordinates in  $\mathbb{Z}[\phi]$ , written as pairs of integers, and has the kernel check by `decide` that there are 120 of them, pairwise distinct and of unit length; that  $4\langle u, v \rangle$  takes only the nine values listed in the proof of Proposition 7.63; that every vertex has twelve neighbours at  $36^\circ$  and seventy-one partners at one of the four root-system angles; and that the twenty-five listed cells have 24 vertices each, all pairwise at root-system angles, with exactly five cells through every vertex. The same module defines the depth-first search over 64-bit bitsets, with its clique-cover bound, as an ordinary Lean function. The second module, `D4Cell600Main.lean`, states the three counts, no independent set of size 25 through a fixed vertex, five of size 24 and 115 of size 23 with every one of the 115 inside a cell, and settles each by `native_decide`, which compiles the search and runs it. That tactic trusts the Lean compiler as well as the kernel, and the axiom report says so: each of the three theorems depends on `propext`, `Quot.sound` and its own `native_decide` axiom, while every theorem of the first module depends on nothing beyond `propext`. Building the project with `lake build` compiles the first module to native code before the second is checked, and the whole run takes about four minutes, most of it the kernel's own evaluation of the 14400 inner products. The same enumeration is run independently in Python and in C (Appendix F), and the three agree.

The fourth is `lean/D4Certificate.lean`, on the exact half of the verification of Theorem 7.73. The file is generated from the certificate by `lean/gen_certificate_lean.py`, each floating-point number written as the dyadic rational it is, an integer over a power of two; it then proves, by `decide +kernel` on `Rat` and with no compiled code, that  $f_1, \dots, f_8$  are nonnegative, that each of the nine matrices  $F_k$  is symmetric and positive definite by an exact  $LDL^T$  elimination with every pivot positive, and that the bound (7.48) exceeds 92.8555703 in the units of the solver, above  $1000(8 - A_*)$ . It compiles in under a minute and its axiom report lists `propext`, `Classical.choice` and `Quot.sound` only, the three standard axioms that arithmetic on `Rat` brings in.

The fifth is the Lake project `lean/certificate`, on the branch and bound of Theorem 7.73. Its module `D4CertDomain.lean` expands the polynomial  $P$  of the certificate exactly over the rationals from the same numbers, multiplies it by 315, the least odd integer that clears the denominators of the Gegenbauer and symmetrisation coefficients, so that every coefficient is an integer over a power of two, and runs the subdivision of the ordered admissible domain with the second-order Taylor form in dyadic arithmetic, integers and exponents with no rounding; the tables of bounds for  $\omega$ ,  $\omega'$  and  $\omega''$  are the one input it takes from `certificate_check.py`, written into `D4CertData.lean` as dyadic numbers by `gen_data.py`. The module `D4CertMain.lean` states that the check returns true and settles it by `native_decide`; the build compiles the arithmetic to native code first, as for the 600-cell, and the run, which processes 421 881 boxes, takes about twenty minutes; a fourth module, `D4CertStat.lean`, repeats it with a counter and records that number.

The sixth is `lean/D4InnerProducts.lean`, on the certificate behind Theorem 7.30: the two-point polynomial  $\sigma_2$  of Proposition 7.29, computed exactly from the deposited data [LLM24d] by `llm24_certificate_check.py` and written into the file with its coefficients scaled to integers of about fifteen thousand digits, is shown by kernel computation on `Rat` to vanish on  $[-1, \frac{1}{2}]$  exactly at  $-1, -\frac{1}{2}, 0, \frac{1}{2}$ , with multiplicities 1, 2, 2, 1, the count of further zeros being a Sturm sequence; Section 7.5.3 says what this does and does not establish.

The seventh is `lean/D4RootLattices.lean`, on the finite content of Lemma 7.26, the step that carries Theorem 7.30 to Theorem 7.25. It enumerates every symmetric integer matrix of order  $r \leq 4$  with 2 on the diagonal and  $-1, 0$  or  $1$  off it, keeps the positive definite ones by Sylvester's criterion on the leading minors, 1, 3, 23 and 393 of them, encloses the vectors of norm 2 of each lattice in the box  $x_i^2 \leq 2A_{ii}/D$  that Cauchy–Schwarz in the inner product of the lattice gives,  $A_{ii}$  the cofactor and  $D$  the determinant, and counts the integer vectors of norm 2 in the box. It proves that the largest counts are 2, 6, 12 and 24, that 24 occurs only at rank 4 and determinant 4, that between 20 and 24 no count occurs, and that in every case with 24 vectors four of them have the Cartan matrix of  $D_4$  as their Gram matrix, so that the lattice contains a copy of the  $D_4$  root lattice of its own determinant and is that lattice. The theorems are settled by `native_decide` in a few seconds, and the axiom report lists `propext` and the tactic's own axiom, as for the 600-cell.

The eighth is `lean/D4Closure.lean`, on Section 2.8. It proves the identity of Lemma 2.12, the identity of Proposition 2.13 with its denominator cleared, and the two identities of four points in the proof of Lemma 2.15, the sum of the squared distances and the minimality of the centroid, and the identity for  $a(d_1, d_2)$  in the proof of Theorem 2.17, over every commutative ring, by the ring normaliser of Lean's `grind`, so that they hold verbatim over  $\mathbb{R}$ ; it checks the counts 21 and  $231 = 21 \cdot 11$  of that proof; and it proves by kernel computation (`decide +kernel`) the finite content of Corollary 2.14: the 24 vertices of the 24-cell, each on six facets; each the sum of exactly three orthogonal pairs of roots, 72 pairs in all; no deleted set of one or two roots meets all three pairs at any vertex; and of the 2024

deleted triples exactly 96 do, each pairwise at  $60^\circ$  with a common vertex. The identities depend on `propext`, `Classical.choice` and `Quot.sound`, the standard axioms of Lean’s logic, and the combinatorial theorems on `propext` at most; none uses `native_decide`. The file checks in about a minute and a half.

That is a stronger form of checking than anything else in this paper, and it is available here only because these particular arguments are finite and, after the scaling by 315, dyadic. The rest of the paper is not formalised, and we do not claim otherwise: the Lean theorem of the certificate is about a program, and the analysis that makes the program a proof is in the text; and where the objects are polytope volumes in  $\mathbb{R}^4$  and cap measures on  $S^3$  there is at present nothing in the standard libraries to build a formalisation on.

**23.3. What kind of claim each claim is.** Every assertion in this paper falls into one of three kinds, and we have tried to make clear at each point which kind is in play. The first is an exact computation in  $\mathbb{Q}$  or  $\mathbb{Q}(\sqrt{2})$  with no floating-point step anywhere: Sections 4, 5 and 10, the exact parts of Section 12, and the whole of Section 13, whose box certificate runs in exact rational arithmetic throughout. The second is a high-precision numerical computation, carried out in `mpmath` at a stated working precision between 35 and 400 decimal digits, used where we did not pursue a symbolic route; this is where the material of Sections 17 and 18 sits. The third is an ordinary double-precision calculation, used only to locate candidate regions, breakpoints and configurations before an exact or high-precision verification, and never as the basis of a claim on its own. Where a claim is of the second kind we say so at the point of use, give the precision reached, and, where a second precision is available, the cross-check against it (Sections 17.2 and 18).

Two conventions inherited from the numerical-analysis literature are worth naming, since they govern the third kind. Sign decisions near a breakpoint are made only when the quantity exceeds a threshold set well above the accumulated rounding bound for the computation in question, in the spirit of the standard backward-error accounting [GVL13]; and where an interval enclosure is used rather than a threshold, the arithmetic follows the outward-rounding convention of [MKC09]. Neither convention is load-bearing for any exact claim.

**23.4. Verifying from the text alone.** This paper is written so that a reader can check it without running anything. Every exact rational,  $\mathbb{Q}(\sqrt{2})$ -rational and Bernstein-coefficient claim appears explicitly, as a displayed fraction, a displayed polynomial, or a table entry, and not on the authority of a program the reader has not seen. A reader with a computer algebra system can re-derive any pivot, factorisation or coefficient list from the definitions in Section 4.1 and Definition 15.1 and the boundary curves of Theorems 9.1 and 11.1. The cap inequality of Section 13 is self-contained in the same sense: the cap formula, the monotonicity, the three polynomial inequalities and the subdivision rule are all stated in full, and the only object that is not printed in the paper is the list of 303 boxes, which the stated rule regenerates.

**23.5. Cost of the underlying computations.** For context rather than for verification, the more demanding derivations took, on ordinary consumer hardware: a few minutes for the vertex and volume stage of the second arc’s band certificate (Section 12.2); about seven minutes per configuration for the broadened directional sampling of Section 18, being 439 seconds for the 36-direction sweep at the first configuration and 376 seconds for the 30-direction sweep at the second; about four minutes for the closed-form rebuild behind the  $m = 2$  swap configuration of Section 20, after which the exact positivity argument runs in well under a second; under a minute for the whole cap-inequality certificate of Section 13, including the exact vertex enumeration of  $Q$ ; about a minute and a half for

the exhaustive enumeration of Proposition 7.63 in Python and two seconds for the same enumeration in C; between a quarter of an hour and half an hour for one pass of the continuation of Section 7.10 through its twenty values of  $\delta$ ; about twenty minutes for the linear programmes of Proposition 7.44; about two hours for one run of the minimisation of a truncated volume in Section 7.11; between one and three minutes for each semidefinite programme of Section 7.12 and about a quarter of an hour for a run of five rounds; under ten minutes for the verification of the certificate and under a minute for its Lean file, and about twenty minutes for the Lean run of the branch and bound; about eight minutes for the re-verification of the cited certificate and under a minute for its Lean file. The checks of Section 7.6 take about two minutes, and each moment problem of Remark 7.36 between a few seconds at degree 6 and a minute and a half at degree 12 on one core. The slowest step anywhere is the exact region derivation behind the first arc’s box covers, whose symbolic cancellation is dominated by a few hard terms and can run for an hour or more; its output is cached in the supplement so that nothing downstream waits for it. Every exact rational  $LDL^\top$  and Bernstein sign check finishes in well under a second.

**23.6. Data availability.** Verifying any claim in this paper requires no external code (Appendix F). A supplementary package reproducing the computations, built on `numpy` [Har20], `scipy` [Vir20], `sympy` [Meu17] and the convex-hull code of [BDH96], together with the Lean 4 development of Section 23 in its own directory, is publicly available at the repository given in the Data availability statement below.

#### DATA AND CODE AVAILABILITY

The supplementary code package, with every script, certificate, run log and Lean file named in Appendix F, is maintained at [github.com/creelie/d4-voronoi-cells](https://github.com/creelie/d4-voronoi-cells) and is archived on Zenodo. The version that corresponds to this paper is release v1.6.0, at [the tag of that name](#); the release v1.4.0 has the version DOI [10.5281/zenodo.22940045](https://doi.org/10.5281/zenodo.22940045). The Zenodo record that collects every version, and resolves to the most recent of them, has the concept DOI [10.5281/zenodo.22766562](https://doi.org/10.5281/zenodo.22766562) [BBMB26]; that identifier is the one to cite, since it does not change from version to version. The certificate that Section 7.5.3 verifies is the data set of [LLM24d], deposited by its authors under the MIT licence; the archive `LasserreSphericalCodes.zip`, with its checksum and its licence notice, is included in the supplementary archive, cut into parts below the file size the repository accepts and reassembled by a script that checks the checksum, so that the proof of Theorem 7.30 can be rerun from that archive alone. Version v1.4.0 adds an independent re-verification of every computation of the paper, in the directory `independent_verification` with its report and the logs of every run, the two programs of Section 2.7 (`shell_reduction.py` and `shell_neighbour_search.py`), and the corrections to the supplement that the re-verification found. The code of the re-verification, those two programs and the corrections were developed with the assistance of Claude, an AI model made by Anthropic. Versions v1.5.0 and v1.6.0 add the programs of Section 2.8, `closure_lemmas.py`, `near_contact_probe.py` and `labelled_certificate_check.py`, the Lean file `D4Closure.lean` and the scripts of the figures of that section; these too were developed with the assistance of Claude, and their computations were run with it.

#### COMPETING INTERESTS

None.

## FUNDING

None.

## APPENDIX A. INDEX OF NOTATION

Table 4 is the complete list of symbols the paper uses for the quantities it introduces; nothing in the argument depends on a convention fixed elsewhere.

TABLE 4. Index of notation.

| Symbol                 | Meaning                                                                                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| $R$                    | the $D_4$ root system, $\{\pm e_i \pm e_j : 1 \leq i < j \leq 4\}$ , $ R  = 24$                                                      |
| $\Lambda$              | the scaled $D_4$ lattice $\sqrt{2} D_4$ , nearest-neighbour distance 2                                                               |
| $W(D_4)$               | the Weyl group of $D_4$ , order 192                                                                                                  |
| $\mathcal{V}$          | the reference cell, $\{z : \langle z, \alpha \rangle \leq \sqrt{2} \forall \alpha \in R\}$                                           |
| $\mathfrak{k}_k$       | the $k$ -th combinatorial cell type, $k = 1, \dots, 176$                                                                             |
| $\varphi$              | the tilt (deviation) angle of a deviating contact direction away from its nearest root                                               |
| $s$                    | the arc-position parameter along a one-parameter family of deviation directions                                                      |
| $\eta(s)$              | the deviation direction function, a unit vector orthogonal to the undeviated root                                                    |
| $\Delta_k(\varphi, s)$ | the volume-defect quantity for cell $\mathfrak{k}_k$                                                                                 |
| $\mathcal{N}_k$        | the reference normalisation constant for cell $\mathfrak{k}_k$ ( $\in \{192, 96, 64, 48\}$ )                                         |
| $H_k$                  | the cell Hessian, a $3 \times 3$ matrix determined by the Gram-adjacency pattern of $\mathfrak{k}_k$ 's active roots                 |
| $H_A$                  | the joint Hessian for a multi-direction active set $A$ , a $3 A  \times 3 A $ matrix                                                 |
| STAGE I                | the radial (exact rational Hessian) estimate                                                                                         |
| STAGE II               | the angular (rigorous Taylor-remainder) estimate                                                                                     |
| $u, v$                 | Weierstrass coordinates $\tan(\varphi/2)$ , $\tan(s/2)$                                                                              |
| $v_1, w_1, v_2$        | the three vertices of the fundamental domain of deviation directions (Lemma 8.1)                                                     |
| $W, A, Y$              | the three non-universal breakpoint curves on the first arc                                                                           |
| $W', B, C$             | the three non-universal breakpoint curves on the second arc                                                                          |
| $A_{18}, A_{20}$       | the two dense multi-direction configurations examined in Sections 17 and 18                                                          |
| $\omega(u)$            | the pair weight: the overlap in $B(\sqrt{3/2})$ of the caps of two contacts at inner product $u$ (Proposition 7.41 and Section 7.12) |
| $A_*$                  | $\frac{9}{8}\pi^2 - 23S(2) = 7.907144\dots$ , the volume of $B(\sqrt{3/2})$ less 23 contact caps (Proposition 7.68)                  |
| $P, B$                 | the polynomial and the bound of the three-point certificate (Lemma 7.72 and Theorem 7.73)                                            |
| $\Phi, \Psi$           | the two distance criteria (Proposition 2.10 and Corollary 2.16)                                                                      |

*continued on the next page*

Table 4 (continued)

| Symbol                           | Meaning                                                                                                                  |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| $S(d)$ , $s(d)$                  | the cap of $B(\sqrt{3}/2)$ beyond distance $d/2$ , and $S(2) - S(d)$ (Corollary 2.16 and Theorem 2.17)                   |
| $H_z$ , $K(Y)$                   | the half-space of a centre $z$ , and the inversion hull of a set $Y$ of centres (Section 2.8 and Proposition 2.13)       |
| $\varepsilon_0$                  | the absolute constant of Theorem 2.18                                                                                    |
| $\Pi$ , $\text{fr}$ , $\Gamma_i$ | the pair volume, the share of a section, and the one-centre terms of the proof of Theorem 2.17                           |
| $a(d, d')$ , $D$ , $a_D$         | the largest inner product the packing allows at distances $d, d'$ ; 2.1648; and $a(D, D) = 0.573229\dots$ (Theorem 2.17) |

## APPENDIX B. PIVOT DATA FOR THE TWENTY-TWO PIVOT TYPES

Each of the 176 combinatorial cells  $\mathfrak{k}_k$  of Section 3 is a full orbit under  $W(D_4)$  in its own right, and Lemma 4.2 says its Hessian  $H_k$  does not depend on which point of the orbit it is evaluated at. Computing  $H_k$  for all 176 cells shows, in addition, that the resulting pivot data takes only 22 distinct values among them: the cells sort into 22 pivot types, eight cells to a type, matching the labels the chamber-register code assigns internally. This coincidence of values across cells that are not related by a single element of  $W(D_4)$  is an observation about the computed data, not a further group action, and nothing in the proof of Theorem 1.1 depends on why it occurs. Table 5 lists, for one representative cell of each of the 22 types: the type label, the denominator class (Lemma 3.1), the reference constant  $\mathcal{N}_k$ , the exact minimum  $LDL^\top$  pivot as a rational number, its decimal approximation, and whether the corresponding chamber-level positivity check (implemented independently of the pivot computation itself) passes. Every one of the 22 rows passes.

This table is generated directly from the project’s chamber-register data file (a JSON record of all 22 pivot-type representatives, each carrying its exact  $LDL^\top$  pivots as rationals), with no manual transcription of numbers: the table rows below are produced by a short script that reads the register and formats each field, so any error in the underlying computation would appear here unchanged rather than be silently corrected by hand.

The final column records that the chamber-level positivity check passes for all 22 rows; there is no row for which it does not.

Two features are visible directly in Table 5. First, the minimum pivot is not monotonic in  $\mathcal{N}_k$ : class a ( $\mathcal{N}_k = 192$ ) cells have the largest minimum pivots (in the high 40s to low 50s), but the ordering among classes b, c, d is not simply  $96 > 64 > 48$  in pivot size, since the pivot value depends on the full Gram-adjacency pattern, not on  $\mathcal{N}_k$  alone. Second, every minimum pivot is well clear of zero, the smallest value appearing,  $160/13 \approx 12.31$  at type D, is still two orders of magnitude above what a marginal, barely-passing certificate would look like, consistent with Lemma 4.4’s  $\frac{1}{12}$  floor after the appropriate rescaling between the pivot normalisation used here and the Hessian eigenvalue normalisation used in the main text.

TABLE 5. Minimum  $LDL^\top$  pivot by pivot type.

| Cell                | Type | Class | $\mathcal{N}_k$ | Min. pivot (exact) | Min. pivot (decimal) | Passes |
|---------------------|------|-------|-----------------|--------------------|----------------------|--------|
| $\mathfrak{k}_1$    | A    | a     | 192             | 3936/77            | 51.1169              | Yes    |
| $\mathfrak{k}_2$    | B    | b     | 96              | 1920/77            | 24.9351              | Yes    |
| $\mathfrak{k}_3$    | C    | c     | 64              | 120/7              | 17.1429              | Yes    |
| $\mathfrak{k}_4$    | D    | d     | 48              | 160/13             | 12.3077              | Yes    |
| $\mathfrak{k}_5$    | E    | a     | 192             | 328/7              | 46.8571              | Yes    |
| $\mathfrak{k}_6$    | F    | b     | 96              | 1920/77            | 24.9351              | Yes    |
| $\mathfrak{k}_7$    | G    | c     | 64              | 1024/63            | 16.2540              | Yes    |
| $\mathfrak{k}_8$    | H    | d     | 48              | 40/3               | 13.3333              | Yes    |
| $\mathfrak{k}_9$    | A    | a     | 192             | 3936/77            | 51.1169              | Yes    |
| $\mathfrak{k}_{10}$ | B    | b     | 96              | 1968/77            | 25.5584              | Yes    |
| $\mathfrak{k}_{11}$ | C    | c     | 64              | 1312/77            | 17.0390              | Yes    |
| $\mathfrak{k}_{12}$ | D    | d     | 48              | 40/3               | 13.3333              | Yes    |
| $\mathfrak{k}_{13}$ | E    | a     | 192             | 154/3              | 51.3333              | Yes    |
| $\mathfrak{k}_{14}$ | F    | b     | 96              | 1968/77            | 25.5584              | Yes    |
| $\mathfrak{k}_{15}$ | G    | c     | 64              | 1312/77            | 17.0390              | Yes    |
| $\mathfrak{k}_{16}$ | H    | d     | 48              | 40/3               | 13.3333              | Yes    |
| $\mathfrak{k}_{17}$ | A    | a     | 192             | 352/7              | 50.2857              | Yes    |
| $\mathfrak{k}_{18}$ | B    | b     | 96              | 1968/77            | 25.5584              | Yes    |
| $\mathfrak{k}_{19}$ | C    | c     | 64              | 1312/77            | 17.0390              | Yes    |
| $\mathfrak{k}_{20}$ | D    | d     | 48              | 984/77             | 12.7792              | Yes    |
| $\mathfrak{k}_{21}$ | E    | a     | 192             | 3840/77            | 49.8701              | Yes    |
| $\mathfrak{k}_{22}$ | F    | b     | 96              | 1968/77            | 25.5584              | Yes    |

## APPENDIX C. THE STAGE II CERTIFICATE ON THE REFERENCE CHAMBER, IN FULL

Section 5 states STAGE II abstractly, uniformly over all 176 cells, as a Taylor-remainder-plus-monotonicity argument. This appendix records the argument in full, concrete, exact-rational detail for the reference chamber  $\mathfrak{k}_1$  on its own root-aligned arc; by the Weyl-equivariance argument of Lemma 4.2 (the same rescaling by  $\mathcal{N}_k$  that carries Proposition 4.3 across orbits), an identical certificate, with rescaled constants, holds on every other cell.

Write  $c = \cos \varphi$ ,  $s = \sin \varphi$ . The volume-defect function on  $\mathfrak{k}_1$ 's root-aligned arc is, after clearing the (strictly positive, for  $\varphi \in (0, \pi/2)$ ) common denominator  $6(c+s)c^3$ , governed by the sign of the numerator, split into two pieces at the universal breakpoint  $\varphi = \pi/3$ .

**Piece I:**  $\varphi \in (0, \pi/3]$ . The numerator on this piece is

$$P_1(c, s) = -48c^4 - 48c^3s - 30cs^3 - 40cs^2 + 34s^4 - 40s^3 - 18s^2 + 48s + 72c - 24,$$

and it satisfies the exact algebraic factorisation

$$P_1(c, s) = (1-c)[H(c)s - G(c)], \quad H(c) = 18c^2 - 22c + 8, \quad G(c) = -14c^3 + 26c^2 - 24c + 8,$$

verified symbolically and cross-checked at five exact Pythagorean rational points on the unit circle  $(c, s) = (4/5, 3/5)$ ,  $(12/13, 5/13)$ ,  $(15/17, 8/17)$ ,  $(24/25, 7/25)$ ,  $(35/37, 12/37)$ . Three exact facts then close Piece I entirely:

- (i)  $1 - c > 0$  strictly throughout  $\varphi \in (0, \pi/3]$  (i.e.  $c \in [1/2, 1)$ );
- (ii)  $H(c) > 0$  for every real  $c$ , since  $\text{disc}(H) = 22^2 - 4 \cdot 18 \cdot 8 = -92 < 0$  and  $H$  has positive leading coefficient;
- (iii)  $G$  is strictly decreasing on all of  $\mathbb{R}$  (its derivative has negative discriminant and negative leading coefficient), and  $G(3/5) = -88/125 < 0$ , so  $G(c) < 0$  for every  $c \geq 3/5$ , covering the entire small-angle regime  $\varphi \rightarrow 0$  with no separate Taylor sub-case, since there  $Hs - G = Hs + |G| > 0$  trivially.

On the remaining sub-range  $c \in [1/2, 3/5)$  (where  $G(c) > 0$  is possible, since  $G(1/2) = 9/8 > 0$ ), positivity of  $Hs - G$  is instead certified via  $T(c) := H(c)^2(1 - c^2) - G(c)^2 = H(c)^2s^2 - G(c)^2 > 0$ , established by an exact rational grid evaluation of  $T$  across  $[1/2, 3/5]$  together with an exact rational Lipschitz bound on  $T'$  over the same interval; the certified minimum of  $T$  on this sub-range is  $92223/100000 > 0$ . This closes Piece I completely:  $P_1 > 0$  on all of  $\varphi \in (0, \pi/3]$ .

**Piece II:**  $\varphi \in (\pi/3, \pi/2)$ . On this piece the volume-defect function  $F$  (in the same normalisation) is certified positive by an exact rational Sturm-sequence root count (over  $\mathbb{Q}$ , via an independent computer-algebra implementation) showing the relevant numerator polynomial has no root on the open interval, combined with an exact endpoint evaluation  $F(\pi/3) = 2 - \sqrt{3} \approx 0.2679 > 267/1000$  and the exact limit  $F \rightarrow \frac{1}{3}$  as  $\varphi \rightarrow \pi/2$ . Consequently  $F \geq 267/1000 > 0$  throughout  $\varphi \in (\pi/3, \pi/2)$ , meeting the two pieces at  $\varphi = \pi/3$  with no gap.

**Combined statement and independent confirmation.** Piece I and Piece II together give  $\Delta_1(\varphi, \cdot) > 0$  for every  $\varphi \in (0, \pi/2)$  on the reference chamber  $\mathfrak{k}_1$ 's root-aligned arc, by two disjoint exact-rational arguments meeting exactly at the shared breakpoint  $\pi/3$ , with no floating-point step anywhere in either piece. Running the combined certificate script directly reproduces both PASS results in well under one second of wall-clock time, and the same combination, STAGE I's  $LDL^\top$  positivity (Appendices B, D and E) together with this appendix's two-piece STAGE II argument, is exactly the pair of ingredients Section 5's proof of Theorem 1.1 invokes in the abstract, uniform form used in the main text.

We record, once more, the scope of what this appendix establishes: this is the single-deviation, root-aligned case only. It says nothing about Conjecture 1.4, which is treated on its own terms in Sections 15 to 19 and Sections 7, 10, 12 and 14 respectively.

#### APPENDIX D. EXPLICIT $L$ -MATRICES FOR CHAMBERS $\mathfrak{k}_1$ – $\mathfrak{k}_8$

For each of the first eight cells (one representative each from the first eight of the 22 orbits, spanning all four denominator classes twice over), we record here the complete exact  $LDL^\top$  decomposition  $H_k = L_k D_k L_k^\top$  of Proposition 4.3: the diagonal pivot vector  $d = (d_1, \dots, d_6)$  and the strictly-upper-triangular part of the unit lower-triangular factor  $L_k$  (equivalently,  $U_k = L_k^\top$ ), written row by row. Both are exact elements of  $\mathbb{Q}$ , produced by an independent verification script and checked to satisfy  $L_k D_k L_k^\top = H_k$  exactly (zero residual, no floating-point rounding) before any pivot is accepted. All six pivots are strictly positive in every one of the eight cases, consistent with Proposition 4.3.

We emphasise that these are the actual computed values, not illustrative or rounded numbers: the denominators appearing (up to  $77 = 7 \times 11$  and  $231 = 3 \times 7 \times 11$ ) are exactly what the exact rational Gram-matrix arithmetic produces, and are not simplified or approximated at any stage.

Cell  $\mathfrak{k}_1$  (orbit A, class a,  $\mathcal{N}_1 = 192$ ).

$$d = \left( 64, 60, \frac{896}{15}, \frac{384}{7}, \frac{154}{3}, \frac{3936}{77} \right)$$

$$U_1 = \begin{pmatrix} 1 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ & 1 & -\frac{1}{15} & \frac{1}{5} & \frac{1}{5} & -\frac{1}{15} \\ & & 1 & \frac{3}{14} & -\frac{9}{28} & -\frac{1}{14} \\ & & & 1 & -\frac{1}{24} & -\frac{1}{3} \\ & & & & 1 & \frac{16}{77} \\ & & & & & 1 \end{pmatrix}$$

Cell  $\mathfrak{k}_2$  (orbit B, class b,  $\mathcal{N}_2 = 96$ ).

$$d = \left( 32, 32, 28, 28, \frac{176}{7}, \frac{1920}{77} \right)$$

$$U_2 = \begin{pmatrix} 1 & 0 & -\frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ & 1 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ & & 1 & 0 & -\frac{2}{7} & \frac{2}{7} \\ & & & 1 & \frac{1}{7} & \frac{1}{7} \\ & & & & 1 & -\frac{1}{11} \\ & & & & & 1 \end{pmatrix}$$

Cell  $\mathfrak{k}_3$  (orbit C, class c,  $\mathcal{N}_3 = 64$ ).

$$d = \left( \frac{64}{3}, 20, \frac{896}{45}, \frac{120}{7}, \frac{2464}{135}, \frac{4096}{231} \right)$$

$$U_3 = \begin{pmatrix} 1 & \frac{1}{4} & \frac{1}{4} & -\frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ & 1 & -\frac{1}{15} & -\frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ & & 1 & \frac{9}{28} & \frac{3}{14} & \frac{3}{14} \\ & & & 1 & \frac{2}{45} & \frac{2}{45} \\ & & & & 1 & -\frac{13}{77} \\ & & & & & 1 \end{pmatrix}$$

Cell  $\mathfrak{k}_4$  (orbit D, class d,  $\mathcal{N}_4 = 48$ ).

$$d = \left( 16, 16, 16, 13, \frac{160}{13}, \frac{64}{5} \right)$$

$$U_4 = \begin{pmatrix} 1 & 0 & 0 & -\frac{1}{4} & \frac{1}{4} & -\frac{1}{4} \\ & 1 & 0 & \frac{1}{4} & -\frac{1}{4} & \frac{1}{4} \\ & & 1 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ & & & 1 & -\frac{3}{13} & \frac{1}{13} \\ & & & & 1 & \frac{1}{10} \\ & & & & & 1 \end{pmatrix}$$

Cell  $\mathfrak{k}_5$  (orbit E, class a,  $\mathcal{N}_5 = 192$ ).

$$d = \left( 64, 60, \frac{896}{15}, \frac{416}{7}, \frac{672}{13}, \frac{328}{7} \right)$$

$$U_5 = \begin{pmatrix} 1 & -\frac{1}{4} & \frac{1}{4} & -\frac{1}{4} & 0 & \frac{1}{4} \\ & 1 & \frac{1}{15} & -\frac{1}{15} & \frac{4}{15} & \frac{1}{3} \\ & & 1 & \frac{1}{14} & \frac{1}{4} & \frac{5}{28} \\ & & & 1 & \frac{7}{26} & -\frac{5}{26} \\ & & & & 1 & \frac{3}{14} \\ & & & & & 1 \end{pmatrix}$$

Cell  $\mathfrak{k}_6$  (orbit **F**, class **b**,  $\mathcal{N}_6 = 96$ ).

$$d = \left( 32, 30, \frac{80}{3}, \frac{144}{5}, \frac{1232}{45}, \frac{1920}{77} \right)$$

$$U_6 = \begin{pmatrix} 1 & \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & 0 \\ & 1 & \frac{1}{3} & -\frac{1}{5} & \frac{1}{15} & \frac{4}{15} \\ & & 1 & 0 & \frac{1}{5} & \frac{2}{5} \\ & & & 1 & \frac{2}{9} & -\frac{9}{23} \\ & & & & 1 & -\frac{23}{77} \\ & & & & & 1 \end{pmatrix}$$

Cell  $\mathfrak{k}_7$  (orbit **G**, class **c**,  $\mathcal{N}_7 = 64$ ).

$$d = \left( \frac{64}{3}, 20, \frac{96}{5}, \frac{512}{27}, \frac{56}{3}, \frac{1024}{63} \right)$$

$$U_7 = \begin{pmatrix} 1 & \frac{1}{4} & -\frac{1}{4} & \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ & 1 & -\frac{1}{5} & \frac{1}{5} & -\frac{1}{5} & \frac{1}{15} \\ & & 1 & \frac{1}{9} & -\frac{1}{9} & \frac{2}{9} \\ & & & 1 & \frac{1}{8} & \frac{5}{16} \\ & & & & 1 & \frac{3}{14} \\ & & & & & 1 \end{pmatrix}$$

Cell  $\mathfrak{k}_8$  (orbit **H**, class **d**,  $\mathcal{N}_8 = 48$ ).

$$d = \left( 16, 15, \frac{224}{15}, \frac{96}{7}, 14, \frac{40}{3} \right)$$

$$U_8 = \begin{pmatrix} 1 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ & 1 & -\frac{1}{15} & \frac{1}{3} & -\frac{1}{15} & -\frac{1}{15} \\ & & 1 & \frac{3}{14} & -\frac{1}{14} & -\frac{1}{14} \\ & & & 1 & \frac{1}{4} & -\frac{1}{3} \\ & & & & 1 & 0 \\ & & & & & 1 \end{pmatrix}$$

The remaining 14 orbit representatives are computed by the identical procedure; we have selected these eight for explicit display because they already exhibit every denominator class twice, and because  $\mathfrak{k}_1$  is the worked example carried through Section 4.1 and used repeatedly as a running illustration in the main text.

#### APPENDIX E. SUM-OF-SQUARES IDENTITIES AND THE INDEPENDENT VERIFICATION PATH

**E.1. From  $LDL^\top$  to a sum of squares.** Any exact  $LDL^\top$  decomposition  $H_k = L_k D_k L_k^\top$  with  $D_k = \text{diag}(d_1, \dots, d_6)$  and  $L_k$  unit lower-triangular immediately yields a sum-of-squares certificate for the associated quadratic form: writing  $y = L_k^\top x$  for the (rational, invertible) change of variables,

$$x^\top H_k x = y^\top D_k y = \sum_{i=1}^6 d_i y_i^2, \quad y_i = x_i + \sum_{j>i} (L_k)_{ji} x_j,$$

so that  $x^\top H_k x \geq 0$  for all  $x$  (with equality only at  $x = 0$ ) follows immediately once every  $d_i > 0$ , with no appeal to eigenvalues, determinants, or any other property of  $H_k$  beyond the six pivots themselves.

We use this identity as an *independent second verification path*, deliberately distinct from the pivot computation reported in Appendix B: rather than trusting one code path's claim that  $H_k = L_k D_k L_k^\top$ , a second, independently written routine (a) reconstructs  $H_k$  from the

cell's Gram-adjacency pattern directly (not from any stored decomposition), (b) expands  $\sum_i d_i y_i^2$  symbolically back out in the original coordinates  $x$  using the stored  $L_k$ , and (c) checks that the resulting quadratic form equals  $H_k$  exactly, entry by entry, as rationals, a zero residual is required, not a small-tolerance numerical match. For all eight cells reported in Appendix D (and, by the orbit argument of Lemma 4.2, for all 176 cells), this residual is exactly zero.

**E.2. Summary of the independent run.** Running this second verification path directly (rather than trusting a cached result) confirms, for chambers  $\mathfrak{k}_1$ – $\mathfrak{k}_8$ : the reported Hessian  $H_k$  has the expected structural form in every case; the  $LDL^\top$  pivots reconstructed by this independent routine agree, to the last digit, with the pivots recorded in Appendix D; the sum-of-squares residual,  $x^\top H_k x - \sum_i d_i y_i^2$ , is identically zero as a polynomial identity in each case; and every one of the eight chambers passes overall. The global minimum pivot across all eight chambers checked here is  $160/13 \approx 12.307692$ , attained at  $\mathfrak{k}_4$  (orbit D, class d,  $\mathcal{N}_4 = 48$ ), consistent with Lemma 4.4's statement that the eigenvalue floor is attained on the most symmetric denominator class.

**E.3. What this does and does not add.** This appendix is deliberately narrow in scope: it does not introduce any new mathematical claim beyond Proposition 4.3, and its purpose is solely to record that two independently written pieces of software, taking different routes (one via Sylvester's criterion applied to a stored decomposition, one via an explicit sum-of-squares expansion checked against a freshly reconstructed Hessian), reach the same exact rational answer on every pivot for every cell tested. The agreement is evidence against a transcription or implementation error in the proof of Proposition 4.3, and nothing more: running the same mathematics through two code paths is not an independent proof. Proposition 4.3 is proved by the exact  $LDL^\top$  decomposition together with Sylvester's criterion, and this appendix corroborates that.

## APPENDIX F. COMPUTATIONAL METHODOLOGY

Every exact and numerical claim in the main text is reproduced in full, as an explicit rational number, an explicit polynomial or Bernstein coefficient list, an explicit matrix, or an explicit high-precision decimal with its stated precision, in Appendices B to E, G and H. Checking any claim therefore means checking arithmetic that is printed here, not running a program. This appendix records how those numbers were obtained, for a reader who wants to redo the derivations independently.

**Four kinds of computation.** Every displayed number in this paper belongs to exactly one of four categories, and the category is named at the point of use.

*Exact rational arithmetic.* The  $LDL^\top$  pivots of Appendix B (the factorisation itself is the standard one, as in [GVL13, §4.1]), the algebraic factorisations of Appendix C, the matrix entries of Appendix D, the sum-of-squares data of Appendix E, and the whole certificate of Section 13 are computed in exact rational arithmetic, or in exact  $\mathbb{Z}[\sqrt{2}]$  where a  $\sqrt{2}$  appears, with no floating-point step. Each was carried out twice, once with plain rational arithmetic and no computer algebra system and once with `sympy` [Meu17] on a separate code path, and the two agreed to the last digit before either was reported. Bernstein sign certificates (Sections 10 and 12) are exact in the same sense: once numerator and denominator are written in the Bernstein basis over a box, positivity of every coefficient is a finite check on exact integers or exact elements of  $\mathbb{Z}[\sqrt{2}]$ , and the basis's variation-diminishing property [Far12] is what makes that check decisive and not merely suggestive.

*Arbitrary-precision numerical evaluation.* Where we did not pursue a closed form, most extensively in the high-precision volume method of Section 17.2, the directional sampling of Section 18, and the obstacle discussed in Section 20.2, values are computed with `mpmath` at a stated precision between 35 and 400 significant digits, and cross-checked at a second precision before being reported. Every such value is labelled as numerical evidence and never used as a proof step.

*Numerical semidefinite programming.* The Gram-matrix check of Section 18.3 is a floating-point semidefinite program (`cvxpy`, solver `CLARABEL`) run over a least-squares fit of directional samples; the underlying method is the one of [Par03, Las01]. Its output is numerical evidence about a fitted approximant, and nothing more. Before it was applied to any of our data it was run on two quartics of known status, a perfect square and the Choi–Lam polynomial, and returned the right answer on each; Section 18.3 records both runs.

*Double-precision search.* Double precision is used only to locate candidate breakpoints, candidate worst-case directions and candidate active sets before an exact or high-precision verification, for instance in the pattern-stability scans of Sections 9 and 11. No claim rests on a double-precision value by itself.

**A pitfall worth recording.** One methodological finding is easy to reproduce by accident, so we record it. During the derivation behind Section 20.5, a general-purpose symbolic algebra package’s default determinant routine, Bareiss fraction-free elimination, returned an outright wrong value on a particular symbolic matrix: not a slow computation, a wrong number, silently. It was caught because the result was cross-checked against Berkowitz’s method and against direct cofactor expansion, which agreed with each other and not with the first. The caution is general. An exact symbolic determinant computed by one algorithm in one implementation should not be trusted without a second, independently implemented check, however routine the computation looks.

The same caution is what `swap_defect_curves.py` is for. It recomputes the three functions of Section 20 from the cell itself, enumerating the local region of the deviated configuration with `scipy`’s `HalfspaceIntersection` and `ConvexHull` and subtracting 8, at 211 angles across  $(0, \theta_*)$ , by a route that shares no code and no method with the closed forms. For the hexagon, where a closed form is printed in Section 20.6, the two agree to  $9 \times 10^{-9}$  across the interval, which also identifies the six-cycle the closed form belongs to; the script finds that cycle by testing every six-cycle of roots at inner product  $\frac{1}{2}$  against the printed formula. It supplies Figure 32, and it reproduces the two numbers of Section 20.4 independently, 0.002474 against 0.000834 at  $\theta = 0.05$  and the crossover at  $\theta = 0.6492$ .

**The box-covering certificate.** The box coverings used in Sections 10 and 12, and the 303-box certificate of Theorem 13.6, follow one algorithm, stated here since the main text refers to it repeatedly.

```

FUNCTION CERTIFY(region  $R$ , tolerance  $\varepsilon$ ):
    stack  $\leftarrow [R]$ 
    certified  $\leftarrow [ ]$ 
    WHILE stack is not empty:
        box  $\leftarrow$  stack.pop()
         $b \leftarrow$  exact bound for the target quantity over box
        IF  $b$  has the required sign:
            certified.append(box)
        ELIF width(box)  $< \varepsilon$ :
            RETURN FAIL

```

```

ELSE:
    (box1, box2) ← bisect(box) along its longest side
    stack.push(box1); stack.push(box2)
RETURN certified

```

Two features of this control flow matter for how results are reported. Every successful call returns a finite list of sub-boxes whose union is exactly  $R$ , with no gaps and no overlaps, verified by exact rational interval arithmetic on the box boundaries rather than by construction alone; so a quoted box count is exact, not an estimate. And the algorithm has one failure mode, reaching the tolerance  $\varepsilon$  with a leaf box whose sign is still undetermined, which returns FAIL rather than a partial certificate. The tolerance is fixed before the run and never loosened afterwards. In the arc certificates the exact bound  $b$  is the pair of Bernstein coefficient ranges of numerator and denominator; in Theorem 13.6 it is the corner value supplied by the monotonicity of Lemma 13.5. The bound used is in both cases an outward-rounded enclosure in the sense of [MKC09, Ch. 5].

**The cap-inequality certificate as a single script.** The chain of Section 13 is short enough to run end to end in one program, and the supplementary package contains it as `cap_inequality_certificate.py`. It performs, in order: the exact rational vertex enumeration of  $Q$ , confirming 25 vertices and  $\text{vol}(Q) = \frac{25}{3}$ , with the pyramid of volume  $\frac{1}{3}$  recovered as the difference from the 24-cell; the exact check that every vertex of  $Q$  has  $\ell^1$  norm at most  $\sqrt{2}$  in the frame, which is Lemma 13.3, together with the observation that the bound is attained; a comparison of the closed form of Lemma 13.4 against directly computed cap volumes at random directions; the symbolic verification that  $g'''(t^2) = 3(20t^2 - 3)/(2t^5)$  and that  $g'''$  vanishes at  $a = \frac{1}{4}$ , which is Lemma 13.5; the exact real-root isolation showing that each of the three polynomials of Theorem 13.6 has no root in its interval; the adaptive subdivision producing the 303 boxes and their corner bounds, entirely in `fractions.Fraction`; and finally, as an independent check of the conclusion rather than a step in it, the direct computation of cell volumes at random tilts and deviation directions. The script prints a pass or fail line for each of its twenty-one checks and exits nonzero if any fails.

**The multi-cap reformulation.** Proposition 7.5 and Remark 7.3 are supported by a second short script, `multi_cap_reformulation.py`. It performs, in order: an exhaustive enumeration showing that  $Q_D$  is bounded for every one of the 24, 276, 2024, 10626 and 42504 packing-valid sets  $D$  of size 1 through 5; the corresponding enumeration at size 6, finding exactly 24 failures among 134596 sets and confirming that the complement of the first of them lies in a closed half-space; a check of the identity (7.3) against directly computed polytope volumes on random packing-valid configurations; a check that  $\text{vol}(Q_D) - 8$  equals  $m/3$  when  $D$  contains no adjacent pair and exceeds it when it does; and the search recorded in Remark 7.3, which finds  $\text{vol}(E_j)$  both above and below its single-deviation value. The first, second and fourth of these are exhaustive over the stated ranges; the third and fifth are random searches and are labelled as such in the output. The script prints a pass or fail line for each of its eleven checks.

**The polar and boundary reformulations.** The material of Section 7.3 is checked by a third script, `polar_surface_reformulation.py`, in eighteen parts, among them the square-antiprism configuration (7.9) of Remark 7.12: that its nine directions are packing-valid, that its facet has 3-volume  $16\sqrt{2} - \frac{64}{3}$  against the octahedron's  $\frac{4}{3}$ , and that the resulting ceiling  $96\sqrt{2} - 128$  falls below both 8 and the covering bound. It verifies that the cell is the polar dual of the convex hull of the contact directions, by computing both volumes at the root configuration (2 and 8, product 16) and by confirming on random contact configurations that

adjoining a point of the hull leaves the cell unchanged; the boundary identity (7.6), exactly at the root configuration, where the twenty-four facets each have 3-volume  $\frac{4}{3}$  and the total is 32, and to fourteen digits on configurations obtained by dropping four roots, perturbing the rest and separating them again; the monotonicity of  $\rho$  and the value  $\rho(\frac{1}{2}) = 1/\sqrt{3}$ ; the inball claim of Lemma 7.11, by computing the inradius of every facet directly; the arithmetic of Remark 7.12; the tight structure at a root facet, in exact rational arithmetic, finding eight tight neighbours with projected Gram values in  $\{-1, -\frac{1}{3}, \frac{1}{3}\}$  and the regular octahedron of volume  $\frac{4}{3}$  as the region they cut out; the monotonicity of the volume under enlarging the configuration, which is Proposition 7.8(ii); and the two evaluations of Remark 7.13, the Jensen bound 7.7351... and the Mahler-type product 16 against  $\frac{32}{3}$ . The two integrals over  $S^3$  are Monte Carlo estimates at four million samples and are labelled as such in the output; every other part is exact or is double-precision polytope arithmetic on an exactly specified configuration. The separation step used to generate test configurations is ordinary pairwise repulsion and enters no argument.

**The covering bound.** Section 7.4 is checked by a fourth script, `covering_bound.py`, in nineteen parts: the cap-area formula  $C(r) = \pi(2r - \sin 2r)$ , against both its value at  $r = \pi$  and direct sampling at four radii; the radial identity (7.10) at the root configuration, where it returns 8 and recovers the covering radius  $45^\circ$ , and on random configurations; the layer-cake rewriting used in the proof, against the direct form; the closed form  $\frac{\pi m}{3} \tan^3 r_m$  against numerical quadrature of the same estimate, agreeing to  $10^{-14}$ ; the whole of Table 2, including the monotonicity in  $m$  that the proof of Corollary 7.17 uses and the fact that 22 is exactly the largest  $m$  at which the bound reaches 8; and the absence of any violation of (7.12), at the root configuration, at one hundred of its subsets of sizes 20 through 23, and at fifty random packing-valid configurations, where the smallest ratio of true volume to bound observed is 1.05. It then checks Proposition 7.18: the closed form  $\phi'(s) = \frac{3}{4\pi} \sec^4(C^{-1}(s))$  against numerical differentiation at five areas, the convexity of  $\phi$  over its whole range, and the fact that equal cell areas reproduce the global bound exactly and minimise the per-cell sum, against fifteen hundred random splittings of the total measure. It checks Proposition 7.21, both the averaging step behind it and the bound itself against directly computed covering radii, and confirms that it moves the volume bound only in the fifth decimal. Finally it reproduces the split of the shortfall at  $m = 24$  into its overlap and tail parts, and the total overlap contributed by the ninety-six pairs of the root system at  $60^\circ$ . The integrals over  $S^3$  are Monte Carlo and are labelled as such in the output; the table, the closed forms and the polytope volumes are not.

**The extendability criterion.** Section 7.5 is checked by a fifth script, `extendability.py`, in six parts: that the three forms of Proposition 7.38 agree, on the root system, on the root system minus one root, and on random configurations; that the root configuration has circumradius  $\sqrt{2}$  and  $g = 1/\sqrt{2}$  and is therefore saturated, as a 24-point kissing configuration must be; that the root system minus one root has circumradius exactly 2, cell volume exactly  $\frac{25}{3}$ , and is extended by the deleted root, so it sits on the boundary of the criterion; that the Gram values of the root configuration are  $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$ , which is what Theorem 7.30 asserts of every 24-point configuration; the numerical chain of Corollary 7.39; and a search, over the twenty-four one-root deletions, greedy random packings and annealed random starts, for a saturated 23-point configuration, which found none. The last of these is exploration and is labelled as such in the output; it is not evidence that none exists.

**The cited certificate, re-verified.** Lemma 7.26 is checked by `root_lattices_rank4.py`, an exact census of the 393 positive definite Gram matrices of order 4 with 2 on the diagonal

and  $-1, 0, 1$  off it, described in its proof, and again by `lean/D4RootLattices.lean`. The certificate behind Theorem 7.30 is verified by `llm24_certificate_check.py`, which reads the deposited data set [LLM24d] (included in the package under `third_party/llm24-certificate`, with its MIT licence notice; the script takes the path of its `proofs/4_24` folder) and shares no code with the authors' Julia package. It repeats five of the seven steps of their verification, as Section 7.5.3 describes: the format of the data, with the block of every signature  $\lambda$  checked to have exactly as many rows as there are admissible index tuples for  $\lambda$ ; the positive definiteness of all 127 blocks by Cholesky in the ball arithmetic of Arb [Joh17] at 256 bits, every pivot a ball inside the positive reals, and of the 81 blocks of size at most 16 by an exact rational  $LDL^T$  as well; the nonnegativity structure of the 125 sum-of-squares prefactors; the objective, exactly 24; and the two-point polynomial  $p_2$ , of degree 16, with its zeros on  $[-1, \frac{1}{2}]$  located by an exact Sturm sequence. It writes the exact coefficients of  $p_2$ , from which `lean/gen_innerproducts.lean.py` generates `lean/D4InnerProducts.lean`, and `p2_zeroeset_check.py` locates the same zeros a third time, with the `Fraction` type of the standard library alone. The run takes about eight minutes and its log is in the package.

The other two steps are repeated by the scripts in `zonal/`, whose `README.md` sets out the method described in Section 7.5.3. `o4.py` integrates monomials over  $O(n)$  by the recursion of Gorin and López [GL08] in exact rationals and carries the self-test; `g12.py` builds the  $GL(2)$  matrix coefficients and the scalars of the authors' Section 3.1; `ps_build.py` writes the three factors of the integrand for each of the 490 entries; `psker.c`, in C with GMP, walks the monomial triples and accumulates  $P(S)$ ; `zonal.py` reduces  $P(S)$  modulo  $S^T S = I$  and reads off an entry of  $Z_\lambda$  at given inner products; and `verify45.py` assembles the four constraint polynomials and checks that they vanish, with `merge4.py` and `combine4.py` adding the pieces of the four-point one, which is too large to hold in one process. `ps_ref.py` recomputes the integral the slow way for  $|\lambda| \leq 5$  and agrees with the kernel exactly, and `check_psd.py` tests the positive semidefiniteness of the zonal matrices on random point sets.

**The overlaps put back.** Lemma 7.22 and Proposition 7.23 are checked in four parts by the script `second_order_estimate.py`, with the quadratures carried out at 30 digits: the circumradius bound for a triple of contact directions, both by the Rayleigh-quotient argument of the proof and against forty thousand random admissible triples; that  $r_{23}$  and  $r_{24}$  both lie below  $\arccos \sqrt{2/3}$ , so that the caps meet only in pairs below  $r_m$ ; the closed form (7.16) for the lens measure against direct sampling on  $S^3$ ; and the value (7.18) at the deletion configuration, together with 7.858738... at the root system.

**The two local decompositions.** Remark 7.19 is checked by `local_cell_obstruction.py`, in five parts: that the mean of  $\sec^4$  over a spherical Voronoi cell of the root configuration is  $16/\pi^2$ , so that (7.14) is sharp; that nine unit vectors of  $\mathbb{R}^3$  with pairwise inner products at most  $\frac{1}{3}$  can be produced, so that nine contacts at  $60^\circ$  are admissible, and that the resulting cell has mean 1.5732..., still 1.5954... when the nine are moved out to  $61^\circ$ ; that on a 22-point configuration there are directions whose nearest contact lies outside the Delaunay cell containing them, by more than 0.1 in cosine, so that the Delaunay decomposition does not localise the integrand; the closed form (7.15) at the regular simplex, against direct sampling at edge  $60^\circ$  and  $62^\circ$ ; and that the Delaunay cells of  $D_4$  are 24 congruent spherical octahedra of circumradius  $45^\circ$  while no four roots are pairwise at  $60^\circ$ , so the regular simplex occurs nowhere in  $D_4$  although it does occur as a

Delaunay cell of a contact configuration. The integrals over  $S^3$  are Monte Carlo and the script says so; the margins reported are percentages.

**Rigidity of the deletion configuration.** Proposition 7.48 is checked by `rigidity23.py`, in five parts, all in exact integer arithmetic on the unnormalised roots: that deleting one root leaves 23 directions and 88 tight pairs, with the four values of  $\langle a_i, a_0 \rangle$  occurring 1, 8, 6, 8 times; that the six pair types listed in the proof are the only ones that occur; that the weights  $y_{ij} \in \{1, 2, 3\}$  are strictly positive on every tight pair; that they satisfy the equilibrium relation (7.33) exactly, the  $23 \times 4$  residual matrix being zero; and that the space of motions holding all 88 pairs at equality has dimension exactly 6 and is spanned by the infinitesimal rotations. There is no floating-point step anywhere in the script. Corollary 7.49 is checked in the same way by `rigidity24.py`, in four parts: that the root system has 24 directions and 96 tight pairs, every direction in 8 of them; that the constant stress, weight 1 on every tight pair and  $-4$  on the diagonal, is positive on the pairs and in equilibrium, the  $24 \times 4$  residual being zero; and that the space of motions holding all 96 pairs at equality has dimension 6 and is spanned by the rotations.

The radius that Theorem 7.51 attaches to that rigidity needs one number beyond the kernel, the smallest nonzero singular value of the rigidity operator, and `rigidity_spectrum.py` computes the whole spectrum to get it, again without a floating-point step. It assembles the integer matrix  $\Lambda'$  of the 96 differentiated tight inner products and the rational projection  $P$  onto the tangent space, forms  $4N = (2P)\Lambda'^T\Lambda'(2P)$ , which is integral with entries bounded by 16, verifies the matrix identity  $4N(4N - 8)(4N - 20)(4N - 24)(4N - 32) = 0$  in `int64` arithmetic, where the bound on the entries keeps every partial product exact, and computes the ranks of  $4N - cI$  for  $c = 0, 8, 20, 24, 32$  over a field of positive characteristic, getting 66, 67, 88, 75, 88. Those ranks bound the ranks over  $\mathbb{Q}$  from below, the five multiplicities they give sum to 96, so each bound is attained; the eigenvalues of  $N$  are therefore 0, 2, 5, 6, 8 with multiplicities 30, 29, 8, 21, 8 and the singular values of  $\Lambda$  on the tangent space are 0, 1,  $\sqrt{5/2}$ ,  $\sqrt{3}$ , 2 with multiplicities 6, 29, 8, 21, 8. `D4Stress.lean` repeats the finite part of this inside Lean's kernel with no library beyond core Lean: the root list, the eight tight neighbours of every root and their sum, and the count of 96 tight pairs, each by `decide`. The matrix identity itself is a product of six  $96 \times 96$  integer matrices and is not formalised; the elimination that settles it in the script is fraction-free over  $\mathbb{Z}$ , so it is exact as it stands.

**How much of a root system a configuration can keep.** Propositions 7.55 and 7.56 are proved in the text and checked independently by two scripts. `root_deletions_exact.py` carries out the underlying question in exact integer arithmetic and settles no part of it by sampling. Writing the roots as integer vectors of squared length 2 turns the cell of a subset  $S$  into  $\sqrt{2}P(S)$  with  $P(S) = \{y : \langle y, r \rangle \leq 1, r \in S\}$ , whose facet normals are integral and whose right-hand sides are 1; every vertex of  $P(S)$  is therefore the solution  $m/e$  of a four-by-four integer system, obtained by Cramer's rule with  $m$  integral and  $e$  a positive integer, and the two tests that matter,  $|m|^2 \leq 2e^2$  for the circumradius and  $|m|^2 = 2e^2$  for a direction that can be added, are comparisons of integers. Running it over all 24, 276 and 2024 ways of removing one, two and three roots reproduces Proposition 7.55 and the classification in the proof of Proposition 7.56: the circumradius is exactly 2 except at the 96 triples that are pairwise at  $60^\circ$ , where it is  $\sqrt{6}$ , and the directions that can be added are the removed roots and nothing else.

`root_meet.py` checks the combinatorial and coordinate statements the proofs actually use: that no pair of roots destroys a whole couple; that 512 of the 2024 triples do, in eight support patterns, four stars and four triangles; that the symmetry group of the root system,

generated here and found to have order 1152, is transitive on the 96 tight triples; and, for the standard star, that every direction of the doorway has first coordinate at least  $1/\sqrt{2}$  and the other three nonnegative, and that no two of them are as much as  $60^\circ$  apart unless both are removed roots. The last part is sampling and is reported as such; the first three are exact. Remark 7.58 is the same script run one rung lower, at four removed roots, where the answer is a minimisation with a margin rather than a proof.

**The pairwise budget at twenty-three contacts.** Proposition 7.41 and Remark 7.42 are checked by `pair_budget.py`, in five parts: the per-pair weight  $\omega(\gamma)$  and its collapse away from  $60^\circ$ , which is where the quadrature sits; the recovery of (7.18) from the 88 tight pairs of the deletion configuration; the count of pairs at  $60^\circ$  that would carry (7.17) past 8, with the values at 90 and at 91 either side of the target; the elementary degree bound on  $S^2$  and the ceiling of 115 it gives; and that the target lies strictly between the two. The quadratures are adaptive and the rest is arithmetic; nothing here is Monte Carlo.

**How far the pair angles reach.** The script `covering_multiplicity.py` checks Propositions 7.43 and 7.44, in two parts. The first evaluates both sides of (7.31) at  $m = 23$  and at the deletion configuration, giving 155.172054... against 162.566121.... The second sets up the linear programme of Proposition 7.44 on a grid of 601 angles, with the Bonferroni constraint at 24 radii between  $r_{23}$  and  $45^\circ$  and the Gegenbauer constraint at the first 12 degrees, and reports its optimum, 0.041573648.... It also checks that the pair angles of the deletion configuration are feasible for that programme, so that the bound really is a relaxation of the truth and not an accident of the grid. Refining the grid, adding radii or adding degrees leaves the optimum where it is.

**The cell inside a ball.** Propositions 7.44 and 7.68 and Theorem 7.69 and the  $r_*$  values of Propositions 7.23 and 7.41 are supported by `three_point_reduction.py`, in three parts. The first evaluates the constants of the pair-only bound at the two integration limits  $r_{23}$  and  $r_*$ , the bracket, the weight of a pair at  $60^\circ$ , the value at a deletion and the number of pairs at  $60^\circ$  that carry the bound past 8; the second evaluates the three-point bound at  $r_4$ , with the triple measure of an equilateral  $60^\circ$  triangle by Monte Carlo, at a deletion and at the root system, against the direct quadrature; the third solves the pair-angle relaxation of Proposition 7.44 for both weights, on a grid of 1201 angles with 60 radii and 24 Gegenbauer degrees, and prints the minimising measure. `truncated_volume.py` evaluates  $\text{vol}(V_c \cap B(R))$  by a fixed quasi-random quadrature on  $S^3$  of 400000 points, checks it against the exact volumes at the root system and at a deletion, and then minimises it over configurations with pairwise inner products at most  $\frac{1}{2} + \delta$ , for a decreasing schedule of  $\delta$ , by the trust-region programme of `inradius_search.py` with the analytic gradient of the quadrature; it takes the truncation radius as an argument, and the runs at  $R = \sqrt{3/2}$  and  $R = \sqrt{8/5}$  are recorded in the package. The identities are exact; the quadrature, the Monte Carlo triple measure and the minimisation are numerical and the output says so.

**The certificate.** Theorem 7.73 rests on two scripts and one Lean file. `three_point_sdp.py` sets up the programme of Lemma 7.72: the Gegenbauer polynomials of  $S^3$  by the Chebyshev recurrence, the matrices  $Y_k$  of (7.46) from the Legendre coefficients with the half-integer powers cancelled symbolically, their symmetrisation  $S_k$  over the six orderings, and the condition (7.47) imposed at a grid of admissible triples  $u \leq v \leq t$ , denser near  $\frac{1}{2}$ , with the weight  $\omega$  interpolated from the quadrature of `three_point_reduction.py` and scaled by 1000. It is solved through `cvxpy` [DB16] with the interior-point solver Clarabel [GC26]. In its first mode the bound is maximised, the solution is checked on a fine grid

and a random sample of about 1.15 million triples, the worst 3000 of these are added to the constraints, and the cycle is repeated; the printed value is the sampled bound less  $\binom{23}{3}/21$  times the largest violation on the check, and the two-point programme, with the matrices removed, is solved alongside for comparison. In its second mode the bound is fixed at a given value, here 0.0929, the least slack of (7.47) over the sample is maximised with  $F_k - 10^{-7}I$  and  $f_k - 10^{-7}$  constrained to be positive semidefinite and with a little static regularisation of the interior-point method, which the degenerate maximin problem needs, and the result is written to `continuation_out/certificate_d8.npz`, with a plain-text copy beside it in which every number is printed exactly. `certificate_check.py` is the verification described in the proof of Theorem 7.73, and it shares no code with the solver: the certificate is read as exact rationals, the matrices are tested by exact  $LDL^T$ , the polynomial  $P$  is expanded exactly from its own implementation of the three recurrences, the closed forms of  $A_*$ ,  $\omega$ ,  $\omega'$  and  $\omega''$  are differentiated with `sympy` and evaluated in the interval arithmetic of `mpmath`, and the branch and bound is run in an interval arithmetic on IEEE doubles written in `numpy` with outward rounding after every operation. It stops at the first failure and says where: a box below the minimum width on which (7.47) could not be decided, or an admissible triple at which (7.47) is negative. The run recorded in the package verifies the certificate in 44 levels and under ten minutes, and a second recorded run, on the certificate with  $10^{-5}$  added to  $f_0$ , is rejected with the triple  $(-\frac{1}{16}, -\frac{1}{16}, -\frac{1}{16})$ . The Lean file is described under the formalised part below.

**The covering radius as the objective.** Proposition 7.46 and Remark 7.47 are checked by `saturation_search.py`. Where `extendability.py` and `spherical_code_23.py` look for a code of large minimal angle and read off the covering radius afterwards, this script takes the covering radius as the objective from the outset. It uses the fact that  $g(W) = \min_{\theta} \max_i \langle \theta, w_i \rangle$  is the support function of  $\text{conv}(W)$  at its minimum, hence the distance from the origin to the nearest facet hyperplane of  $\text{conv}(W)$ , so it is read off a convex hull exactly, not sampled. Four parts: the root system, where  $g = 1/\sqrt{2}$ ; all twenty-four deletions, where  $g = \frac{1}{2}$  with zero spread; the 528 one-variable linear programmes of Proposition 7.46, together with the least violation 0.0740002839... computed at 60 digits; and a projected-gradient minimisation of  $\max_I |z_I|$ , with the closed-form gradient  $\partial |z_I|^2 / \partial w_i = -2a_i z_I$  for  $a = G_I^{-1} \mathbf{1}$ , started from perturbed deletions, from random configurations and from partially completed root systems. The first three parts are proof; the fourth is exploration and the script says so in its output.

**The formalised part.** Eight parts of the paper are also verified in Lean 4: the six files `D4Stress.lean`, `D4Meet.lean`, `D4Certificate.lean`, `D4InnerProducts.lean`, `D4RootLattices.lean` and `D4Closure.lean` in the directory `lean`, and the Lake projects in `lean/cell600` and `lean/certificate`. All are independent of everything else in the supplement and of every Lean library: the first two use only integer arithmetic on the unnormalised roots and settle each statement by kernel computation, so a reader who compiles them is checking the arithmetic against Lean’s kernel rather than against a Python program. The first covers the equilibrium relation (7.33) and the combinatorics around it, which is the arithmetic of Proposition 7.48. The second covers the finite half of Propositions 7.55 and 7.56: that two removed roots always leave a whole couple of complementary supports, that exactly 48 ordered triples of supports do not and that these are the four stars and four triangles taken in every order, that 96 triples of roots are pairwise at  $60^\circ$  and no four roots are, that the Hadamard map is an isometry permuting the roots and carries the standard triangle to the standard star, and that the standard star

leaves the twelve roots of the lower block and three of the four roots at each remaining support. The third covers Proposition 7.63: the 600-cell's vertices in  $\mathbb{Z}[\phi]$ , their inner products, the twelve-regular graph of pairs at  $36^\circ$ , the twenty-five inscribed copies of  $R$  with five through each vertex, all by kernel computation, and then the three counts of the enumeration, which are settled by `native_decide` on the compiled search and therefore trust the compiler as well as the kernel, as Section 23 explains. The fourth, `lean/D4Certificate.lean`, is written by `lean/gen_certificate_lean.py` from the certificate file, with every floating-point number converted to the exact dyadic rational it denotes, an integer over a power of two, so that the Lean source and the Python verification hold the same numbers; it proves by kernel computation on `Rat`, with `decide +kernel` and no compiled code, that  $f_1, \dots, f_8$  are nonnegative, that each of  $F_0, \dots, F_8$  is symmetric and has positive pivots throughout an exact  $LDL^\top$  elimination, and that the bound of (7.48) exceeds 92.8555703 in the scaled units, that is  $0.0928555703 > 8 - A_*$ ; the check takes under a minute and the axiom report lists only the three standard axioms. The fifth, the Lake project `lean/certificate`, is the branch and bound of Theorem 7.73 in exact dyadic arithmetic, as Section 23 describes: `D4CertData.lean` holds the certificate and the tables, `D4CertDomain.lean` the polynomial arithmetic, the expansion of  $P$ , the dyadic intervals and the subdivision, and `D4CertMain.lean` the theorem, by `native_decide`. The sixth, `lean/D4InnerProducts.lean`, is the last step of the verification of the certificate: the two-point polynomial  $\sigma_2$  of Proposition 7.29, computed exactly from the published data by `llm24_certificate_check.py` and written into the file with its coefficients scaled to integers, is shown by kernel computation on `Rat`, with `decide +kernel`, to vanish at  $-1, -\frac{1}{2}, 0, \frac{1}{2}$ , to be exactly divisible by  $(u+1)(u+\frac{1}{2})^2u^2(u-\frac{1}{2})$ , and to have a quotient of degree 10 that does not vanish at those points and whose Sturm sequence has the same number of sign changes at  $-1$  as at  $\frac{1}{2}$ ; the check takes under a minute and the axiom report lists only the three standard axioms. The seventh, `lean/D4RootLattices.lean`, is the finite content of Lemma 7.26: the census of positive definite Gram matrices of norm-2 bases in rank at most 4, the enclosure of the norm-2 vectors of each lattice in an exact box, the counts 2, 6, 12, 24, and a  $D_4$  basis among the roots of every lattice that has 24, by `native_decide` in a few seconds. The eighth, `lean/D4Closure.lean`, is the exact content of Section 2.8: the identities behind Lemmas 2.12 and 2.15, Proposition 2.13, and Theorem 2.17, proved over every commutative ring by the ring normaliser of `grind`, and the orthogonal root pairs and deleted sets of Corollary 2.14, by `decide +kernel`; it checks in about two minutes, and its axiom report lists the three standard axioms at most. The theorems in each file are listed in `lean/README.md`, together with the axiom report that each file prints when it is checked. Nothing else in the paper is formalised.

**Contact configurations as spherical codes.** Proposition 7.59 and Remark 7.62 are supported by `spherical_code_23.py`. The script does one thing, in four sizes: it minimises the Riesz energy  $\sum_{i < j} |w_i - w_j|^{-s}$  on  $(S^3)^m$  with  $s$  running through 4, 16, 64, 256, 1024, projecting back to the sphere after each step, and then reduces the largest inner product directly by a softmax descent with a shrinking step. The three calibration sizes are run first and are the reason to trust the fourth:  $m = 24$ , where the answer is known to be  $\frac{1}{2}$ ;  $m = 22$ , where a code with room to spare exists; and  $m = 25$ , where no contact configuration exists at all. It then runs  $m = 23$  from 1500 independent random starts, records the smallest largest inner product reached, counts the starts that fall below  $\frac{1}{2}$ , and fingerprints every outcome that comes within  $5 \times 10^{-3}$  of feasibility by the multiset of its Gram entries, which is invariant under  $O(4)$  and therefore separates configurations that are not related by an

isometry. Everything in the script is floating point and it is exploration, not proof; the output says so, and Remark 7.62 repeats it.

**The volume identity at twenty-four contacts.** Section 7.6 is checked by `m24_exact_reduction.py` in five parts. The first derives in `sympy` the closed forms of the cap volume  $c$ , of  $\omega_2(\frac{1}{2})$  and of  $\omega_3(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$  and  $\tau$  from (7.26). The second estimates the cap, pair and triple volumes inside the ball of radius  $\sqrt{2}$  by Monte Carlo in  $\mathbb{R}^4$ , independently of the closed forms. The third checks the finite content of Proposition 7.34 in integer arithmetic: the only positive inner product between distinct roots, no four roots pairwise at inner product 1 among the 10626 quadruples, the determinant 5, the 96 pairs and 96 triangles, and the 24 vertices of norm  $\sqrt{2}$ . The fourth is an independent quadrature of the triple intersection, in polar coordinates on the three-dimensional span with Richardson extrapolation. The fifth reads the degree-6 pseudo-configuration of Remark 7.36 and checks its normalisation, the admissibility of its triples, the two-point moments, the positive definiteness of every moment matrix on its range, and its value with  $\omega_3$  by direct quadrature. The computations behind Remark 7.36 are `m24_volume_functional.py` (the functions  $c$ ,  $\omega_2$ ,  $\omega_3$ ), `m24_omega3_table.py` (a table of  $\omega_3$  on  $[0, \frac{1}{2}]^3$  at step  $\frac{1}{40}$ ), `m24_dual_sdp.py` (the certificate problem), `m24_primal_sdp.py` and `m24_primal_blocks.py` (the moment problem, rescaled on the range of each block), `m24_descent.py` (the tangent cone at the root configuration) and `m24_verify_direction.py` (the recheck of a descent direction by actual steps). These take the grid and the degrees on the command line: the runs quoted here are `m24_dual_sdp.py 6 3 1`, `m24_primal_blocks.py 12 8 6`, `m24_primal_sdp.py 12 8 1e-6 6`, `m24_descent.py 12 8 6` and `m24_verify_direction.py 6`, the two integers being the number of grid points in the pair variables and in the third; each of the five takes those same values by default, so running it with no arguments repeats the run quoted here. The primal run writes `continuation_out/m24_primal_d6.npy`, which the fifth part of `m24_exact_reduction.py` reads; that file and `m24_dir_d6.npy` are in the package, so the reduction script runs on its own. The fifth part and these scripts are floating point; the first four parts are the proof-relevant checks.

**The 600-cell, enumerated.** Proposition 7.63 is established by `cell600_exact.py`, with `cell600_enum.c` as an independent second implementation and the Lean project as a third. The Python script builds the 120 vertices with doubled coordinates as pairs  $(a, b)$  standing for  $a + b\phi$  and computes every inner product in  $\mathbb{Z}[\phi]$ , so that no floating-point number appears anywhere; it checks that the nine values listed in the proof are the only ones, builds the 12-regular graph of pairs at  $36^\circ$  and the 71-regular graph of pairs at root-system angles, finds the twenty-five inscribed copies of  $R$  as the 24-cliques of the latter, and enumerates the independent sets of the former through a fixed vertex, of sizes 25, 24 and 23, by depth-first search with the clique-cover bound. It reports 0, 5 and 115, checks that each of the 115 lies in one of the five cells through the vertex, and writes the graph and the cell list to a text file. The C program reads that file and repeats the enumeration of the 23-sets with the same bound on 64-bit words, in under two seconds; the Python enumeration takes about a minute and a half. Neither is sampling. The earlier script `cell600.py`, which drew two million maximal independent sets by greedy growth along random orders and is where the question was first looked at, is kept in the package for the record. `octahedral48_exact.py` does the same for the 48 unit quaternions of the binary octahedral group, in  $\mathbb{Z}[\sqrt{2}]$ , and finds the two root systems and their forty-eight deletions and nothing else.

**The inradius as the objective, and the continuation in the slack.** Section 7.10 rests on two scripts. `inradius_search.py` maximises the inradius  $g(W)$  of  $\text{conv}(W)$  directly,

subject to the contact constraints, by a sequential linear programme with a trust region: at each step the facet offsets and the pairwise inner products are linearised in a tangent move of bounded size, the linear programme maximising the smallest linearised offset is solved, and the move is accepted only if the true inradius, read from the facet equations of a fresh convex hull, has improved without any contact constraint being violated. The facet list is refreshed after every accepted move. Starts are of three kinds, taken in rotation: random points brought to feasibility by a relaxed-then-tightened schedule of the same programme; deletions of a root with each direction rotated through a random angle; and twenty random roots with three random directions. Every feasible endpoint is recorded with its inradius and the multiset of its inner products rounded to  $10^{-6}$ , which separates configurations not related by an isometry. `slack_continuation.py` runs the same optimiser at each  $\delta$  of a decreasing schedule from 0.05 to 0, starting from the best configurations of the previous level, restored to the new constraint by a feasibility programme of the same kind, together with fresh random starts at every level, and prints the table that Section 7.10 reproduces. Both scripts are double precision throughout and both are exploration; their output says so. `symmetric_search.py` runs the same continuation inside the classes of configurations invariant under a rotation of order at most 12 or under one of the two improper involutions, one orbit structure at a time, with the orbit representatives as the parameters and sequential least-squares programming in place of the linear programme; it prints, for each of the 73 structures, how many starts reached a feasible endpoint at  $\delta = 0$ , the largest inradius among them, the largest at  $\delta = 0.01$ , and whether the best endpoint is a deletion.

**Neighbours that do not touch.** Section 2.7 rests on two scripts. `shell_reduction.py` checks, in the ball arithmetic of Arb [Joh17] through `python-flint`, the covering radius  $45^\circ$  of the root directions behind Lemma 2.11; the value of  $\Phi$  for  $m$  contacts, above 8.044 for every  $m \leq 22$ ; the thresholds of Table 1; and the two examples that falsify the earlier forms of Lemmas 2.7 and 2.9. Each value of  $\Phi$  is a lower Riemann sum in which the first factor of the integrand is taken at the right end of each interval and the second at the left, both factors being monotone, so every printed bound is a bound. `shell_neighbour_search.py` is the search of Section 2.7; it is floating point and exploration, and its output says so.

Section 2.8 rests on three more. `closure_lemmas.py` checks the identities behind Lemmas 2.12 and 2.15 and Proposition 2.13 symbolically; the combinatorics of Corollary 2.14 in integer arithmetic, the 72 orthogonal pairs at the 24 vertices and every set of at most four deleted roots; and, in ball arithmetic,  $S(2)$ , the bound  $\frac{9}{8}\pi^2 - 22S(2) > 8.046$  of Corollary 2.16, the thresholds of  $\Psi$  beside those of  $\Phi$ , and the angle  $\arccos(\sqrt{6}/4)$  quoted after Theorem 2.17. `labelled_certificate_check.py` is the verification of Theorem 2.17: it rebuilds the certificate of Theorem 7.73 and the bound  $B$  exactly, rechecks the positivity of the matrices  $F_k$ , evaluates  $s(D)$ ,  $a_D$ ,  $\text{fr}(1, 1, \frac{1}{2})$ ,  $r$ ,  $\kappa$  and  $c$  in ball arithmetic, reruns the branch and bound of (7.47), and runs the two new ones, over 30 051 and 6 482 boxes, with the tables of the  $\Gamma_i$  in ball arithmetic and outward rounding throughout; with the rerun it takes about six minutes. `near_contact_probe.py` is floating point and exploration: the inversion hull beside the volume, and the minimisations near the root system quoted after Theorem 2.18. So is `truncated_search.py`, the minimisation of the truncated volume with exactly  $M$  centres within  $\sqrt{6}$  quoted there; the closed form it uses for the pair terms is checked against quadrature before it starts.

**The independent re-verification.** The directory `independent_verification` of the supplement records a re-run of every computation of this paper from a fresh copy of the package: the Lean verifications (seven at the time; `D4Closure.lean` was added in v1.5.0),

the certificates for 23 and 24 contacts, including steps 3 and 5 of Section 7.5.3 with all fifty sum-of-squares blocks of the four-point constraint, every script with its output compared against the shipped logs, the figures, and the manuscript itself. Its report lists what was found and corrected.

**Software.** The computations were carried out in Python, with exact rational arithmetic from the standard-library `fractions` module, `sympy` [Meu17] for symbolic manipulation, exact algebraic numbers and Sturm root counting, `mpmath` for the arbitrary-precision evidence, and `numpy` [Har20] and `scipy` [Vir20] for double-precision vertex enumeration and convex hull volumes, the latter through the Quickhull implementation of [BDH96]. These are standard and widely available; the semidefinite programmes of Section 7.12 are modelled in `cvxpy` [DB16] and solved by Clarabel [GC26], both open source, and the re-verification of the cited certificate uses the exact rational matrices and polynomials of FLINT and the ball arithmetic of Arb [Joh17] through `python-flint`. No specialised computer algebra system and no commercial solver was used, the one proof assistant is Lean 4 as described above, and nothing reported here needed more than a few minutes on ordinary consumer hardware apart from the continuation of Section 7.10, which takes up to half an hour per pass, the symmetric search, which takes about an hour, the minimisation of the truncated volumes in Section 7.11, which takes about two hours per run, and the semidefinite programmes of Section 7.12, which take between one and three minutes per solve and about a quarter of an hour for a run of five rounds.

## APPENDIX G. NUMERICAL TABLES

This appendix collects, in one place, the numerical evidence quoted at various points in the main text for Conjecture 1.4. Every table entry is reproduced directly from a script run recorded in Appendix F; where a value is double-precision numerical evidence rather than an exact rational, we say so at the entry.

### G.1. Chain configurations (Lemma 15.3).

| Chain length $m$ | Richardson-extrapolated $\lambda_{\min}(H_A)$ | Raw values at $h = 0.04, 0.02, 0.01$ |
|------------------|-----------------------------------------------|--------------------------------------|
| 3                | 0.275801                                      | 0.276304, 0.275929, 0.275833         |
| 4                | 0.180435                                      | 0.181187, 0.180632, 0.180486         |
| 5                | 0.168532                                      | 0.169316, 0.168736, 0.168585         |
| 6                | 0.167689                                      | 0.168474, 0.167894, 0.167742         |
| 7                | 0.167650                                      | 0.168424, 0.167847, 0.167700         |
| 8                | 0.167637                                      | 0.168411, 0.167833, 0.167687         |

All values strictly positive; the floor plateaus at  $\approx 0.1676$  rather than continuing to decrease past chain length 5–6, and 8 is the true maximum induced-path length in the Gram- $\frac{1}{2}$  adjacency graph on  $R$  (found by exhaustive search from every starting vertex, not merely tested up to an arbitrary cutoff). These figures are double-precision, Richardson-extrapolated finite-difference estimates, not exact rationals; Lemma 15.3 states only the qualitative conclusion (a floor near 0.168, strictly positive) that they support.

### G.2. Dense (non-chain) configurations (Sections 15.4 and 17).

| $m$        | Active-root indices (of 24)                                            | Extrap. $\lambda_{\min}(H_A)$                  |
|------------|------------------------------------------------------------------------|------------------------------------------------|
| 9 (star)   | one root + all 8 Gram- $\frac{1}{2}$ neighbours                        | 0.083336 ( $\approx \frac{1}{12} = 0.083333$ ) |
| 9 (greedy) | 0, 4, 5, 8, 9, 12, 13, 16, 17                                          | 0.083342                                       |
| 12         | 0, 4, 5, 8, 9, 12, 13, 16, 17, 21, 22, 23                              | 0.073070                                       |
| 15         | 0, 2, 4, 5, 8, 9, 11, 12, 13, 16, 17, 20, 21, 22, 23                   | 0.044694                                       |
| 18         | 0, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 16, 17, 20, 21, 22, 23         | -0.000013 (noise floor)                        |
| 20         | 0, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17, 19, 20, 21, 22, 23 | 0.000017 (noise floor)                         |
| 22         | (22 of the 24 roots)                                                   | 0.000023 (noise floor)                         |
| 23         | (23 of the 24 roots)                                                   | 0.000008 (noise floor)                         |

The greedy dense-growth procedure (at each step adding whichever remaining root has the most neighbours already active) drives the minimum eigenvalue down monotonically and strictly through  $m = 15$ ; from  $m = 18$  onward the extrapolated value is smaller in magnitude than double-precision finite-difference noise can resolve (an  $O(10^{-10})$  absolute volume precision divided by  $h^2$ ), which is exactly why Sections 17 and 18 turn to arbitrary-precision arithmetic instead of attempting further double-precision refinement.

**G.3. Broadened sampling at the exact singularity (Section 18).** At the  $m = 18$  configuration above, the near-null space of  $H_A$  is rigorously confirmed 4-dimensional (basis vectors  $w_0, w_1, w_2, w_3$ , via three-step-size eigenvalue scaling: eigenvalues 0–3 shrink by a factor of  $\approx 4$  per halving of the step size, matching a true zero to that order, while eigenvalue 4  $\approx 0.075$  does not shrink and is a real positive eigenvalue). Sampling the quartic coefficient  $a_4$  of  $\Delta$  restricted to each of 36 directions spanning this near-null space (the 4 basis vectors, all  $\binom{4}{2} = 6$  normalised pairwise sums/differences among them, and 20 uniformly random unit combinations), using arbitrary-precision (`mpmath`) volume evaluation instead of double precision:

| Quantity                       | Value    |
|--------------------------------|----------|
| Directions tested              | 36       |
| Minimum $a_4$ found            | 0.008779 |
| Maximum $a_4$ found            | 0.313708 |
| Directions with negative $a_4$ | 0 of 36  |

A six-direction subsample, re-evaluated at a second step size ( $s = 0.0075$  versus  $s = 0.015$ ), gives ratios of 0.954 to 0.991 (expected  $\approx 1$  for a quartic leading term), confirming the quartic scaling directly instead of assuming it.

At an independent  $m = 20$  configuration (active indices 0, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17, 19, 20, 21, 22, 23, distinct from the  $m = 20$  row of the dense-growth table above, which used a different greedy path), the near-null space is confirmed 5-dimensional; sampling 30 directions:

| Quantity                       | Value    |
|--------------------------------|----------|
| Directions tested              | 30       |
| Minimum $a_4$ found            | 0.007628 |
| Maximum $a_4$ found            | 0.313731 |
| Directions with negative $a_4$ | 0 of 30  |

with a four-direction re-confirmation at the second step size giving ratios of 0.952 to 0.991. As Sections 18 and 19 state plainly, this is real, cross-validated computational evidence at two independent dense configurations, spanning their full (respectively 4- and 5-dimensional)

near-null subspaces with both structured and random directions, but it is evidence at two configurations only, out of a combinatorially vast space of possible dense active sets, and does not constitute a proof of Conjecture 1.4.

#### G.4. Breakpoint crossing values on the second arc (Sections 11 and 12).

| Crossing | s-value                                | Curves meeting                                      |
|----------|----------------------------------------|-----------------------------------------------------|
| $s_1$    | $t_{\text{cross}} = 0.169918$          | curve C meets the universal curve $\varphi = \pi/3$ |
| $s_2$    | $t_* = \arctan(1/2) = 0.463648$        | curve B meets curve C                               |
| $s_3$    | $t_3 = \arcsin(1/\sqrt{3}) = 0.615480$ | curve B meets the universal curve $\varphi = \pi/3$ |

These three values are exact (two are elementary closed forms in  $\arctan$ ,  $\arcsin$ ; the first is located numerically to the displayed precision as the intersection of curve C with the constant curve  $\pi/3$ ) and partition the band immediately below region top into the three sub-parts discussed in Sections 11.4 and 12.2:  $(0, s_1)$  bounded below by C,  $(s_1, s_3)$  bounded below by the universal curve (certified in Theorem 12.2), and  $(s_3, \pi/4)$  bounded below by B. Only the middle sub-part carries a direct certificate; the other two are covered by the identification of Proposition 8.5, which carries the first arc’s certificates onto the whole of the second arc.

#### APPENDIX H. EXPLICIT COORDINATES AND A DIRECT NUMERICAL CHECK OF THE REFERENCE CELL

Lemmas 2.3 and 2.5 are stated and proved in Section 2 by an exact algebraic argument. This appendix records an independent, purely numerical cross-check of the same facts, carried out directly (via `scipy.spatial.ConvexHull`) on the explicit coordinate list, for a reader who wants to reproduce the polytope’s basic invariants in a few lines of code before trusting the exact argument.

**The vertex list.** The 8 axis vertices are  $\pm\sqrt{2}e_i$  for  $i = 1, \dots, 4$ ; the 16 sign vertices are  $\frac{1}{\sqrt{2}}(\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4)$  for every choice of signs  $\epsilon_i \in \{\pm 1\}$ . Both families lie exactly on the sphere of radius  $\sqrt{2}$ : direct computation confirms  $|\pm\sqrt{2}e_i| = \sqrt{2}$  and  $|\frac{1}{\sqrt{2}}(\pm 1, \pm 1, \pm 1, \pm 1)| = \frac{1}{\sqrt{2}}\sqrt{1+1+1+1} = \sqrt{2}$  for all 24 points, matching Lemma 2.3 exactly (verified here to full floating-point precision on all 24 points, not merely spot-checked).

**Direct convex-hull check.** Feeding these 24 points directly into a convex-hull routine (with no other input, in particular, without assuming any of the half-space description of Definition 2.2) gives:

| Quantity                      | Computed value                                                         |
|-------------------------------|------------------------------------------------------------------------|
| Number of vertices            | 24 (all 24 input points are extreme)                                   |
| Hull volume                   | 7.999999999999984 (matches $\text{vol}(\mathcal{V}) = 8$ to 14 digits) |
| Number of triangulated facets | 96                                                                     |

The facet count of 96 is exactly consistent with, not merely close to, the classical combinatorics of the regular 24-cell: it has 24 octahedral three-dimensional facets, and a general-purpose convex-hull routine triangulates each octahedron into 4 tetrahedra internally ( $24 \times 4 = 96$ ), so this is a confirmation of the expected cell structure, not a discrepancy to explain away.

**Support function check.** Evaluating  $h(u) = \max_{z \in \mathcal{V}} \langle z, u \rangle$  directly (as a maximum over the 24 vertices, with no appeal to Definition 2.2’s half-space description) at all 24 unit root directions  $\alpha/\sqrt{2}$ ,  $\alpha \in \mathbb{R}$ , gives  $h = 1.0$  at every one of them (minimum and maximum both exactly 1.0 to floating-point precision), and at all 8 coordinate directions  $\pm e_i$  gives  $h = \sqrt{2} \approx 1.41421356$  at every one, both figures matching Lemma 2.5 exactly, and obtained here by a direct, independent computational path (brute-force maximisation over the vertex list) rather than the exact algebraic argument given in the main text.

We emphasise the limited role of this appendix: it is a numerical sanity check computed directly by us for this paper, useful as an independent cross-validation and as a minimal reproducible starting point for a reader exploring the polytope computationally, but it is not a substitute for, and adds no logical strength beyond, the exact proof of Lemmas 2.3 and 2.5 already given in Section 2.

## APPENDIX I. AN ILLUSTRATIVE TOY MODEL OF THE BERNSTEIN-BASIS CERTIFICATE

The Bernstein-basis sign certificate used at scale throughout Sections 5, 10 and 12 is a general technique for certifying that a polynomial is non-negative on a closed interval by an exact, finite computation. Since the actual cell polynomials  $N_k, D_k$  of Section 3 are large enough (degree up to 12 in each of two variables, Lemma 3.3) that writing one out in full would obscure the method rather than illustrate it, we give here a small, entirely self-contained model problem worked in full, with no connection to any specific cell of the atlas: its only purpose is to make the mechanics of the certificate, and one of its limitations, concrete for a reader meeting the technique for the first time.

**I.1. The certificate: a worked instance.** Consider, purely as an illustration, the polynomial

$$p(t) = t^2 - t^3 + \frac{1}{4}t^4, \quad t \in [0, 1],$$

chosen to mimic the qualitative shape of a volume-defect function  $\Delta_k$  restricted to a root-aligned arc: it vanishes to exactly second order at the left endpoint ( $p(0) = p'(0) = 0$ ,  $p''(0) = 2 \neq 0$ ) and we would like to certify  $p(t) \geq 0$  throughout  $[0, 1]$  without appeal to calculus or root-finding, purely by an algebraic sign check.

Write  $p$  in the degree-4 Bernstein basis on  $[0, 1]$ ,  $B_{k,4}(t) = \binom{4}{k} t^k (1-t)^{4-k}$ ,  $k = 0, \dots, 4$ : a direct change of basis (matching coefficients of  $t^j$  on both sides, a purely mechanical linear computation) gives

$$p(t) = \sum_{k=0}^4 b_k B_{k,4}(t), \quad (b_0, b_1, b_2, b_3, b_4) = \left(0, 0, \frac{1}{6}, \frac{1}{4}, \frac{1}{4}\right).$$

Every Bernstein basis function  $B_{k,4}(t)$  is manifestly non-negative on  $[0, 1]$  (a product of non-negative factors there), so a polynomial with every Bernstein coefficient non-negative is itself non-negative on  $[0, 1]$ , termwise, with no further argument needed. Here all five coefficients are non-negative (two are exactly zero, matching the double root at  $t = 0$  exactly, and the remaining three are strictly positive), certifying  $p(t) \geq 0$  on  $[0, 1]$ , with equality only at  $t = 0$ , entirely by inspection of five rational numbers. This is the pattern used, at far larger degree and in two variables at once, throughout Sections 10 and 12: reduce a transcendental-looking positivity question on a bounded interval or box to a finite list of rational numbers with the right positivity property.

**I.2. A limitation: the certificate is sufficient, not necessary.** The Bernstein sign test just illustrated is a *sufficient* condition for non-negativity, not a necessary one, and this distinction is not merely academic; it is the reason a small number of sub-regions in this paper’s own atlas required the box-covering refinement of Sections 10 and 12 rather than a single direct Bernstein check. A classical example makes the limitation vivid. Take  $q(t) = (2t - 1)^2 = 4t^2 - 4t + 1$ , manifestly non-negative on  $[0, 1]$  (indeed everywhere), with a double zero at the *interior* point  $t = \frac{1}{2}$ . Its degree-2 Bernstein coefficients are  $(1, -1, 1)$ , the middle coefficient is negative, so the direct test fails to certify a polynomial that is, in fact, non-negative. Nor does raising the degree rescue the test here: elevating  $q$  to the degree-6 Bernstein basis gives coefficients

$$(1, \frac{1}{3}, -\frac{1}{15}, -\frac{1}{5}, -\frac{1}{15}, \frac{1}{3}, 1),$$

still with three negative entries, and no finite degree elevation produces an all-non-negative Bernstein representation of  $q$ : a classical fact about the Bernstein (equivalently, Polya) positivity certificate is that it can certify non-negativity on a closed interval only when the polynomial has no zero in the *interior* of that interval (zeros only at the endpoints, as in the  $p(t)$  example above, are compatible with eventual all-non-negative Bernstein coefficients at high enough degree; an interior zero, as in  $q(t)$ , is not). This is precisely why an interior zero of a cell’s own boundary curve, rather than a zero confined to the domain’s endpoint, is the recurring source of difficulty across this paper’s box-covering sub-regions: where the direct Bernstein test meets such a zero, the paper falls back to covering the domain by finitely many sub-boxes, each certified separately once restricted to a range not containing the interior zero (Sections 10 and 12), rather than abandoning the method. The one place in this paper where a direct certificate was not reached, the B-bounded sub-part of Section 12.4, failed at an earlier stage than this: the symbolic volume formula for that sub-part was never obtained, so no Bernstein test was run there at all. It is covered instead by the symmetry of Proposition 8.5.

**I.3. Why this is not merely a numerical convenience.** Every number appearing in this appendix, the change-of-basis coefficients  $b_0, \dots, b_4$ , the degree-2 and degree-6 Bernstein coefficients of  $q$ , is an exact rational, obtained from an exact rational input by exact rational linear algebra (the Bernstein change of basis is a fixed, exactly-invertible linear map at each degree, independent of the particular polynomial). No floating-point step, and no numerical tolerance, enters anywhere in this appendix, exactly as none enters the corresponding computations at full scale in Sections 5, 10 and 12.

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